STITCH: ROBUST APPEARANCE-INVARIANT VISUAL PLACE RECOGNITION FOR ANCHORING SOCIAL AR CONTENT IN THE REAL WORLD

A DISSERTATION

by

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November, 2026

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Candidate's Declaration

I, **PRONOB BHATTACHARJEE**, present the dissertation entitled "**Stitch: Robust Appearance-Invariant Visual Place Recognition for Anchoring Social AR Content in the Real World**" in partial fulfilment of the requirements for the award of the degree of **MASTER OF TECHNOLOGY** in Artificial Intelligence, submitted to the Mehta Family School of Data Science and Artificial Intelligence, Indian Institute of Technology Roorkee. The report documents the implementation, executed experiments and attributed prior research for the Stitch project, prepared under the guidance of Dr. R. Balasubramanian.

All experimental results are linked to the accompanying commands, datasets and software artifacts. Published methods, public datasets and automated assistance are acknowledged in the report. This declaration is prepared for review and signature by the candidate.

Date: 6 November 2026 (PRONOB BHATTACHARJEE)
Place: Roorkee

CERTIFICATION

The dissertation submitted by PRONOB BHATTACHARJEE (Enrollment No. 25565017) is presented for review under the guidance of Dr. R. Balasubramanian. Final certification and acceptance are to be completed by the supervisor and the institution after review.

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Signature: _____

Dedicated to

my family, teachers

and mentors

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The report and presentation adopt the institutional artwork and layout references supplied by the author. The experiments, scripts and provenance records accompany the dissertation so that its claims can be reviewed and reproduced. Automated implementation and writing assistance formed part of artifact preparation; the author is responsible for reviewing the final submission.

I am sincerely grateful to my supervisor, Dr. R. Balasubramanian, for the guidance, encouragement and feedback provided throughout this internship. I also thank my family and friends for their constant support.

Abstract

Stitch proposes a social augmented-reality interface in which digital posts are associated with physical places and recovered when a later visitor observes the same scene. Its original implementation matches SIFT features to one reference image and projects content using a planar homography. This thesis investigates whether pretrained representations improve recognition and anchoring under appearance change while retaining a reproducible path toward mobile deployment.

A modular implementation compares SIFT with FLANN, SuperPoint with LightGlue, CosPlace global retrieval, and a hierarchical combination of retrieval, local matching, geometric verification, and rendering. Experiments use a documented HPatches subset, a custom cross-time subset of real Tokyo 24/7 query photographs, and seeded photometric transformations of repository and benchmark images. Original scripts are executed through a bounded file-input adapter. All reported measurements, plots and tables are derived from preserved run outputs.

The evaluation distinguishes retrieval, geometric acceptance and ground-truth homography accuracy. Tokyo metadata supports place labels but does not establish metric camera pose or planar anchoring error. Short contrastive adaptation trains only a projection head on disjoint HPatches places. Model export and integer quantisation are tested using actual ONNX inference. The report documents accuracy losses, rejected anchors, and the limits of narrow galleries and synthetic illumination changes.

The resulting artifact demonstrates a reproducible planar-AR research pipeline. It does not establish a city-scale AR cloud, a six-degree-of-freedom localisation benchmark result, or phone-device performance. The deployment analysis specifies the mapping, calibration, tracking, privacy and evaluation work needed before persistent public AR content can be supported responsibly.

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Nomenclature

$H_{r \to q}$	Homography mapping reference pixels to query pixels
AR	Augmented reality
CI	Confidence interval
CNN	Convolutional neural network
FLANN	Fast library for approximate nearest neighbours
GeM	Generalised mean pooling
GPS	Global positioning system
ONNX	Open Neural Network Exchange
PnP	Perspective-n-point
QDQ	Quantise–dequantise graph representation
RANSAC	Random sample consensus
RSS	Resident set size
SfM	Structure from motion
SIFT	Scale-invariant feature transform
SLAM	Simultaneous localisation and mapping
VPR	Visual place recognition

Chapter 1

Introduction

1.1 Motivation and application context

Stitch begins with an application idea: a social feed whose organising surface is the physical world. A visitor could leave a photograph on a building facade, attach a note to a meeting point, or place a virtual object beside a familiar landmark. A subsequent visitor would recover that content by pointing a camera at the same scene. The repository describes this ambition through the metaphor of a social wall distributed across real surroundings [1]. This thesis uses "Facebook in the real world" as a motivating description, rather than as a statement of affiliation or of implemented social-network functionality.

The idea creates a correspondence problem before it creates a graphics problem. The system must decide which stored place, if any, is visible. It must then determine how content associated with that place should project into the current camera image. A visually similar building is insufficient evidence for rendering a post on a particular window. Likewise, retrieving the correct street does not identify a stable coordinate frame for a three-dimensional object. Recognition and geometric alignment therefore require separate outputs and separate evaluation criteria.

The original software provides a compact demonstration of this distinction. It detects SIFT features in a reference image and a camera frame, matches their descriptors, estimates a homography, and warps an overlay into the frame. The latest script adds optical-flow tracking and smoothing. These operations are useful for a locally planar target already represented by a reference photograph. They do not by themselves provide a place database, a city-scale search index, or a persistent world coordinate system. The engineering task is to extend that working demonstration while retaining its inspectable geometric basis.

An appearance change challenges the first correspondence step. Daylight produces different shadows, contrast and visible detail from artificial illumination at night. Seasonal vegetation may remove or reveal structures. Weather can introduce reflections or reduce contrast. These changes interact with viewpoint: a correctly recognised scene may have little shared visible surface with the reference. Long-term visual-localisation benchmarks explicitly examine these difficulties [2]. A method suitable for Stitch must therefore be assessed on more than an unchanged photograph or a visually convincing overlay video.

1.2 Problem statement

Let a place database contain reference images $\mathcal{D} = \{I_i\}$, place identifiers p_i, and content records $\mathcal{C}_{p_i}$. Given a query image I_q, the system should return a place hypothesis, evidence supporting that hypothesis, a geometric transformation where identifiable, and the associated content. A rejection is a valid result when the evidence is inadequate. The planar research prototype studied here returns

$$\mathcal{F}(I_q, \mathcal{D}) = (\widehat{p}, \widehat{H}_{r \to q}, \mathcal{C}_{\widehat{p}}, s), \tag{1.1}$$

where s contains diagnostic matching and verification information. When no candidate passes verification, $\widehat{p}$ and $\widehat{H}$ are absent and no anchored content is rendered.

The desired invariance is conditional. A representation should ignore exposure and other nuisance changes while retaining evidence that distinguishes places. It cannot discard all local structure and still determine the intended surface. Nor can it restore image information destroyed by underexposure. "Appearance-invariant" in the project title therefore identifies the research objective; it does not claim mathematical invariance or universal robustness. The experiments measure how particular pretrained representations behave under documented changes.

The resource constraint is also explicit. The available host is a laptop with an Apple M2 processor and a limited-memory configuration, recorded in the environment artifact. The implementation uses CPU inference because the recorded runtime did not expose an available accelerator. A mobile-class design is investigated through compact descriptors, bounded feature counts, model export and quantisation. A laptop CPU timing cannot establish performance on a phone, and the thesis does not make that substitution.

1.3 Objectives

The first objective is a reproducible classical reference. The original scripts must be inspected and executed on their bundled video without requiring an unavailable private camera stream. A refactored implementation should retain the SIFT–FLANN–RANSAC structure while making image coordinates, failures and rendering masks explicit. This baseline anchors the discussion in measured behaviour rather than an assumed weakness of handcrafted features.

The second objective is a controlled comparison of learned components. SuperPoint with LightGlue supplies local correspondences; CosPlace supplies an image-level descriptor. Each has a different role. Their combination is evaluated alongside exhaustive local matching and global-only retrieval. Identical query lists and geometry settings are used within each protocol so that a benefit is not explained by an easier test subset or a more permissive RANSAC threshold.

The third objective is to connect recognition to content. A place record must support reference photographs and posted assets, and a query must retrieve the content associated with its verified place. The implemented output is an image overlay attached to a reference-plane polygon. Video capture and rendering are included in the demonstration. The more ambitious multi-user service and three-dimensional anchor lifecycle are specified as architectural extensions,

with implementation boundaries stated explicitly.

The fourth objective is a thesis whose evidence can be audited. Raw predictions and pair statistics are retained. Plotting and table generation consume those files, and prose-level headline values use generated LaTeX macros. Dataset and checkpoint hashes identify the inputs. Commands and exit codes identify the actions that produced results, including failed download and environment checks. The report should make an unmeasured quantity visibly unavailable rather than filling a requested table with an invented estimate.

1.4 Research questions

1. Does pretrained local matching improve correspondence and accepted homography accuracy under the measured illumination and viewpoint changes?
2. How does global retrieval affect the candidate search needed for reliable place recognition, and when does its shortlist discard a geometrically useful reference?
3. How often does a plausible retrieval result lead to an accepted and correct place, and how does that differ from ground-truth planar alignment accuracy?
4. Does short contrastive adaptation of a compact projection head improve retrieval on held-out places and real cross-time images?
5. What accuracy, size and latency trade-offs follow from an executed descriptor export and INT8 quantisation attempt?

Each question maps to an observable artifact. Pair statistics address the first; ranked predictions and candidate-count ablations address the second and third; the seeded training records address the fourth; native-versus-ONNX descriptors and timed inference address the fifth. Metric six-degree-of-freedom localisation is excluded from these answers because the downloaded evaluation subsets do not supply the required calibrated poses and three-dimensional map.

1.5 Scope of the empirical contribution

The public-data contribution comprises a deterministic HPatches subset and a custom Tokyo cross-time protocol. HPatches contributes real illumination and viewpoint changes with image-to-image homographies [3]. Tokyo contributes photographs captured at different times with place and orientation labels [4]. The latter protocol uses a small reference gallery drawn from the query collection; it is deliberately named a cross-time subset rather than the official Tokyo street-view retrieval benchmark. Its results support a narrow comparison under matched labels, not a claim of benchmark-leading performance.

Synthetic queries serve a different purpose. They provide known geometric maps while varying brightness, gamma, colour balance, noise and contrast. Since all transformations are seeded, the same source identity can be followed through a controlled perturbation. However, altered

pixels retain the original scene, visibility and much of its texture. Their performance is therefore interpreted as a stress test of an implementation, not as a substitute for real night imagery or evidence about winter conditions.

The methodological contribution is the integration and evaluation of available pretrained components in the Stitch setting. No new foundation model, feature detector, or neural matcher is claimed. The short training experiment changes only a projection head on frozen global descriptors. Its purpose is to expose the practical consequences of limited adaptation data and descriptor compression. Treating a neutral or negative outcome as a valid answer is necessary to avoid selecting a favourable training seed after observing the test set.

The software contribution includes modular extraction, matching, geometry, rendering, database and I/O modules; a command-line interface; configuration and tests; data/model acquisition scripts; evaluation and demonstration code; and a generated-report workflow. The reference database stores content alongside place identifiers so that the experiment exercises retrieval of an actual posted asset rather than merely drawing an arbitrary polygon after matching. Persisting that record on disk establishes local repeatability, while cross-device consistency remains future work.

1.6 Assumptions and boundaries

The reference photograph is assumed to represent a surface on which a planar overlay is meaningful. Within that model, an estimated homography maps reference pixels to query pixels. A street scene is generally not planar, so an accepted Tokyo homography is evidence of a consistent image region, not proof that every part of the street obeys the same map. This distinction motivates the separate reporting of geometric acceptance and true homography success.

The gallery is treated as known and locally available during the experiment. Reference descriptors and local features may be cached before a query arrives. Timings state whether they include this setup, image decoding, rendering and content-file access. The production design would need to constrain cache size and fetch latency. Those costs are analysed symbolically and through measured component sizes; they are not represented as measured cloud-service costs.

The work assumes that dataset access does not imply unrestricted redistribution or commercial permission. Software and imagery have separate provenance. Official documentation is retained where available, and manual-fetch instructions identify larger benchmarks whose image and mapping archives were not downloaded. A future product must resolve additional content rights, user consent and location privacy before reusing a research pipeline in public spaces.

1.7 Organisation of the thesis

Chapter 2 reviews the representations, geometric models, datasets and anchor services relevant to the project. Chapter 3 analyses the original scripts and their recorded execution. Chapter 4 describes the implemented modules and the proposed multi-user service boundary. Chapter 5

defines the downloaded subsets and all preparation rules. Chapter 6 develops the mathematical formulation and algorithms. Chapter 8 specifies hardware, metrics and experimental controls. Chapter 10 reports the measured comparisons and validity threats. Chapter 14 examines export, quantisation and scaling. Chapter 17 treats privacy and abuse as system requirements. Chapter 18 answers the research questions and defines a staged roadmap. The appendices provide reproduction commands, directory structure, qualitative evidence and key source listings.

The expanded technical treatment in Chapters 7, 9 and 11 derives coordinate transformations, separates decision metrics and traces stage-specific failures. Chapters 12 and 13 provide the complete scene-level evidence for every selected sequence and site. Chapters 15 and 16 document executable verification contracts and a proposed revision-aware service design. Appendix E retains all oracle-pair outcomes, including failures. These additions expose more of the existing evidence without presenting it as a larger independently acquired experimental population.

Chapter 2

Background and Literature Review

2.1 Recognition, localisation and anchoring

Visual place recognition asks which previously observed place is depicted in a query. Image retrieval implements this as ranking a gallery using visual descriptors. A retrieved image can supply an approximate location or a candidate region, but its similarity score does not determine the query camera's geometric pose. Visual localisation instead estimates the camera transformation relative to a map. In a calibrated three-dimensional setting that transformation has rotation and translation components. The established long-term benchmark distinguishes retrieval-like recognition from six-degree-of-freedom localisation under changing conditions [2].

Anchoring adds an application coordinate convention. A post needs a position or surface in the map, and the current device needs a transformation from that map to its view. A planar reference image can provide such a convention for image overlays: content coordinates map to reference-image pixels and then to query-image pixels. General three-dimensional objects require a world or surface coordinate frame whose scale and orientation are known. These outputs impose different observability requirements and should not share a single undifferentiated "localisation success" score.

This separation guides the experimental design. Global-only retrieval is evaluated as retrieval. A local matcher is evaluated on image pairs whose relationship is known independently. A complete candidate-search pipeline is evaluated both for its ranked place result and for whether its selected candidate passes verification. On HPatches, an additional ground-truth comparison checks the estimated mapping. This arrangement permits a matcher to be good even when retrieval misses its correct candidate, and permits a retrieval method to be good even when a query lacks enough shared surface for anchoring.

2.2 Handcrafted local features

2.2.1 SIFT

SIFT constructs local descriptors around characteristic image structures selected across scale. Orientation assignment and gradient-distribution normalisation provide robustness to several nuisance transformations. Lowe's account also makes clear that robustness is bounded: perspective distortion, severe appearance change and absent detail remain relevant [5]. Stitch uses the OpenCV implementation rather than reimplementing the detector.

For this application, a local descriptor provides more than an image-level match. Each descriptor remains associated with a pixel location, so accepted correspondences constrain a geometric map. This is the central reason a global embedding cannot directly replace the original pipeline. The comparison tests whether learned keypoints and descriptors provide better correspondences while preserving that association.

2.2.2 Approximate matching and the ratio test

FLANN selects approximate nearest neighbours using indexing strategies suited to descriptor search [6]. In the upstream scripts, a randomised tree index and a bounded search effort retrieve the first two neighbours for each reference descriptor. A ratio criterion rejects a match when its best neighbour is insufficiently separated from the second-best candidate. This ambiguity test is useful for repeated textures, although it cannot establish physical scene identity by itself.

The implementation must also handle cases omitted by compact demonstrations. A query with no descriptors produces no candidate correspondences. A reference descriptor may receive fewer than two neighbours if the target set is too small. Neither condition should raise an unhandled tuple-unpacking error or lead to a reused anchor being interpreted as a new recognition. The refactor makes empty sets and rejection first-class outputs.

2.3 Learned local representations and matching

2.3.1 SuperPoint

SuperPoint jointly predicts interest points and descriptors from an image. Its training framework uses synthetic supervision and homographic adaptation to improve repeatability without requiring manually annotated natural-image keypoints [7]. The present work uses publicly released pretrained weights. Neither the detector nor its descriptor backbone is trained on the evaluation subset.

The practical hypothesis is that joint learning may preserve useful correspondence evidence when raw contrast or local gradient distributions change. That hypothesis remains empirical. The learned extractor can still miss features in severe low light, repeated patterns or regions with little texture. A bounded keypoint count is used for both local methods, so the learned method is not granted an unlimited correspondence budget.

2.3.2 SuperGlue and contextual assignment

SuperGlue conditions correspondence decisions on both feature sets through a graph neural network with attention and a differentiable assignment formulation. It explicitly models unmatched features, allowing occlusion and non-repeatability to be handled within matching [8]. This provides the conceptual motivation for comparing a learned matching stage with nearest-neighbour filtering. SuperGlue itself is reviewed but not executed in this artifact.

2.3.3 LightGlue

LightGlue revisits sparse learned matching with adaptive computation. Its official implementation supports SuperPoint and several other feature families and exposes mechanisms that adjust work to pair difficulty [9, 10]. The experiments use SuperPoint-compatible pretrained weights, CPU inference and the recorded confidence settings. Flash-attention acceleration is disabled so the tested execution path is explicit.

Adaptivity matters for Stitch because a difficult night query may require more work than an easy repeated view. A single average latency can conceal that variation. The artifact consequently records per-pair times and separately profiles complete queries. However, a laptop inference result does not demonstrate the same adaptive performance in a mobile runtime: export operators, memory allocation and backend support must also be verified.

2.3.4 LoFTR and the detector-free alternative

LoFTR establishes correspondences through coarse-to-fine dense feature interaction rather than relying on a separately detected sparse keypoint set [11]. Its larger contextual field motivates it as an alternative when repeated keypoints are scarce. It is not included in the executed comparison, which focuses on a sparse pipeline compatible with cached per-reference features. A claim about LoFTR's accuracy or mobile cost on these subsets would require an additional run.

2.4 Global descriptors for place retrieval

2.4.1 NetVLAD

NetVLAD makes residual aggregation trainable and uses weak place supervision to learn image embeddings for retrieval [12]. Its broader lesson is that visual similarity can be learned from place relationships rather than from generic object categories alone. In Stitch, this role is coarse candidate selection; it reduces the number of local reference-query comparisons before geometric verification.

2.4.2 CosPlace

CosPlace addresses large-scale visual geo-localisation through a classification-based training formulation and releases compact pretrained retrieval models [13, 14]. The artifact selects a ResNet18

model with a 512-dimensional output. The choice provides a smaller encoder than the larger released backbones and a straightforward convolutional export path. It is an engineering selection for the available host, rather than a demonstrated optimum among all modern retrieval systems.

2.4.3 MixVPR and foundation features

MixVPR aggregates feature maps through feature mixing and is a relevant alternative to residual aggregation [15]. DINOv2 provides self-supervised visual features that can also motivate retrieval pipelines [16]. These approaches are discussed to position the selected model within the literature. They are not silently substituted for CosPlace, and the thesis attributes no locally measured performance to them.

2.4.4 Candidate retrieval as a bottleneck

If a correct reference is absent from the selected shortlist, a downstream matcher cannot recover it without expanding the search. A small shortlist can therefore improve latency while reducing the ceiling on successful anchoring. Increasing its size may recover relevant references but also adds expensive nonmatching pairs and more opportunities for a false geometric acceptance. The stage ablation varies shortlist size on exactly the same queries to expose this trade-off.

A retrieval score also has no universal probability interpretation. Cosine similarity measures proximity in the selected descriptor space. Its distribution changes with gallery composition, camera conditions and training domain. Treating a high similarity as permission to display content would bypass the local evidence needed to identify an anchoring surface. The no-geometry ablation is retained as a diagnostic, not as the recommended rendering policy.

2.5 Projective geometry and robust estimation

Image correspondences constrain different geometric models depending on scene structure and calibration. Homographies describe plane-induced image mappings and pure-rotation cases. Essential matrices describe calibrated two-view epipolar relationships. PnP estimates camera pose from known three-dimensional points and their image projections [17, 18]. Stitch's available assets and HPatches supply a direct planar formulation; the downloaded Tokyo metadata does not supply the calibrated 2D–3D relationships needed for PnP.

RANSAC estimates a model from subsets of tentative correspondences while rejecting gross outliers [19]. It does not transform an inappropriate model into a physically correct one. A street scene may admit many correspondences on one facade while foreground objects remain inconsistent with the same homography. Similarly, repeated windows may support a projective map to the wrong location. Inlier support is therefore necessary evidence for this prototype, but it is not complete evidence of place identity or three-dimensional consistency.

2.6 Hierarchical visual localisation

Coarse-to-fine localisation combines global hypotheses with local geometric evidence. HF-Net demonstrates this principle using jointly predicted local and global representations [20]. The present system adopts the hierarchy but uses separate pretrained networks and a planar verifier. Its proposed combination is an integration choice, not a claim to originate hierarchical localisation.

Separating the stages remains valuable for a small research artifact. Each component can be replaced and each failure can be attributed to a recorded intermediate output. A wrong shortlist, an inadequate correspondence set and an inaccurate accepted homography become different diagnoses. The resulting modularity also exposes costs that a monolithic end-to-end timing would hide, such as reference extraction and content asset loading.

2.7 Benchmarks and domain coverage

HPatches offers image sequences with illumination or viewpoint variation and known planar relationships [3, 21]. It is suited to evaluating image alignment, but it does not establish geographic recognition over a city or metric pose accuracy. A small distractor gallery assembled from its sequence references is an explicit custom retrieval task.

Tokyo 24/7 was designed around changes in time of day and viewpoint, with camera-phone queries and a street-view database [4]. Using its accessible query photographs as both cross-time references and queries creates a useful paired recognition experiment but changes the task. The benchmark's original database scale and ground-truth conventions must not be carried over to this subset's results.

Aachen Day-Night, RobotCar Seasons and Extended CMU Seasons extend evaluation toward long-term metric localisation [2, 22]. MSLS provides broader geographic, seasonal and sequence variation [23]. Those capabilities motivate future validation, while their absence from the executed image set limits what this thesis can conclude about seasonal robustness and map-level localisation.

2.8 Persistent AR services and lifecycle

ARCore Cloud Anchors uses hosted scene information and an anchor identifier to support later resolution by devices observing the environment [24]. ARKit world maps support saving spatial state and attempting relocalisation in a later session [25]. These systems demonstrate that persistence involves mapping and coordinate recovery, not only content storage. Their documentation informs the proposed integration boundary; neither service is called by the experiments.

Azure Spatial Anchors is relevant historically, but Microsoft states that the service was retired in November 2024 [26]. It is therefore discussed as background rather than recommended as a new dependency. This distinction illustrates why service status must be checked independently of the publication date of an AR architecture proposal.

2.9 Research gap addressed by the artifact

The literature already provides strong representations, matchers and hierarchical localisation designs. The immediate gap in Stitch is the absence of an evaluated connection between those components and its reference-plane rendering demonstration. The artifact addresses that gap by preserving the classical behaviour, introducing explicit place records, executing controlled comparisons, and making rejection and uncertainty visible. The remaining gap is larger: metric map construction, cross-device anchor management and realistic phone validation are still required for the original social-AR ambition.

Chapter 3

Analysis of the Existing System

3.1 Inspected source snapshot

The upstream repository was cloned and inspected at the revision recorded in the bibliography and model/data provenance. Its executable Python content comprises `tep.py`, `tep2.py` and `tep3.py`. Bundled media include a reference image, a test image, an overlay, a short video and an animated walkthrough. No labelled place database, learning code or downloadable benchmark is included [1]. The original scripts remain present and unmodified in the working artifact.

The inspection distinguishes what is implied by comments from what the code actually does. In particular, the latest script is not simply a frame-by-frame SIFT matcher: it includes optical flow, adaptive detection intervals, forward–backward checking and homography smoothing. Consequently, comparing its tracking speed directly with a learned re-recognition pass would confound different workloads. Both workloads are useful, but they answer different questions.

3.2 The file-processing script

3.2.1 Initialisation

`tep.py` reads `reference.jpg` in grayscale and `overlay.png` in colour. It constructs a SIFT extractor with the upstream feature budget, extracts reference descriptors once, and resizes the overlay to the reference dimensions. A FLANN matcher uses the recorded tree and search settings. The input is `video.mp4`; the output writer uses a reduced playback frame rate. That output frame rate is a presentation choice and must not be interpreted as measured inference throughput.

Reference caching is an important strength. It avoids repeating work that is unchanged across frames. The same principle is preserved in the modular system, where local reference features and global gallery descriptors are prepared before an online query. A fair runtime report must therefore state whether this preparation is included. The upstream bounded execution includes initialisation, whereas the later warm profiles isolate query-time processing.

3.2.2 Frame processing and verification

Each frame is converted to grayscale, blurred and enhanced with CLAHE. SIFT descriptors are extracted from the enhanced image. The first two descriptor neighbours are compared with the script's ratio threshold. A tentative homography is estimated only after enough matches survive; it is used only after an additional inlier-count test. These operations are conceptually sensible but encode thresholds and preprocessing directly in a monolithic script.

The reference and query are not processed identically. The reference is loaded directly as grayscale, whereas the query receives blur and local contrast enhancement. This asymmetry is part of the preserved upstream behaviour. The controlled A–D experiment instead uses a common image-resizing convention and shared geometry thresholds, with the difference documented. The latter is a normalised comparison of methods, not a claim of byte-for-byte reproduction of every upstream preprocessing decision.

3.2.3 Persistence and compositing

When a new homography is unavailable, `tep.py` can reuse the previous one for a bounded number of frames. This can maintain visual continuity during a short detection failure. It can also display an old transformation after camera motion, because the reused map is not updated by a geometric motion model. A rendered frame is therefore not equivalent to a successful fresh recognition.

The overlay mask is inferred from warped image intensity. Pixels that are sufficiently dark are treated as transparent. This is simple for demonstration assets, but it makes legitimate black content disappear and does not represent semi-transparency. The refactored renderer uses an explicit alpha plane and premultiplied colour. A unit test verifies that an opaque black overlay remains opaque and that a fully transparent one leaves the frame intact.

3.3 The first live-stream variant

`tep2.py` follows the same broad matching and rendering sequence but reads a stream and displays a window. In the inspected snapshot, the example URL definition is commented out while the variable is subsequently used. Native execution therefore lacks a required input value. The script also continues when a frame read fails, without a finite retry policy or an EOF distinction.

These issues are I/O failures rather than evidence about the SIFT algorithm. The execution adapter supplies a file source and suppresses display calls so that the vision code can be exercised in this environment. It does not invent a live phone experiment. The code path is labelled "bounded file-input adapter" in the result record, and the original stream requirement is retained in the source analysis.

3.4 The latest detection-and-tracking variant

3.4.1 Detection and scheduling

`tep3.py` reads a hard-coded private DroidCam URL and resizes frames to a fixed working shape. During detection it downsamples the grayscale query, extracts SIFT descriptors, matches them to the full reference descriptors, and rescales matched query points back to working-frame coordinates. This coordinate correction is necessary: fitting the homography in a mixture of downsampled and full-frame pixels would corrupt the overlay location.

Detection is attempted at an interval rather than on every frame. Failed detections increase that interval up to a cap. A successful detection resets the schedule and initialises tracking. This reduces repeated expensive searches when a target is absent, but it also introduces recognition delay. A frame-count average can look favourable partly because some frames perform little or no recognition work. The bounded execution therefore reports decoded frames, homography calls and warp calls separately.

3.4.2 Optical flow and consistency checks

Once points are available, pyramidal Lucas–Kanade optical flow propagates their positions between frames. A periodic backward pass checks whether tracking approximately returns to the previous point positions. A geometric filter further rejects motions that differ substantially from median motion. These checks target local tracking errors; they do not compare a newly visible place with a database.

The source contains implementation limitations worth preserving in the audit. It does not fully validate all optional optical-flow outputs before indexing them, and the backward-status array is not incorporated into every acceptance decision. A feature set may therefore be lost abruptly rather than producing a clean diagnostic rejection. The modular recognition path handles empty feature sets and model-estimation failures explicitly. It does not claim that every tracking behaviour has been ported or improved.

3.4.3 Homography smoothing

The tracking correspondences estimate a fresh homography. A determinant-based validity check rejects nonfinite or implausibly scaled matrices. The script may lock the prior transformation when average point motion is small, or blend matrix entries with the new estimate when motion is larger. This can reduce visible jitter in the demonstration.

However, a homography is defined only up to scale. Entrywise averaging is not generally invariant to that representation, and a smooth matrix sequence need not correspond to the intended surface motion. The determinant is also affected by multiplying the whole matrix by a scalar. The new verifier normalises the matrix representation, checks rank and numerical conditioning, and examines the projected reference boundary. More principled temporal filtering in a calibrated pose or surface-coordinate space remains a deployment extension.

3.5 Execution protocol and measured baseline

The three scripts were executed through `scripts/run_originals.py`. The adapter redirects video capture to the bundled file, supplies the missing URL variable where needed, suppresses GUI display, and terminates the run after the same decoded-frame budget. It wraps homography and warping calls to count them without changing their arguments or results. The exact command, elapsed time, exit state and raw output are preserved in the command journal.

Table 3.1: Measured upstream-script execution on the same bounded file input. Timing includes initialisation and file I/O.

Script	Frames	H estimates	Warps	ms/frame
tep.py	90	90	90	55.70
tep2.py	90	90	90	56.45
tep3.py	90	87	87	11.31

Table 3.1 reports the recorded measurements. The latest script's lower average time is consistent with its design: many frames track an existing target rather than recomputing complete SIFT detection and matching. The earlier scripts repeatedly detect and describe features. The table cannot isolate an algorithmic speedup independent of this workload change, nor does it establish behaviour on a real day/night stream.

The bundled video provides a functional regression input. It tests decoding, feature processing, homography calls and output writing on the media supplied with the project. It lacks a ground-truth pose, framewise homography annotations and independent place labels. The number of nonnull estimates or warps therefore describes execution, not accuracy. Later benchmark experiments supply the missing ground-truth comparison where available.

3.6 Failure mechanisms motivating the redesign

The first limitation is a fixed reference identity. Every successful match is interpreted relative to one target image. There is no mechanism to select among stored places or attach multiple users' posts to a shared identity. Adding a gallery without explicit rejection would introduce false-place overlays, because even a nonmatching frame can have nearest neighbours.

The second limitation is that temporal continuity is used to bridge a recognition failure. Reuse and tracking help when the correct target has already been found, but they do not solve re-entry under a different appearance. A visitor arriving at night needs a fresh correspondence to the stored representation. That task motivates the learned local and global components evaluated here.

The third limitation is implicit state. Input configuration, recognition thresholds, cached geometry and rendering all live in script-level variables. A failure in one stage can be difficult to attribute. Separating extraction, matching, verification and compositing makes it possible to store the evidence supporting an anchor and to distinguish a wrong retrieval from a bad homography.

The fourth limitation is the absence of experimental accounting. Without fixed query lists, ground truth, saved predictions and runtime definitions, a demonstration can support only a narrow claim that an overlay appeared on selected footage. The redesign adds this accounting before drawing broader conclusions. Its aim is to establish which parts of the original idea are supported by measured evidence and which remain a research or deployment requirement.

Chapter 4

System Design and Architecture

4.1 End-to-end responsibilities

The implemented system divides a query into capture or decoding, preprocessing, place hypothesis generation, local correspondence, geometric verification, content lookup and rendering. The division follows the different evidence required at each stage. Capture supplies pixels; retrieval supplies candidate identities; geometry supplies a mapping; the database supplies content associated with the verified identity. A later stage is not allowed to turn an absent earlier result into an apparently successful anchor.

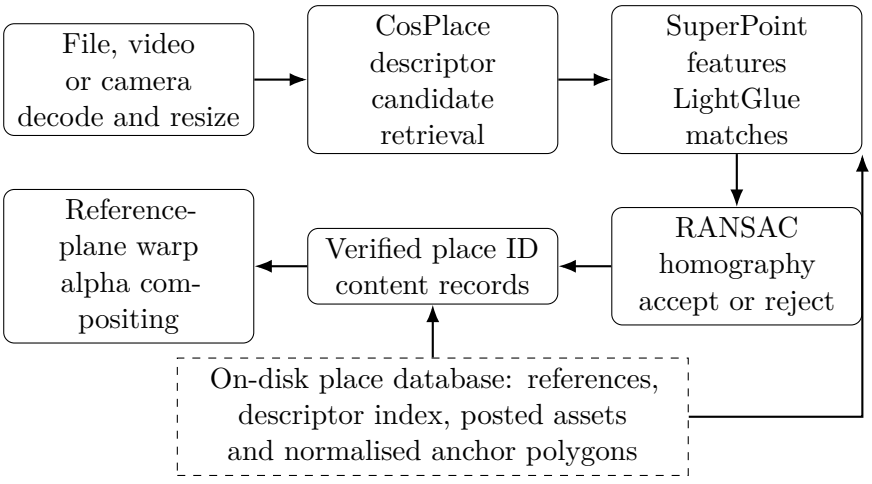

Figure 4.1: Implemented hierarchical recognition and planar content-rendering flow. The diagram is an architectural specification, not a measured throughput result.

Figure 4.1 depicts the implemented path. The classical mode replaces the global shortlist and learned correspondence stage with exhaustive SIFT–FLANN verification. The learned-local mode removes global retrieval but retains exhaustive SuperPoint–LightGlue comparisons. Keeping those alternatives behind the same query interface makes stage removal explicit rather than changing the application data model for each method.

4.2 Module contracts

4.2.1 Image I/O and coordinates

The I/O module returns a decoded BGR image and the transformation from original image coordinates to the working resolution. This second output is essential for evaluation: ground-truth homographies are expressed in the original images, whereas inference uses resized images. The module preserves aspect ratio and records separate horizontal and vertical scale values after integer rounding. It rejects unreadable files instead of returning an empty array that fails later in feature extraction.

The video iterator accepts a file path, numeric webcam index or stream URL. It stops on EOF or a failed read and releases the capture resource. An explicit frame limit supports bounded experiments. A production live-stream client could add a retry state with a deadline; that policy should be kept outside geometry and should not silently reuse the last camera frame as a new observation.

4.2.2 Local features and correspondences

A local-feature record contains pixel coordinates, descriptor arrays and any model-specific tensors required by the matcher. The shared interface returns two coordinate arrays and the descriptor-index pairs that produced them. Empty output has a stable shape. This allows geometry to reject insufficient support without depending on the internal format of either OpenCV or LightGlue.

Learned features are associated with the resized image, and internal extractor resizing is disabled for the controlled comparison. Reference features are cached when the pipeline is constructed. Query features are extracted once and reused across candidates. This division avoids measuring repeated extraction merely because a shortlist contains several references.

4.2.3 Geometry and verification

The geometry module returns the estimated homography, an inlier mask, count and ratio, and a separate acceptance flag. Retaining a rejected matrix for diagnostics is useful, but rendering checks the acceptance decision through the pipeline's selected prediction. A matrix existing in memory does not itself authorise an overlay. The interface also exposes corner projection and camera-pose error utilities for tests and future calibrated evaluations.

The acceptance policy combines sufficient correspondence support with numerical and boundary checks. This policy is common to the normalised local-method comparison. It is intentionally distinct from the original scripts' hard-coded thresholds, whose unmodified results are reported separately. The distinction prevents a software refactor from being mistaken for a perfectly controlled reproduction of every upstream heuristic.

4.2.4 Global retrieval

The global module loads the official CosPlace weights into the compatible ResNet18 and pooling architecture. Random ImageNet initialisation weights are not downloaded because the complete pretrained state is supplied by the selected checkpoint. A fixed preprocessing path resizes to the recorded descriptor input shape and applies channel normalisation. Gallery and query descriptors follow the same path.

Ranking uses the dot product of normalised vectors with stable tie ordering. Exact search is appropriate for the small galleries in this artifact and makes the evaluation easy to inspect. An approximate index would introduce another source of candidate recall loss; it should be evaluated separately before a larger deployment. The descriptor index stores image fingerprints so stale references cannot be silently matched against an obsolete cached vector.

4.3 Place-database schema

The database document has a schema version and a list of places. Each place has a unique identifier, one or more reference-image paths, and a list of posted-content records. Paths resolve relative to the database document so the artifact can be moved as a directory. The schema keeps a place identity distinct from a particular reference view: adding a second photograph need not duplicate all associated posts.

Listing 4.1: Implemented place-record structure with illustrative identifiers. The numerical anchor coordinates are configuration values, not experimental results.

```
{
  "schema_version": 1,
  "places": [{
    "id": "place_identifier",
    "references": ["reference_view.jpg"],
    "content": [{
      "id": "post_identifier",
      "kind": "image",
      "asset": "posted_image.png",
      "author": "example-user",
      "created_at": "2026-11-05",
      "anchor_normalized": [
        [0.2, 0.2], [0.8, 0.2],
        [0.8, 0.65], [0.2, 0.65]
      ],
      "opacity": 0.8
    }]
  }]
}
```

Listing 4.1 shows the implemented fields. Normalised polygon coordinates preserve the intended reference region when an image is resized. They describe a quadrilateral in the reference image, not a metric plane pose or a three-dimensional volume. The asset can retain its own alpha channel. The current renderer consumes image assets; the schema's content-kind field provides a point for future video, note and mesh renderers without claiming they are implemented.

The index is stored in a companion compressed NumPy file. It contains descriptors, reference

paths, image hashes, the encoder checkpoint hash and preprocessing settings, including the image resize limit. Loading checks the index version, reference list, fingerprint count, configuration and finite descriptor matrix. A changed reference image, checkpoint or resize limit requires rebuilding. Enforcing cache-size limits remains a service implementation requirement.

Each content polygon is associated with one reference photograph. An optional `reference` field selects that photograph; when omitted, the first reference is used. A post is rendered only through its associated reference. Adding a second photograph does not transfer polygon coordinates between views; that requires a validated geometric relation.

4.4 Implemented query behaviour

A query extracts its local features once. In full mode it also extracts a global descriptor and ranks references, selecting the configured shortlist. Each candidate receives local matching and RANSAC verification. Accepted candidates are ranked by inlier count and then inlier ratio. If none is accepted, the returned place identifier is absent. Otherwise, the selected reference and the content records belonging to that reference's place are returned together.

Rendering loads each selected asset, maps its corners to its reference-anchor polygon, composes that map with the reference-to-query homography, and alpha-composites into the query frame. This implements a complete local recognition-to-content path. The content is not selected by manually choosing an overlay after observing a query result. It is retrieved through the matched place record.

Candidate diagnostics accompany the decision. They record place identifiers, support counts, ratios and acceptance flags. These diagnostics make it possible to explain why a familiar scene did not display a post: the correct reference may be outside the shortlist, or its local support may be inadequate. A user interface could convert those states into a prompt to move the camera or improve lighting. The research CLI reports the evidence directly.

4.5 Proposed multi-user posting and retrieval flow

Design scope. The following specification describes an engineering extension; it is not a claim that a deployed multi-user service was measured.

A production posting transaction would begin with an authenticated user observing an already mapped place or requesting creation of a new place. The client would submit a reference observation and a proposed anchor. A server would validate the observation, associate it with a map revision and persist the content record only after the identity and permissions are established. An unresolved observation should remain a draft rather than create a potentially incorrect public anchor.

A later visitor would first obtain nearby place candidates, using an approximate location prior if the user permits it. Visual retrieval and geometric verification would establish a local map relation. The service would then return only content visible to that visitor under access and

moderation rules. Content delivery and anchor resolution are separate operations: resolving a place must not expose a private post merely because its author once used that map.

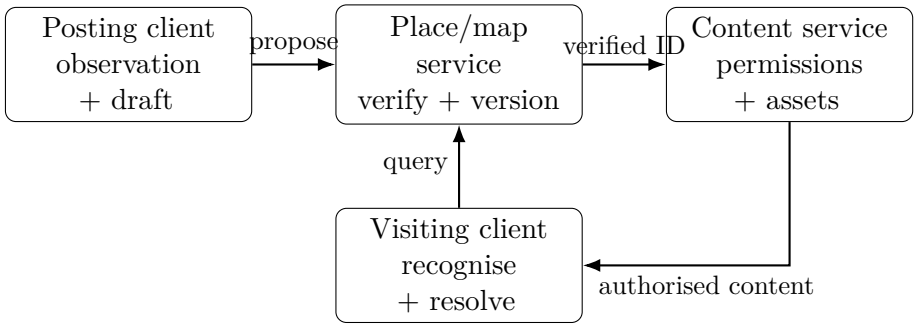

Figure 4.2: Proposed multi-user service responsibilities. These network transactions are not implemented or load-tested in the local artifact.

Figure 4.2 separates the responsibilities needed to make the original social concept credible. A map service controls coordinate consistency and reference revisions. A content service controls visibility, asset retrieval and moderation state. The client controls capture, local inference and presentation. This division prevents an image-matching success from being overloaded as proof of authorisation, current map validity and safe content.

4.6 API and consistency sketch

An API could expose a place-creation operation, an observation-verification operation, content posting, content retrieval and anchor retirement. Each response should include a place ID and map revision, not only a raw pose or homography. Posting should be idempotent under a client-generated request key so a network retry cannot create duplicate objects. Updating a map should create a new revision and retain an explicit relation to existing anchors.

Anchor migration is a geometric operation. If an old reference plane is replaced, old polygon coordinates cannot simply be copied into an unrelated photograph. The service would need a validated transformation or a human-assisted reattachment. Until that relation exists, affected posts should be marked unresolved. This policy is more defensible than rendering an obsolete anchor with a stale cached homography.

Content deletion should remove the public record and invalidate cached asset access, while map deletion should address references and derived indexes. An audit record may retain a minimal event description without retaining the original image indefinitely. These design choices are discussed as requirements rather than as implemented guarantees. The local database is intentionally small and file-based, which supports inspection but does not provide concurrent transactions or distributed deletion.

4.7 Server and device allocation

The implemented artifact runs entirely on the host CPU. For a mobile deployment, global retrieval could remain on device for a geographically limited gallery, while a server supplies additional map candidates. Local verification could also run on device after downloading sparse reference features. The trade-off depends on model size, bandwidth, privacy requirements and the measured phone execution backend; those quantities cannot be decided from desktop throughput alone.

A viable interaction loop should separate occasional recognition from high-frequency tracking. The original `tep3.py` already illustrates this principle for a known target. Learned recognition could establish or refresh the map relation, after which the native AR tracking system updates the view. This architecture would not require executing the full hierarchical recogniser at the display refresh rate. Nevertheless, tracking drift and recovery latency would need their own measurements before the system could be described as persistent.

4.8 Testability and failure containment

The module boundaries support tests with known transforms and synthetic outliers, independently of network downloads. The database can be inspected without loading a neural model. Rendering can be tested with an opaque black asset and a transparent asset. I/O failures can be reproduced through bounded files. These checks protect coordinate and state conventions that otherwise produce convincing but incorrect overlays.

The main containment rule is rejection on insufficient evidence. A missing descriptor set, failed estimate, singular transform or absent verified candidate returns no anchor. Qualitative demonstrations retain these failures visibly. The resulting behaviour may display less content than a permissive prototype, but it provides an interpretable basis for studying whether a future recognition improvement actually increases correctly supported anchors.

Chapter 5

Datasets and Data Preparation

5.1 Selection principles and evidence levels

The repository contains demonstration media but no labelled research dataset. Data acquisition was therefore part of the experiment. Selection prioritised reachable public files, variation relevant to the application, ground truth appropriate to the measured task, and a subset size compatible with the available host. Two complementary public collections were selected: HPatches for planar geometric evaluation and Tokyo query photographs for real cross-time place recognition. A generated set adds controlled photometric perturbations.

These collections supply different evidence levels. A labelled place identity establishes whether a retrieved record refers to the intended location. A homography establishes a planar pixel mapping. A calibrated camera pose and three-dimensional map would establish metric localisation error. The first two are present in the selected data; the third is not. The report does not infer missing ground truth from visual plausibility, GPS labels or a fitted matrix.

Table 5.1: Downloaded and generated evaluation collections. Pair counts refer to this artifact, not full public datasets.

Collection	Images	Pairs / queries	Gallery
HPatches subset	144	120	24
Tokyo cross-time subset	96	96	24
Synthetic transformations	90	90	26

Table 5.1 summarises the actual evaluation collections. The downloaded and generated image counts come from the data and preparation manifests. The HPatches archive was downloaded in full from the author's reachable mirror, while only selected sequences were extracted for evaluation. Tokyo images were retrieved as selected ZIP members using byte ranges; their complete metadata records were retained to make the selection rule inspectable. Download failures and the bounded full-archive attempt are preserved in the command journal.

5.2 HPatches subset

5.2.1 Acquisition and deterministic selection

The original HPatches host failed DNS resolution during the recorded download check. The dataset author's Hugging Face mirror was reachable and supplied the full sequence archive. The manifest records its URL, file hash, extracted-image hashes and per-sequence file counts. This distinguishes a verified local copy from a statement that a URL happened to exist when the thesis was written.

The selected evaluation set consists of the first twelve sequence names in lexicographic order for each of the illumination and viewpoint prefixes. Every selected sequence contributes its reference image, all five targets and their corresponding homography files. Thus 144 images and 120 reference-target pairs are evaluated. The choice uses names and metadata before inference, rather than selecting pairs on which one method appears stronger.

The images are resized only when loaded by the common I/O module. Original homographies remain unmodified on disk. Their transformation into the working-image coordinate convention is computed during evaluation. This avoids creating a second ambiguous ground-truth directory whose matrices might no longer correspond to the original files. The exact selected names are listed in Appendix B.

5.2.2 Illumination and viewpoint protocols

The two prefix groups are evaluated separately because they probe different failure mechanisms. Illumination sequences change the appearance of a scene while retaining a related view. Viewpoint sequences change the projective relationship and visible detail. The official dataset description motivates this distinction [3, 21]. No attempt is made to relabel every illumination pair as a day/night pair: the prefix is the authoritative task label available for this experiment.

For matching, each target is paired with its own sequence reference. For the custom retrieval task, the gallery contains one reference from every selected sequence, and a target's correct identity is its sequence name. This provides distractors and a reproducible image-retrieval exercise. It does not add geographic coverage or convert HPatches into a city-scale VPR benchmark. Related targets from one sequence also remain statistically dependent, which motivates clustered uncertainty estimates.

5.2.3 Rights and source attribution

HPatches combines sequences with source-specific attribution described by its maintainers. The repository's software licence should not be assumed to grant a blanket commercial licence for all imagery. The artifact retains source links and uses the data as a research input. Any distribution of extracted photographs or a public product built from them requires review of the original source terms. This limitation is documented rather than replaced with an unsupported "open data" label.

5.3 Tokyo cross-time subset

5.3.1 Available archive and labels

The accessible Tokyo archive contains camera-phone query images and CSV metadata. The metadata identifies image filenames, geographic positions, viewing-direction labels and capture-condition labels. The original dataset was introduced for place recognition under time-of-day and viewpoint change [4]. The public query URL was confirmed through the dataset-formatting documentation [27].

A direct full-archive transfer exceeded the chosen bounded timeout and was not treated as a successful dataset download. Instead, the downloader read the ZIP central directory and retrieved required members with HTTP byte ranges. It checked the decompressed size and the archive's CRC for each member before writing it. SHA256 fingerprints were then recorded for future reproduction. Partial archive files are not used as valid complete archives.

All available CSV members were downloaded before subset selection. Records were grouped by their exact latitude-longitude site key. A complete selected site needed a front-facing daytime image, a left-facing daytime image, a front-facing night image and a front-facing sunset image. The first 24 complete site keys in lexicographic order were retained. That produces 96 image files with a transparent, metadata-only selection rule.

Figure 5.1 illustrates the resulting condition coverage. Visual inspection confirms that the retained night examples are real night photographs rather than images created by gamma correction. It also shows that a left-facing daytime observation can depict substantially different scene content from the front-facing gallery image. This makes the day-to-day split a viewpoint stress test, not necessarily an easier control than the aligned cross-time pairs.

5.3.2 Reference-to-query direction

Condition names throughout the artifact follow the arrow from reference to query. Day-to-night uses a daytime front reference and a night front query. Night-to-day reverses those roles. Day-to-sunset uses a daytime front reference and a sunset front query. Day-to-day uses a daytime front reference and a daytime left query. There is one query per selected site in each of these protocols.

The gallery has one reference per selected site. Place correctness uses the stored site identity, not a distance threshold inferred after observing predictions. The reverse protocol is useful because extraction and matching may not behave symmetrically when query and reference roles change. It nevertheless reuses the same physical sites, so the protocols are related observations rather than independent geographic test sets.

5.3.3 Difference from the official benchmark

The official Tokyo retrieval task uses a street-view image database. The executed protocol uses query photographs as a small cross-time reference gallery. This changes database scale, appearance distribution, viewpoint alignment and distractor difficulty. Results are therefore labelled "Tokyo

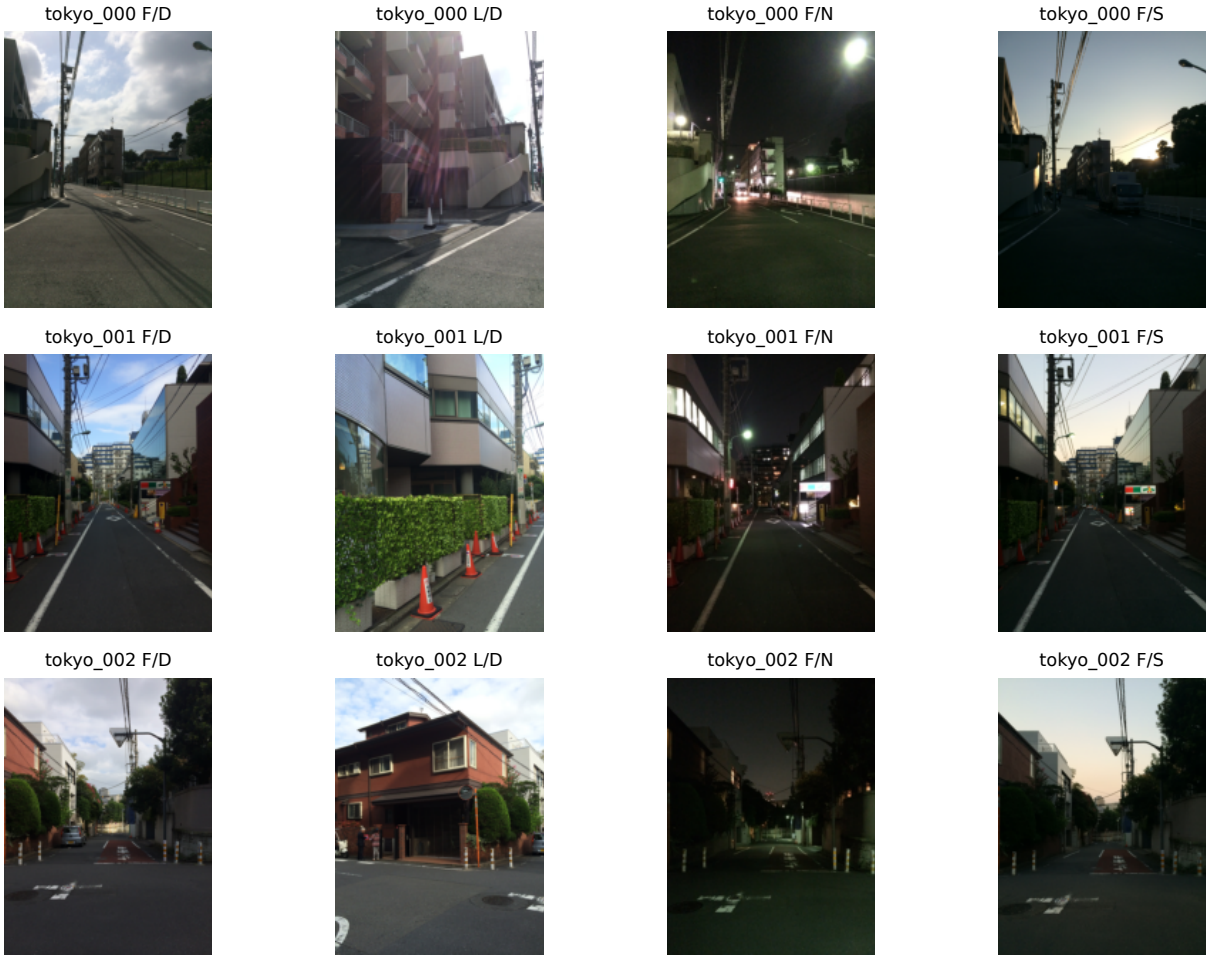

Figure 5.1: Actual downloaded photographs from the first selected Tokyo sites. Columns follow the retained front/day, left/day, front/night and front/sunset records. Orientation and condition labels come from the source metadata.

cross-time subset" throughout the thesis. They must not be compared directly with published official benchmark recall values.

The restriction is scientifically important when recall saturates. A small gallery with aligned front-facing views can be easy even though night recognition is difficult in a larger operational setting. Conversely, a deliberately different viewing direction can reduce overlap despite unchanged time of day. The experiment identifies these task properties rather than treating all appearance labels as a fixed difficulty ordering.

The query archive does not provide the calibrated camera poses or per-pair homographies needed for metric localisation or true planar-accuracy scoring. GPS site labels support recognition evaluation only. Geometric acceptance is reported, but a successful fitted homography is not assigned a metric translation error. The missing pose criteria are explicitly listed as unavailable in the results.

5.4 Synthetic appearance and camera perturbations

5.4.1 Source images and identity semantics

The synthetic set contains transformations of both repository images and selected HPatches references. Repository filenames do not establish independent real-world places. Their labels are therefore source-image identities. Retrieving a transformed `reference.jpg` means recognising its source image, not proving that the system has generalised to a new geographic location.

Six source images are transformed under five conditions and three fixed seeds, giving 90 generated queries. The gallery includes additional HPatches references as distractors. Every generated query has a known source identity and a known reference-to-query homography. The preparation manifest records the generated file hashes, seeds and benchmark-manifest hash so the exact test inputs can be regenerated and checked.

5.4.2 Photometric operations

The daytime condition changes brightness within a bounded range. Night conditions apply a gamma curve, reduced gain and additive noise. The severe condition uses a stronger attenuation; a warm-night condition additionally changes channel gains. The fog-like condition blends a bright low-contrast component and applies mild blur. These are explicit stressors, not calibrated physical camera or weather models.

Figure 5.2 provides an audit of their visual effect. The operations preserve scene geometry and do not introduce night-specific light sources, glare, moving objects or seasonal occlusion. Additive noise is not a full sensor noise model. A good result here establishes tolerance to these implemented perturbations only. Real Tokyo images provide a separate, narrower check under natural temporal appearance change.

Figure 5.2: Actual seeded synthetic queries generated from the two repository images. Columns show the evaluated photometric conditions. “Fog” denotes a contrast-and-blur simulation, not a real weather capture.

5.4.3 Known geometry and deterministic seeds

Before appearance modification, a small random projective warp perturbs the image corners. The seed determines both this warp and the subsequent photometric randomness. The homography is stored explicitly, including the scale mapping from the original reference to the working-resolution source. This detail prevents a synthetic experiment from accidentally comparing an estimate in resized coordinates with a ground-truth matrix in original coordinates.

The three query seeds support repeatability and sensitivity analysis. They do not create three independent physical scenes. Confidence statements must therefore preserve source identity as a grouping variable. Likewise, increasing the number of augmentations does not compensate for the small number of base photographs. The thesis reports those dependencies as validity constraints.

5.5 Training and evaluation separation

The optional adaptation experiment reserves the first four illumination and first four viewpoint sequences for training the projection head. Remaining selected sequences form the held-out HPatches retrieval evaluation. Training never uses Tokyo photographs. Additional synthetic training perturbations use a separate fixed seed list. The backbone remains frozen, and no test-based checkpoint selection is performed.

The main pretrained comparison may evaluate all selected HPatches sequences because none is used to train the supplied models within this artifact. The adaptation table uses only its held-out identities. Its numbers should therefore be compared with the pretrained descriptor reevaluated on that same held-out subset, not with a main-table average over a different set. The scripts

preserve both denominators to prevent that drift.

5.6 Open-set diagnostic

An additional diagnostic uses HPatches images as unknown queries against a Tokyo gallery. These images have no valid place identity in that gallery. A retrieval system still returns a ranking, but an anchoring system should reject all candidates unless there is sufficient evidence. The diagnostic reports acceptance as an error condition; retrieval recall is not meaningful when no correct gallery identity exists.

The collection-level difference makes this a limited open-set test. It does not represent a difficult nearby look-alike building, malicious image replay, or two nearly identical store fronts. Its purpose is to verify that the pipeline can return no anchor and to expose gross false acceptance. More realistic negative data are required before using the measured rejection behaviour as a safety assurance.

5.7 Larger benchmarks not included

Official Aachen, RobotCar and Extended CMU documentation and download locations were inspected. Their image and mapping archives were not downloaded for the executed comparisons. The choice reflects the already bounded laptop experiment and the additional mapping, calibration and evaluation work needed to report pose metrics correctly. They are not described as universally inaccessible or as requiring registration when the recorded sources were publicly reachable.

MSLS was inspected as a route to broader seasonal and sequence coverage. Its official repository points to its current download site and describes relevant subtasks [23]. No MSLS images were included. Pittsburgh, SF-XL, Oxford/Paris and Google Landmarks were candidate alternatives rather than attempted downloads in this run. The artifact makes no statement that these untested candidates failed to download. Exact manual locations and expected layouts for the principal missing benchmarks are provided in Appendix A.

5.8 Integrity and idempotent acquisition

The downloader validates both archive-derived member properties and locally recorded hashes. Repeated execution checks existing files and skips a valid collection. Missing data can be restored and checked against the pinned manifest; an existing corrupt file produces an explicit error instead of silently changing the input. Archive paths are not extracted blindly: selected basenames and sequence directories determine destinations.

File-count checks enforce the expected image and homography structure. The final `benchmark.json` freezes protocol definitions before inference, and its hash is recorded with the run configuration. These controls do not prove that the original benchmark labels are error-free, but they establish exactly which input bytes and label rules produced the thesis results.

Chapter 6

Methodology

6.1 Notation and coordinate conventions

An original reference image is denoted I_r^o and an original query I_q^o. Their working-resolution forms are I_r and I_q. Pixel coordinates use the conventional horizontal-then-vertical ordering (x, y). A homogeneous point is $\widetilde{\mathbf{x}} = (x, y, 1)^\top$, and the inhomogeneous projection is $\pi(a, b, c) = (a/c, b/c)$ when $c \neq 0$. Every estimated homography in the implementation maps reference pixels into query pixels.

Let S_r and S_q map original coordinates into their resized images. An original ground-truth map H^o is converted by

$$H_{r \to q} = S_q H^o S_r^{-1}. \tag{6.1}$$

This follows by substituting $\widetilde{\mathbf{x}}_r^o = S_r^{-1}\widetilde{\mathbf{x}}_r$ into the original mapping and then applying S_q. The implementation records actual rounded image dimensions, so S contains the realised horizontal and vertical ratios rather than a nominal scalar alone.

This convention also applies to generated queries. Their saved homography includes the original-reference-to-generation scale. Ground truth and estimated points are compared only after both have been expressed in the same working coordinate frame. A numerical unit test verifies a known projective transformation with outliers; the reproduction tests do not depend on benchmark-model downloads.

6.2 Method A: classical local matching

6.2.1 Feature extraction

For an image I, SIFT returns coordinates and descriptors

$$\mathcal{L}_A(I) = \{(\mathbf{x}_j, \mathbf{d}_j)\}_{j=1}^K, \qquad \mathbf{d}_j \in \mathbb{R}^{128}. \tag{6.2}$$

The representation is based on local gradient structure and scale/orientation normalisation [5]. The controlled experiment limits the requested feature budget through the common configuration.

The original script budgets and preprocessing remain in the separate upstream execution.

The normalised comparison converts each working image to grayscale and extracts features once. Empty descriptor sets are represented explicitly. No claim is made that the common budget yields an identical number of points for the two detectors: each detector selects its own repeatable structures, and SIFT may return multiple oriented features at related locations. The recorded feature and match counts describe the realised outputs.

6.2.2 Descriptor matching

For each reference descriptor $\mathbf{d}_i$, approximate search identifies its first and second query neighbours, $\mathbf{e}_{j_1}$ and $\mathbf{e}_{j_2}$. The tentative correspondence survives when

$$\|\mathbf{d}_i - \mathbf{e}_{j_1}\|_2 < \rho \, \|\mathbf{d}_i - \mathbf{e}_{j_2}\|_2 \,, \tag{6.3}$$

with ρ fixed in the configuration. FLANN settings match the recorded index strategy [6]. The matcher guards against fewer than two returned neighbours. Its output is a coordinate correspondence set, not a place decision.

To extend the one-reference baseline to place recognition, Method A verifies every gallery reference. Candidates are ranked by RANSAC inlier support and then by inlier ratio, with stable order for remaining ties. The best accepted candidate supplies the anchor. A ranking is still retained when no candidate is accepted so retrieval quality and anchoring rejection can be analysed separately.

6.3 Method B: pretrained learned local matching

SuperPoint supplies keypoint positions, descriptors and the feature tensors required by LightGlue [7]. The pretrained networks are loaded in evaluation mode and executed without gradient recording. The input is the same working-resolution grayscale image used by the comparison; additional internal resizing is disabled. The model is not fine-tuned in the main A–D experiment.

LightGlue returns a partial set of descriptor-index correspondences. Its contextual interaction can be written schematically as

$$(\widehat{\mathcal{D}}_r, \widehat{\mathcal{D}}_q) = \Phi_\theta(\mathcal{D}_r, \mathcal{D}_q, \mathcal{X}_r, \mathcal{X}_q), \tag{6.4}$$

followed by learned assignment and filtering. This notation describes the role of the pretrained matcher rather than rederiving all of its layers. The official model supplies the adaptive depth and feature-pruning behaviour [9]. Recorded configuration values identify the executed path.

Method B is evaluated in two ways. Oracle-pair evaluation supplies the correct reference pair so local correspondence quality can be isolated. Exhaustive gallery evaluation performs learned matching against every reference, allowing retrieval and verification to be compared with Method A on identical queries. Exhaustive matching is an informative accuracy reference but may have unacceptable query cost as the gallery grows.

6.4 Method C: global retrieval

6.4.1 Descriptor construction

The global encoder is the released CosPlace ResNet18 model with generalised mean pooling, a learned projection and final normalisation [13, 14]. If $F_{c,u,v}$ denotes a channel activation, the pooling operation is

$$g_c = \left(\frac{1}{UV} \sum_{u,v} \max(F_{c,u,v}, \epsilon)^p \right)^{1/p}. \tag{6.5}$$

The pretrained parameter p is loaded with the checkpoint. The global descriptor is

$$\mathbf{z}(I) = \frac{W\mathbf{g} + \mathbf{b}}{\|W\mathbf{g} + \mathbf{b}\|_2}, \qquad \mathbf{z} \in \mathbb{R}^{512}, \tag{6.6}$$

with the implementation retaining the official architecture's normalisation placement. Input channel means and scales are fixed, and gallery and query images use the same descriptor input size.

6.4.2 Ranking and identity

For a gallery matrix Z whose rows are normalised descriptors, similarity to a query is

$$s_i = \mathbf{z}(I_i)^\top \mathbf{z}(I_q). \tag{6.7}$$

Sorting these scores yields a complete reference ranking. A reference's place identifier is then used for correctness. With multiple references per place, a production evaluator would need an explicit reference-to-place aggregation policy; the implemented database supports multiple references, while the frozen benchmark galleries use one per identity.

Method C alone does not estimate a homography or render an anchored post. Its anchor-accuracy columns are consequently unavailable rather than zero-performance geometric estimates. The no-geometry ablation additionally treats its top retrieval as accepted, solely to expose the consequence of replacing verification with similarity. That variant does not produce a valid projective transform.

6.5 Robust homography estimation

6.5.1 Linear constraints and minimal evidence

A planar correspondence obeys

$$\widetilde{\mathbf{x}}_q \sim H\widetilde{\mathbf{x}}_r, \qquad H \in \mathbb{R}^{3\times 3}, \quad \operatorname{rank}(H) = 3. \tag{6.8}$$

The equivalence is up to a nonzero scalar. Expanding the cross-product constraint

$$\widetilde{\mathbf{x}}_q \times H\widetilde{\mathbf{x}}_r = \mathbf{0} \tag{6.9}$$

gives two independent linear constraints per point pair. A nondegenerate minimal sample has four correspondences. Collinear or nearly collinear evidence cannot constrain a stable general planar map. These projective principles motivate the rank and boundary checks used after robust fitting [17].

The comparison uses OpenCV's RANSAC homography estimation with a fixed pixel threshold, confidence and iteration cap. For a candidate H, forward transfer error is

$$e_j = \|\mathbf{x}_{q,j} - \pi(H\widetilde{\mathbf{x}}_{r,j})\|_2 \,. \tag{6.10}$$

The inlier mask indicates correspondences compatible with the fitted model at the configured threshold. The thesis retains this support even if the final acceptance policy rejects the matrix.

6.5.2 Sampling interpretation

For an idealised inlier fraction w, minimal sample size s and desired sampling confidence c, the familiar RANSAC relation is

$$N \geq \frac{\log(1-c)}{\log(1-w^s)}. \tag{6.11}$$

It follows by requiring the probability of never selecting an all-inlier sample, $(1-w^s)^N$, to be no larger than $1-c$ [19]. The relation illustrates why descriptor ambiguity can raise geometric-estimation cost. It is a theoretical interpretation, not a claim that each recorded call used exactly this number of iterations.

6.5.3 Acceptance policy

The verifier requires a minimum inlier count, a minimum inlier ratio, a finite full-rank map and an admissible projected boundary. After normalising the homogeneous scale, it rejects severe numerical conditioning, collapsed projected area and a reference boundary crossing the projective horizon. The latter is checked through denominator signs at the corners. Convexity of the projected quadrilateral supplies an additional sanity check.

Homogeneous scale normalisation precedes numerical threshold checks: multiplying a valid map by a small nonzero constant must not change its validity, projected points or rendered content. Anchor polygons are independently checked for finite, ordered, convex corners. Stored normalised coordinates must lie within the reference image; crossed or collapsed polygons are rejected before warping.

These conditions reduce obvious geometric failures but do not certify a correct physical anchor. A wrong repeated facade can produce a consistent planar map. A correct place can contain several depth layers that violate a single homography. A large region can project outside the visible query. The acceptance flag is therefore treated as a decision made by the prototype, while true homography success requires independent ground truth.

Algorithm 1 applies unchanged to the two normalised local methods. This is important for attributing an improvement to correspondence rather than to a weaker geometric acceptance rule.

Algorithm 1 Shared local verification

Require: Reference/query features, reference image shape, fixed thresholds
1: Match descriptors and recover coordinate pairs $(\mathbf{x}_{r,j}, \mathbf{x}_{q,j})$
2: **if** fewer than four pairs exist **then**
3: **return** rejection with empty support
4: **end if**
5: Estimate H and an inlier mask with RANSAC
6: Count inliers and compute their fraction of tentative matches
7: Normalise H and test finiteness, rank, conditioning and boundary validity
8: **return** H, mask, support statistics, and the combined acceptance decision

The implementation stores failures as rows, so a method cannot improve its reported accuracy by excluding difficult pairs from the denominator.

6.6 Method D: hierarchical recognition and anchoring

The full pipeline ranks the gallery globally, selects a fixed-size shortlist, and performs Method B's local verification on that shortlist. It selects the accepted candidate with the strongest inlier support, breaking ties by ratio. An absent accepted candidate yields no place and no rendered post. The primary configuration uses the recorded shortlist size; additional runs vary that size and remove stages.

Algorithm 2 Hierarchical place and content query

Require: Query I_q, cached reference features, descriptor matrix Z, place database
1: Compute global query descriptor $\mathbf{z}_q$ and a ranked candidate list
2: Compute local query features once
3: **for** each of the first k reference candidates **do**
4: Run Algorithm 1 against the cached reference features
5: Retain diagnostics and any accepted transformation
6: **end for**
7: **if** no candidate is accepted **then**
8: **return** no anchor and the candidate diagnostics
9: **end if**
10: Select the strongest accepted reference and its place identifier
11: Retrieve that place's posted-content records
12: Compose content-to-reference and reference-to-query transformations
13: Render content with explicit alpha and return the verified place

Algorithm 2 implements the hierarchy reviewed in Chapter 2. Global retrieval limits search, while geometry supplies the map needed by the renderer. It does not provide an independent world-scale estimate. Reference caching and one-time query extraction reduce duplicated work but do not remove the cost of matching several candidates.

6.6.1 A decomposition of failure

Let G denote that a usable correct reference enters the shortlist, L that sufficient correct local evidence is recovered, and V that the verifier accepts an adequate map. Then the desired event is bounded by the upstream stages:

$$P(G \cap L \cap V) = P(G)P(L \mid G)P(V \mid G, L). \tag{6.12}$$

This is a probability identity, not an independence assumption. It explains why increasing local accuracy cannot recover a reference excluded by retrieval, and why perfect retrieval cannot ensure anchoring when the images do not overlap. The ablation tables expose these failure pathways using recorded intermediate decisions.

False acceptance is a different event: a candidate may pass V without being the correct place or without satisfying ground-truth planar accuracy. Reporting only the acceptance rate would reward such failures. The results therefore pair acceptance with correctly verified identity and, where available, actual corner-error success.

6.7 Content coordinates and rendering

A content asset has its own pixel domain. Let $A_{c\to r}$ map its four corners to a normalised reference-anchor quadrilateral converted into reference pixels. The composed transform is

$$H_{c\to q} = H_{r\to q}A_{c\to r}. \tag{6.13}$$

This order follows directly from applying the content-to-reference mapping first. A reversed order would generally produce an unrelated projection and is not interchangeable even though both matrices are 3×3.

For premultiplied content colour αC and warped alpha α', compositing is

$$I_{\text{out}} = \text{warp}(\alpha C, H_{c\to q}) + (1 - \alpha')I_q. \tag{6.14}$$

An asset without an alpha channel receives an explicit opaque mask before the configured opacity is applied. This preserves black pixels as valid content. Rendering is skipped when the selected transformation is invalid or no candidate is accepted. The current application supports planar image posts; world-space occlusion and illumination-consistent three-dimensional rendering are future requirements.

6.8 Calibrated alternatives and their prerequisites

For calibrated image points $\mathbf{u}_r$ and $\mathbf{u}_q$, two-view geometry can constrain an essential matrix through $\mathbf{u}_q^\top E\mathbf{u}_r = 0$. Relative translation remains scale-ambiguous without additional informa-

tion. A known 3D map instead supplies correspondences satisfying

$$\widetilde{\mathbf{x}}_q \sim K(R\mathbf{X} + t), \tag{6.15}$$

which supports PnP and metric map-relative pose estimation [18]. The thesis presents these equations to define an integration route; neither essential-matrix nor PnP results are claimed for the uncalibrated selected subsets.

A calibrated extension should return camera-to-map transforms with a documented convention. If R, t are world-to-camera parameters, the camera centre is $-R^\top t$, not t itself. The supplied pose-error utility and its unit test make this convention explicit. Actual metric error reporting still requires independent ground-truth poses, a common map scale and correctly associated camera intrinsics.

6.9 Short appearance-adaptation experiment

6.9.1 Restricted learning scope

The optional experiment freezes the CosPlace backbone and learns a bias-free projection $W_h \in \mathbb{R}^{128\times512}$. The output is

$$\mathbf{u}(I) = \frac{W_h\mathbf{z}(I)}{\|W_h\mathbf{z}(I)\|_2}. \tag{6.16}$$

This is both descriptor compression and adaptation. Its comparison cannot isolate those effects completely, so the report avoids describing it as full-network day/night fine-tuning. The training-place list, fixed seeds and final head checkpoints are retained.

6.9.2 Triplet loss and sampling

For anchor, positive and negative descriptors from the selected training places, the implemented loss is

$$\mathcal{L} = \frac{1}{B}\sum_{b=1}^{B}\max\bigl(0, d(\mathbf{u}_a, \mathbf{u}_p) - d(\mathbf{u}_a, \mathbf{u}_n) + m\bigr), \qquad d(\mathbf{a}, \mathbf{b}) = 1 - \mathbf{a}^\top\mathbf{b}. \tag{6.17}$$

The positive shares the source-place label and excludes the same sample index; the negative comes from another training place. The margin, optimiser, step count and learning rate are fixed in the script. One condition uses available real sequence images only; the other adds synthetic night variants. No held-out score selects a checkpoint or adjusts the step count.

Algorithm 3 is repeated for three fixed initialisations. Reporting the mean, variability and small-sample interval prevents selecting a favourable seed. A low training loss is not treated as evidence of better place recognition outside the training domain. The chosen experiment is deliberately short and interpretable; its limitations are part of the result.

Algorithm 3 Frozen-descriptor projection training

Require: Disjoint training/test places, fixed seed, frozen encoder, step budget
1: Extract training descriptors and labels; optionally add seeded night variants
2: Initialise a 512-to-128 linear head from the specified seed
3: **for** each optimiser step **do**
4: Sample anchors, different positive samples from the same place, and negative places
5: Normalise projected descriptors and compute the triplet loss
6: Update only the projection head
7: **end for**
8: Save the final head and loss trace
9: Evaluate held-out HPatches places and all selected Tokyo protocols once

6.10 Computational accounting

For M references and descriptor dimension d, exact global scoring uses a matrix-vector product with cost proportional to Md. Exhaustive local verification repeats pair matching M times, while the hierarchy repeats it k times after retrieval. Query extraction is shared across those comparisons. These expressions describe scaling of the chosen implementation; adaptive matcher behaviour and image size still determine actual measured latency.

The evaluation stores individual feature and pair timings so stage costs can be inspected. Their sums are labelled stage sums, not independently timed complete queries. A separate isolated profiler executes query and rendering paths on predecoded images, with cached gallery features, to measure actual wall-clock behaviour and peak process memory. Keeping these definitions separate prevents cached evaluation work from producing an implausible online throughput claim.

Chapter 7

Projective Geometry: Derivation and Worked Examples

7.1 Why the coordinate model deserves a separate analysis

The complete Stitch query connects three image domains: the posted asset, a stored reference photograph and the visitor's camera frame. A visually plausible result can still be geometrically wrong if any boundary uses the inverse transformation or a different resizing convention. This chapter derives those boundaries and explains the numerical checks used in the implementation. The numerical examples are algebraic illustrations, not additional benchmark measurements. The projective model follows the standard image-geometry treatment in Hartley and Zisserman [17]; the experiment remains a planar image-to-image study.

A point in a photograph has two pixel coordinates. A persistent physical point would additionally require information about camera calibration and scene depth. The distinction is easy to lose because both a homography and a pose can be represented by matrices. Their dimensions, domains and observable constraints are different. Here, the output of local verification is a relation between two image planes. It cannot be interpreted as metres of displacement or as the visitor's absolute orientation.

7.2 Homogeneous representation and scale freedom

Represent an image point as $\widetilde{\mathbf{x}} = (u, v, 1)^\top$. For a nonsingular 3×3 matrix H, the corresponding point is

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = H\widetilde{\mathbf{x}}, \qquad \pi(H\widetilde{\mathbf{x}}) = (a/c, b/c)^\top. \tag{7.1}$$

Multiplying H by any nonzero scalar leaves the result unchanged wherever $c \neq 0$. Consequently, matrix entries cannot be interpreted independently of their homogeneous scale. A numerical validity check must not reject $10^{-12}H$ while accepting H solely because its unnormalised entries

are smaller.

The implementation normalises an admissible map before applying absolute denominator tolerances. For the rectangular reference domain used here, the upper-left point is included. A zero bottom-right entry maps that point onto the projective horizon and is therefore rejected by the boundary policy. This is an application-domain restriction, not a statement that every projective matrix with a zero bottom-right entry is globally meaningless.

Normalisation can itself overflow if the entries span a range beyond the floating-point representation. The corrected verifier checks finiteness again after division and rejects an unrepresentable map before invoking matrix decompositions. Rejection is preferable to returning an exception from an otherwise routine candidate comparison. The regression suite exercises both very small homogeneous scales and a matrix whose normalisation cannot be represented reliably.

7.3 The correspondence equations

For a reference point (u, v) and query point (u', v'), the reference-to-query mapping implies

$$u'(h_{31}u + h_{32}v + h_{33}) = h_{11}u + h_{12}v + h_{13}, \tag{7.2}$$

$$v'(h_{31}u + h_{32}v + h_{33}) = h_{21}u + h_{22}v + h_{23}. \tag{7.3}$$

Each correspondence therefore contributes two linear constraints on the nine homogeneous matrix entries. Four correspondences provide eight constraints, which is the minimum count for a general eight-degree-of-freedom homography when the configuration is nondegenerate. More correspondences do not automatically provide more independent information: repeated points or a nearly collinear arrangement can still leave the estimate unstable.

Stitch delegates robust estimation to OpenCV rather than implementing a separate direct linear solver. The equations nevertheless explain two important rejection cases. A blank image supplies no constraints, so it must yield no map. A set of points along one edge may supply many tentative matches but insufficient two-dimensional support. An absolute inlier count cannot diagnose either case by itself; matrix and projected-boundary checks remain necessary.

7.4 Resizing as a change of coordinate system

Let the original reference size be (w_r, h_r) and the working size be $(\bar{w}_r, \bar{h}_r)$. The implemented pixel transform is

$$S_r = \operatorname{diag}(\bar{w}_r/w_r, \bar{h}_r/h_r, 1). \tag{7.4}$$

There is a separate transform S_q for the query. If H_o maps original reference coordinates to original query coordinates, the working-resolution relation is

$$H_w = S_q H_o S_r^{-1}. \tag{7.5}$$

This equation follows by substituting $\widetilde{\mathbf{x}}_r = S_r^{-1}\widetilde{\mathbf{x}}_{r,w}$ into the original mapping and then applying S_q. It is a coordinate conversion; it does not estimate a new physical relation.

Consider an illustrative reference reduced from 1000×800 to 500×400, and a query retained at its original size. An original reference point $(200, 100)$ becomes $(100, 50)$ in the working reference. If the original relation is a translation of $(20, 10)$ pixels, directly applying that translation matrix to $(100, 50)$ gives $(120, 60)$, which is wrong for the unchanged query domain. The conjugated relation produces $(220, 110)$, as required. The translation did not change physically; its input coordinate unit changed.

Integer rounding makes horizontal and vertical scale values slightly different for some dimensions. The implementation records the realised output size rather than assuming an ideal shared scale. This matters when corner-error thresholds are only a few working pixels. A resizing approximation that is negligible to a global descriptor can still affect a geometric pass/fail boundary.

7.5 Forward and backward transfer error

The reference-to-query direction is fixed throughout the prototype. A forward transfer residual for a correspondence is

$$r_f = \left\| \pi(H\widetilde{\mathbf{x}}_r) - \mathbf{x}_q \right\|_2 . \tag{7.6}$$

Its unit is a query pixel. A backward residual uses H^{-1} and has reference-pixel units. Adding the two without accounting for image scale can mix different units. A future symmetric error policy should therefore specify the coordinate systems and normalisation explicitly, rather than describing an unspecified bidirectional error as universally stricter.

The reported corner metric is a different quantity. It compares the estimated and ground-truth maps at the same four reference corners. It is not the average RANSAC correspondence residual, and it can remain large when a small interior cluster fits well. This is why the report retains both local support statistics and independent corner-error success on collections that provide ground truth.

On Tokyo, no independent corner relation is available for the retained cross-time pairs. Measuring the residual of the very correspondences used to estimate the map would not replace that missing ground truth. Such a residual could describe self-consistency, but it would not establish correct surface alignment. The site atlas consequently reports identity and decision states without inventing a Tokyo corner-error score.

7.6 RANSAC success probability and its limits

In an idealised sampling calculation, let w be the probability that a tentative correspondence is a genuine inlier and let a minimal sample contain s correspondences. If samples are independent, the probability of at least one all-inlier sample after N attempts is

$$P = 1 - (1 - w^s)^N. \tag{7.7}$$

Solving for a desired probability gives

$$N \geq \frac{\log(1-P)}{\log(1-w^s)}. \tag{7.8}$$

For a homography, $s = 4$ in the generic minimal case. The expression explains why a lower inlier fraction can increase robust-estimation work sharply [19]. It is an analytical sampling model, not a guarantee that a selected sample is nondegenerate or that the scene satisfies a planar relation.

Repeated windows illustrate the distinction. A matcher may return many mutually consistent correspondences between different but similar structures. RANSAC can fit those correspondences and return strong support. Its success at finding a consensus does not prove the gallery identity or the physical presence of the user. The open-set diagnostic tests that broader decision problem and records wrong acceptances rather than reclassifying them as successful geometry.

7.7 The projective horizon and boundary policy

The projective denominator at a reference point is the affine function $d(u, v) = h_{31}u + h_{32}v + h_{33}$. On a rectangular domain, its extrema occur at vertices. If all four vertex values have the same strict sign, the denominator does not cross zero anywhere inside the rectangle. Opposite signs imply a horizon crossing and potentially unbounded image coordinates. This supplies an inexpensive domain-level validity check before rendering.

The verifier also checks projected convexity and a minimum projected area. These are practical filters for a stable reference boundary. They do not establish that the entire physical scene is planar, that occluders are absent or that a target surface is safe for displaying content. A highly foreshortened but genuine target can be rejected by the area policy; the resulting loss should be counted as rejection rather than hidden by changing the denominator of an accuracy metric.

A condition-number threshold serves a related numerical purpose. It identifies matrices for which inversion or perturbation may be sensitive. Its value depends on the coordinate representation, so it should be treated as an implementation policy evaluated at the recorded working resolution. It is not a universal physical plausibility score or a calibrated probability of anchor correctness.

7.8 Content coordinates and alpha compositing

The asset has its own corners, which generally differ from reference-image corners. A content-to-reference map A is constructed from the asset corners and its stored reference polygon. The rendered content map is HA. The multiplication order is essential: AH applies a reference/query relation in the wrong input domain. Normalised anchor coordinates are first converted using the actual working reference width and height.

An anchor polygon must contain four finite corners in a consistent boundary order and form a noncollapsed convex quadrilateral. A crossed polygon is not a harmless alternative ordering: it can induce a folded or unbounded warp. The renderer rejects that input, and database validation

additionally checks that stored normalised coordinates lie in $[0, 1]$. Associating the polygon with a specific photograph prevents an anchor defined on a front view from being copied onto an unrelated side view.

Alpha compositing uses premultiplied colour. If the asset colour is C, its opacity mask is α, and the frame is F, the compositor evaluates the warped premultiplied colour plus $F(1-\alpha_w)$. Warping unpremultiplied colour separately can mix invisible border colours into partially transparent edge pixels. Explicit alpha also preserves opaque black content, which would be lost by treating every black pixel as background.

7.9 What these derivations validate

The derivations justify coordinate conjugation, transform direction, homogeneous scale handling and explicit alpha. The numerical regression tests validate concrete examples of those contracts. Neither form of validation substitutes for independent real-world alignment measurements. The HPatches corner metric provides one such measurement within its image-plane scope, while metric pose and long-term persistent-world tests remain future work.

The practical outcome is a narrower and more defensible interface. A query either returns a supported reference-to-query map with an associated reference-bound post, or returns no anchor. It does not return a nominal world pose merely because a matrix was computed. This keeps subsequent deployment design aligned with the evidence the implemented estimator can actually provide.

Chapter 8

Experimental Setup

8.1 Recorded execution environment

The experiments ran on the available Apple M2 laptop with the memory configuration recorded in `results/environment.json`. The software environment is a clean project-local CPython virtual environment. The initial default interpreter was unsuitable for the pinned dependency set, so the available system CPython interpreter was used. Dependency versions were installed and recorded rather than assumed from pre-existing user packages.

The runtime reported neither CUDA nor an available MPS backend for these runs. All reported inference therefore uses the CPU. Both PyTorch and OpenCV are configured to use two threads in the controlled scripts. This restriction makes the workload explicit and avoids silently changing thread counts between methods. It does not establish that this is the fastest possible host configuration.

The primary installed versions are listed in the environment artifact and dependency specification. The reproducibility bundle also records the official LightGlue and CosPlace code revisions and SHA256 hashes of all downloaded checkpoints. Model parameter counts and checkpoint sizes are obtained from actual loaded models and files. They are not inferred by counting the number of layers in a paper diagram.

Table 8.1: Measured parameter counts and downloaded checkpoint sizes. Checkpoints include serialisation metadata.

Model	Parameters	Checkpoint MB
CosPlace	11439169	45.83
SuperPoint	1300865	5.21
LightGlue	11851601	47.50

Table 8.1 separates trainable-parameter count from serialised checkpoint size. Checkpoint bytes include format metadata and any stored buffers. Runtime memory is larger because it includes model objects, image tensors, local features, library imports and allocator state. A separate complete-process peak measurement is consequently reported in the deployment analysis.

8.2 Configuration and fixed controls

Table 8.2: Executed comparison configuration. Values are explicit experimental settings, not measured outcomes.

Setting	Value
Working-image maximum side	640 pixels
Requested local-feature budget	512
SIFT ratio threshold	0.70
RANSAC transfer threshold	5 working pixels
RANSAC confidence / maximum iterations	0.999 / 3000
Minimum inliers / inlier ratio	12 / 0.20
Primary full-pipeline shortlist	3 references
Shortlist ablations	1 and 5 references
CosPlace input / descriptor	224×224 / 512 dimensions
PyTorch / OpenCV threads	2 / 2
Main seed	17
Synthetic-query and training-head seeds	17, 23, 42

Table 8.2 states the settings shared across compared methods. The feature budget and RANSAC thresholds are not tuned separately on each test split. Reference extraction is cached. Query local features are extracted once per method and reused across the gallery. Global-only retrieval follows its own smaller input shape, which is documented because global descriptors and pixel correspondences have different computational needs.

PyTorch, NumPy and OpenCV random states are set. Candidate matching receives a deterministic seed derived from its method and image-path identities so repeated comparisons do not depend on the order in which a cache is populated. This improves repeatability within the installed environment. Floating-point kernels and library changes can still produce differences on another machine, so exact input hashes and environment versions remain necessary.

8.3 Evaluation populations

The frozen benchmark contains 306 labelled positive reference-query pairs, which produce 306 measurements per local method and therefore twice that many oracle-pair rows. The query protocols additionally include unknown-place diagnostics. The executed exhaustive comparison produced 2226 query-method prediction rows. These counts are generated from the saved preparation and result files.

Table 8.3: Frozen evaluation protocols. Reference-to-query direction is used in condition names.

Protocol	Gallery	Queries
hpatches_illumination	24	60
hpatches_viewpoint	24	60
tokyo_day_to_day	24	24

Continued on next page

Table 8.3 (continued)

Protocol	Gallery	Queries
tokyo_day_to_night	24	24
tokyo_night_to_day	24	24
tokyo_day_to_sunset	24	24
synthetic_day	26	18
synthetic_night	26	18
synthetic_night_severe	26	18
synthetic_warm_night	26	18
synthetic_fog	26	18
open_set	24	12

Table 8.3 gives the denominator for each protocol. The same query list is evaluated by A, B, C and the full pipeline. Stage ablations use those same lists. A query failure remains in its denominator; it is not removed because a model produced no descriptors or a homography could not be estimated.

The principal unit of evidence is a place or source image rather than an individual pixel correspondence. HPatches targets from one sequence share a reference and scene. Synthetic variants share a source image. Tokyo condition splits reuse physical sites. These dependencies limit the interpretation of aggregate counts, especially when uncertainty intervals are calculated from a small, selected set.

8.4 Retrieval metrics

Let Q be a query set and $\mathcal{P}(q)$ the valid gallery-place identities for query q. With ranked identities $r_1(q), r_2(q), \ldots$, recall is

$$R@k = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}\{\mathcal{P}(q) \cap \{r_1(q), \ldots, r_k(q)\} \neq \varnothing\}. \tag{8.1}$$

The implemented benchmark has one correct place identity per positive query. Recall@1, @5 and @10 share a query denominator and use the same ordered result list. Larger values of k are necessarily easier in these small galleries, so high Recall@10 is not interpreted as high-precision anchoring.

For exhaustive local methods, the ranking uses inlier support even if the top candidate is not accepted. This permits a correct retrieval with insufficient evidence to render. For the full pipeline, shortlisted references are reranked locally and the remaining global ranking is retained as an unverified tail. Its Recall@5 or @10 may therefore include references that were never locally verified. Correctly verified-place rate is reported separately to show actual decisions.

The global-only method does not declare an anchor. It is inappropriate to compare its absence of a geometric result with a zero homography success rate as if a model had attempted and failed to fit a transform. Tables mark that field unavailable. The no-geometry ablation instead

explicitly makes an unverified acceptance decision, allowing its false-acceptance consequences to be measured.

8.5 Matching and planar geometry metrics

8.5.1 Correspondence support

For M tentative matches and N RANSAC inliers, inlier ratio is N/M when $M > 0$ and zero otherwise. Median inlier counts and mean ratios are computed across all evaluated positive pairs, including failures. A high ratio from a very small correspondence set is not equivalent to a well-supported surface map, which is why the verifier also requires an absolute count.

Geometric acceptance rate is the fraction of pairs satisfying the configured support and validity conditions. On Tokyo it is the strongest available geometric decision metric, but it is not ground-truth alignment accuracy. On HPatches and synthetic pairs, independent ground-truth homographies make a stronger evaluation possible.

8.5.2 Corner-error success

Let $\mathbf{c}_1, \ldots, \mathbf{c}_4$ be the working reference corners. Mean corner error is

$$e_H = \frac{1}{4} \sum_{i=1}^{4} \left\| \pi(\widehat{H}\widetilde{\mathbf{c}}_i) - \pi(H^{\mathrm{gt}}\widetilde{\mathbf{c}}_i) \right\|_2 . \tag{8.2}$$

An oracle-pair homography success requires acceptance and $e_H \leq 5$ working pixels. Invalid estimates are failures. For cumulative error curves, they remain at infinite error so the curve denominator includes all queries. A conditional median over finite errors is interpreted only alongside the failure rate.

End-to-end planar success additionally requires the selected place to be correct. A highly accurate homography to a wrong reference is not counted. This stricter metric measures the chain from candidate selection to anchoring, while oracle-pair success isolates local geometry when the correct reference is supplied. Their difference reveals search-stage losses.

8.6 Metric localisation criteria

The requested position-and-orientation thresholds require calibrated ground-truth camera poses. For world-to-camera parameters (R, t) and (R_g, t_g), the corresponding errors would be

$$e_t = \left\| -R^\top t + R_g^\top t_g \right\|_2 , \tag{8.3}$$

$$e_R = \arccos\left(\mathrm{clip}\left(\frac{\mathrm{tr}(RR_g^\top) - 1}{2}, -1, 1 \right) \right) . \tag{8.4}$$

The utility function returns camera-centre distance and angle in degrees. Neither utility execution on a unit test nor GPS site metadata supplies the real query poses needed for a benchmark mea-

surement. The result table therefore records the three requested criteria as unavailable. Official map-based benchmarks provide the proper route for future evaluation [22].

8.7 Timing, memory and model-size protocols

The upstream measurements time bounded script execution including initialisation and file I/O. The main evaluation records feature-extraction and matching-plus-geometry times, reusing cached reference computation. Its query stage sum adds the realised query extraction and selected pair costs; it is useful for accounting but excludes some application work and is not a direct end-to-end stopwatch measurement.

The separate runtime profiler starts an isolated process for each method. It builds the reference cache, warms the path, and times actual query and rendering calls on a fixed small set of pre-decoded Tokyo images with repeated execution. It reports median, mean and upper-tail latency, and stores every timing sample. Asset loading during rendering remains inside the measured call. Camera decoding, network transfer and display are excluded.

Peak RSS is obtained from the operating system's process resource accounting, with platform-specific units converted into bytes. It includes imports, model weights, gallery caches and transient allocation. It is not presented as the exclusive tensor memory of the selected model. Serialised model size is measured from file length and reported separately. No phone power, thermal, battery or neural-accelerator measurement is claimed.

8.8 Uncertainty and statistical interpretation

For a binary site-level outcome with k successes among n sites, the report computes a Wilson interval. With $\widehat{p} = k/n$ and a normal quantile z, its centre and half-width are

$$\mu_W = \frac{\widehat{p} + z^2/(2n)}{1 + z^2/n}, \tag{8.5}$$

$$h_W = \frac{z}{1 + z^2/n}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n} + \frac{z^2}{4n^2}}. \tag{8.6}$$

The interval remains nonzero even when every selected query succeeds. It describes finite-sample uncertainty under an independent-site approximation, not a correction for the metadata-based geographic selection or the changed Tokyo benchmark task.

HPatches uncertainty is also computed with a cluster bootstrap that samples sequence identities and includes their associated targets. Synthetic variants are discussed as repeated observations of a few sources, not independent places. For projection training, three fixed seeds yield a mean, sample standard deviation and a small-sample t interval. With so few seeds the interval is unstable; it communicates uncertainty rather than establishing a precise population effect.

No significance claim is based on selecting the best seed, dropping rejected queries or treating thousands of pixel matches as independent successful places. No hyperparameters are chosen

using the test-set result tables. The design is an exploratory reproducible comparison, with the limited data coverage made explicit.

8.9 Ablations and failure inspection

The shortlist-size ablation compares one, three and five candidates. Exhaustive learned matching removes the global stage, and global-only retrieval removes local geometric evidence. A no-geometry acceptance diagnostic shows the distinction between a candidate rank and a supported anchor. Projection training compares available sequence images with and without extra synthetic night variants. Export validation compares native, float ONNX and QDQ INT8 descriptors.

Qualitative selections follow documented rules: prefer a case where learned local verification succeeds while SIFT rejects, otherwise use a fixed fallback. Correspondence coordinates and masks are saved by an actual rerun, and overlay figures use saved estimates. This is a deliberately illustrative selection rather than a random sample of success. Failure examples and aggregate tables prevent the selected demo from standing in for overall accuracy.

8.10 Reproducibility controls

Every substantive execution command is recorded with its argument vector, working directory, start time, output-log path, elapsed time and exit code. Failed actions remain in the journal. Build scripts derive tables and figures from measured files and record input hashes. A final checker scans the PDF build log for errors and unresolved citations/references, checks data-to-table consistency, and verifies that required demonstration files can be decoded.

These controls do not make future results bitwise identical across all hardware. They do make the experiment inspectable: a reader can determine the source images, exact model bytes, protocol definitions, configuration and command that produced a reported value. The appendices provide those routes without presenting an unexecuted command as an observed result.

Chapter 9

Decision Analysis and Statistical Interpretation

9.1 The unit of success

An evaluation of Stitch can count descriptors, tentative matches, image pairs, queries, sites or user sessions. Those units answer different questions. A system may improve the number of inliers without improving the number of users who see a correctly attached post. The present evaluation makes the query the decision unit and retains site or sequence identities for interpreting dependence. Pixel correspondences provide evidence within a decision; they are not independent repetitions of the application task.

This distinction also limits claims about large sample counts. The 90 synthetic queries originate from six source images. Increasing the number of seeded transformations makes the stress test more repeatable and exposes variability within those sources, but it does not add 90 independently sampled physical places. Likewise, the five HPatches targets for a sequence remain related observations of the same source. The scene atlases preserve these groupings rather than presenting every row as independent geographic coverage.

9.2 A decision tree for one query

The first branch asks whether the correct place is represented in the gallery. The known-place protocols answer yes by construction, while the open-set diagnostic answers no. A retrieval ranking is meaningful in the first case. In the second, an anchoring system must decide whether to reject despite having a nearest neighbour. Every finite gallery can produce a top-ranked image; the existence of that rank does not establish a valid match.

For a known query, the next branch asks whether candidate generation retains a useful reference. The full method considers only its shortlist for local verification. A correct reference outside the shortlist cannot be rescued by improving RANSAC on the candidates already selected. An exhaustive learned-local method removes that search restriction but incurs more pair comparisons.

The two methods therefore expose a cost–coverage trade-off rather than an unconditional ordering in quality.

The final branches distinguish rejection, wrong-place acceptance and correct-place acceptance. On collections with independent homography ground truth, correct-place acceptance is subdivided into accurate and inaccurate planar alignment. These leaves explain why retrieval recall, correctly verified-place rate and end-to-end H5 appear separately. Combining them into one unnamed accuracy number would hide the stage at which an apparent improvement occurs.

9.3 Coverage, precision and conditional denominators

Let N be the number of queries, A the number of accepted decisions, C the number of correctly accepted identities and $W = A - C$ the wrong accepted identities. Acceptance coverage is A/N, correctly verified-place rate is C/N, and accepted-identity precision is C/A when $A > 0$. The identity

$$\frac{C}{N} = \frac{A}{N}\frac{C}{A} \tag{9.1}$$

shows why a highly selective system can achieve strong conditional precision while returning few useful anchors. The report's primary verified rate uses the full query denominator. A future application study should additionally report coverage and precision together, including an explicit convention for the no-acceptance case.

On homography datasets, define G as correctly accepted queries whose corner error is within the declared tolerance. Then G/N measures the full recognition-to-alignment chain. Reporting G/C alone would answer how often an already correct accepted identity has accurate geometry. That conditional measure can be informative, but omitting C/N would conceal search and rejection failures. The atlas includes both stages so the reader can inspect this separation at sequence level.

9.4 Paired comparisons and discordant cases

Two methods evaluated on the same query should be compared as paired outcomes. Partition the queries into both correct, neither correct, method A only correct and method B only correct. If the two discordant counts are n_{10} and n_{01}, the difference in full-denominator success rates is

$$\Delta = \frac{n_{01} - n_{10}}{N}. \tag{9.2}$$

The two methods can have equal aggregate scores while succeeding on different images. That pattern matters operationally: combining reference coverage or improving a particular failure mode may be more useful than choosing one method globally.

Each HPatches atlas entry reports learned-only and SIFT-only oracle successes for its five targets. These counts describe the observed paired evidence. They are not presented as independent significance tests for every sequence. Testing many small subsets after inspecting their

results would invite selection effects, especially when the same images appear in several method comparisons and ablations.

The Tokyo site atlas uses an analogous paired view across conditions. A site's changed-direction daytime query and front-facing night query share a physical identity but differ in overlap and acquisition conditions. Their comparison is valuable for diagnosing reference coverage. It does not isolate an illumination treatment because view direction is not held fixed between those two protocols.

9.5 Confidence intervals and saturated scores

A score of 100 percent on 24 selected sites is a statement about those 24 decisions. It is not a proof that the system succeeds on all future night queries. The Wilson intervals in the results chapter retain finite uncertainty at the boundary. Their assumptions remain limited by the subset's metadata-based selection and geographic coverage. A wider nominal interval cannot repair a gallery that excludes difficult scene types before evaluation.

For repeated targets within HPatches sequences, a sequence-level bootstrap respects more of the grouping than resampling individual targets independently. In each bootstrap replicate, whole sequence identities are sampled and all their associated outcomes contribute. This preserves within-sequence dependence in the resampling design. It still cannot create new locations, camera models or weather conditions that were absent from the selected data.

The three-seed projection experiment measures a different source of variability. Its seed standard deviation concerns the short optimisation procedure on a fixed train/test partition. It does not measure uncertainty over all possible partitions, all training collections or all urban environments. The corresponding small-sample interval is therefore interpreted as a description of this bounded adaptation exercise, with no broad claim of statistical superiority.

9.6 Selection bias and the reference gallery

The gallery is part of the task definition. A small gallery of distinct scenes can make global retrieval easier than a large gallery containing adjacent views and repeated structures. Adding distractors changes the ranking problem even when the query photographs remain identical. Consequently, the custom Tokyo scores cannot be transferred to the official street-view task by changing the dataset name in a caption.

The subset requires complete front/day, left/day, front/night and front/sunset observations. That rule supports controlled paired comparisons, but it selects the sites for which all required metadata combinations are available. It may exclude sites whose observations are incomplete or otherwise difficult to group. The chosen lexicographic ordering further fixes a reproducible subset without making it a random sample of the entire city.

HPatches contributes independent image-to-image homographies, which strengthens the geometric evaluation. Its cropped or scene-specific image pairs nevertheless differ from arbitrary

visitor camera motion and world-map persistence. A successful H5 result supports the stated planar relation; it does not establish performance for nonplanar scenes or for an AR session that continues after the recognised surface leaves the frame.

9.7 Threshold tuning and locked evaluation

The current support and geometry thresholds are shared across the compared local methods. A future deployment may reasonably tune a decision policy, but that tuning should use a separate validation population. Selecting thresholds on the final test rows and then reporting the resulting rates as an untouched evaluation would make the apparent improvement partly a consequence of selection.

A useful future protocol would reserve sites for threshold calibration, keep test sites unseen, and evaluate the entire policy at a declared operating point. Rejection costs and wrong-anchor costs should be chosen from product requirements before examining the final test outcome. The current report does not claim to have completed such a policy-calibration study. Its fixed-threshold comparison is an exploratory research baseline.

9.8 Latency as a distribution of defined work

The isolated profiler records actual warm query and rendering calls. Its median and upper quantile summarise a small set of predecoded queries repeated under a fixed host configuration. Those statistics should not be interpreted as a city-scale service latency distribution. The six chosen queries and their repetitions provide evidence about this local workload; they do not include network contention, camera startup, phone thermals or concurrent users.

The evaluation stage sum has an even narrower definition. It reuses measured extraction and cached pair costs to account for selected work. It is useful for comparing candidate counts, but it excludes parts of the complete application call. Mixing that sum with an end-to-end runtime from another path would produce an unfair comparison. Tables retain the distinction and state whether references, decoding and rendering are included.

9.9 A claim-to-evidence reading rule

Every empirical claim should be connected to its denominator, input population, output artifact and metric definition. For example, a statement about day-to-night verified places refers to the custom Tokyo protocol, 24 selected sites, accepted identity decisions and the saved prediction rows. A statement about model size refers to produced file lengths. A statement about future native tracking is a design proposal and has no measured result in this artifact.

This reading rule is especially important in an expanded report. More pages provide room for derivations, failure inspection and reproducibility records, but they do not increase the experimental population. The additional scene-level chapters reorganise existing evidence so that

readers can inspect all selected cases. They do not multiply the sample size or convert a narrow observation into a general benchmark claim.

Chapter 10

Results and Discussion

10.1 Reading the evidence

The results are derived from the saved predictions, pair statistics, training records, export outputs and runtime profiles. All empirical tables and plots in this chapter are generated by `scripts/generate_artifacts.py`. The generation step records hashes of its input result files. Headline values in the prose use generated LaTeX macros so a rerun cannot silently leave those values inconsistent with the tables.

Three outcomes are kept separate throughout. Retrieval asks whether the correct identity appears in a ranked list. Correct verification asks whether the pipeline accepts the correct identity. Planar success additionally asks whether the accepted transformation agrees with an independent ground-truth homography. A result may satisfy one outcome without satisfying the next. This distinction is especially important for Tokyo, where no true planar or metric-pose error can be computed from the downloaded metadata.

10.2 Oracle-pair local matching

Table 10.1: Oracle-pair matching under shared geometry settings. Inliers are medians, ratios are means over all pairs; H_5 requires accepted geometry and mean corner error at most five resized pixels. Tokyo has no homography ground truth.

Protocol	Method	n	Inliers	Ratio (%)	Accept (%)	H_5 (%)
HP illum.	SIFT	60	58.5	83.7	83.3	81.7
HP illum.	SP+LG	60	317.0	94.6	100.0	100.0
HP view.	SIFT	60	97.5	89.8	93.3	86.7
HP view.	SP+LG	60	218.0	88.9	90.0	86.7
Tokyo D–D	SIFT	24	9.0	51.7	41.7	—
Tokyo D–D	SP+LG	24	39.0	61.1	54.2	—
Tokyo D–N	SIFT	24	8.5	64.3	33.3	—

Continued on next page

Table 10.1 (continued)

Protocol	Method	n	Inliers	Ratio (%)	Accept (%)	H_5 (%)
Tokyo D–N	SP+LG	24	169.0	84.7	100.0	—
Tokyo N–D	SIFT	24	7.0	64.9	33.3	—
Tokyo N–D	SP+LG	24	169.5	86.0	100.0	—
Tokyo D–S	SIFT	24	35.0	73.7	66.7	—
Tokyo D–S	SP+LG	24	204.0	84.2	100.0	—
Synth. day	SIFT	18	255.0	95.5	100.0	100.0
Synth. day	SP+LG	18	416.5	96.5	100.0	100.0
Synth. night	SIFT	18	130.5	82.0	94.4	88.9
Synth. night	SP+LG	18	388.5	95.3	100.0	100.0
Synth. severe	SIFT	18	10.0	41.8	33.3	33.3
Synth. severe	SP+LG	18	188.0	87.6	100.0	100.0
Synth. warm	SIFT	18	101.0	79.5	88.9	88.9
Synth. warm	SP+LG	18	375.0	96.0	100.0	100.0
Synth. fog	SIFT	18	76.0	84.5	100.0	100.0
Synth. fog	SP+LG	18	402.0	96.3	100.0	100.0

Table 10.1 compares SIFT–FLANN and SuperPoint–LightGlue on identical positive image pairs with common geometry settings. Supplying the correct reference removes candidate-search failure from this comparison. The inlier statistics describe each method's own tentative matches, so a higher count is interpreted alongside acceptance and corner-error success rather than as a standalone proof of accuracy.

On the selected HPatches illumination pairs, SIFT achieves 81.7% accepted five-pixel homography success, while the learned combination achieves 100.0%. This supports an illumination-robustness benefit on this subset. The result remains conditional on the selected scenes, resolution and verification policy. It does not show that every illumination condition is solved or that the matcher can reconstruct detail absent from an underexposed image.

On the viewpoint pairs, the success values are 86.7% for SIFT and 86.7% for the learned combination. The learned method therefore does not dominate every geometric condition in the tested set. In particular, accepted estimates can still extrapolate poorly to reference corners outside the supported region. This observation cautions against using a greater number of matches as a substitute for evaluated alignment accuracy.

Figure 10.1 emphasises the different illumination and viewpoint outcomes. The illumination gain is consistent with the motivation for learned features, while the viewpoint result provides a negative control against an unconditional improvement claim. Different subsets or thresholds may change the comparison, and the artifact makes those choices explicit rather than treating its rates as universal model properties.

Figure 10.2 provides more information than a single threshold. A method whose finite-error estimates are precise can still have a low maximum curve value because many pairs fail. Conversely, accepted estimates with large corner error contribute only at loose thresholds. Keeping failures in the denominator avoids presenting a conditional accuracy curve as if it represented all evaluated

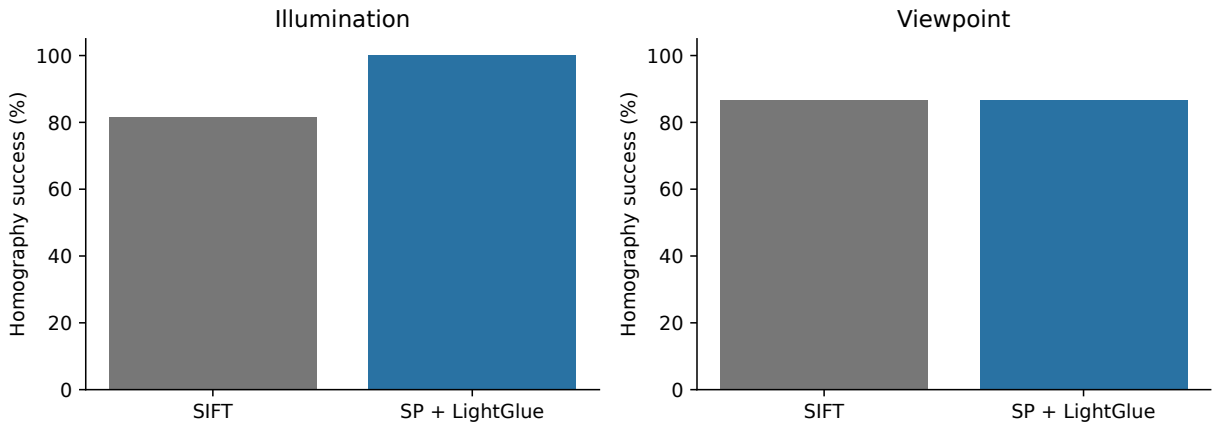

Figure 10.1: Measured oracle-pair homography success on the selected real HPatches sequences. A success requires accepted geometry and mean corner error no greater than five working pixels.

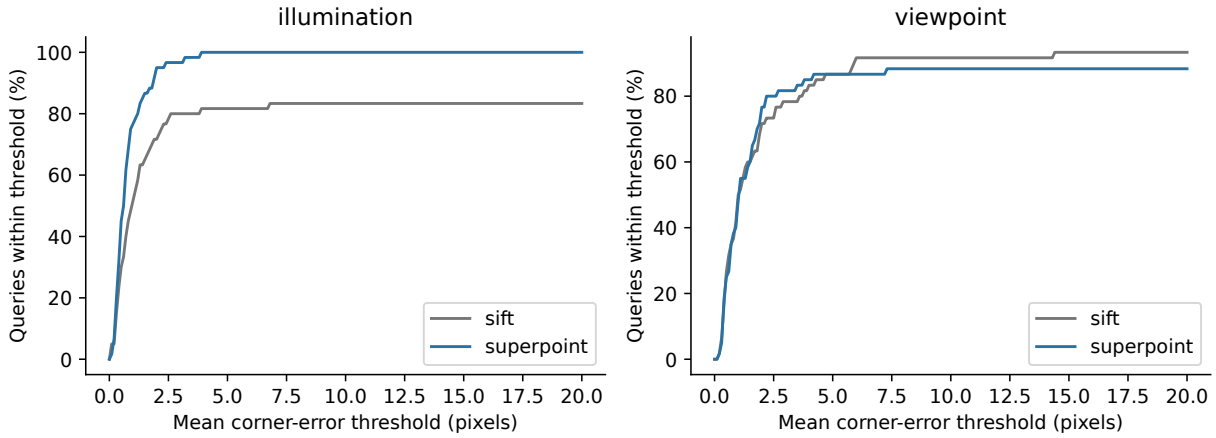

Figure 10.2: Cumulative accepted-homography accuracy against corner-error threshold. Failed or rejected estimates remain outside the curve at finite thresholds; all pairs remain in each denominator.

queries.

10.3 Place retrieval on real Tokyo photographs

Table 10.2: Retrieval and correctly verified place rates (percent) on Tokyo protocols. Global-only retrieval does not produce an anchor. Full-pipeline recall uses its reranked complete list, with the unverified tail retained.

Split	Method	R@1	R@5	R@10	Verified
day_to_day	A: SIFT	54.2	66.7	75.0	41.7
day_to_day	B: SP+LG	58.3	70.8	75.0	54.2
day_to_day	C: CosPlace	41.7	62.5	75.0	—
day_to_day	D: Full ($k = 3$)	41.7	62.5	75.0	37.5
day_to_night	A: SIFT	58.3	91.7	91.7	33.3
day_to_night	B: SP+LG	100.0	100.0	100.0	100.0
day_to_night	C: CosPlace	100.0	100.0	100.0	—
day_to_night	D: Full ($k = 3$)	100.0	100.0	100.0	100.0
night_to_day	A: SIFT	50.0	70.8	91.7	33.3
night_to_day	B: SP+LG	100.0	100.0	100.0	100.0
night_to_day	C: CosPlace	100.0	100.0	100.0	—
night_to_day	D: Full ($k = 3$)	100.0	100.0	100.0	100.0
day_to_sunset	A: SIFT	83.3	95.8	100.0	66.7
day_to_sunset	B: SP+LG	100.0	100.0	100.0	100.0
day_to_sunset	C: CosPlace	100.0	100.0	100.0	—
day_to_sunset	D: Full ($k = 3$)	100.0	100.0	100.0	100.0

Table 10.2 gives the custom cross-time retrieval results. Day-to-night Recall@1 is 58.3% for exhaustive SIFT, 100.0% for exhaustive learned local matching, and 100.0% for CosPlace. The full hierarchy correctly verifies 100.0% of those selected queries. The corresponding SIFT correctly verified-place rate is 33.3%, which is lower than its retrieval recall because some correct ranked candidates lack enough accepted local evidence.

The saturation of learned and global recall on this split is a property of the selected protocol. It uses aligned front-facing observations and a gallery of 24 selected sites. It does not use the official street-view database. Small-sample intervals and the task distinction are therefore necessary qualifications, not minor reporting details. Perfect observed recall in this setting cannot establish perfect recognition over a city or across arbitrary camera directions.

The daytime viewpoint-change result exposes a complementary weakness. CosPlace Recall@1 falls to 41.7%, and the primary full pipeline correctly verifies 37.5% of queries. Exhaustive SIFT correctly verifies 41.7%, and exhaustive learned matching verifies 54.2%. The hierarchy therefore sacrifices some recoverable cases when its shortlist excludes a useful reference. This is a measured negative result for the chosen global stage and shortlist policy.

Figure 10.3 shows that the split label alone does not determine difficulty. Appearance change

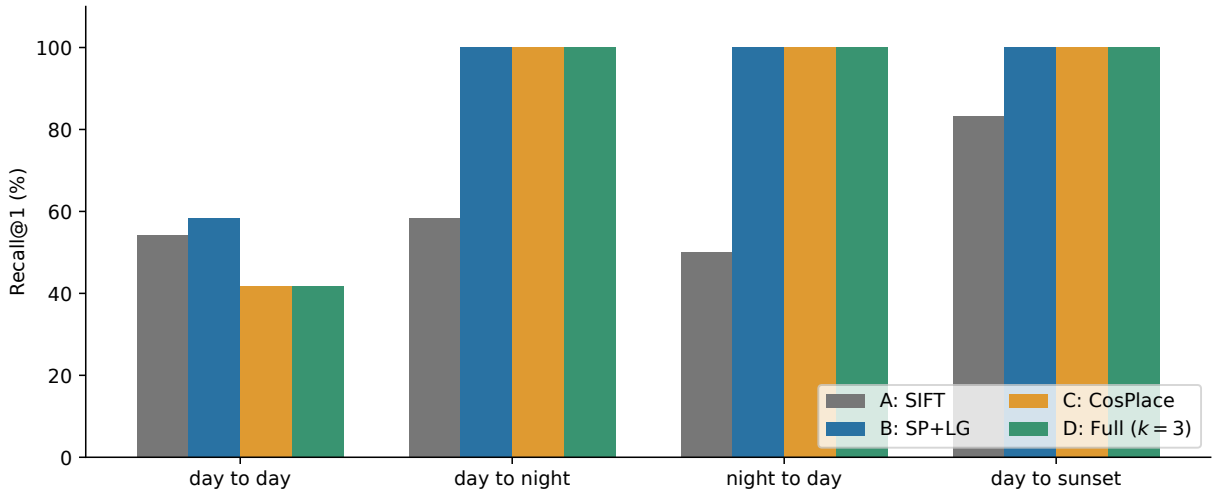

Figure 10.3: Recall@1 on the actual selected Tokyo query photographs. Day-to-day uses a different labelled viewing direction; the other splits use the retained front-facing records across time.

with similar viewpoint can be easier than unchanged daylight with reduced overlap. This supports retaining both kinds of variation in future validation. It also suggests that collecting several reference directions may be more useful than adding another photometric variant of one view, although that proposed collection strategy has not been tested here.

Night-to-day and sunset results are reported separately in the table because descriptor extraction and matching are direction-sensitive. They should not be counted as independent additional cities or sites. Their shared identities and reference observations limit any claim about geographic generalisation. The evidence supports behaviour under the recorded cross-time pairings.

10.4 HPatches retrieval and end-to-end planar success

Table 10.3: Retrieval and correctly verified place rates (percent) on HPatches protocols. Global-only retrieval does not produce an anchor. Full-pipeline recall uses its reranked complete list, with the unverified tail retained.

Split	Method	R@1	R@5	R@10	Verified
illumination	A: SIFT	90.0	98.3	98.3	81.7
illumination	B: SP+LG	98.3	100.0	100.0	98.3
illumination	C: CosPlace	100.0	100.0	100.0	—
illumination	D: Full ($k = 3$)	98.3	100.0	100.0	98.3
viewpoint	A: SIFT	96.7	98.3	100.0	93.3
viewpoint	B: SP+LG	91.7	93.3	93.3	90.0
viewpoint	C: CosPlace	71.7	98.3	100.0	—
viewpoint	D: Full ($k = 3$)	83.3	98.3	100.0	78.3

Table 10.3 evaluates retrieval among the selected HPatches reference images. Its scene-identity

labels are sequence labels, not geographic locations. The task is useful for checking whether a coarse descriptor identifies a plausible reference before geometric verification, but it remains a custom distractor gallery rather than a standard place-recognition leaderboard.

Table 10.4: End-to-end planar success: the selected place must be correct, its geometry accepted, and mean corner error at most five working pixels. This differs from the oracle-pair matching table.

Protocol	Method	n	Correct place (%)	H_5 (%)
HP illum.	A: SIFT	60	81.7	80.0
HP illum.	B: SP+LG	60	98.3	98.3
HP illum.	D: Full ($k = 3$)	60	98.3	98.3
HP view.	A: SIFT	60	93.3	86.7
HP view.	B: SP+LG	60	90.0	86.7
HP view.	D: Full ($k = 3$)	60	78.3	78.3
Synth. day	A: SIFT	18	100.0	100.0
Synth. day	B: SP+LG	18	100.0	100.0
Synth. day	D: Full ($k = 3$)	18	100.0	100.0
Synth. night	A: SIFT	18	94.4	88.9
Synth. night	B: SP+LG	18	100.0	100.0
Synth. night	D: Full ($k = 3$)	18	100.0	100.0
Synth. severe	A: SIFT	18	33.3	33.3
Synth. severe	B: SP+LG	18	100.0	100.0
Synth. severe	D: Full ($k = 3$)	18	100.0	100.0
Synth. warm	A: SIFT	18	88.9	88.9
Synth. warm	B: SP+LG	18	100.0	100.0
Synth. warm	D: Full ($k = 3$)	18	100.0	100.0
Synth. fog	A: SIFT	18	100.0	100.0
Synth. fog	B: SP+LG	18	100.0	100.0
Synth. fog	D: Full ($k = 3$)	18	100.0	100.0

Table 10.4 adds the complete planar-success criterion. A correctly selected identity may still fail the five-pixel condition. Likewise, good oracle-pair accuracy does not guarantee that the end-to-end system finds that reference. This table is the appropriate evidence when the claim concerns an accurately anchored planar post rather than recognition alone.

The comparison also illustrates a property of geometric verification: it can reject a correct place for application-valid reasons. A query may show a small region of the scene with insufficient spatial extent for stable extrapolation to the full reference plane. The appropriate response for the current renderer is to withhold the overlay, even though a broader semantic recogniser might confidently name the street. Future region-specific anchor evaluation could distinguish this case from complete recognition failure.

10.5 Synthetic appearance stress tests

Table 10.5: Retrieval and correctly verified place rates (percent) on synthetic protocols. Global-only retrieval does not produce an anchor. Full-pipeline recall uses its reranked complete list, with the unverified tail retained.

Split	Method	R@1	R@5	R@10	Verified
day	A: SIFT	100.0	100.0	100.0	100.0
day	B: SP+LG	100.0	100.0	100.0	100.0
day	C: CosPlace	100.0	100.0	100.0	—
day	D: Full ($k = 3$)	100.0	100.0	100.0	100.0
night	A: SIFT	100.0	100.0	100.0	94.4
night	B: SP+LG	100.0	100.0	100.0	100.0
night	C: CosPlace	100.0	100.0	100.0	—
night	D: Full ($k = 3$)	100.0	100.0	100.0	100.0
night_severe	A: SIFT	61.1	83.3	83.3	33.3
night_severe	B: SP+LG	100.0	100.0	100.0	100.0
night_severe	C: CosPlace	100.0	100.0	100.0	—
night_severe	D: Full ($k = 3$)	100.0	100.0	100.0	100.0
warm_night	A: SIFT	88.9	100.0	100.0	88.9
warm_night	B: SP+LG	100.0	100.0	100.0	100.0
warm_night	C: CosPlace	100.0	100.0	100.0	—
warm_night	D: Full ($k = 3$)	100.0	100.0	100.0	100.0
fog	A: SIFT	100.0	100.0	100.0	100.0
fog	B: SP+LG	100.0	100.0	100.0	100.0
fog	C: CosPlace	100.0	100.0	100.0	—
fog	D: Full ($k = 3$)	100.0	100.0	100.0	100.0

Table 10.5 reports the seeded transformations as their own collection. Learned methods retain strong source-image recognition under the implemented night stressors, while severe attenuation reduces SIFT's supported decisions. This is consistent with the implementation-level motivation, but the dependence on a small set of base images remains substantial.

Synthetic night retains the exact original scene content and applies a known warp. It cannot reproduce all changes in visibility, illumination direction, light-source placement or sensor response. The fog-like transformation likewise does not establish weather robustness. The actual input examples in Figure 5.2 define the scope of the claim more precisely than the condition names alone.

The three seeds vary random perturbations, not the physical scene population. Treating all generated frames as independent places would overstate evidence. For this reason, the report does not combine synthetic and real-night queries into one headline "night robustness" score. Such an aggregate would obscure both the shared-source dependence and the different ground-truth availability.

10.6 Stage ablations

Table 10.6: Stage ablations on identical queries. Exhaustive learned matching removes global retrieval; the no-geometry variant accepts the highest global result without an estimated transform.

Protocol	Variant	R@1	Correct (%)	Accept (%)
Tokyo D–D	B: SP+LG	58.3	54.2	75.0
Tokyo D–D	Full ($k = 1$)	41.7	37.5	41.7
Tokyo D–D	D: Full ($k = 3$)	41.7	37.5	50.0
Tokyo D–D	Full ($k = 5$)	41.7	41.7	54.2
Tokyo D–D	No geometry	41.7	41.7	100.0
Tokyo D–N	B: SP+LG	100.0	100.0	100.0
Tokyo D–N	Full ($k = 1$)	100.0	100.0	100.0
Tokyo D–N	D: Full ($k = 3$)	100.0	100.0	100.0
Tokyo D–N	Full ($k = 5$)	100.0	100.0	100.0
Tokyo D–N	No geometry	100.0	100.0	100.0
Synth. severe	B: SP+LG	100.0	100.0	100.0
Synth. severe	Full ($k = 1$)	100.0	100.0	100.0
Synth. severe	D: Full ($k = 3$)	100.0	100.0	100.0
Synth. severe	Full ($k = 5$)	100.0	100.0	100.0
Synth. severe	No geometry	100.0	100.0	100.0

Table 10.6 compares candidate limits and stage removal. Exhaustive learned matching provides the no-global-stage reference. Increasing the shortlist permits more comparisons but does not guarantee more correct accepted places: additional candidates can also create competing or incorrect geometric hypotheses. The measured viewpoint split shows that candidate recall and local support must be considered jointly.

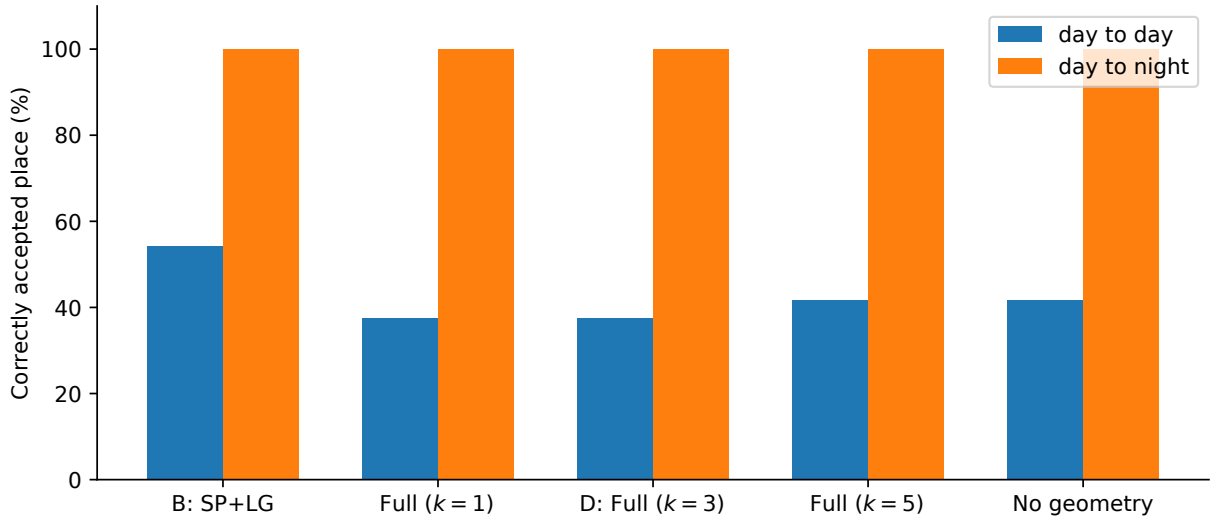

Figure 10.4: Measured correctly accepted-place rates for retrieval and geometry ablations. "No geometry" is a diagnostic acceptance policy without an anchored transform.

Figure 10.4 makes the hierarchy's main trade-off visible. When the global representation

already places the correct reference near the top, a small shortlist retains the observed success of exhaustive learned matching. When viewpoint changes move the useful reference down the ranking, exhaustive search can recover cases missed by the primary hierarchy. The runtime comparison in Chapter 14 quantifies the cost of that additional search.

10.6.1 Unknown places and false acceptance

Table 10.7: Measured open-set diagnostic against a disjoint-collection gallery. Global-only ranking is omitted because ranking alone makes no acceptance decision.

Variant	Unknown queries	False acceptances	Rate (%)
A: SIFT	12	0	0.0
B: SP+LG	12	2	16.7
Full ($k = 1$)	12	0	0.0
D: Full ($k = 3$)	12	0	0.0
Full ($k = 5$)	12	1	8.3
No geometry	12	12	100.0

Table 10.7 reports the unknown-query diagnostic. It is possible for learned local matching to accept a geometrically plausible wrong candidate even when the image collection is disjoint. Increasing the candidate count exposes more such opportunities. A no-geometry policy necessarily accepts a top-ranked candidate for every unknown query in this implementation.

These failures matter for social AR because a false acceptance can display a post at the wrong physical location. The small diagnostic cannot establish a production false-positive bound, particularly for nearby look-alike buildings or adversarial replay. It does demonstrate why retrieval similarity and homography support cannot be treated as complete identity assurance. The discussion of abuse in Chapter 17 follows directly from this measured limitation.

10.7 Short projection training

Table 10.8: Short frozen-backbone projection training, three fixed seeds. Rates are percentages. HPatches rows use only held-out place identities. Intervals are clipped to the feasible percentage range and are unstable with only three seeds.

Protocol	Descriptor	Mean R@1	SD	95% t interval
HP illum.	Pretrained 512	100.0	0.0	100.0–100.0
HP illum.	Head, no night aug.	100.0	0.0	100.0–100.0
HP illum.	Head + night aug.	100.0	0.0	100.0–100.0
HP view.	Pretrained 512	77.5	0.0	77.5–77.5
HP view.	Head, no night aug.	75.8	5.2	62.9–88.8
HP view.	Head + night aug.	74.2	1.4	70.6–77.8
Tokyo D–D	Pretrained 512	41.7	0.0	41.7–41.7

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Table 10.8 (continued)

Protocol	Descriptor	Mean R@1	SD	95% t interval
Tokyo D–D	Head, no night aug.	30.6	2.4	24.6–36.5
Tokyo D–D	Head + night aug.	37.5	7.2	19.6–55.4
Tokyo D–N	Pretrained 512	100.0	0.0	100.0–100.0
Tokyo D–N	Head, no night aug.	94.4	6.4	78.6–100.0
Tokyo D–N	Head + night aug.	94.4	4.8	82.5–100.0
Tokyo D–S	Pretrained 512	100.0	0.0	100.0–100.0
Tokyo D–S	Head, no night aug.	97.2	2.4	91.2–100.0
Tokyo D–S	Head + night aug.	98.6	2.4	92.6–100.0
Tokyo N–D	Pretrained 512	100.0	0.0	100.0–100.0
Tokyo N–D	Head, no night aug.	93.1	4.8	81.1–100.0
Tokyo N–D	Head + night aug.	94.4	4.8	82.5–100.0

Table 10.8 compares the pretrained descriptor with the short projection-head experiments on identical held-out queries. The main outcome is that the selected adaptation does not establish a consistent improvement over the pretrained representation. Some individual seeds recover or improve particular held-out cases, but that is insufficient reason to replace the original descriptor in the primary pipeline.

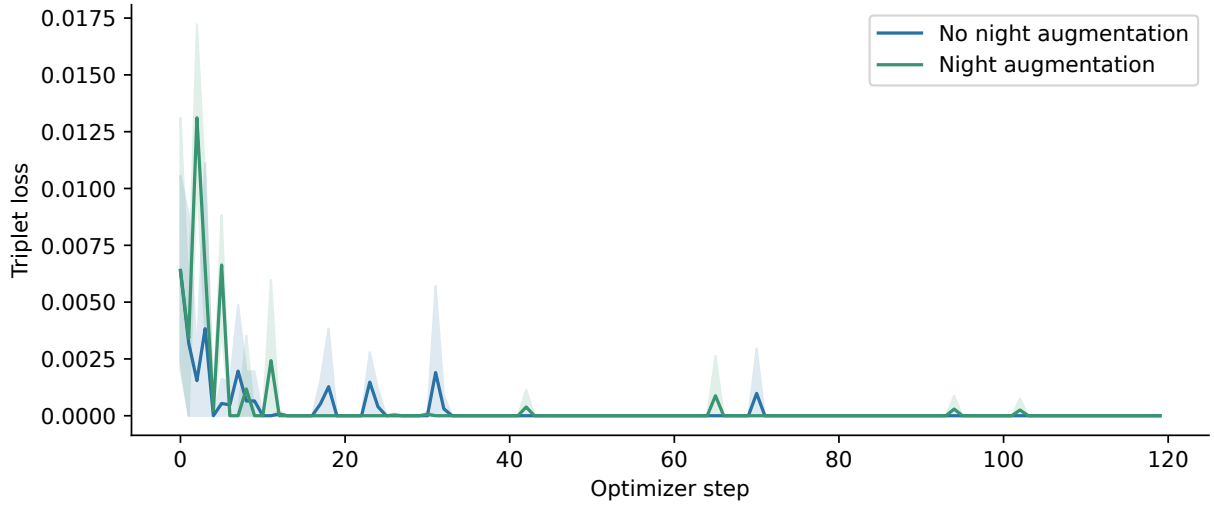

Figure 10.5: Actual triplet-loss traces from the executed frozen-backbone projection experiment. Curves average fixed training seeds and shaded regions show their observed range. Lower training loss is not a held-out retrieval result.

Figure 10.5 shows that the small head can optimise the selected training objective. That does not imply that its compressed embedding preserves all ranking information required by Tokyo or the held-out viewpoint sequences. The experiment changes both dimensionality and training distribution; those effects are not separately identified. A future study should compare an untrained fixed compression control and longer, geographically broader adaptation before attributing changes specifically to synthetic night pairs.

The reported intervals use only three fixed seeds and are therefore broad where the outcomes vary. Deterministic pretrained scores repeated across seed rows have zero seed variance, which does not mean zero data uncertainty. The table separates training randomness from the unmeasured variability of collecting different places. No best-seed checkpoint was selected for the deployment claim.

10.8 Qualitative correspondences and overlays

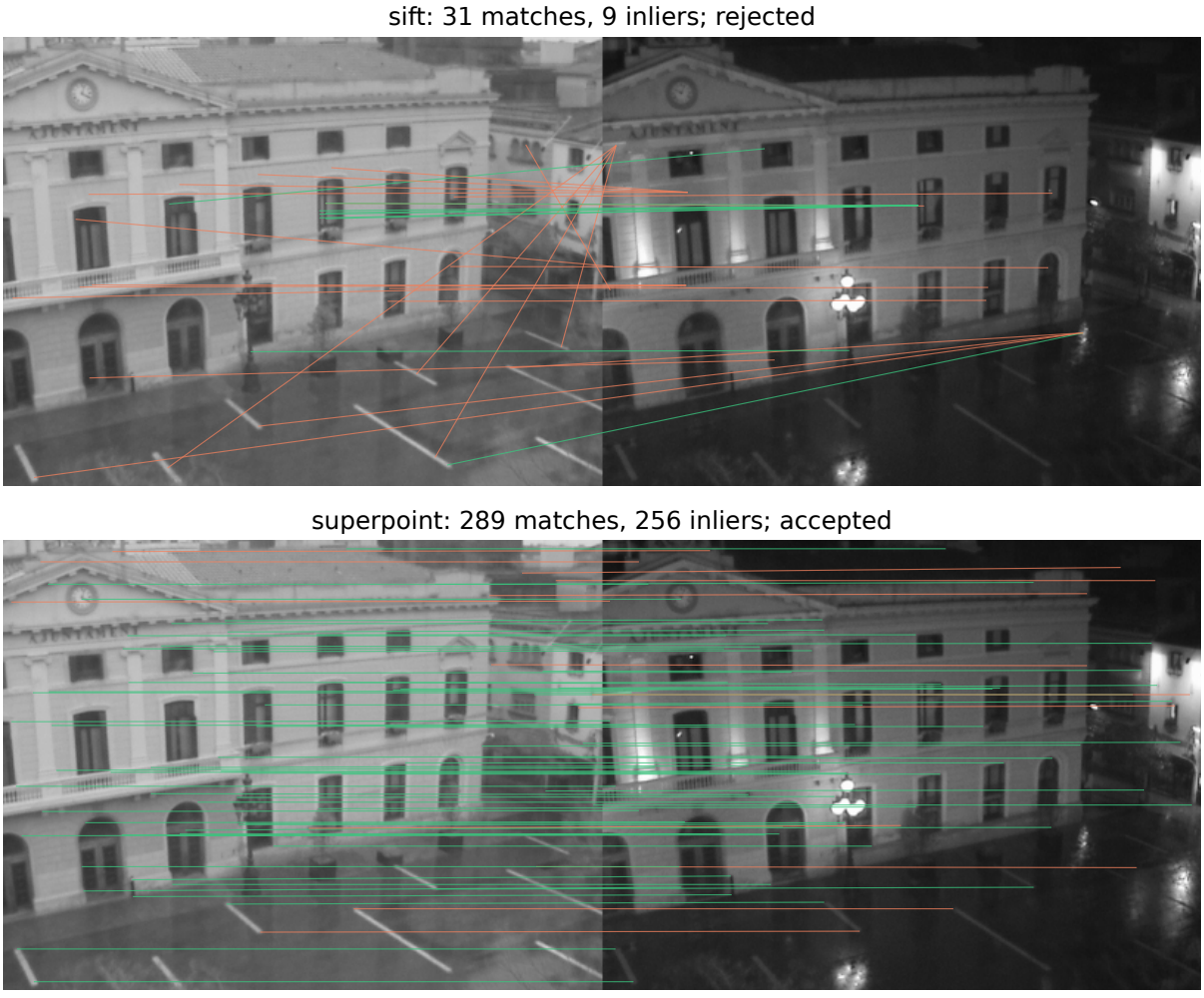

Figure 10.6: Executed correspondence visualisation for the documented illumination example. At most eighty actual tentative correspondences are drawn per panel; green denotes the saved RANSAC inlier mask and orange denotes other tentative matches.

Figure 10.6 shows how the numerical support arises on a selected example. The displayed lines are a deterministic subset of actual correspondences, not an illustration invented independently of inference. Their colour reflects RANSAC compatibility, which should not be confused with an independent manual label for every match.

Figure 10.7 links the geometry to the original application. An accepted map can support an

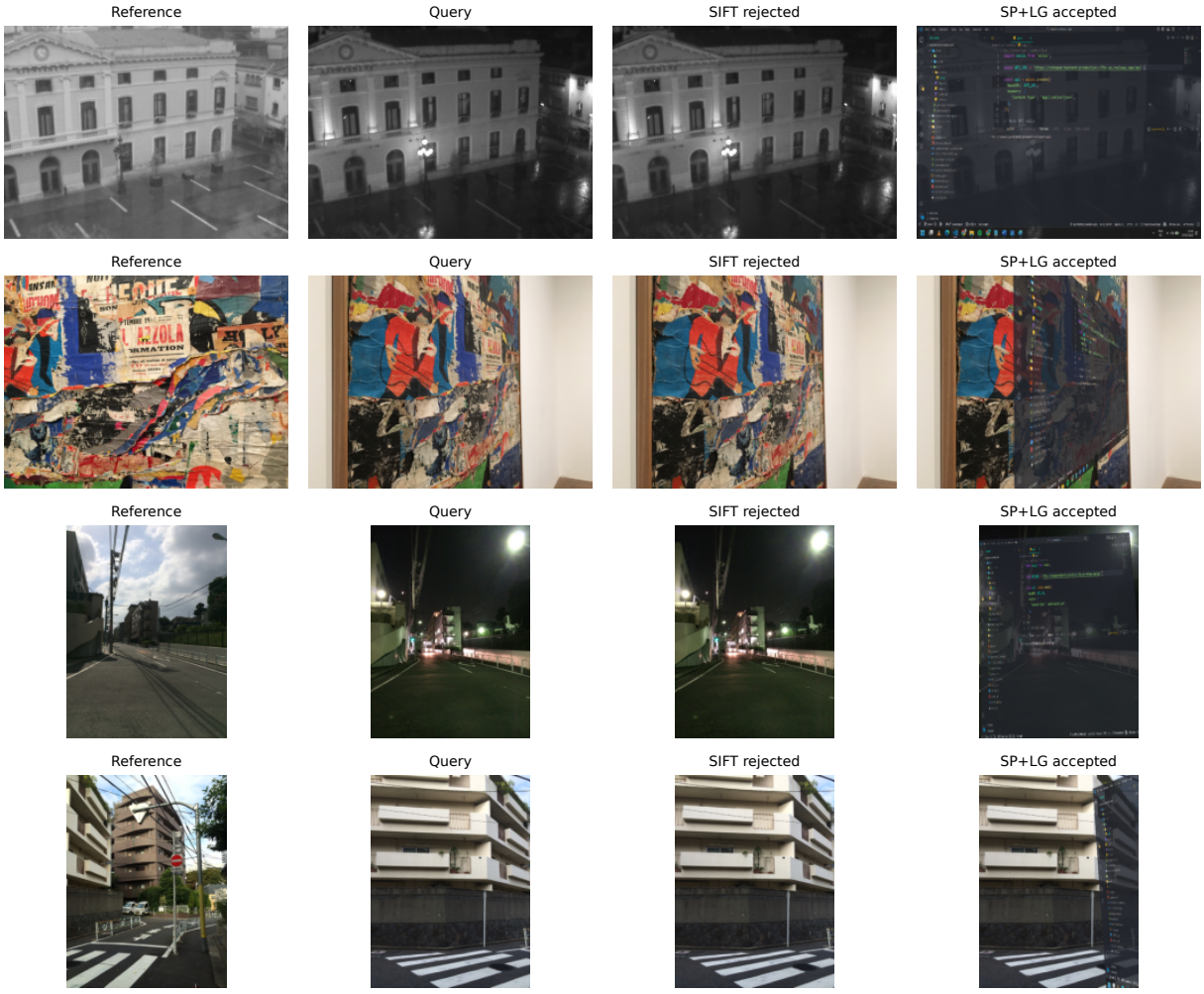

Figure 10.7: Saved-transform overlay examples across illumination, viewpoint and Tokyo protocols. Rejected geometry leaves the query unchanged. Accepted Tokyo overlays remain planar demonstrations without ground-truth pose or full-scene planarity.

attached image, and rejection remains visible rather than being replaced with a ground-truth or manually aligned overlay. A plausible street overlay is still not proof of correct depth, occlusion, or world-scale persistence. Additional match visualisations and their selection rule appear in Appendix C.

The demo uses real selected day and night photographs with generated small two-dimensional camera motion. It is explicitly labelled as generated motion rather than real temporal video captured in those conditions. Both complete pipelines execute on each generated query frame. The bundled source video is separately used for the original-script execution and CLI smoke test. These demonstrations establish implemented data flow and rendering, not high-frequency real-world tracking performance.

10.9 Uncertainty and unavailable measurements

Table 10.9: Finite-sample uncertainty for correctly verified places. These intervals describe this selected set under an independent-site approximation; they do not compensate for geographic selection bias.

Protocol	Method	Correct	95% Wilson interval (%)
Tokyo D–D	A: SIFT	10/24	24.5–61.2
Tokyo D–D	B: SP+LG	13/24	35.1–72.1
Tokyo D–D	D: Full ($k = 3$)	9/24	21.2–57.3
Tokyo D–N	A: SIFT	8/24	18.0–53.3
Tokyo D–N	B: SP+LG	24/24	86.2–100.0
Tokyo D–N	D: Full ($k = 3$)	24/24	86.2–100.0
Tokyo N–D	A: SIFT	8/24	18.0–53.3
Tokyo N–D	B: SP+LG	24/24	86.2–100.0
Tokyo N–D	D: Full ($k = 3$)	24/24	86.2–100.0

Table 10.9 prevents perfect observed rates from being read as certainty. The Wilson intervals remain finite-sample statements about the chosen site set under the stated approximation. Cluster-bootstrap intervals for HPatches are also retained in the generated statistics artifact. Neither procedure removes the selection bias of a small geographic subset or repairs missing calibration.

Table 10.10: Metric localisation results are unavailable, rather than zero. Neither GPS site labels nor pixel homographies establish these errors.

Requested pose criterion	Measured rate	Reason
$(0.25\,\text{m}, 2°)$	—	No calibrated query pose/map
$(0.5\,\text{m}, 5°)$	—	No calibrated query pose/map
$(5\,\text{m}, 10°)$	—	No calibrated query pose/map

Table 10.10 records a requested but unmeasured result. No metric translation-and-rotation rates are inferred from GPS identities or homography corner errors. Likewise, there is no real

seasonal benchmark result, phone power measurement or city-scale service cost. The report distinguishes these absences from actual negative results such as wrong accepted places or reduced retrieval after adaptation.

10.10 Threats to validity

10.10.1 Internal validity

The normalised local comparison shares preprocessing dimensions, feature budgets and geometry settings, but it cannot equalise the internal training data or detector semantics of pretrained models. Approximate descriptor search and floating-point kernels can introduce small run differences. Fixed seeds, exact model hashes and cached pair accounting reduce ambiguity, while the separate original-script runs avoid conflating the refactor with unchanged upstream behaviour.

Some evaluation stage timings were recorded while independent preparation tasks were also active. They are therefore treated as accounting diagnostics. The isolated complete-query profiles are the principal latency evidence. Even those profiles use a small fixed query selection and a warmed process, so they do not describe cold mobile startup or worst-case thermally constrained execution.

10.10.2 Construct validity

Inlier support is not ground-truth correspondence accuracy, accepted geometry is not true alignment, and place recognition is not metric localisation. The thesis reports those constructs separately. The planar corner metric itself measures extrapolation to the reference boundary, which may penalise a locally useful surface match. That is appropriate for a full-reference overlay but should be reconsidered for a smaller user-defined anchor region.

The custom Tokyo gallery changes the benchmark problem. Its perfect aligned cross-time results should be read alongside its viewpoint failures and small denominator. Synthetic night and fog labels describe transformations, not natural capture conditions. The proposed mobile architecture is not an implemented phone deployment. These boundaries are retained in captions and prose to prevent a demonstration from being used as evidence for a stronger construct.

10.10.3 External validity

The selected scenes, geographic region and reference directions do not cover the intended worldwide application. There are no measured seasonal transitions, construction changes, crowds, adversarial displays or long-term map revisions. A production gallery would include many more near-duplicate structures and unknown places. Extrapolating the measured false-acceptance rate or retrieval saturation to that setting would be unjustified.

10.10.4 Statistical conclusion validity

The sample is small and correlated. Three training seeds cannot characterise the full optimisation distribution, and a larger number of synthetic pixels does not create independent physical places. Intervals are presented with their grouping assumptions, not as proof of a universal method ranking. The conclusions emphasise supported direction and failure mechanisms rather than claiming statistical superiority across all deployment conditions.

10.11 Implications for the research questions

The measured evidence supports learned local matching for the selected illumination changes, while providing no uniform viewpoint advantage under the planar criterion. Global retrieval can retain aligned cross-time recognition with fewer local comparisons, but it limits success under the tested direction change. Geometry reduces indiscriminate acceptance yet still produces some wrong-place decisions. Short projection adaptation is not a reliable improvement here. The executed export establishes a practical descriptor-only deployment artifact, whose trade-offs are examined next.

Chapter 11

Failure Taxonomy and Diagnostic Reasoning

11.1 Failures as different observable states

A missing overlay can arise from missing input, weak retrieval, insufficient correspondences, invalid geometry or absent content. Those states require different remedies. Treating all of them as a single tracking failure makes debugging difficult and can encourage an unsafe fallback that renders a stale anchor. This chapter describes a diagnostic framework grounded in the implemented module boundaries and the saved result fields. Proposed remedies are identified as future work unless the delivered code already implements them.

The key distinction is between an unsupported anchor and an incorrectly supported identity. A rejection may reduce usefulness, but it avoids assigning a post to a place without sufficient evidence under the current policy. A wrong acceptance can display plausible content at the wrong physical location. The open-set and viewpoint results show why both outcomes must remain visible in the report.

11.2 Acquisition and preprocessing failures

An unreadable image fails at the decoding boundary. A camera or stream that cannot open fails before feature extraction. A video that opens but produces no decodable frame also fails explicitly in the CLI. These are operational failures, not zero-recall test queries, unless an evaluation protocol deliberately includes acquisition failure in its denominator.

A subtler acquisition error occurs when a frame is valid but its coordinate metadata are wrong. Orientation changes, an unexpected crop or a mismatched resize transform can leave useful visual features while invalidating the ground-truth conversion. The current file-based experiments fix the preprocessing route and retain resize transforms. A phone implementation would need additional tests for sensor orientation, camera intrinsics and display cropping.

Appearance enhancement is another boundary to monitor. Changing enhancement only for

cached references or only for later queries alters descriptor comparability. The global index therefore includes the supported preprocessing configuration in its identity. A future adaptive enhancement policy should be evaluated as a separate method rather than introduced silently after inspecting difficult queries.

11.3 Descriptor ambiguity and gallery coverage

A global descriptor compresses an image into a vector and can discard information needed to distinguish nearby repetitive views. When the correct identity ranks outside the shortlist, local verification never sees it. The full hierarchy may then reject or accept a competing reference. This failure is different from a local matcher failing on a supplied true pair, and it cannot be diagnosed from the final inlier count alone.

The Tokyo changed-direction split is particularly relevant because the reference and query can have limited shared visual content. The complete site atlas shows the retained front and left photographs alongside their ranks. Those photographs support inspection of reference coverage, but the report does not infer a numerical overlap percentage from visual judgement. A future study could annotate overlap independently and analyse it as a covariate.

Adding references is a plausible remedy, yet it creates new responsibilities. More references can improve coverage while increasing storage and the number of competing hypotheses. Posts must remain tied to the view on which their polygon was defined, or be transferred through a validated relation. Simply duplicating a place ID across photographs does not make an existing polygon geometrically portable.

11.4 Insufficient or unstable local evidence

Low texture, blur, darkness or strong viewpoint change can reduce the number of repeatable local features. The implementation represents empty feature sets explicitly and returns an empty match list with stable dimensions. Geometry then rejects insufficient support. This behaviour prevents an application exception from being confused with a successful query that happened to return no content.

The ratio filter for SIFT and the learned assignment policy for LightGlue do not produce interchangeable correspondence populations. A larger inlier count for one method can partly reflect the number and distribution of tentative matches it supplies. The pair tables therefore include acceptance and independent corner-error success. They allow a reader to identify a case with strong support but poor alignment without treating support count as a universal quality score.

Spatial concentration is an additional limitation. A dozen inliers in one small repeated region can fit a transformation that extrapolates badly to the reference boundary. The current projected-boundary checks reject some implausible maps, but they are not a calibrated coverage test. A future spatial-support rule should measure point distribution and be validated on separate data

before replacing the fixed comparison policy.

11.5 Consensus at the wrong identity

Geometric verification tests whether tentative correspondences support a model under specified thresholds. It does not directly test the semantic uniqueness of the place. Similar facades, signs or repeated architectural motifs can generate a consistent but wrong consensus. The correct response in evaluation is to retain that outcome as wrong-place acceptance, even when the fitted transformation has many inliers.

The open-set diagnostic is deliberately narrow: HPatches queries are searched against a Tokyo gallery. It establishes that unrelated-collection inputs can still trigger acceptance in some variants. It does not bound the false-acceptance rate for nearby repeated buildings, screenshots of known references or maliciously crafted inputs. Those negative populations would require their own explicitly acquired and labelled test sets.

A production policy could combine geometric support with reference identity margins, multiple observations or a coarse location prior. Each addition introduces new failure modes and privacy choices. A location prior can exclude the correct place when inaccurate, and temporal confirmation can preserve a wrong identity if all frames depict the same repeated structure. These mechanisms should be assessed through whole-session error rates, not assumed to solve the identity problem by construction.

11.6 Correct identity with inaccurate alignment

On HPatches and synthetic data, a query can identify the correct source yet fail the H5 criterion. Possible reasons include noisy correspondences, a biased spatial distribution or an unstable estimate. The end-to-end metric separates that case from a wrong identity and from an outright rejection. The oracle-pair table further indicates whether candidate search was responsible for losing a potentially accurate relation.

On Tokyo, the equivalent alignment claim cannot be measured with the available labels. A correctly verified place is therefore not renamed a successful planar anchor. The site tables preserve that uncertainty. This is a substantive limit of the evidence rather than an omission that can be repaired by drawing a more convincing overlay.

The displayed demo is selected by a documented illustrative rule. Its generated motion is applied to real still images, and actual pipeline outputs are rendered. That makes it useful for inspecting the implemented path. It remains insufficient for measuring temporal drift, recovery after occlusion or alignment under real camera translation, because those phenomena are not supplied by generated two-dimensional motion alone.

11.7 Content and cache failures

A verified place may have no post associated with the selected reference view. Returning an empty content list is then valid database behaviour. Copying another view's polygon to avoid an empty result would create a geometric error. The implemented binding rule defaults a post to the place's first reference unless an explicit reference field selects another photograph.

Descriptor caches can fail more silently. A matrix can have the expected shape while containing values computed from an earlier checkpoint or different resize policy. The corrected index records a checkpoint hash, preprocessing identity, reference fingerprints and format version. It rejects incompatible or nonfinite data. The reference list and fingerprint count are checked before the matrix is used for ranking.

These protections address consistency, not all forms of integrity. A service would need controlled model distribution, authenticated content changes and an atomic update policy. The local JSON/NumPy artifact is intentionally inspectable and does not claim concurrent transactional guarantees. Its failure checks should be understood as research-software contracts rather than a complete multi-user storage system.

11.8 Tracing a failure without changing the benchmark

Begin with the saved query identity and protocol. Check that the image hash and configuration match the frozen run. Inspect the correct identity's global rank, then determine whether it entered the configured shortlist. If it did, inspect the corresponding local pair row, support ratio and acceptance. On ground-truth collections, finally compare corner error with the declared threshold.

This sequence avoids a common diagnostic mistake: changing a threshold or reference after seeing a failure and then presenting the repaired case as if it belonged to the original experiment. Repairs can be useful engineering work, but they require a new configuration and a clearly separated result artifact. The original row should remain available so the reader can understand what changed.

The complete evidence chapters support this workflow without requiring readers to rerun all neural inference. They provide every selected scene, all main-method Tokyo decisions and every oracle-pair result. The associated machine-readable files retain the precise maps and ranked identities when a deeper inspection is needed. Interpretation can then proceed from the recorded evidence rather than from a selected screenshot alone.

Chapter 12

Complete HPatches Sequence Evidence

12.1 How to read this evidence atlas

This chapter covers every selected sequence rather than only a favourable qualitative example. The six photographs in each panel are the actual reference and five evaluated targets. The first table reports saved oracle-pair estimates, while the second reports end-to-end retrieval and accepted decisions. A correct place can still have an inaccurate homography, and an accurate oracle pair can be lost during candidate search. All counts retain rejected queries in their denominators. These are additional views of the same experiments, not independent new trials. HPatches supplies the source homographies [3]; the selected protocol is defined in Chapter 5.

In the pair tables, I is the saved RANSAC inlier count, E is mean four-corner error in working pixels, and H5 requires both accepted geometry and E at most five pixels. A dash denotes an unavailable finite error, not a zero error. End-to-end H5 additionally requires a correctly selected place. The SP+LG label abbreviates SuperPoint with LightGlue.

12.2 Sequence i_ajuntament

This sequence belongs to the illumination subset and contributes five reference-to-target pairs. Its identity is reserved for the projection-head training split; learned components in the main A–D comparison retain their original pretrained weights in either case.

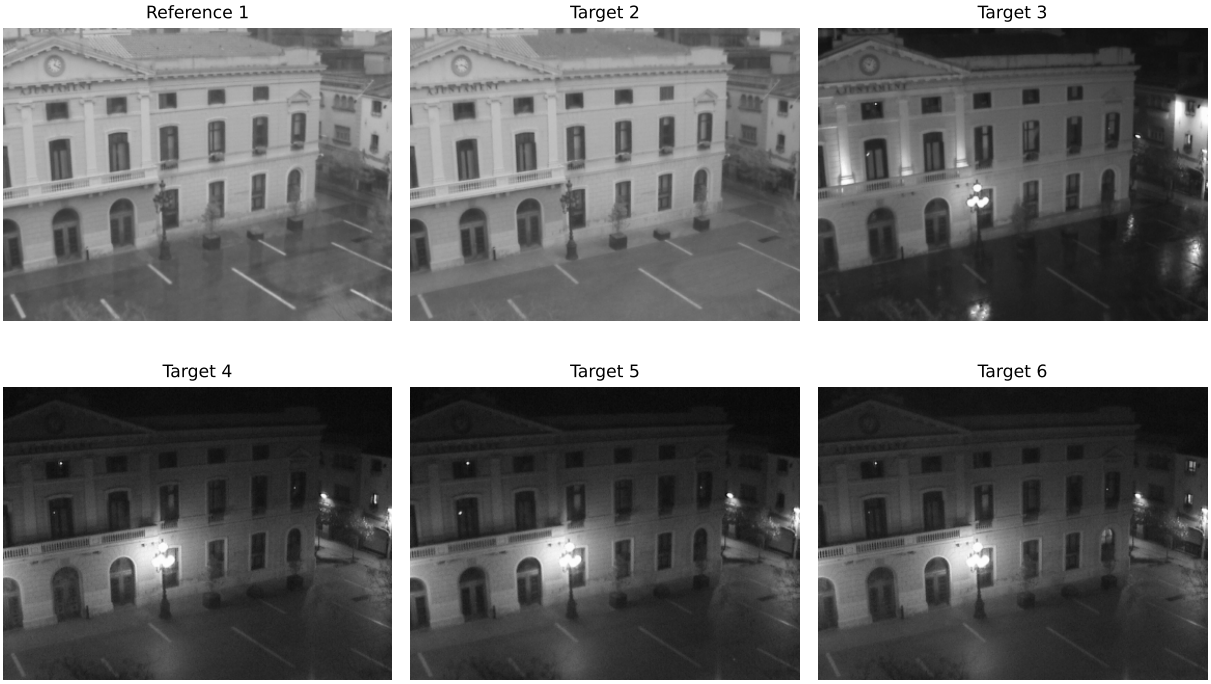

Figure 12.1: Actual reference and all evaluated targets for i_ajuntament.

Table 12.1: Saved oracle-pair support and corner error for i_ajuntament.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	176	0.50	yes	366	0.71	yes
3	9	–	no	256	0.85	yes
4	5	–	no	193	0.93	yes
5	12	–	no	179	3.86	yes
6	7	–	no	193	2.33	yes

SIFT obtains 1 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 4 learned-only successes and 0 SIFT-only successes. Median inlier counts are 9.0 and 193.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.2: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	3/5	1/5	1/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle

table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.3 Sequence i_autannes

This sequence belongs to the illumination subset and contributes five reference-to-target pairs. Its identity is reserved for the projection-head training split; learned components in the main A–D comparison retain their original pretrained weights in either case.

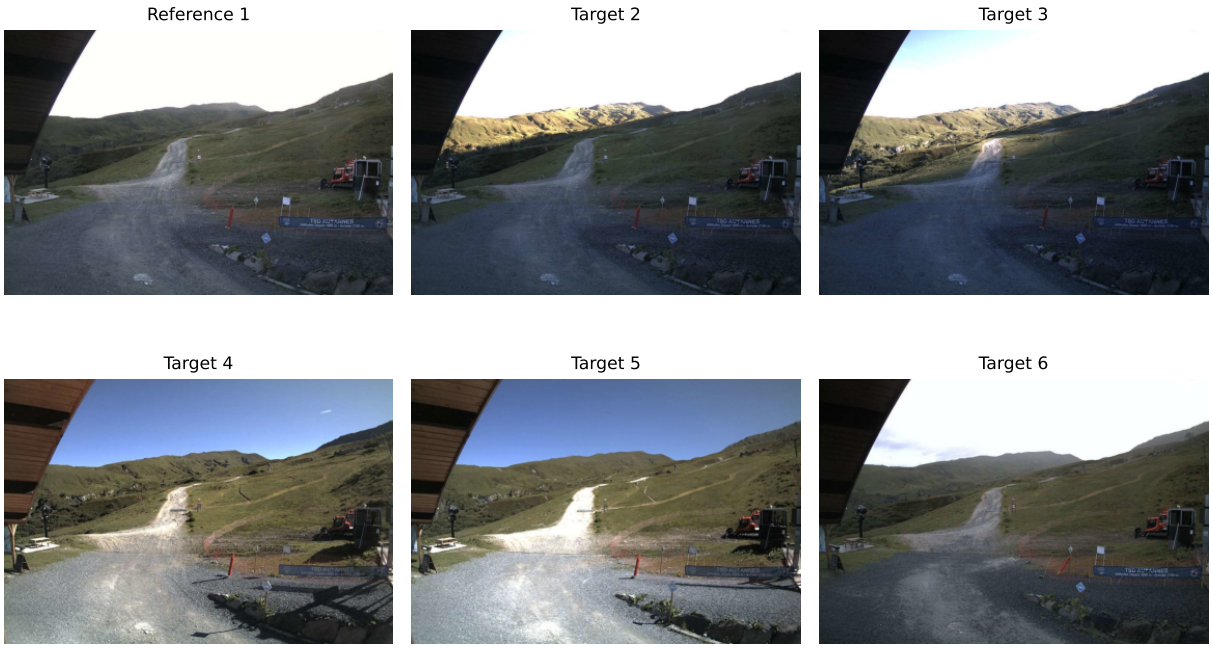

Figure 12.2: Actual reference and all evaluated targets for i_autannes.

Table 12.3: Saved oracle-pair support and corner error for i_autannes.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	122	0.39	yes	335	0.54	yes
3	60	6.79	no	305	0.83	yes
4	19	1.25	yes	173	3.11	yes
5	9	2.38	no	221	1.46	yes
6	57	1.56	yes	315	0.61	yes

SIFT obtains 3 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 2 learned-only successes and 0 SIFT-only successes. Median inlier counts are 57.0 and 305.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.4: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	4/5	3/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.4 Sequence i_bologna

This sequence belongs to the illumination subset and contributes five reference-to-target pairs. Its identity is reserved for the projection-head training split; learned components in the main A–D comparison retain their original pretrained weights in either case.

Figure 12.3: Actual reference and all evaluated targets for i_bologna.

Table 12.5: Saved oracle-pair support and corner error for i_bologna.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	80	0.80	yes	366	0.64	yes
3	43	1.85	yes	349	0.38	yes
4	48	0.99	yes	336	0.65	yes
5	48	2.28	yes	261	1.82	yes
6	35	1.03	yes	308	1.31	yes

SIFT obtains 5 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 0 learned-only successes and 0 SIFT-only successes. Median inlier counts are 48.0 and 336.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.6: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	5/5	5/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.5 Sequence i_books

This sequence belongs to the illumination subset and contributes five reference-to-target pairs. Its identity is reserved for the projection-head training split; learned components in the main A–D comparison retain their original pretrained weights in either case.

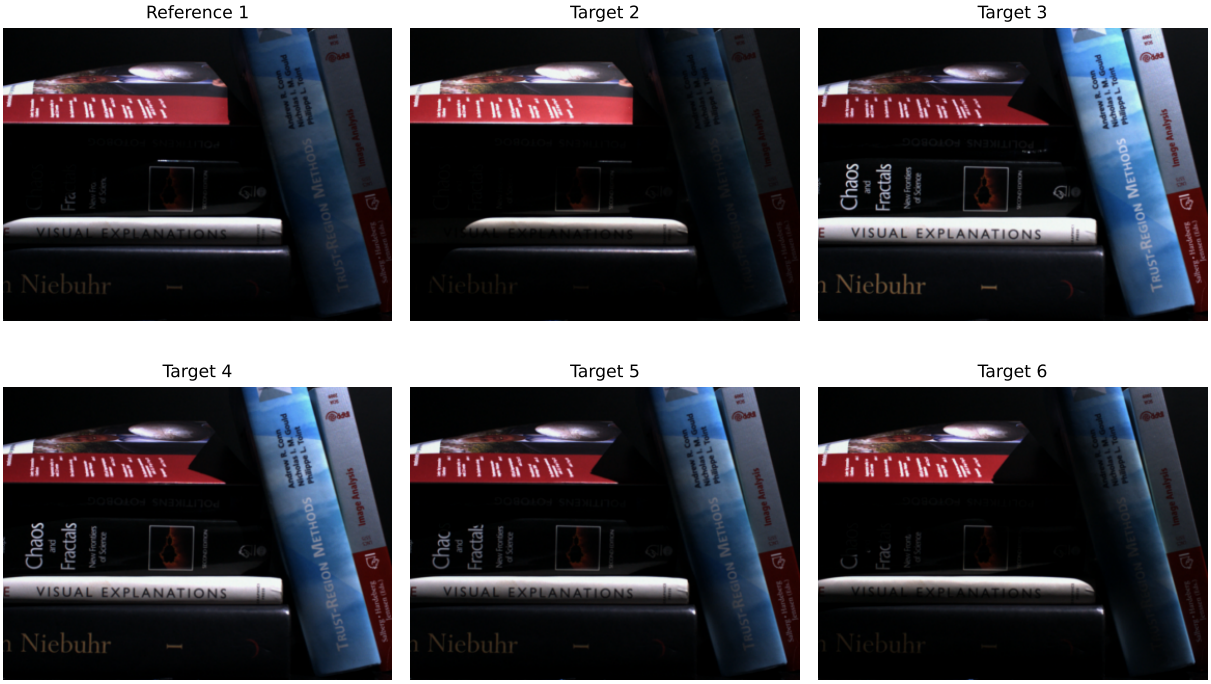

Figure 12.4: Actual reference and all evaluated targets for i_books.

Table 12.7: Saved oracle-pair support and corner error for i_books.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	188	0.76	yes	421	0.20	yes
3	164	0.68	yes	429	0.24	yes
4	202	0.89	yes	438	0.31	yes
5	234	0.09	yes	441	0.21	yes
6	267	0.33	yes	431	0.27	yes

SIFT obtains 5 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 0 learned-only successes and 0 SIFT-only successes. Median inlier counts are 202.0 and 431.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.8: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	5/5	5/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle

table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.6 Sequence i_boutique

This sequence belongs to the illumination subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

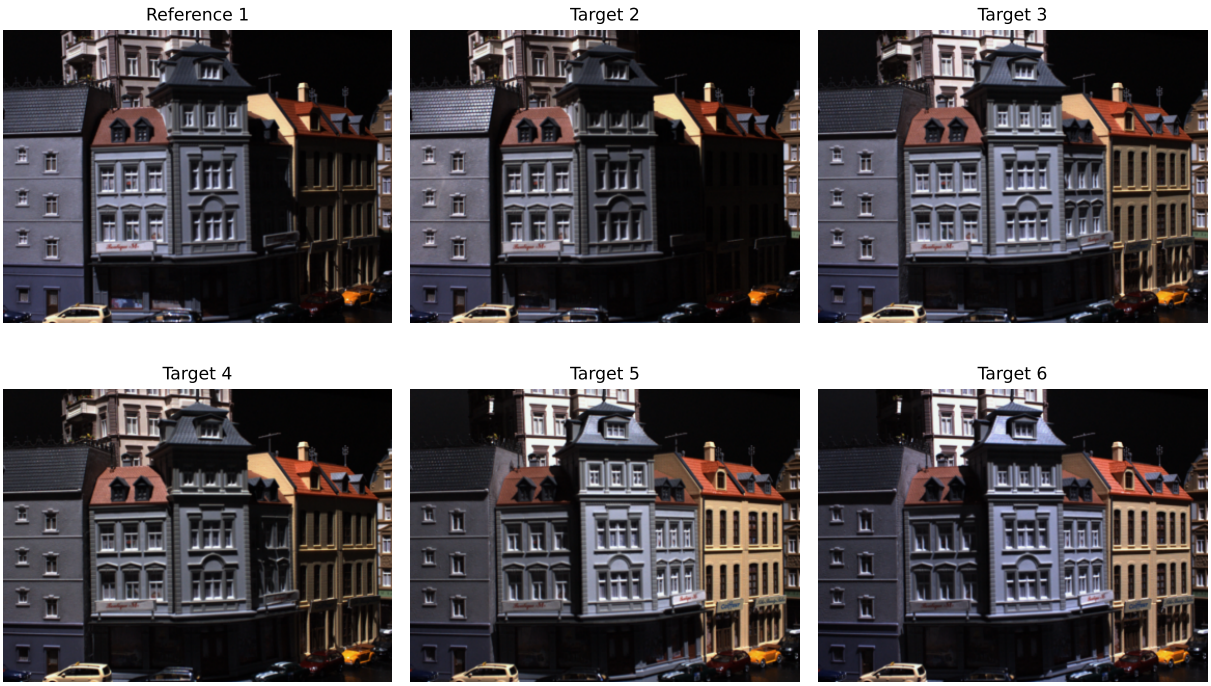

Figure 12.5: Actual reference and all evaluated targets for i_boutique.

Table 12.9: Saved oracle-pair support and corner error for i_boutique.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	157	0.27	yes	371	0.32	yes
3	51	0.29	yes	272	0.45	yes
4	70	0.68	yes	317	0.51	yes
5	7	–	no	248	0.66	yes
6	8	0.57	no	226	0.80	yes

SIFT obtains 3 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 2 learned-only successes and 0 SIFT-only successes. Median inlier counts

are 51.0 and 272.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.10: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	4/5	3/5	3/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.7 Sequence i_bridger

This sequence belongs to the illumination subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

Figure 12.6: Actual reference and all evaluated targets for i_bridger.

Table 12.11: Saved oracle-pair support and corner error for i_bridger.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	127	0.43	yes	339	0.29	yes
3	99	0.42	yes	318	0.45	yes
4	89	0.23	yes	341	0.39	yes
5	56	1.25	yes	308	0.61	yes
6	4	–	no	172	1.27	yes

SIFT obtains 4 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 1 learned-only success and 0 SIFT-only successes. Median inlier counts are 89.0 and 318.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.12: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	4/5	4/5	4/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.8 Sequence i_brooklyn

This sequence belongs to the illumination subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

Figure 12.7: Actual reference and all evaluated targets for i_brooklyn.

Table 12.13: Saved oracle-pair support and corner error for i_brooklyn.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	140	0.71	yes	296	0.48	yes
3	232	1.73	yes	396	1.62	yes
4	15	2.53	yes	295	1.03	yes
5	244	2.00	yes	411	1.96	yes
6	121	2.45	yes	322	1.85	yes

SIFT obtains 5 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 0 learned-only successes and 0 SIFT-only successes. Median inlier counts are 140.0 and 322.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.14: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	5/5	5/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore

a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.9 Sequence i_castle

This sequence belongs to the illumination subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

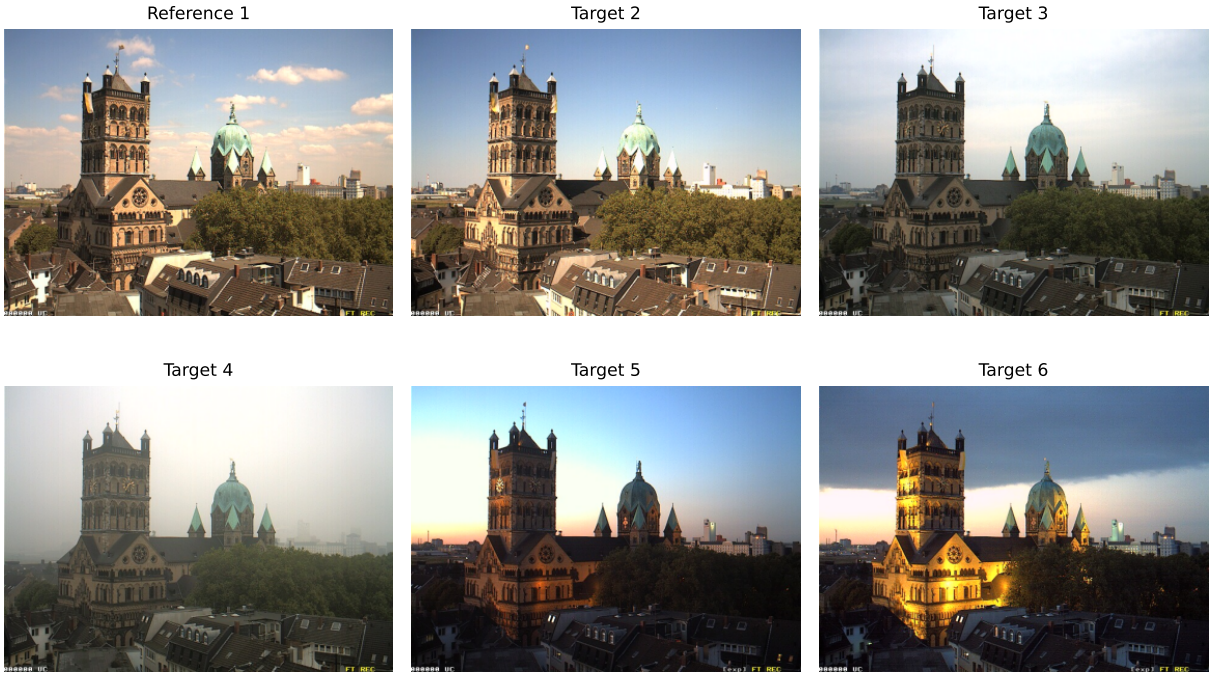

Figure 12.8: Actual reference and all evaluated targets for i_castle.

Table 12.15: Saved oracle-pair support and corner error for i_castle.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	12	3.85	yes	282	1.30	yes
3	49	0.96	yes	359	0.83	yes
4	53	1.19	yes	299	1.14	yes
5	10	5.27	no	295	0.64	yes
6	12	1.02	yes	290	0.48	yes

SIFT obtains 4 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 1 learned-only success and 0 SIFT-only successes. Median inlier counts are 12.0 and 295.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.16: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	4/5	4/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.10 Sequence i_chestnuts

This sequence belongs to the illumination subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

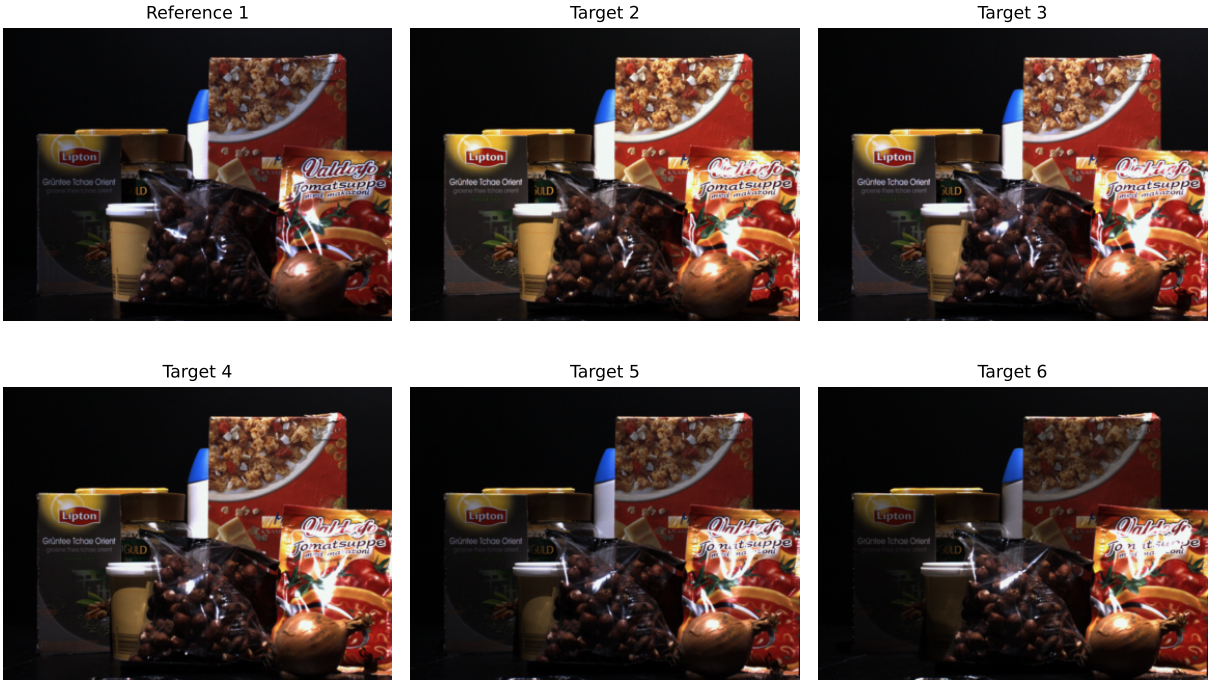

Figure 12.9: Actual reference and all evaluated targets for i_chestnuts.

Table 12.17: Saved oracle-pair support and corner error for i_chestnuts.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	153	0.25	yes	317	0.73	yes
3	150	0.36	yes	370	0.23	yes
4	155	0.25	yes	342	0.28	yes
5	121	0.42	yes	317	0.31	yes
6	56	0.63	yes	273	0.49	yes

SIFT obtains 5 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 0 learned-only successes and 0 SIFT-only successes. Median inlier counts are 150.0 and 317.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.18: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	5/5	5/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.11 Sequence i_contruction

This sequence belongs to the illumination subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

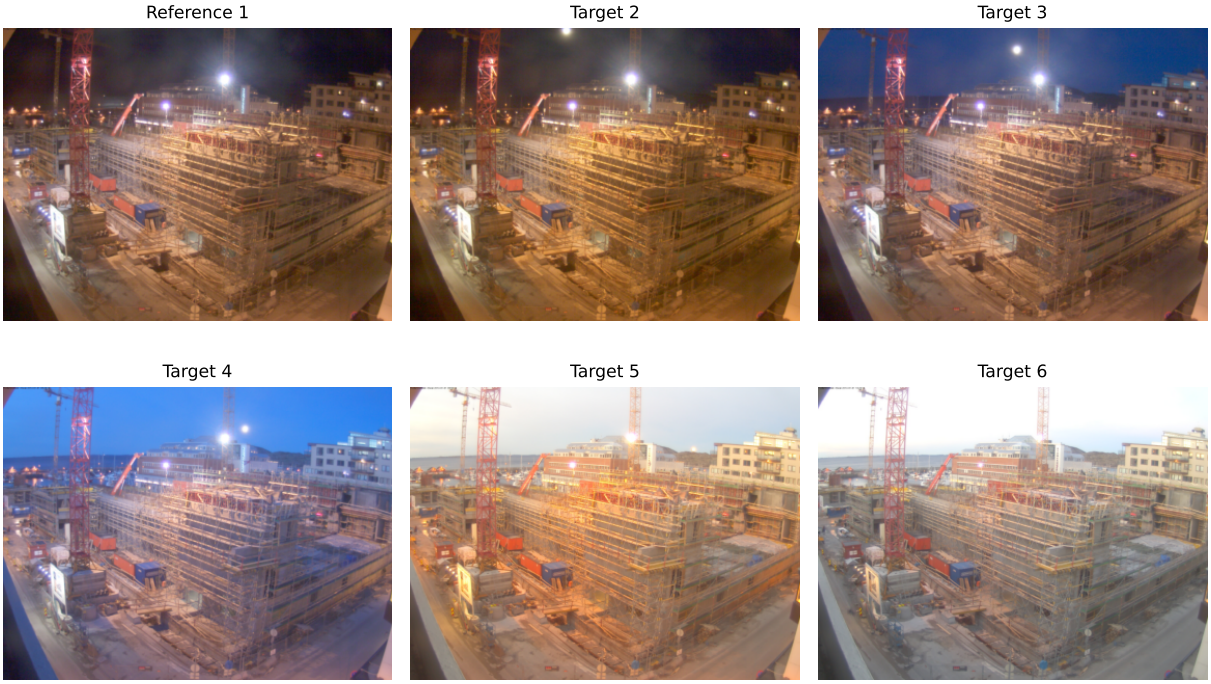

Figure 12.10: Actual reference and all evaluated targets for i_contruction.

Table 12.19: Saved oracle-pair support and corner error for i_contruction.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	424	0.05	yes	462	0.07	yes
3	414	0.05	yes	449	0.28	yes
4	276	0.32	yes	392	0.15	yes
5	70	0.46	yes	253	0.46	yes
6	29	1.11	yes	236	0.54	yes

SIFT obtains 5 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 0 learned-only successes and 0 SIFT-only successes. Median inlier counts are 276.0 and 392.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.20: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	5/5	5/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle

table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.12 Sequence i_crownday

This sequence belongs to the illumination subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

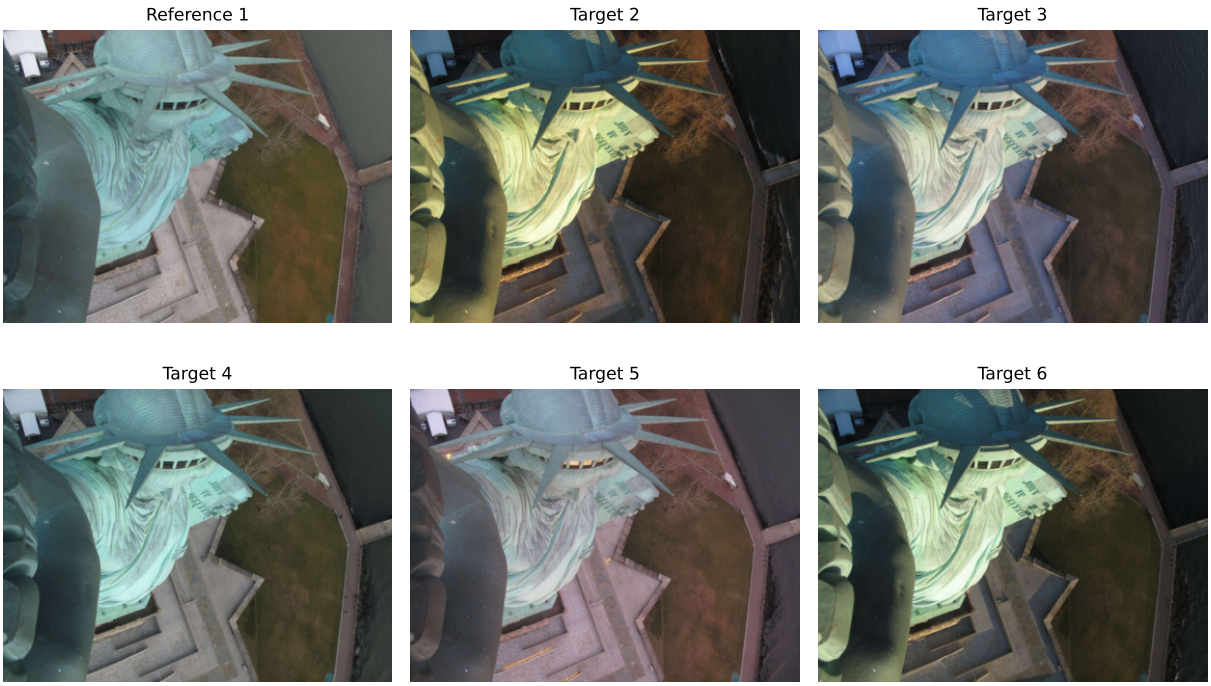

Figure 12.11: Actual reference and all evaluated targets for i_crownday.

Table 12.21: Saved oracle-pair support and corner error for i_crownday.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	20	1.29	yes	272	0.48	yes
3	37	2.11	yes	325	0.63	yes
4	82	0.60	yes	350	0.37	yes
5	181	0.68	yes	373	0.76	yes
6	22	1.60	yes	274	0.29	yes

SIFT obtains 5 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 0 learned-only successes and 0 SIFT-only successes. Median inlier counts

are 37.0 and 325.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.22: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	5/5	5/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.13 Sequence i_crownnight

This sequence belongs to the illumination subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

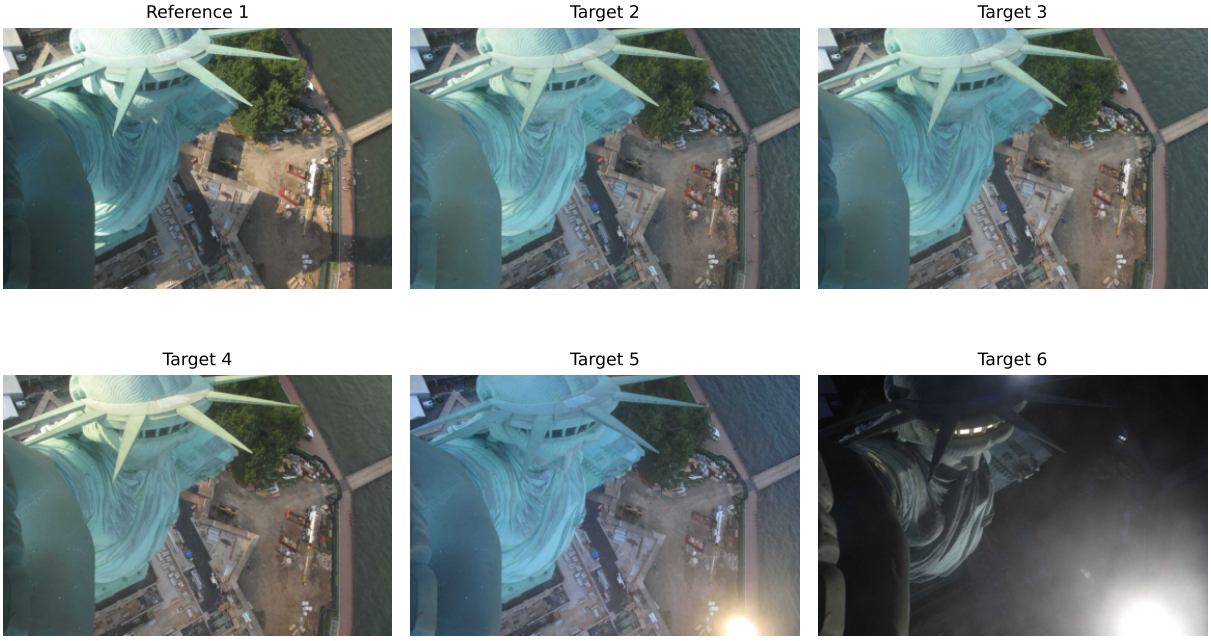

Figure 12.12: Actual reference and all evaluated targets for i_crownnight.

Table 12.23: Saved oracle-pair support and corner error for i_crownnight.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	79	0.28	yes	343	0.47	yes
3	65	0.78	yes	308	0.77	yes
4	54	1.45	yes	327	0.20	yes
5	34	0.33	yes	318	0.37	yes
6	0	–	no	119	1.95	yes

SIFT obtains 4 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 1 learned-only success and 0 SIFT-only successes. Median inlier counts are 54.0 and 318.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.24: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	3/5	3/5	3/5
SP+LG	4/5	4/5	4/5
Full	4/5	4/5	4/5

The full hierarchy correctly verifies 4 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.14 Sequence v_abstract

This sequence belongs to the viewpoint subset and contributes five reference-to-target pairs. Its identity is reserved for the projection-head training split; learned components in the main A–D comparison retain their original pretrained weights in either case.

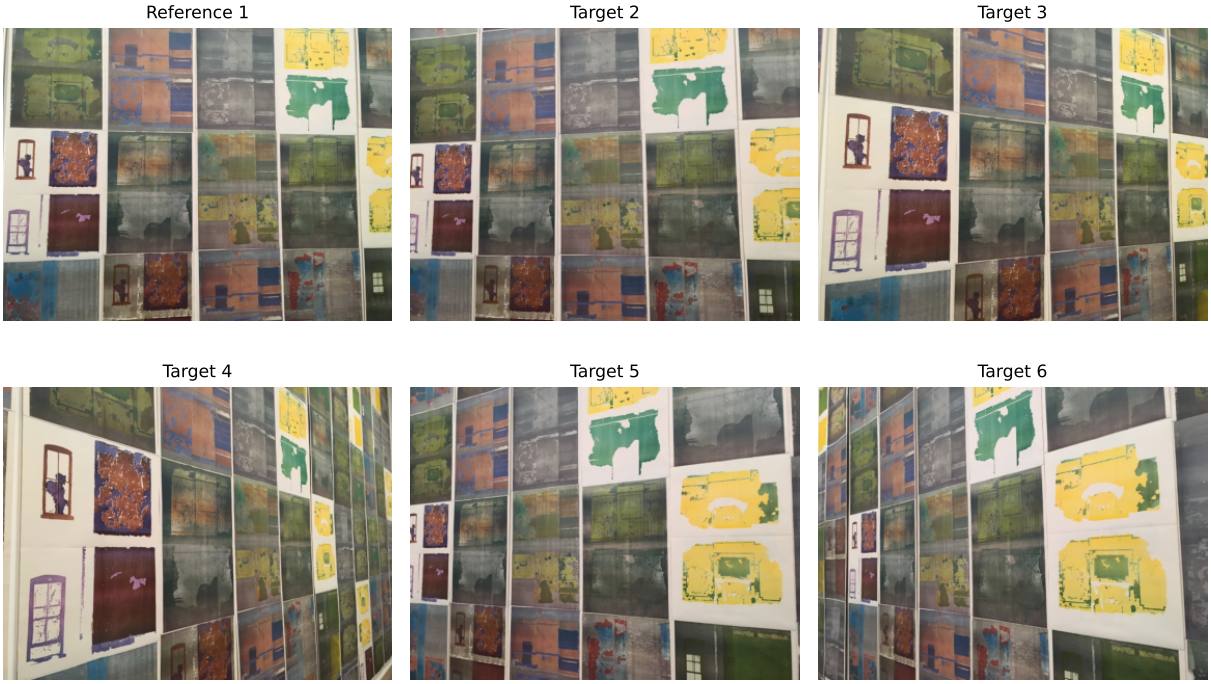

Figure 12.13: Actual reference and all evaluated targets for v_abstract.

Table 12.25: Saved oracle-pair support and corner error for v_abstract.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	226	0.38	yes	373	0.20	yes
3	181	0.61	yes	353	0.37	yes
4	86	2.60	yes	215	1.96	yes
5	130	1.50	yes	279	0.67	yes
6	49	3.80	yes	189	1.80	yes

SIFT obtains 5 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 0 learned-only successes and 0 SIFT-only successes. Median inlier counts are 130.0 and 279.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.26: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	5/5	5/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle

table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.15 Sequence v_adam

This sequence belongs to the viewpoint subset and contributes five reference-to-target pairs. Its identity is reserved for the projection-head training split; learned components in the main A–D comparison retain their original pretrained weights in either case.

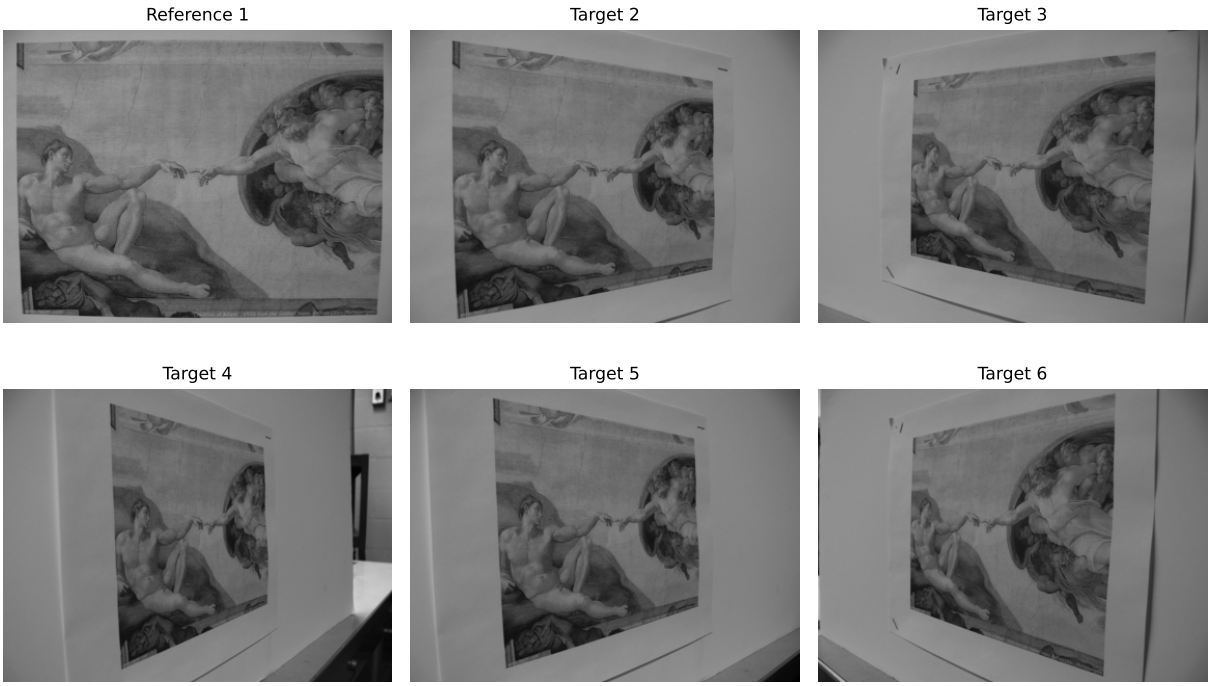

Figure 12.14: Actual reference and all evaluated targets for v_adam.

Table 12.27: Saved oracle-pair support and corner error for v_adam.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	231	0.90	yes	375	0.53	yes
3	173	0.86	yes	377	0.79	yes
4	54	2.58	yes	259	1.08	yes
5	135	1.66	yes	288	1.72	yes
6	102	1.28	yes	320	1.07	yes

SIFT obtains 5 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 0 learned-only successes and 0 SIFT-only successes. Median inlier counts

are 135.0 and 320.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.28: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	5/5	5/5
SP+LG	5/5	5/5	5/5
Full	4/5	4/5	4/5

The full hierarchy correctly verifies 4 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.16 Sequence v_apprentices

This sequence belongs to the viewpoint subset and contributes five reference-to-target pairs. Its identity is reserved for the projection-head training split; learned components in the main A–D comparison retain their original pretrained weights in either case.

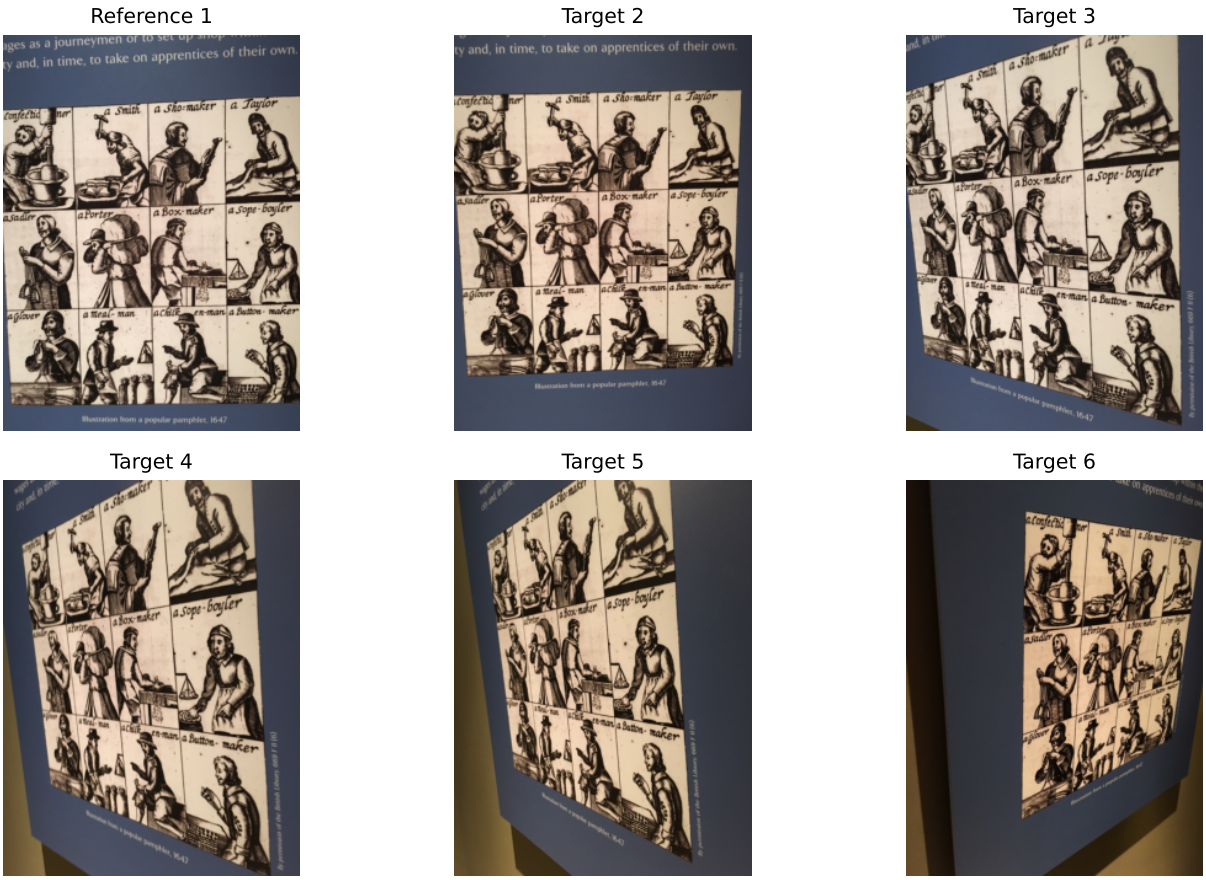

Figure 12.15: Actual reference and all evaluated targets for v_apprentices.

Table 12.29: Saved oracle-pair support and corner error for v_apprentices.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	302	0.38	yes	392	0.22	yes
3	223	0.92	yes	320	0.44	yes
4	104	1.93	yes	282	0.41	yes
5	45	4.00	yes	258	0.34	yes
6	41	0.96	yes	279	0.47	yes

SIFT obtains 5 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 0 learned-only successes and 0 SIFT-only successes. Median inlier counts are 104.0 and 282.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.30: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	5/5	5/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.17 Sequence v_artisans

This sequence belongs to the viewpoint subset and contributes five reference-to-target pairs. Its identity is reserved for the projection-head training split; learned components in the main A–D comparison retain their original pretrained weights in either case.

Figure 12.16: Actual reference and all evaluated targets for v_artisans.

Table 12.31: Saved oracle-pair support and corner error for v_artisans.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	125	0.42	yes	243	0.35	yes
3	87	1.20	yes	209	0.93	yes
4	84	4.22	yes	248	1.54	yes
5	8	–	no	109	–	no
6	116	0.40	yes	260	0.36	yes

SIFT obtains 4 of five successful oracle homographies; SP+LG obtains 4 of five. The paired comparison contains 0 learned-only successes and 0 SIFT-only successes. Median inlier counts are 87.0 and 243.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.32: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	4/5	4/5
SP+LG	5/5	4/5	4/5
Full	4/5	3/5	3/5

The full hierarchy correctly verifies 3 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.18 Sequence v_astronautis

This sequence belongs to the viewpoint subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

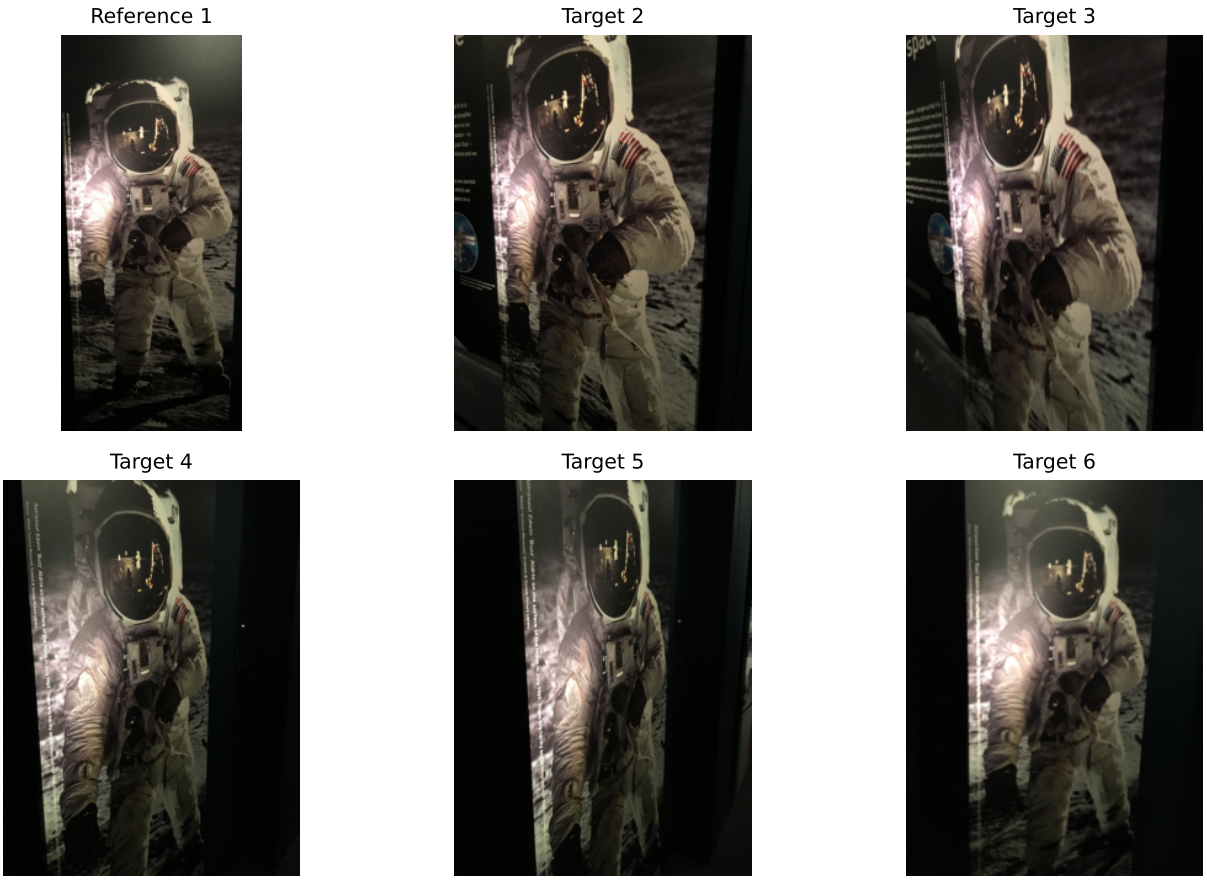

Figure 12.17: Actual reference and all evaluated targets for v_astronautis.

Table 12.33: Saved oracle-pair support and corner error for v_astronautis.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	162	1.35	yes	307	1.02	yes
3	51	5.82	no	236	1.08	yes
4	83	1.10	yes	305	0.72	yes
5	21	2.80	yes	203	7.21	no
6	160	1.12	yes	298	3.44	yes

SIFT obtains 4 of five successful oracle homographies; SP+LG obtains 4 of five. The paired comparison contains 1 learned-only success and 1 SIFT-only success. Median inlier counts are 83.0 and 298.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.34: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	5/5	4/5
SP+LG	5/5	5/5	4/5
Full	3/5	3/5	3/5

The full hierarchy correctly verifies 3 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.19 Sequence v_azzola

This sequence belongs to the viewpoint subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

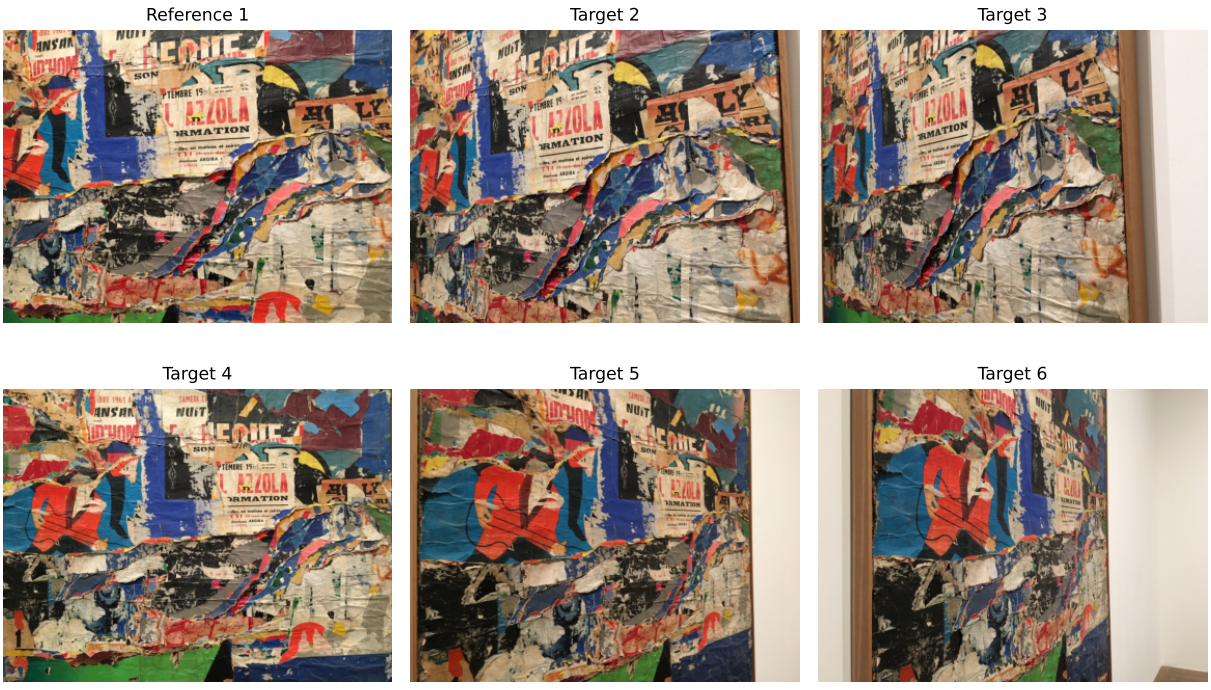

Figure 12.18: Actual reference and all evaluated targets for v_azzola.

Table 12.35: Saved oracle-pair support and corner error for v_azzola.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	164	1.99	yes	217	1.58	yes
3	33	3.56	yes	199	2.63	yes
4	278	0.91	yes	249	0.64	yes
5	60	5.79	no	168	1.83	yes
6	6	20.65	no	107	3.76	yes

SIFT obtains 3 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 2 learned-only successes and 0 SIFT-only successes. Median inlier counts are 60.0 and 199.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.36: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	4/5	4/5	3/5
SP+LG	5/5	5/5	5/5
Full	4/5	4/5	4/5

The full hierarchy correctly verifies 4 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.20 Sequence v_bark

This sequence belongs to the viewpoint subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

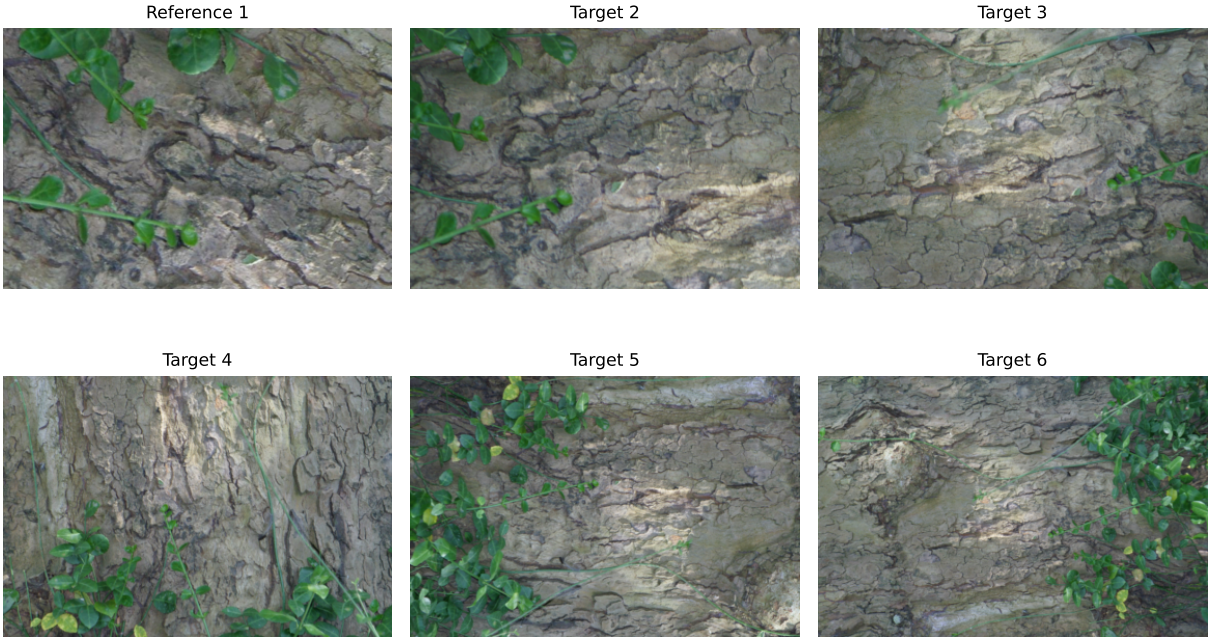

Figure 12.19: Actual reference and all evaluated targets for v_bark.

Table 12.37: Saved oracle-pair support and corner error for v_bark.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	116	0.41	yes	240	0.66	yes
3	73	0.67	yes	4	–	no
4	53	0.41	yes	5	451.46	no
5	47	0.54	yes	41	1.38	yes
6	23	0.46	yes	5	495.20	no

SIFT obtains 5 of five successful oracle homographies; SP+LG obtains 2 of five. The paired comparison contains 0 learned-only successes and 3 SIFT-only successes. Median inlier counts are 53.0 and 5.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.38: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	5/5	5/5
SP+LG	2/5	2/5	2/5
Full	3/5	2/5	2/5

The full hierarchy correctly verifies 2 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore

a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.21 Sequence v_bees

This sequence belongs to the viewpoint subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

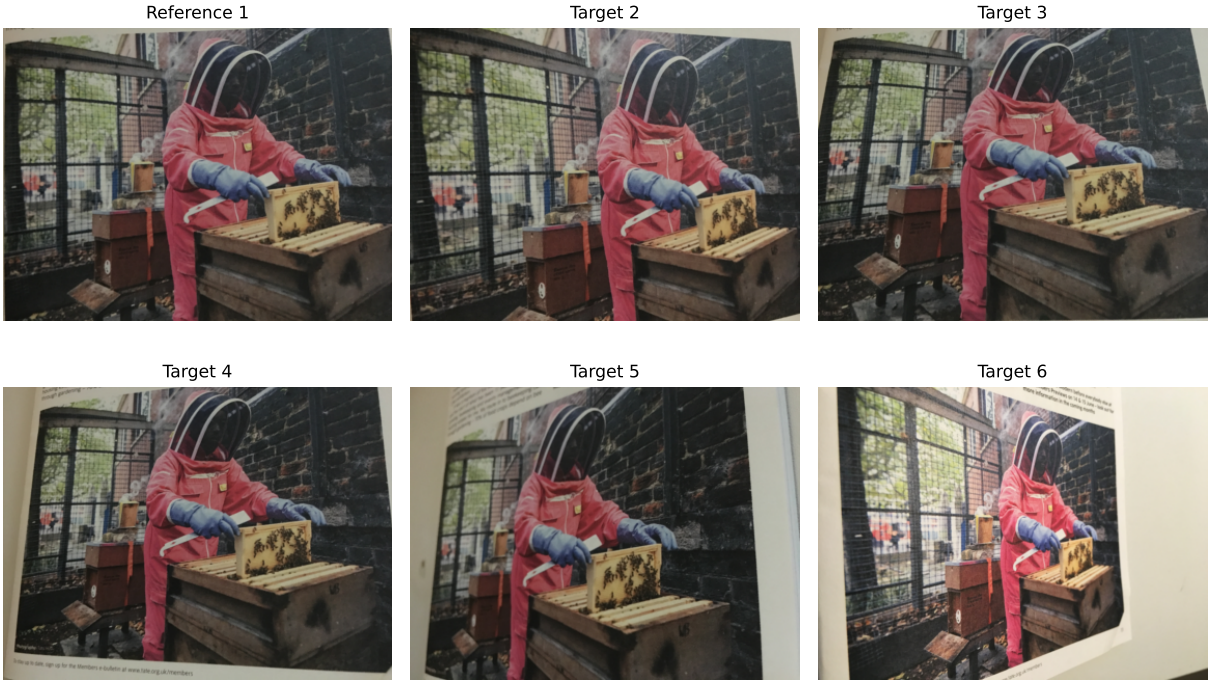

Figure 12.20: Actual reference and all evaluated targets for v_bees.

Table 12.39: Saved oracle-pair support and corner error for v_bees.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	307	0.43	yes	368	0.70	yes
3	341	0.23	yes	401	0.47	yes
4	304	0.44	yes	342	0.90	yes
5	161	0.84	yes	231	2.14	yes
6	166	1.86	yes	325	0.96	yes

SIFT obtains 5 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 0 learned-only successes and 0 SIFT-only successes. Median inlier counts are 304.0 and 342.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.40: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	5/5	5/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.22 Sequence v_beyus

This sequence belongs to the viewpoint subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

Figure 12.21: Actual reference and all evaluated targets for v_beyus.

Table 12.41: Saved oracle-pair support and corner error for v_beyus.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	93	0.58	yes	217	0.98	yes
3	9	43.75	no	189	1.51	yes
4	230	0.23	yes	281	0.32	yes
5	14	5.91	no	158	2.00	yes
6	37	1.80	yes	211	1.34	yes

SIFT obtains 3 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 2 learned-only successes and 0 SIFT-only successes. Median inlier counts are 37.0 and 211.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.42: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	4/5	3/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.23 Sequence v_bip

This sequence belongs to the viewpoint subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

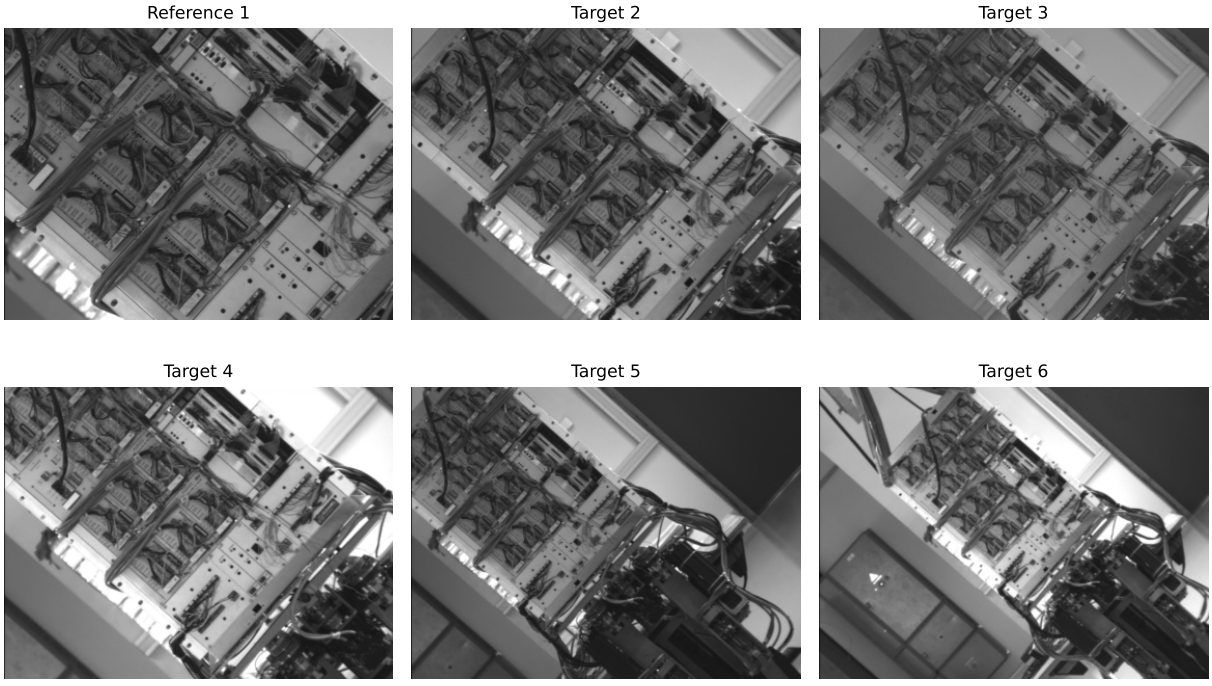

Figure 12.22: Actual reference and all evaluated targets for v_bip.

Table 12.43: Saved oracle-pair support and corner error for v_bip.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	173	0.32	yes	227	0.38	yes
3	140	0.20	yes	180	0.68	yes
4	146	0.38	yes	187	0.36	yes
5	88	1.86	yes	7	5.71	no
6	78	0.56	yes	4	–	no

SIFT obtains 5 of five successful oracle homographies; SP+LG obtains 3 of five. The paired comparison contains 0 learned-only successes and 2 SIFT-only successes. Median inlier counts are 140.0 and 180.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.44: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	5/5	5/5
SP+LG	3/5	3/5	3/5
Full	4/5	3/5	3/5

The full hierarchy correctly verifies 3 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.24 Sequence v_bird

This sequence belongs to the viewpoint subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

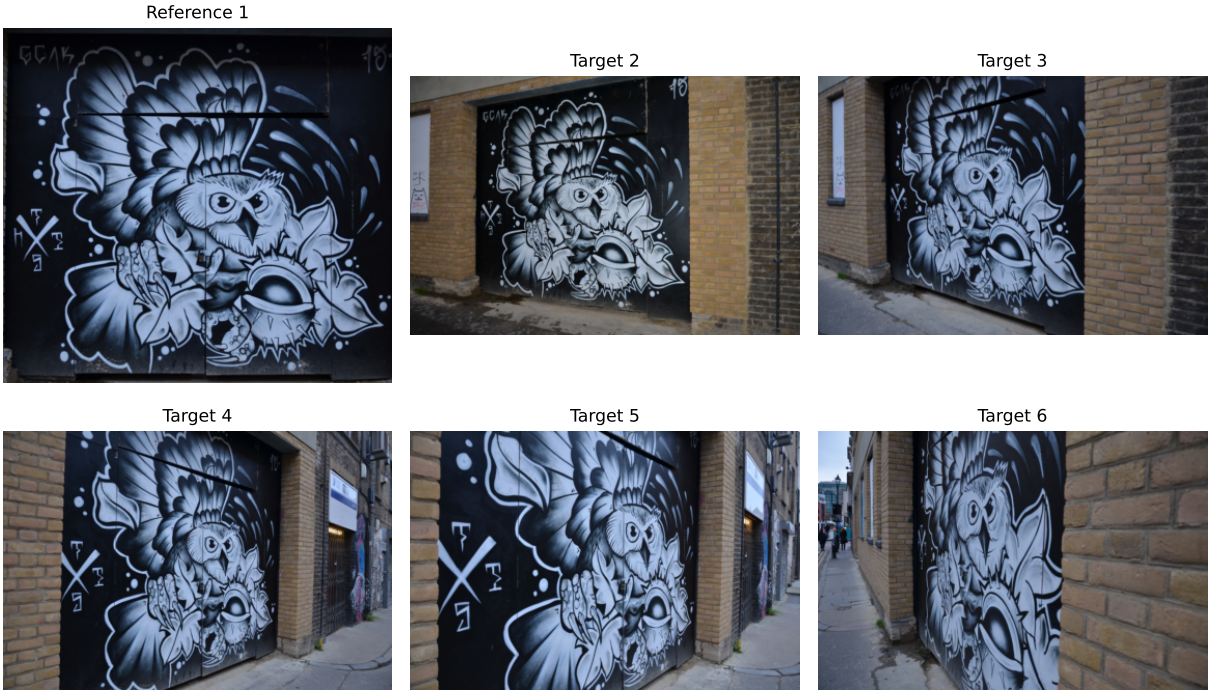

Figure 12.23: Actual reference and all evaluated targets for v_bird.

Table 12.45: Saved oracle-pair support and corner error for v_bird.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	206	0.38	yes	190	0.36	yes
3	129	1.20	yes	173	1.99	yes
4	120	2.20	yes	214	1.65	yes
5	53	14.30	no	172	4.15	yes
6	4	–	no	126	26.82	no

SIFT obtains 3 of five successful oracle homographies; SP+LG obtains 4 of five. The paired comparison contains 1 learned-only success and 0 SIFT-only successes. Median inlier counts are 120.0 and 173.0, respectively. These support counts are computed from different tentative correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.46: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	4/5	4/5	3/5
SP+LG	5/5	5/5	4/5
Full	3/5	3/5	3/5

The full hierarchy correctly verifies 3 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle

table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

12.25 Sequence v_birdwoman

This sequence belongs to the viewpoint subset and contributes five reference-to-target pairs. Its identity is held out from projection-head training; learned components in the main A–D comparison retain their original pretrained weights in either case.

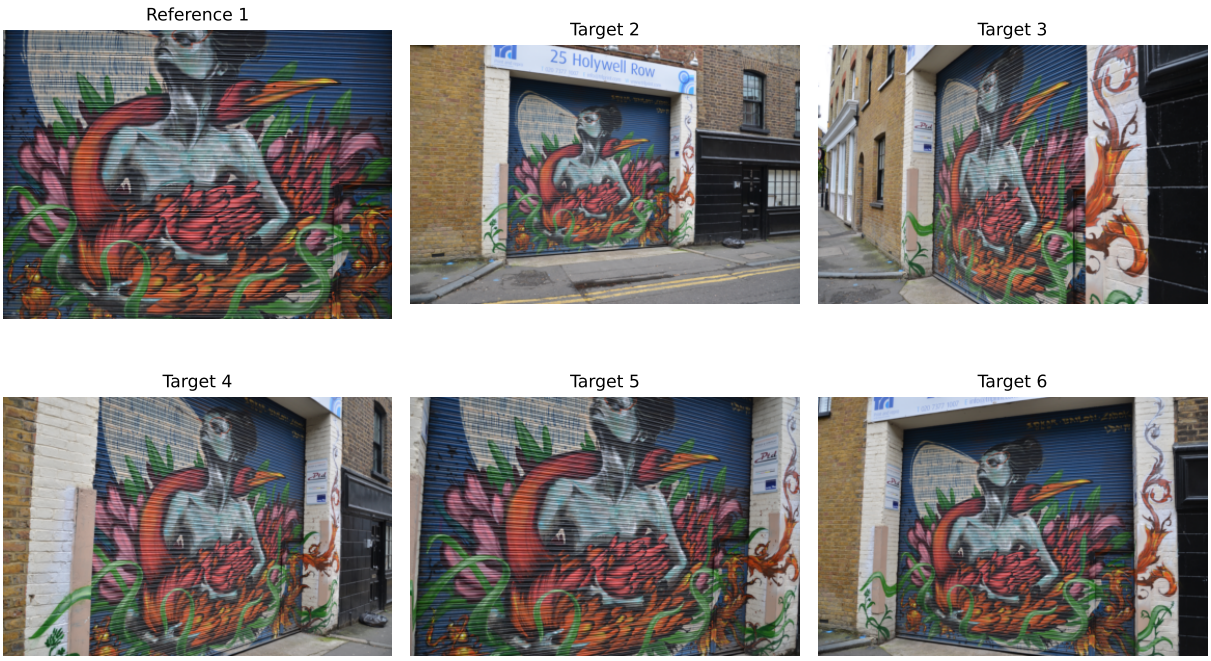

Figure 12.24: Actual reference and all evaluated targets for v_birdwoman.

Table 12.47: Saved oracle-pair support and corner error for v_birdwoman.

Target	SIFT I	SIFT E	SIFT H5	SP+LG I	SP+LG E	SP+LG H5
2	22	0.94	yes	49	1.47	yes
3	26	4.65	yes	82	2.11	yes
4	84	0.79	yes	91	0.91	yes
5	240	0.29	yes	219	0.28	yes
6	50	0.80	yes	106	0.98	yes

SIFT obtains 5 of five successful oracle homographies; SP+LG obtains 5 of five. The paired comparison contains 0 learned-only successes and 0 SIFT-only successes. Median inlier counts are 50.0 and 91.0, respectively. These support counts are computed from different tentative

correspondence sets, so their magnitudes alone do not establish a more accurate mapping.

Table 12.48: Search and anchoring outcomes for the same five targets.

Method	Rank-1 correct	Correctly verified	End-to-end H5
SIFT	5/5	5/5	5/5
SP+LG	5/5	5/5	5/5
Full	5/5	5/5	5/5

The full hierarchy correctly verifies 5 of these five queries after searching the mixed 24-reference gallery. Its end-to-end column includes candidate selection and homography accuracy; the oracle table supplies the true reference directly. Any difference between those columns is therefore a search or decision effect rather than an additional illumination observation. The sequence remains one physical source of correlated targets, so five outcomes should not be treated as five independently sampled places.

Chapter 13

Complete Tokyo Site Evidence

13.1 Site-level observations and decision states

Each section contains the four retained photographs for one selected physical site and every A–D decision for its four cross-time protocols. The panels expose reference coverage directly. Front/day, left/day, front/night and front/sunset metadata define the views; their identity labels do not supply an independently measured planar transform or metric camera pose. The source imagery is from Tokyo 24/7 [4], but these tables belong to the custom query-to-query subset.

R is the rank of the correct identity in the saved complete ranking. Decision states are correct, wrong, rejected and ranking only. A wrong state means geometric acceptance of another place; it does not mean every correspondence is known to be false. Global retrieval has no acceptance or inlier-count claim. Protocol labels D–D, D–N, N–D and D–S mean day to changed-direction day, day to night, night to day and day to sunset.

13.2 Site tokyo_000

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Front / day

Left / day

Front / night

Front / sunset

Figure 13.1: All retained photographs for tokyo_000.

Table 13.1: All main-method decisions for tokyo_000.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	1	correct	62
D–D	SP+LG	1	correct	106
D–D	Global	6	ranking only	–
D–D	Full	6	rejected	0
D–N	SIFT	3	rejected	0
D–N	SP+LG	1	correct	163
D–N	Global	1	ranking only	–
D–N	Full	1	correct	163
N–D	SIFT	3	rejected	0
N–D	SP+LG	1	correct	168
N–D	Global	1	ranking only	–
N–D	Full	1	correct	168
D–S	SIFT	1	correct	31
D–S	SP+LG	1	correct	194
D–S	Global	1	ranking only	–
D–S	Full	1	correct	194

Across these four protocols, SIFT correctly verifies 2, exhaustive SP+LG 4, and the full hierarchy 3 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 1 rejection. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is rejected, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.3 Site tokyo_001

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Front / day

Left / day

Front / night

Front / sunset

Figure 13.2: All retained photographs for tokyo_001.

Table 13.2: All main-method decisions for tokyo_001.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	1	correct	69
D–D	SP+LG	1	correct	129
D–D	Global	1	ranking only	–
D–D	Full	1	correct	129
D–N	SIFT	1	correct	17
D–N	SP+LG	1	correct	199
D–N	Global	1	ranking only	–
D–N	Full	1	correct	199
N–D	SIFT	1	correct	17
N–D	SP+LG	1	correct	176
N–D	Global	1	ranking only	–
N–D	Full	1	correct	176
D–S	SIFT	1	correct	132
D–S	SP+LG	1	correct	280
D–S	Global	1	ranking only	–
D–S	Full	1	correct	280

Across these four protocols, SIFT correctly verifies 4, exhaustive SP+LG 4, and the full hierarchy 4 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 0 rejections. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is correctly verified, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.4 Site tokyo_002

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Front / day

Left / day

Front / night

Front / sunset

Figure 13.3: All retained photographs for tokyo_002.

Table 13.3: All main-method decisions for tokyo_002.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	1	correct	73
D–D	SP+LG	1	correct	105
D–D	Global	1	ranking only	–
D–D	Full	1	correct	105
D–N	SIFT	1	correct	32
D–N	SP+LG	1	correct	176
D–N	Global	1	ranking only	–
D–N	Full	1	correct	176
N–D	SIFT	1	correct	30
N–D	SP+LG	1	correct	202
N–D	Global	1	ranking only	–
N–D	Full	1	correct	202
D–S	SIFT	1	correct	119
D–S	SP+LG	1	correct	315
D–S	Global	1	ranking only	–
D–S	Full	1	correct	315

Across these four protocols, SIFT correctly verifies 4, exhaustive SP+LG 4, and the full hierarchy 4 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 0 rejections. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is correctly verified, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.5 Site tokyo_003

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Front / day

Left / day

Front / night

Front / sunset

Figure 13.4: All retained photographs for tokyo_003.

Table 13.4: All main-method decisions for tokyo_003.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	1	rejected	0
D–D	SP+LG	1	correct	73
D–D	Global	1	ranking only	–
D–D	Full	1	correct	73
D–N	SIFT	1	correct	15
D–N	SP+LG	1	correct	168
D–N	Global	1	ranking only	–
D–N	Full	1	correct	168
N–D	SIFT	1	correct	20
N–D	SP+LG	1	correct	167
N–D	Global	1	ranking only	–
N–D	Full	1	correct	167
D–S	SIFT	1	correct	13
D–S	SP+LG	1	correct	101
D–S	Global	1	ranking only	–
D–S	Full	1	correct	101

Across these four protocols, SIFT correctly verifies 3, exhaustive SP+LG 4, and the full hierarchy 4 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 0 rejections. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is correctly verified, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.6 Site tokyo_004

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Front / day

Left / day

Front / night

Front / sunset

Figure 13.5: All retained photographs for tokyo_004.

Table 13.5: All main-method decisions for tokyo_004.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	1	correct	28
D–D	SP+LG	1	correct	78
D–D	Global	1	ranking only	–
D–D	Full	1	correct	78
D–N	SIFT	2	rejected	0
D–N	SP+LG	1	correct	211
D–N	Global	1	ranking only	–
D–N	Full	1	correct	211
N–D	SIFT	7	rejected	0
N–D	SP+LG	1	correct	202
N–D	Global	1	ranking only	–
N–D	Full	1	correct	202
D–S	SIFT	1	correct	144
D–S	SP+LG	1	correct	328
D–S	Global	1	ranking only	–
D–S	Full	1	correct	328

Across these four protocols, SIFT correctly verifies 2, exhaustive SP+LG 4, and the full hierarchy 4 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 0 rejections. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is correctly verified, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.7 Site tokyo_005

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Figure 13.6: All retained photographs for tokyo_005.

Table 13.6: All main-method decisions for tokyo_005.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	1	correct	60
D–D	SP+LG	1	correct	166
D–D	Global	1	ranking only	–
D–D	Full	1	correct	166
D–N	SIFT	11	rejected	0
D–N	SP+LG	1	correct	172
D–N	Global	1	ranking only	–
D–N	Full	1	correct	172
N–D	SIFT	8	rejected	0
N–D	SP+LG	1	correct	178
N–D	Global	1	ranking only	–
N–D	Full	1	correct	178
D–S	SIFT	1	correct	98
D–S	SP+LG	1	correct	282
D–S	Global	1	ranking only	–
D–S	Full	1	correct	282

Across these four protocols, SIFT correctly verifies 2, exhaustive SP+LG 4, and the full hier-

archy 4 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 0 rejections. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is correctly verified, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.8 Site tokyo_006

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Front / day

Left / day

Front / night

Front / sunset

Figure 13.7: All retained photographs for tokyo_006.

Table 13.7: All main-method decisions for tokyo_006.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	1	correct	40
D–D	SP+LG	1	correct	137
D–D	Global	1	ranking only	–
D–D	Full	1	correct	137
D–N	SIFT	3	rejected	0
D–N	SP+LG	1	correct	243
D–N	Global	1	ranking only	–
D–N	Full	1	correct	243
N–D	SIFT	1	rejected	0
N–D	SP+LG	1	correct	244
N–D	Global	1	ranking only	–
N–D	Full	1	correct	244
D–S	SIFT	1	correct	43
D–S	SP+LG	1	correct	275
D–S	Global	1	ranking only	–
D–S	Full	1	correct	275

Across these four protocols, SIFT correctly verifies 2, exhaustive SP+LG 4, and the full hierarchy 4 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 0 rejections. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is correctly verified, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.9 Site tokyo_007

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Front / day

Left / day

Front / night

Front / sunset

Figure 13.8: All retained photographs for tokyo_007.

Table 13.8: All main-method decisions for tokyo_007.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	1	rejected	0
D–D	SP+LG	1	correct	34
D–D	Global	16	ranking only	–
D–D	Full	16	wrong	13
D–N	SIFT	1	rejected	0
D–N	SP+LG	1	correct	87
D–N	Global	1	ranking only	–
D–N	Full	1	correct	87
N–D	SIFT	3	rejected	0
N–D	SP+LG	1	correct	88
N–D	Global	1	ranking only	–
N–D	Full	1	correct	88
D–S	SIFT	1	correct	37
D–S	SP+LG	1	correct	280
D–S	Global	1	ranking only	–
D–S	Full	1	correct	280

Across these four protocols, SIFT correctly verifies 1, exhaustive SP+LG 4, and the full hierarchy 3 of the four queries. The full hierarchy makes 1 wrong-place acceptance and 0 rejections. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is accepted at the wrong place, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.10 Site tokyo_008

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Front / day

Left / day

Front / night

Front / sunset

Figure 13.9: All retained photographs for tokyo_008.

Table 13.9: All main-method decisions for tokyo_008.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	1	correct	67
D–D	SP+LG	1	correct	163
D–D	Global	1	ranking only	–
D–D	Full	1	correct	163
D–N	SIFT	3	rejected	0
D–N	SP+LG	1	correct	245
D–N	Global	1	ranking only	–
D–N	Full	1	correct	245
N–D	SIFT	15	rejected	0
N–D	SP+LG	1	correct	256
N–D	Global	1	ranking only	–
N–D	Full	1	correct	256
D–S	SIFT	1	correct	37
D–S	SP+LG	1	correct	318
D–S	Global	1	ranking only	–
D–S	Full	1	correct	318

Across these four protocols, SIFT correctly verifies 2, exhaustive SP+LG 4, and the full hierarchy 4 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 0 rejections. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is correctly verified, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.11 Site tokyo_009

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Front / day

Left / day

Front / night

Front / sunset

Figure 13.10: All retained photographs for tokyo_009.

Table 13.10: All main-method decisions for tokyo_009.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	1	correct	100
D–D	SP+LG	1	correct	186
D–D	Global	1	ranking only	–
D–D	Full	1	correct	186
D–N	SIFT	1	correct	12
D–N	SP+LG	1	correct	190
D–N	Global	1	ranking only	–
D–N	Full	1	correct	190
N–D	SIFT	1	correct	13
N–D	SP+LG	1	correct	189
N–D	Global	1	ranking only	–
N–D	Full	1	correct	189
D–S	SIFT	1	correct	76
D–S	SP+LG	1	correct	307
D–S	Global	1	ranking only	–
D–S	Full	1	correct	307

Across these four protocols, SIFT correctly verifies 4, exhaustive SP+LG 4, and the full hierarchy 4 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 0 rejections. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is correctly verified, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.12 Site tokyo_010

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Figure 13.11: All retained photographs for tokyo_010.

Table 13.11: All main-method decisions for tokyo_010.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	2	rejected	0
D–D	SP+LG	1	rejected	0
D–D	Global	18	ranking only	–
D–D	Full	18	rejected	0
D–N	SIFT	2	rejected	0
D–N	SP+LG	1	correct	198
D–N	Global	1	ranking only	–
D–N	Full	1	correct	198
N–D	SIFT	1	rejected	0
N–D	SP+LG	1	correct	185
N–D	Global	1	ranking only	–
N–D	Full	1	correct	185
D–S	SIFT	2	rejected	0
D–S	SP+LG	1	correct	207
D–S	Global	1	ranking only	–
D–S	Full	1	correct	207

Across these four protocols, SIFT correctly verifies 0, exhaustive SP+LG 3, and the full

hierarchy 3 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 1 rejection. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is rejected, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.13 Site tokyo_011

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Figure 13.12: All retained photographs for tokyo_011.

Table 13.12: All main-method decisions for tokyo_011.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	14	rejected	0
D–D	SP+LG	18	wrong	18
D–D	Global	7	ranking only	–
D–D	Full	7	rejected	0
D–N	SIFT	16	rejected	0
D–N	SP+LG	1	correct	141
D–N	Global	1	ranking only	–
D–N	Full	1	correct	141
N–D	SIFT	17	rejected	0
N–D	SP+LG	1	correct	138
N–D	Global	1	ranking only	–
N–D	Full	1	correct	138
D–S	SIFT	3	rejected	0
D–S	SP+LG	1	correct	128
D–S	Global	1	ranking only	–
D–S	Full	1	correct	128

Across these four protocols, SIFT correctly verifies 0, exhaustive SP+LG 3, and the full hierarchy 3 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 1 rejection. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is rejected, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.14 Site tokyo_012

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

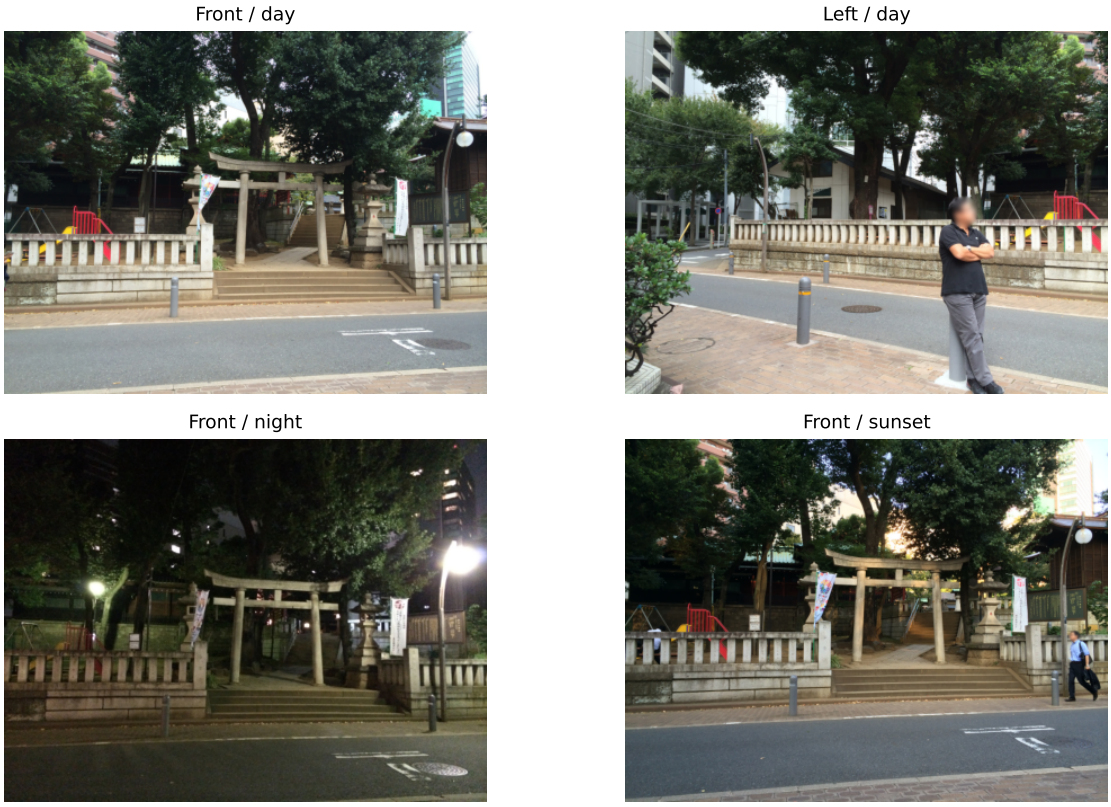

Figure 13.13: All retained photographs for tokyo_012.

Table 13.13: All main-method decisions for tokyo_012.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	1	correct	15
D–D	SP+LG	1	correct	56
D–D	Global	1	ranking only	–
D–D	Full	1	correct	56
D–N	SIFT	1	correct	22
D–N	SP+LG	1	correct	231
D–N	Global	1	ranking only	–
D–N	Full	1	correct	231
N–D	SIFT	1	correct	18
N–D	SP+LG	1	correct	234
N–D	Global	1	ranking only	–
N–D	Full	1	correct	234
D–S	SIFT	1	correct	113
D–S	SP+LG	1	correct	316
D–S	Global	1	ranking only	–
D–S	Full	1	correct	316

Across these four protocols, SIFT correctly verifies 4, exhaustive SP+LG 4, and the full hier-

archy 4 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 0 rejections. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is correctly verified, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.15 Site tokyo_013

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Figure 13.14: All retained photographs for tokyo_013.

Table 13.14: All main-method decisions for tokyo_013.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	1	correct	44
D–D	SP+LG	1	correct	70
D–D	Global	11	ranking only	–
D–D	Full	11	rejected	0
D–N	SIFT	1	rejected	0
D–N	SP+LG	1	correct	170
D–N	Global	1	ranking only	–
D–N	Full	1	correct	170
N–D	SIFT	1	rejected	0
N–D	SP+LG	1	correct	167
N–D	Global	1	ranking only	–
N–D	Full	1	correct	167
D–S	SIFT	1	correct	119
D–S	SP+LG	1	correct	255
D–S	Global	1	ranking only	–
D–S	Full	1	correct	255

Across these four protocols, SIFT correctly verifies 2, exhaustive SP+LG 4, and the full hierarchy 3 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 1 rejection. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is rejected, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.16 Site tokyo_014

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

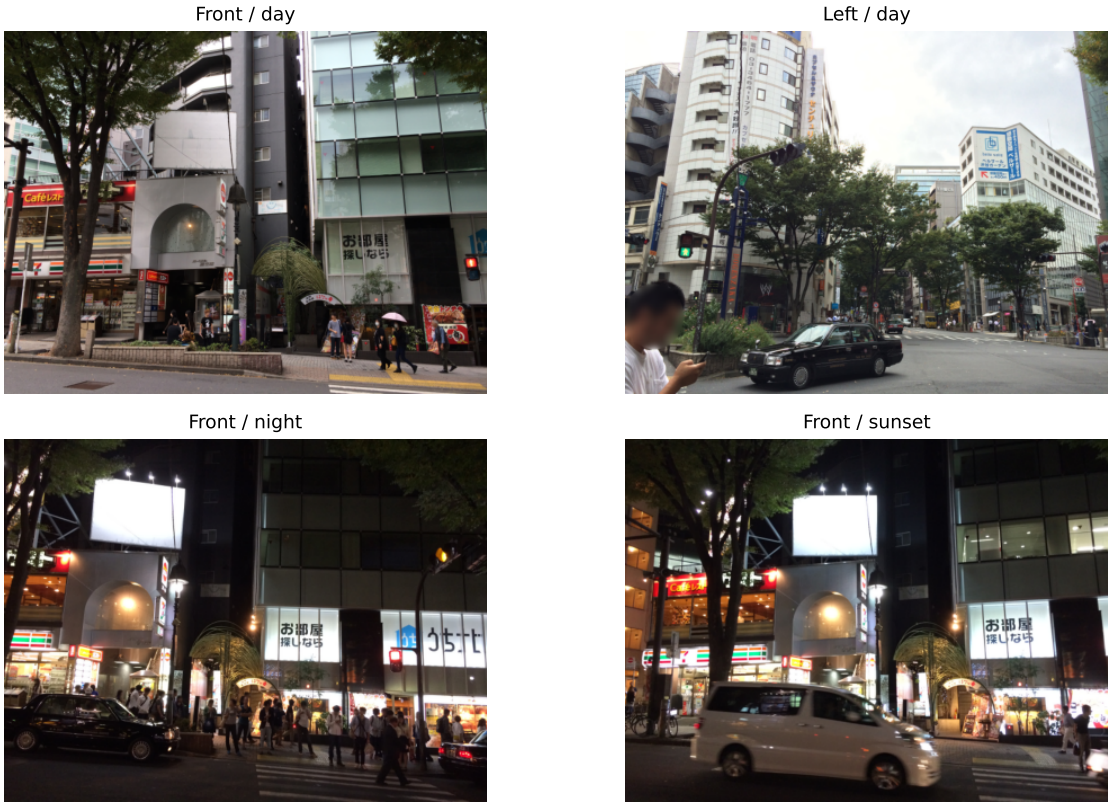

Figure 13.15: All retained photographs for tokyo_014.

Table 13.15: All main-method decisions for tokyo_014.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	18	rejected	0
D–D	SP+LG	16	wrong	17
D–D	Global	18	ranking only	–
D–D	Full	18	rejected	0
D–N	SIFT	2	rejected	0
D–N	SP+LG	1	correct	125
D–N	Global	1	ranking only	–
D–N	Full	1	correct	125
N–D	SIFT	2	rejected	0
N–D	SP+LG	1	correct	140
N–D	Global	1	ranking only	–
N–D	Full	1	correct	140
D–S	SIFT	1	rejected	0
D–S	SP+LG	1	correct	173
D–S	Global	1	ranking only	–
D–S	Full	1	correct	173

Across these four protocols, SIFT correctly verifies 0, exhaustive SP+LG 3, and the full

hierarchy 3 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 1 rejection. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is rejected, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.17 Site tokyo_015

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Front / day

Left / day

Front / night

Front / sunset

Figure 13.16: All retained photographs for tokyo_015.

Table 13.16: All main-method decisions for tokyo_015.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	8	rejected	0
D–D	SP+LG	17	wrong	15
D–D	Global	5	ranking only	–
D–D	Full	5	wrong	15
D–N	SIFT	1	rejected	0
D–N	SP+LG	1	correct	39
D–N	Global	1	ranking only	–
D–N	Full	1	correct	39
N–D	SIFT	8	rejected	0
N–D	SP+LG	1	correct	47
N–D	Global	1	ranking only	–
N–D	Full	1	correct	47
D–S	SIFT	1	rejected	0
D–S	SP+LG	1	correct	69
D–S	Global	1	ranking only	–
D–S	Full	1	correct	69

Across these four protocols, SIFT correctly verifies 0, exhaustive SP+LG 3, and the full hierarchy 3 of the four queries. The full hierarchy makes 1 wrong-place acceptance and 0 rejections. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is accepted at the wrong place, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.18 Site tokyo_016

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Figure 13.17: All retained photographs for tokyo_016.

Table 13.17: All main-method decisions for tokyo_016.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	1	rejected	0
D–D	SP+LG	6	rejected	0
D–D	Global	4	ranking only	–
D–D	Full	4	rejected	0
D–N	SIFT	1	rejected	0
D–N	SP+LG	1	correct	167
D–N	Global	1	ranking only	–
D–N	Full	1	correct	167
N–D	SIFT	1	rejected	0
N–D	SP+LG	1	correct	167
N–D	Global	1	ranking only	–
N–D	Full	1	correct	167
D–S	SIFT	2	rejected	0
D–S	SP+LG	1	correct	110
D–S	Global	1	ranking only	–
D–S	Full	1	correct	110

Across these four protocols, SIFT correctly verifies 0, exhaustive SP+LG 3, and the full

hierarchy 3 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 1 rejection. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is rejected, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.19 Site tokyo_017

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Figure 13.18: All retained photographs for tokyo_017.

Table 13.18: All main-method decisions for tokyo_017.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	21	rejected	0
D–D	SP+LG	2	rejected	0
D–D	Global	8	ranking only	–
D–D	Full	8	rejected	0
D–N	SIFT	2	rejected	0
D–N	SP+LG	1	correct	164
D–N	Global	1	ranking only	–
D–N	Full	1	correct	164
N–D	SIFT	2	rejected	0
N–D	SP+LG	1	correct	164
N–D	Global	1	ranking only	–
N–D	Full	1	correct	164
D–S	SIFT	1	rejected	0
D–S	SP+LG	1	correct	113
D–S	Global	1	ranking only	–
D–S	Full	1	correct	113

Across these four protocols, SIFT correctly verifies 0, exhaustive SP+LG 3, and the full hierarchy 3 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 1 rejection. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is rejected, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.20 Site tokyo_018

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Figure 13.19: All retained photographs for tokyo_018.

Table 13.19: All main-method decisions for tokyo_018.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	20	rejected	0
D–D	SP+LG	16	rejected	0
D–D	Global	2	ranking only	–
D–D	Full	1	rejected	0
D–N	SIFT	1	correct	34
D–N	SP+LG	1	correct	152
D–N	Global	1	ranking only	–
D–N	Full	1	correct	152
N–D	SIFT	1	correct	33
N–D	SP+LG	1	correct	162
N–D	Global	1	ranking only	–
N–D	Full	1	correct	162
D–S	SIFT	1	correct	33
D–S	SP+LG	1	correct	201
D–S	Global	1	ranking only	–
D–S	Full	1	correct	201

Across these four protocols, SIFT correctly verifies 3, exhaustive SP+LG 3, and the full

hierarchy 3 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 1 rejection. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is rejected, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.21 Site tokyo_019

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

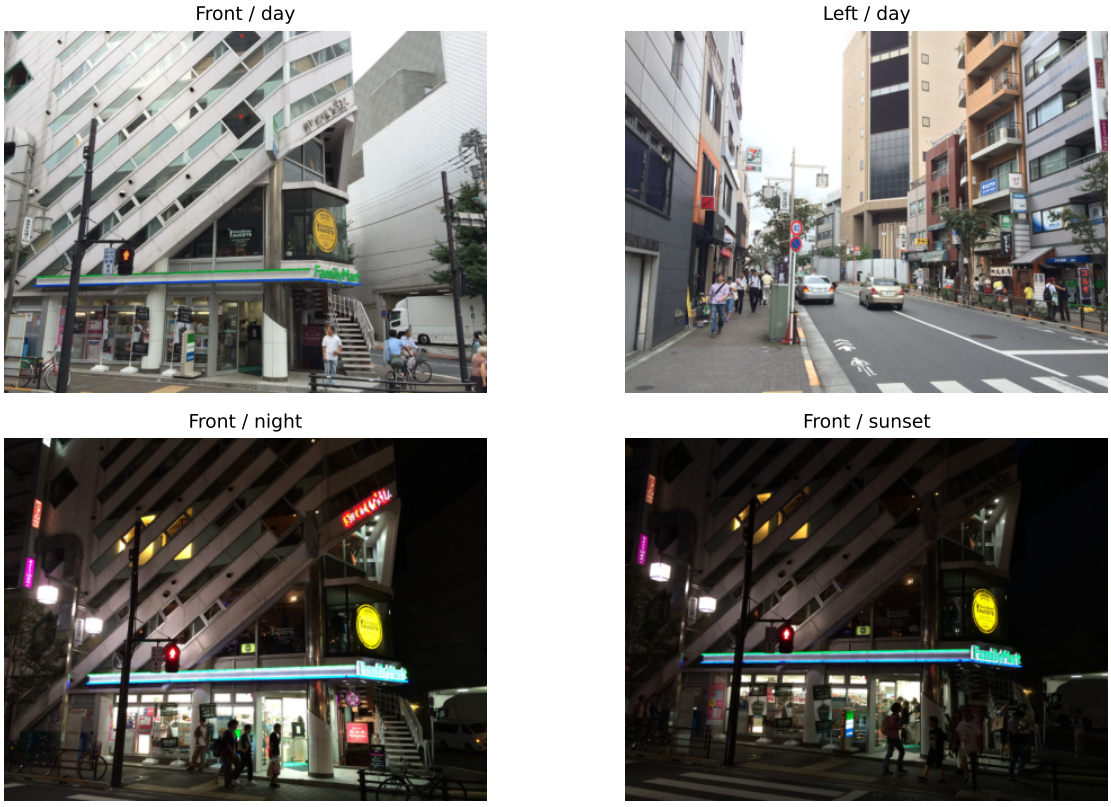

Figure 13.20: All retained photographs for tokyo_019.

Table 13.20: All main-method decisions for tokyo_019.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	9	rejected	0
D–D	SP+LG	11	rejected	0
D–D	Global	13	ranking only	–
D–D	Full	13	rejected	0
D–N	SIFT	1	rejected	0
D–N	SP+LG	1	correct	178
D–N	Global	1	ranking only	–
D–N	Full	1	correct	178
N–D	SIFT	6	rejected	0
N–D	SP+LG	1	correct	180
N–D	Global	1	ranking only	–
N–D	Full	1	correct	180
D–S	SIFT	6	rejected	0
D–S	SP+LG	1	correct	171
D–S	Global	1	ranking only	–
D–S	Full	1	correct	171

Across these four protocols, SIFT correctly verifies 0, exhaustive SP+LG 3, and the full hierarchy 3 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 1 rejection. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is rejected, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.22 Site tokyo_020

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Figure 13.21: All retained photographs for tokyo_020.

Table 13.21: All main-method decisions for tokyo_020.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	4	rejected	0
D–D	SP+LG	1	correct	44
D–D	Global	4	ranking only	–
D–D	Full	4	rejected	0
D–N	SIFT	1	correct	23
D–N	SP+LG	1	correct	139
D–N	Global	1	ranking only	–
D–N	Full	1	correct	139
N–D	SIFT	1	correct	22
N–D	SP+LG	1	correct	140
N–D	Global	1	ranking only	–
N–D	Full	1	correct	140
D–S	SIFT	1	correct	25
D–S	SP+LG	1	correct	140
D–S	Global	1	ranking only	–
D–S	Full	1	correct	140

Across these four protocols, SIFT correctly verifies 3, exhaustive SP+LG 4, and the full

hierarchy 3 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 1 rejection. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is rejected, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.23 Site tokyo_021

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

Figure 13.22: All retained photographs for tokyo_021.

Table 13.22: All main-method decisions for tokyo_021.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	23	rejected	0
D–D	SP+LG	5	wrong	14
D–D	Global	5	ranking only	–
D–D	Full	5	wrong	14
D–N	SIFT	1	correct	62
D–N	SP+LG	1	correct	201
D–N	Global	1	ranking only	–
D–N	Full	1	correct	201
N–D	SIFT	1	correct	60
N–D	SP+LG	1	correct	200
N–D	Global	1	ranking only	–
N–D	Full	1	correct	200
D–S	SIFT	1	correct	103
D–S	SP+LG	1	correct	251
D–S	Global	1	ranking only	–
D–S	Full	1	correct	251

Across these four protocols, SIFT correctly verifies 3, exhaustive SP+LG 3, and the full hierarchy 3 of the four queries. The full hierarchy makes 1 wrong-place acceptance and 0 rejections. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is accepted at the wrong place, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.24 Site tokyo_022

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

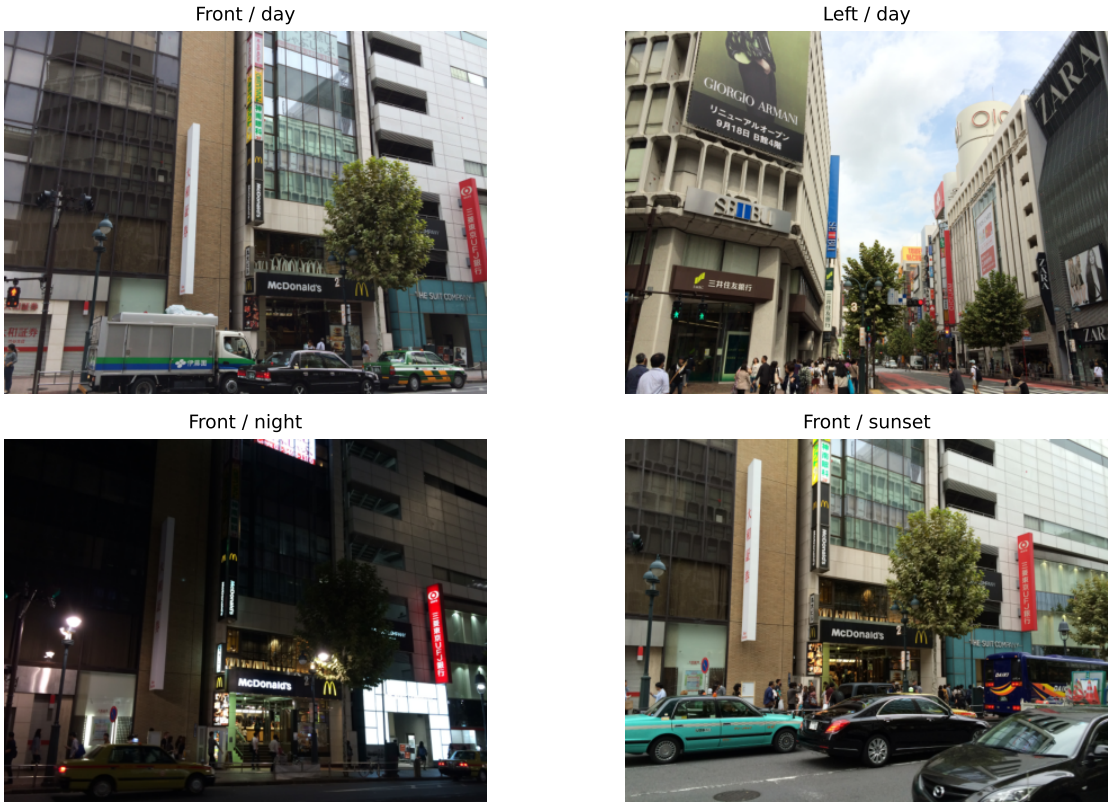

Figure 13.23: All retained photographs for tokyo_022.

Table 13.23: All main-method decisions for tokyo_022.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	2	rejected	0
D–D	SP+LG	5	wrong	13
D–D	Global	21	ranking only	–
D–D	Full	21	rejected	0
D–N	SIFT	1	rejected	0
D–N	SP+LG	1	correct	162
D–N	Global	1	ranking only	–
D–N	Full	1	correct	162
N–D	SIFT	9	rejected	0
N–D	SP+LG	1	correct	171
N–D	Global	1	ranking only	–
N–D	Full	1	correct	171
D–S	SIFT	1	correct	78
D–S	SP+LG	1	correct	198
D–S	Global	1	ranking only	–
D–S	Full	1	correct	198

Across these four protocols, SIFT correctly verifies 1, exhaustive SP+LG 3, and the full

hierarchy 3 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 1 rejection. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is rejected, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

13.25 Site tokyo_023

This site contributes four real photographs and four labelled retrieval queries. Each protocol uses one of the photographs as its query and the condition-specific 24-site gallery defined in the frozen benchmark. Reusing a site across protocols produces paired condition observations, not four new geographic locations.

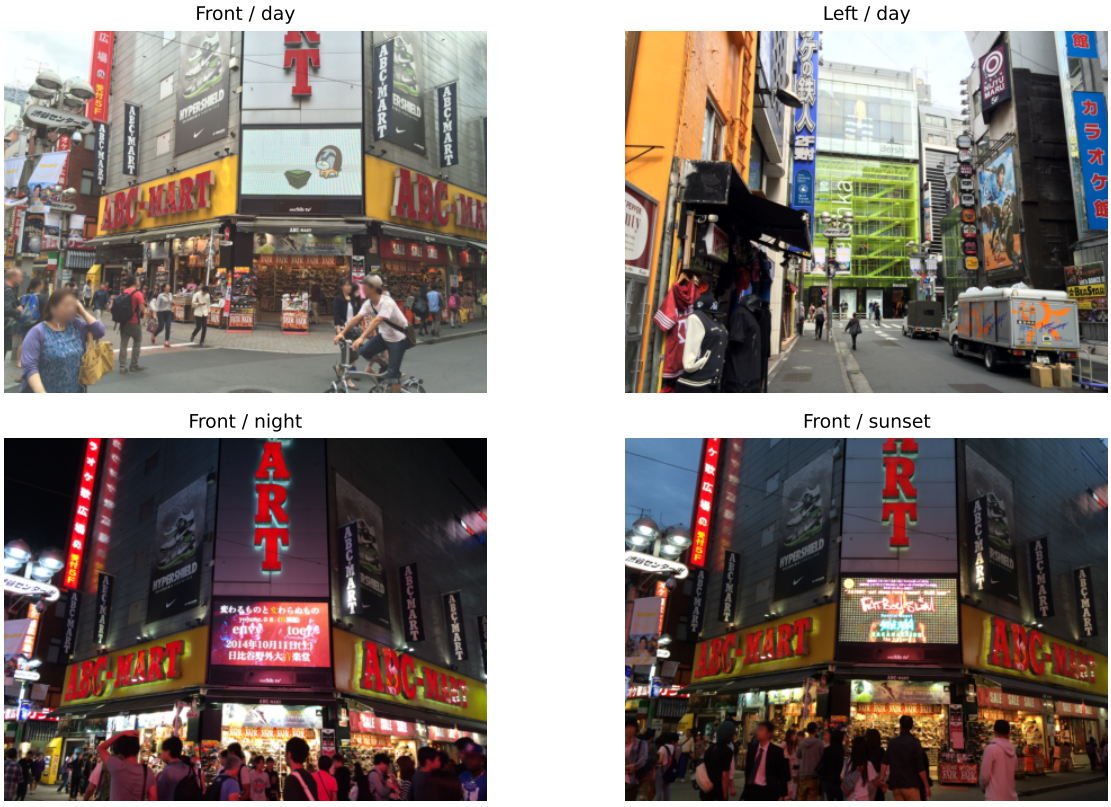

Figure 13.24: All retained photographs for tokyo_023.

Table 13.24: All main-method decisions for tokyo_023.

Protocol	Method	R	Decision	Selected inliers
D–D	SIFT	24	rejected	0
D–D	SP+LG	13	rejected	0
D–D	Global	1	ranking only	–
D–D	Full	3	rejected	0
D–N	SIFT	2	rejected	0
D–N	SP+LG	1	correct	135
D–N	Global	1	ranking only	–
D–N	Full	1	correct	135
N–D	SIFT	3	rejected	0
N–D	SP+LG	1	correct	137
N–D	Global	1	ranking only	–
N–D	Full	1	correct	137
D–S	SIFT	1	rejected	0
D–S	SP+LG	1	correct	149
D–S	Global	1	ranking only	–
D–S	Full	1	correct	149

Across these four protocols, SIFT correctly verifies 0, exhaustive SP+LG 3, and the full hierarchy 3 of the four queries. The full hierarchy makes 0 wrong-place acceptances and 1 rejection. These counts separate a missing anchor from an incorrectly accepted identity; the latter is the more direct location-association failure.

For this site, the full-pipeline changed-direction daytime query is rejected, while its front-facing day-to-night query is correctly verified. This paired observation should be read with the photographs and the rank column. It does not identify the causal effect of illumination alone, because overlap and view direction also change between the protocols. No corner-error or metric-pose value is inferred from a correct place label.

Chapter 14

Deployment on Mobile and Scalability

14.1 What the measured prototype establishes

The artifact executes a complete local pipeline, but its measured host behaviour does not support running full learned recognition on every display frame. The primary full pipeline has an isolated median query-and-render time of 1332.5 ms on the recorded CPU configuration. Exhaustive learned matching takes 3020.4 ms, exhaustive SIFT takes 186.2 ms, and global-only retrieval takes 18.5 ms. These generated values describe different functionality, so the fast global result cannot be substituted for a complete anchored query.

Table 14.1: Isolated warm CPU query and rendering measurements with cached references. RSS is the complete Python process peak, including imports, models, gallery and allocator state.

Pipeline	Median ms	95th pct. ms	Peak RSS MB
classical	186.2	211.3	568.3
local	3020.4	4040.9	1435.5
retrieval	18.5	21.4	537.2
full	1332.5	1523.1	1492.3

Table 14.1 includes upper-tail latency and process peak memory. The full learned path requires substantially more host resources than the classical recogniser. Reference caching reduces repeated work but also retains feature tensors. The process peak includes Python and library imports as well as weights and intermediate buffers; it is not a phone application memory estimate.

Figure 14.1 supports the main deployment conclusion: a useful mobile design should decouple occasional recognition from continuous native tracking. This is a design implication of the measured workload, not a claim that such a phone integration has been completed. The original tracking script's faster frame average in Table 3.1 illustrates the distinction between finding a target and maintaining an already acquired relation.

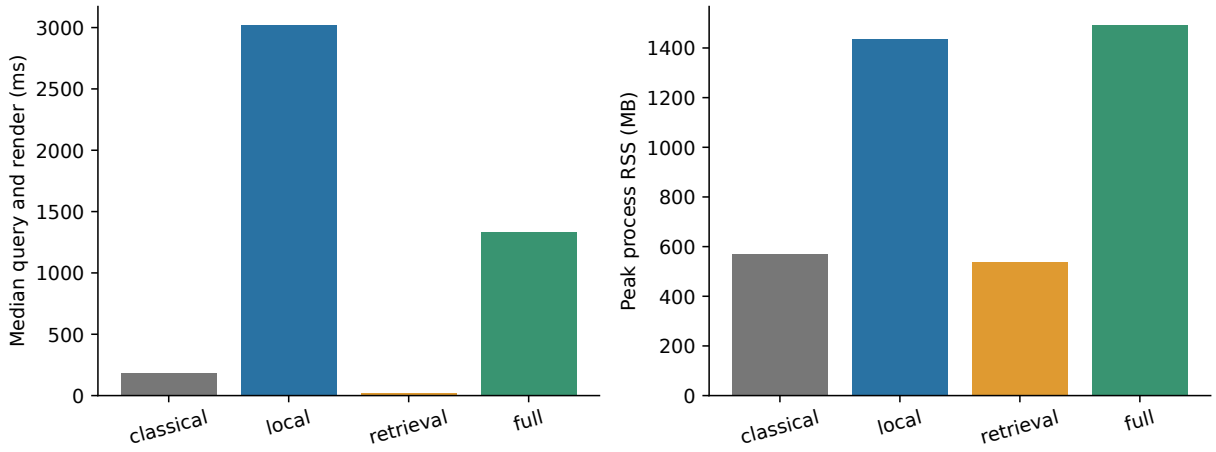

Figure 14.1: Measured isolated host query latency and complete-process peak RSS. Reference features are cached and query images are predecoded. The global-only path does not perform geometric anchoring.

14.2 Executed ONNX export

The CosPlace global model was exported to a fixed-input ONNX graph and checked with the ONNX validator. The graph uses the same learned weights and descriptor normalisation as native PyTorch. Float ONNX outputs were compared with native descriptors on the actual downloaded Tokyo images. The observed maximum absolute difference is recorded in Table 14.2; it is sufficiently small for the float retrieval rankings in this run to match the native results.

Table 14.2: Executed global-descriptor export and quantisation on the host CPU. Parity is measured against native PyTorch across downloaded Tokyo images. Timing excludes preprocessing and is not a phone measurement.

ONNX variant	MB	Median ms	95th pct. ms	Max abs. error
fp32	45.75	24.04	29.71	5.5e-07
int8	11.52	17.32	25.79	0.07

The export scope is deliberately narrow: it covers the global descriptor, not SuperPoint, LightGlue, RANSAC, rendering, or a complete camera application. A successful descriptor graph does not establish that dynamic keypoint operations and adaptive matching export without changes. Nor does graph validation prove that a chosen mobile execution provider supports every operator efficiently. No TFLite, CoreML or phone-native full-pipeline artifact is claimed.

The executed float graph is 45.75 MB and has median descriptor inference of 24.04 ms in the recorded ONNX Runtime CPU test. This timing excludes image preprocessing and gallery search. It also differs from the separate native global query profile because the runtime, timed work and query population differ. The thesis retains these definitions rather than combining the faster number from one test with the broader functionality of another.

14.3 Executed INT8 quantisation

14.3.1 Calibration and graph form

Static quantisation used QDQ graph representation with a calibration reader drawn only from the reserved HPatches training places. Convolution and supported linear operators were selected for quantisation; remaining operations continue in floating point. The model is therefore a mixed graph with INT8-weight and activation paths, not a claim that every computation uses integer arithmetic. ONNX Runtime documents the distinction between graph representations and quantisation settings [28].

In a generic affine quantiser, a real value x is mapped to

$$q = \mathrm{clip}(\mathrm{round}(x/s) + z, q_{\min}, q_{\max}), \qquad \widehat{x} = s(q - z), \tag{14.1}$$

where s is a scale and z a zero point. Calibration estimates ranges for the selected activations. This introduces approximation error and possible saturation; a successful graph conversion alone cannot establish preservation of descriptor rankings. The artifact consequently performs actual quantised inference and retrieval evaluation.

14.3.2 Measured size, latency and accuracy

The executed quantised graph occupies 11.52 MB and has median descriptor inference of 17.32 ms in the recorded CPU test. Both reductions are measured from produced files and runtime calls. Descriptor parity error is larger than for the float graph, as expected from the approximation, but the practical consequence must be read from retrieval rather than from an arbitrary tolerance alone.

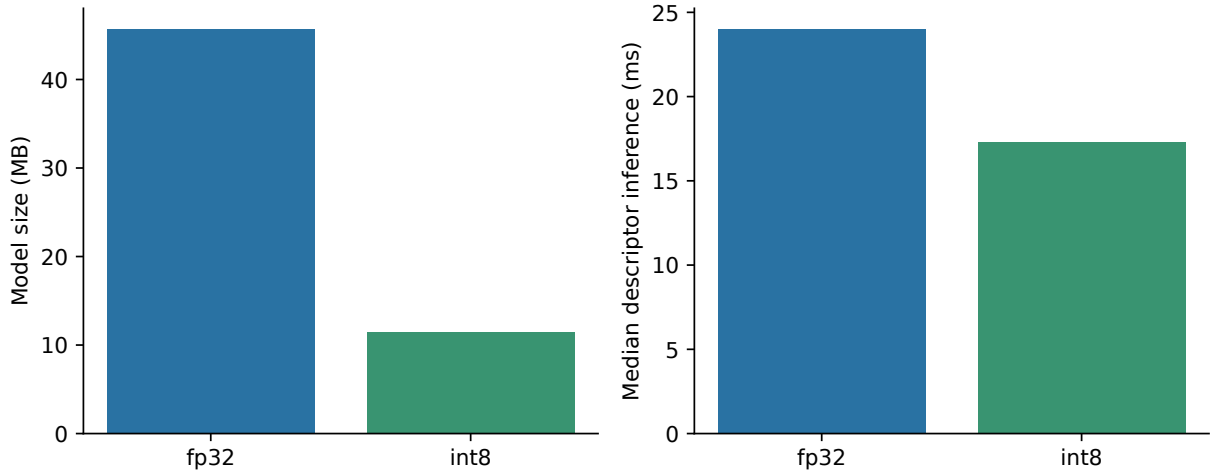

Figure 14.2: Actual global-descriptor ONNX file size and CPU inference after QDQ quantisation. These measurements exclude the local matcher and do not represent an end-to-end phone speedup.

Table 14.3: Retrieval accuracy (percent) after actual ONNX inference. Calibration uses training places, not Tokyo queries.

Protocol	Float R@1	INT8 R@1	Float R@5	INT8 R@5
tokyo_day_to_day	41.7	33.3	62.5	70.8
tokyo_day_to_night	100.0	100.0	100.0	100.0
tokyo_night_to_day	100.0	100.0	100.0	100.0
tokyo_day_to_sunset	100.0	100.0	100.0	100.0

Figure 14.2 and Table 14.3 show the trade-off. The aligned cross-time subsets retain their observed saturated retrieval, but the difficult daytime viewpoint split changes ranking accuracy. Thus quantisation is not accuracy-free even though several protocol scores remain unchanged. Calibration on training scenes also leaves an appearance-domain mismatch with Tokyo, which a broader deployment study should investigate.

The global encoder is only one part of the full query cost. Reducing it cannot remove the much larger local extraction and matching workload observed in the host profile. The descriptor export is still useful because it produces a concrete deployment candidate and reduces storage, but it does not establish that the full learned AR pipeline meets an interactive frame budget.

14.4 Recognition and tracking schedules

Design scope. The following specification describes an engineering extension; it is not a claim that a deployed multi-user service was measured.

A proposed mobile state machine has unlocalised, searching, verified and tracking states. Searching invokes retrieval and local verification. A verified observation establishes a relation to the stored place representation. A native visual-inertial tracker then propagates the camera pose at the platform's tracking rate. Loss of tracking, large uncertainty, or an elapsed refresh policy can trigger recognition again.

This design separates two failure types. Recognition failure means no supported map relation was found. Tracking failure means the relation was previously known but current device motion can no longer be estimated reliably. Reusing the last recognition output indefinitely would conceal the second failure. Instead, anchor visibility should depend on current tracking quality and the age of the last verified relation.

The state machine would require explicit time accounting. If a recognition call takes T_r and is requested at an interval Δ, the average recognition workload scales with T_r/Δ when calls do not overlap. This expression is a scheduling model, not a measured battery result. A real implementation must also account for camera processing, graphics, model warmup, thermal throttling and concurrent native tracking.

14.5 Coordinate integration with AR platforms

A persistent world-space post requires a relation between the place map and the device's current tracking world. Let $T_{w\leftarrow m}$ map map coordinates into the local tracking world, and let $T_{m\leftarrow a}$ define the stored anchor. The rendered anchor relation is

$$T_{w\leftarrow a} = T_{w\leftarrow m}T_{m\leftarrow a}. \tag{14.2}$$

The matrix direction must be documented at the API boundary; otherwise an inverse or axis-convention error can produce a stable but incorrect pose. Camera intrinsics, image rotation and screen-to-camera transforms must also be handled consistently.

The current homography prototype supplies a reference-plane image relation rather than $T_{w\leftarrow m}$. A calibrated plane with known geometry could provide additional constraints, while a sparse map and PnP supply a more general route. Neither follows automatically from a retrieved photograph. This is the key missing integration between the implemented planar demo and the social concept's arbitrary three-dimensional objects.

ARCore Cloud Anchors and ARKit saved world maps provide relevant platform mechanisms for hosting or restoring spatial relations [24, 25]. Integrating either would require its current lifecycle and privacy requirements, plus on-device tests. Azure Spatial Anchors is not proposed as a new dependency because its retirement has been confirmed [26]. The architecture should avoid making long-lived content records depend exclusively on a service identifier that cannot be migrated.

14.6 Reference coverage and GPS priors

The Tokyo viewpoint failure suggests a reference-coverage problem as well as a representation problem. A single front-facing reference may share too little visual content with a side-facing query. A proposed place model should therefore retain several directions or local submaps. Additional references raise storage and matching costs, making the retrieval hierarchy more important. This collection strategy remains a hypothesis until evaluated with an expanded gallery.

If the user permits a coarse location prior, candidate retrieval could first restrict the map region. The visual score could be combined with a spatial prior schematically as

$$S_i = \lambda s_i + (1-\lambda)\log P(p_i \mid \ell), \tag{14.3}$$

where ℓ is the permitted location observation. The parameter λ and prior model would need validation. The current experiments do not use GPS filtering, so their results cannot be attributed to such a prior.

Location filtering creates its own failure mode when a noisy or denied position excludes the correct place. A mobile client should support uncertainty-aware region expansion or a visual-only fallback. It should also avoid logging detailed movement simply to make retrieval convenient. The

privacy analysis treats the prior as an optional input with a distinct retention policy.

14.7 Descriptor and map storage

For M references and global dimension d, a dense descriptor matrix requires approximately Mdb bytes when each value occupies b bytes, before index metadata. Local storage adds keypoint coordinates, local descriptors and any associated 3D-point identifiers. The actual checkpoint and process-memory measurements show that model bytes alone are not the complete footprint.

The 128-dimensional projection experiment demonstrates one compact-descriptor route but does not establish its accuracy benefit. Quantised storage or dimensional reduction should be evaluated through retrieval recall and geometric success, not only compression ratio. Approximate search would add another accuracy-cost parameter. A scalable system should benchmark those parameters independently before combining them into a single efficiency claim.

Reference-feature caches also need an eviction policy. Keeping every reference tensor resident is feasible for this small artifact but not for a global application. A region-aware cache could retain common nearby maps and load others on demand. The implemented global index checks image fingerprints, preprocessing, resize settings and the encoder checkpoint hash. A service should additionally identify extractor code revisions and version local-feature caches, which are held only in memory in this prototype.

14.8 Anchor drift and map revision

An anchor may drift because of tracking error, incorrect map alignment, a changed physical surface, or a stale content-to-map relation. These causes require different remedies. Increasing a smoothing coefficient can reduce visible motion but cannot repair a wrong reference match. Likewise, repeated recognition of an outdated facade cannot recover a surface that has been removed.

A proposed map service should assign revisions and preserve the transform or retirement status of existing anchors. When a new map lacks a validated relation to an old anchor, that anchor should remain unresolved rather than being copied by identifier alone. Confidence and expiry should therefore be attached to the map relation as well as to the social post. Measuring these lifecycle behaviours requires longitudinal data absent from the current experiment.

14.9 Bandwidth and service-cost considerations

The local prototype makes no network-service cost measurement. A symbolic request budget can separate query-image upload, descriptor upload, reference-feature download and content-asset delivery. If N_q requests each transfer B_q bytes and accepted content adds B_c bytes, total transferred volume depends on both request count and acceptance behaviour. Caching and on-device retrieval change that accounting but can introduce stale-state and privacy costs.

Reporting a monetary cost from these expressions would require a selected provider, region, billing date, cache policy and measured workload distribution. None is present, so no fabricated per-user or per-anchor price is supplied. The engineering implication is to instrument those quantities in a future pilot before making an operational feasibility claim.

14.10 Deployment gates supported by the evidence

The present artifact is suitable for reproducible CPU research and a planar content demonstration. The descriptor-only ONNX graph and quantised variant are concrete export outputs. A phone pilot still requires native camera orientation handling, runtime-provider validation, full local-model deployment or a server split, calibrated map alignment, tracking recovery and device power testing. A public multi-user release additionally requires the access, moderation and location-privacy controls developed in the next chapter.

Chapter 15

Software Reliability and Artifact Verification

15.1 Reliability at the boundaries between modules

The research pipeline depends on several libraries with different tensor layouts, image conventions and error behaviour. Reliability therefore comes from explicit boundary contracts rather than from assuming every component returns a useful result. An image decoder can return no pixels, a detector can return no features, a matcher can return too few correspondences, and a model estimator can fail despite receiving arrays of the expected shape. Each state must be represented without accidentally reusing an older successful result.

The delivered package separates image I/O, local features, global retrieval, geometry, database access and rendering. This permits a failure to be tested at its own boundary. A geometry regression does not require network access or a large pretrained checkpoint, while an exported-model replay deliberately uses the real model artifact. The two checks provide different evidence and should not be described as interchangeable forms of model validation.

15.2 Data contracts and representation invariants

Working images are BGR arrays, while global-model preprocessing converts them to RGB and applies the recorded channel normalisation. Local learned extraction uses grayscale at the working resolution with additional internal resizing disabled. A feature record supplies points, descriptors and model-specific tensors. A match record supplies two equally sized coordinate arrays and their index pairs. These conventions constrain the meaning of every later matrix.

The geometry interface interprets its source points as reference pixels and destination points as query pixels. Failure returns an unaccepted verification record with a defined inlier-mask shape. A successful identity decision includes the selected reference, which is necessary to resolve both image dimensions and reference-bound content. The renderer cannot infer those dimensions from the query frame, because the two images may differ in size.

The database schema distinguishes its document version from the companion descriptor-index version. A content record can be valid without a precomputed descriptor matrix, while an old matrix may be incompatible with the current encoder. Keeping those versions separate allows an index rebuild without pretending that all posted content has changed. Conversely, a geometry-changing reference revision needs more than a descriptor rebuild if existing anchor polygons must be migrated.

15.3 Cache identity and invalidation

An index is valid only for the image bytes, preprocessing and model that generated it. File names alone are insufficient: replacing an image in place can preserve its path while changing all descriptors. Likewise, loading another checkpoint with the same architecture preserves the vector dimension while changing its meaning. The index therefore records SHA256 image fingerprints and the encoder checkpoint fingerprint alongside the reference list and preprocessing configuration.

The supported configuration includes descriptor input size, enhancement mode and working-image resize limit. A preprocessing-version field provides an explicit place to invalidate caches after a deliberate implementation change. The current service-free artifact records the encoder type and checkpoint identity; future distributed implementations should additionally bind the executable preprocessing revision and any local-feature cache format to their own versioned artifacts.

Loading checks the number of fingerprints, reference order, index version, configuration and finite two-dimensional descriptor matrix. These checks catch several structurally plausible stale caches. They do not establish that the checkpoint was acquired from a trusted distributor or that the database has not been maliciously rewritten. Those are separate software-supply and access-control responsibilities outside the local research demonstration.

15.4 Regression tests that target a meaningful failure

The geometry tests use a known projective map with injected outliers and verify both acceptance and corner accuracy. Empty and collinear correspondence sets exercise rejection. Homogeneous-scale tests ensure that multiplying a map by a nonzero scalar does not change the projection or rendered image. A separate extreme matrix checks that an unrepresentable normalisation is rejected before a decomposition raises an exception.

The rendering tests preserve opaque black pixels and fully transparent content, support a grayscale asset, and reject crossed or collapsed anchor quadrilaterals. These tests target visible application failures rather than copying the implementation's arithmetic into an identical assertion. A place/reference selection regression uses two places sharing a reference-path string but different cached image dimensions, ensuring that rendering resolves the full identity pair.

Database tests change a reference image after building an index, vary the encoder fingerprint or resize setting, and damage structural index fields. Each operation must make loading fail with a rebuild instruction. Duplicate place identities and empty databases are rejected. Invalid

normalised anchor polygons are rejected before neural inference, so an application does not spend a full query discovering that a stored post cannot be warped safely.

CLI checks also matter. A decoder that opens but produces no frame must not create an apparently successful empty video result. An image write that reports failure must not be followed by a success message. The regression suite covers these states with controlled test doubles. Separate model-backed image and video checks exercise the actual path through descriptors, verification, content lookup and output decoding.

15.5 Independent reconstruction of experimental decisions

The research audit does more than verify file presence. It reconstructs each saved ranking and selected candidate from the recorded global descriptors and candidate-pair evidence. It checks correct-identity ranks, accepted identities, correctly verified outcomes and selected homographies against the prediction file. Oracle-pair rows are cross-checked against the same underlying evidence rather than treated as a second independent source of results.

Candidate acceptance is also recomputed from support counts, ratios, configuration and the current projective-validity checks. This is useful after a numerical bug fix: it establishes whether the corrected predicate changes any recorded decision. If it did, the old reported statistics would need a clearly identified revision. In the reviewed artifact, the saved experiment outcomes remain consistent with the corrected implementation.

The audit recomputes end-to-end homography success using the frozen query labels, ground-truth coordinate conversions and saved selected maps. It then checks the CSV aggregates and generated statistics. This chain can detect a summary-table error even when the underlying neural inference is left untouched. It cannot discover an incorrectly labelled public image solely from hashes and arithmetic, which is why image-level inspection and dataset provenance remain necessary.

15.6 Replaying saved model artifacts

The short adaptation experiment produces six projection-head checkpoints: three seeds with and without the additional night augmentation. Verification loads each saved head, applies it to the retained native descriptors, rebuilds the held-out rankings and compares every recorded training-result row. This validates the connection between checkpoint files and reported adaptation accuracy without rerunning optimisation or selecting another training seed.

The exported-model audit validates the produced ONNX graphs and executes both float and INT8 graphs on the actual Tokyo images. It compares their retrieval scores with the export record and checks output differences against the retained native descriptors. This is stronger than checking that a graph file can be opened. It remains narrower than a phone deployment test because the execution provider is the recorded host CPU runtime.

Timing aggregates are independently recomputed from the stored per-call samples. That

catches a mistaken median, mean or upper quantile without pretending to obtain a fresh latency measurement. The distinction is explicit: replaying accuracy, recomputing aggregates and rebenchmarking runtime are separate activities. The final audit states which activities were executed.

15.7 Document builds as a dependency graph

The report consumes generated tables and figures, and the presentation consumes a subset of those same generated artifacts. Consequently, a parallel build must finish report-data generation before presentation-data generation. Otherwise, a successful compiler invocation could still produce slides containing the previous version of a figure or number. The build now enforces report-before-presentation order even when Make is invoked with parallel jobs.

Author metadata and the copied code listings are refreshed before the report is compiled. The LaTeX process stops on errors, and post-build checks reject unresolved citations, references, overfull boxes and oversized floats. Compiler success alone is not sufficient evidence of a correct report: metadata, empirical inputs, reference-layout measurements and final deliverable hashes are checked separately.

The source packages are tested after extraction into newly created temporary directories. Reusing an old build directory could hide a missing source file or bibliography artifact, because the compiler might find a stale generated dependency. Fresh extraction prevents that particular false pass. The standalone Makefiles compile only the bundled LaTeX snapshot; they do not require datasets, model downloads or the project's Python environment.

15.8 Provenance across report and presentation formats

The canonical report PDF and presentation PDF are copied into the deliverables folder, and a manifest records file lengths and SHA256 values. The source checker compares bundled files with their current project counterparts. These checks detect an archive that still contains an older chapter even when the main PDF has already been rebuilt. The reference PDFs supplied by the author remain outside the set of files edited for the Stitch report.

The PowerPoint is an image-based export of the LaTeX slides, with extracted slide text in notes. This preserves the mathematics and original institutional artwork while making the LaTeX source the editable master. It does not provide separately editable PowerPoint text boxes. The limitation is documented in the package so reviewers do not discover it only after attempting a late presentation edit.

15.9 What the final check can and cannot establish

Passing tests and artifact audits establish consistency for the implemented contracts and retained experiment files. They do not prove that no software defect remains under every possible input.

The reviewed system still has the documented scientific limits: a small selected gallery, planar image relations, synthetic appearance stressors and host-only runtime measurements. Reliability checks strengthen the trustworthiness of those bounded results; they do not enlarge their empirical scope.

The most useful next reliability step is therefore targeted coverage. A future failure should be reduced to a reproducible input and a contract-level regression, followed by a separately identified experiment if the decision policy changes. This keeps fixes reviewable and prevents the report from drifting away from the executable implementation it describes.

Chapter 16

Reference Revisions and Persistent-AR Transactions

16.1 Scope of this service design

Design scope. The following specification describes an engineering extension; it is not a claim that a deployed multi-user service was measured.

The implemented artifact is a local planar recogniser with JSON place records and image posts. It has no authenticated posting server, distributed map service or native tracking integration. This chapter develops a concrete transaction and revision design to explain what would be required to turn a verified image relation into a maintainable social-AR system. The proposed states and messages are specifications for future implementation, not evidence of a deployed backend.

The design begins with an important restriction. A post is not attached to a place name alone; it is attached through a particular coordinate representation and revision. If that representation changes, the post needs a validated migration relation or an explicit unresolved state. A system that retains only a place identifier cannot determine whether an old polygon is still geometrically meaningful after reference replacement.

16.2 Separate identities for places, references and anchors

A place identifier represents a semantic location used for retrieval and content grouping. A reference identifier represents an observation or map fragment with its own image fingerprint, calibration metadata if available and revision history. An anchor identifier represents a content attachment expressed in that reference or map domain. These identifiers should remain separate even when an initial prototype contains only one reference and one post per place.

For the current planar model, an anchor can be represented conceptually as (p, r, v, A): place p, reference r, reference revision v and content-to-reference map A. A query supplies a verified $H_{r \to q}$ for the same reference revision. Rendering is permitted only when the revision and coordinate interpretation agree. Merely finding another photograph with the same place label is insufficient

to compose the stored map.

A future three-dimensional representation would replace A with an anchor pose in a map coordinate system and would additionally require calibrated localisation and a relation to the device tracking world. The transaction principles still apply, but the present homography is not silently promoted into that pose. The mapping and validation work would need to be executed and measured separately.

16.3 A proposed posting transaction

The client first creates a draft containing content metadata and a candidate attachment. The draft is not immediately public. A verification step resolves a supported reference or map revision and checks that the attachment lies within the allowed coordinate domain. The server then checks permissions, content visibility and whether the referenced revision remains active before committing the post.

The commit should use an idempotency key generated for the user's logical posting action. If a network timeout causes the client to retry, the same logical request should return the already created record rather than create a second post. The key must be scoped to the user and operation, with a documented lifetime. An arbitrary repeated image is not a suitable idempotency key because a user may intentionally post two different records using the same asset.

The commit also needs a revision precondition. A client that verified revision v should not accidentally attach its draft to revision $v + 1$ if the map changed while the asset was uploading. The server can reject the stale precondition or retain an unresolved draft. It should not reinterpret coordinates without a validated transfer relation. This prevents an ordinary concurrent update from becoming a silent anchor-placement error.

16.4 Observation verification and content delivery

A visiting client sends an observation or descriptor under a declared privacy policy and receives candidate information appropriate to its permissions. Local or server-side verification resolves a reference identity and revision. Only after that relation is supported does the content service return visible posts bound to that revision. Recognition and authorisation remain distinct: visual access to a facade is not permission to read every private post associated with it.

The response should carry a bounded validity interval or revision token so the client can determine whether cached content remains current. The exact interval is a product policy requiring measurement, not a value derived from the desktop benchmark. A client can cache image assets independently of anchor validity, but it must stop rendering them when the corresponding anchor relation becomes invalid or access is revoked.

For a planar implementation, the response must preserve the reference-specific attachment. Posts attached to another view can remain unresolved until that view is verified or a validated reference relation becomes available. The local database's optional content-reference field is a

limited version of this principle. It avoids inventing a cross-view transfer merely to display more content.

16.5 Reference replacement and migration

Suppose an old reference r_o is replaced by a new reference r_n, and a validated image relation $M_{o\to n}$ is available for the relevant plane. A content-to-old-reference map A_o could be transferred as

$$A_n = M_{o\to n} A_o. \tag{16.1}$$

This composition is valid only within the relation's geometric scope. If the old post lies on a different depth layer, an occluded region or a changed physical object, a single plane relation may not preserve its intended attachment. Migration therefore needs an explicit validity policy rather than an assumption that all polygons can be copied through any successful image match.

If no trustworthy relation exists, the service should retain the original revision for history and mark the active attachment unresolved or retired. A user-assisted reattachment can create a new anchor revision while preserving the content's identity and audit history. This design distinguishes content deletion from coordinate repair: a geometrically obsolete attachment need not imply that its text or media must be silently discarded.

Rollback also needs a declared rule. Restoring an earlier reference revision does not automatically mean every later post should be attached to it. Each anchor carries the revision on which it was created or migrated. A rollback operation must evaluate those references explicitly and avoid generating a state in which the database label says one revision while cached geometry assumes another.

16.6 State transitions for a displayed post

A useful proposed lifecycle includes draft, attached, unresolved, retired and deleted states. Draft means no public attachment has been committed. Attached means a valid coordinate relation and visibility policy exist. Unresolved means the content exists but its current geometric relation cannot be established. Retired preserves a deliberate historical record without active rendering, while deleted removes the content under the service's deletion policy.

These states should not be inferred from whether an asset file is still present in a cache. A cached image may remain available after its anchor becomes unresolved or its visibility changes. Rendering decisions need the authoritative anchor state and access token. Otherwise, an offline cache can preserve an attachment that the map or content service already invalidated.

State transitions should record the reason and revision involved. A failure to recognise a place during one query is not necessarily a permanent unresolved state for every post there. Conversely, a retired map revision should not become active merely because a future frame happens to match its old photograph. Query uncertainty and persistent lifecycle changes are different events.

16.7 Tracking and recognition as separate clocks

Recognition can be expensive relative to display refresh, as the recorded CPU measurements demonstrate. A future client should therefore treat recognition and native tracking as processes with different update rates. Recognition establishes or refreshes a map relation; tracking propagates a previously supported relation through camera motion. Losing tracking must invalidate the presentation state even when the last recognition result was correct.

The client should retain timestamps and revision identities for both relations. A mathematically composable transform can still be stale if its map revision or observation age is no longer acceptable. A recovery policy might request new recognition after occlusion, a tracking reset or a large uncertainty increase. Those triggers need device measurements and session-level evaluation before they can be recommended as fixed operating values.

The current demo does not test this state machine. Its still-image motion is generated and its comparison displays actual per-frame recogniser outputs. It is therefore evidence that the local recognition and rendering modules execute together, not a measurement of inertial tracking quality or recovery after a real device relocalisation.

16.8 Consistency and partial failure

Uploading an asset, committing metadata and publishing an attachment are distinct operations that can fail independently. A future service needs a cleanup policy for uploaded assets whose post commit never succeeds. It also needs a way to return a committed post after the client loses the acknowledgement. Idempotent commits and explicit draft identifiers address different sides of this problem.

A map update can race with content delivery. The response should identify the revision used to verify the query and the revision required by each returned anchor. The client must reject a mismatched combination rather than combining a new reference hash with an old content-to-reference matrix. This is the distributed counterpart of the local descriptor-index configuration checks.

Deletion can race with caching and replication as well. A service design should specify which events invalidate anchor records, access tokens and cached asset references. This chapter does not assign a legal retention period or claim a globally immediate deletion guarantee. It identifies the technical dependencies that an implemented policy would need to control and test.

16.9 A testable implementation roadmap

The first extension should retain a small local gallery while adding explicit reference and anchor revisions. Tests can then simulate a replaced photograph, an idempotent retry, a revoked post and a query using a stale revision. The expected behaviour is concrete: either a compatible attachment is returned or rendering is withheld with a traceable state. No city-scale infrastructure is required to validate those contracts initially.

The next stage would connect calibrated localisation and native tracking on a real device, using an evaluation set that measures alignment error, recovery and wrong-place display over time. Only then would broader geographic indexing and multi-user load tests provide meaningful deployment evidence. Expanding the server before validating the coordinate lifecycle would scale an unresolved geometric ambiguity rather than resolve it.

Chapter 17

Ethics, Privacy and Safety

17.1 Why the application changes the risk model

Stitch associates digital content with physical places and retrieves it through camera observations. That combination can reveal more than a conventional image-processing demonstration. A query may expose where a person is standing, which private space they are visiting, or which content they are permitted to see. A persistent post can also affect a person who never installed the application if it is attached to their home or workplace.

The present artifact is a local research prototype. It does not upload user camera frames to a service, publish posts, authenticate users or enforce production moderation. This boundary reduces the actions performed during the experiment but does not resolve the requirements of the proposed application. The following analysis treats those requirements as design obligations and identifies where the current software supplies no guarantee.

17.2 Captured faces, vehicles and private details

Reference and query images can include people, number plates, documents, screens and interiors. Even when these details are irrelevant to place recognition, they may remain in stored imagery or be recoverable from distributed assets. The research therefore uses documented public datasets and the supplied repository media rather than collecting new unconsented live footage. Dataset availability is not treated as proof that every depicted person's privacy concern has disappeared.

A deployment should minimise the imagery retained for localisation and review whether redaction is needed before storage or sharing. Face and number-plate filtering itself can fail, so an implementation should not label a map "anonymous" merely because a detector ran. Human review, a reporting channel and bounded retention may still be appropriate. No face or plate detector is evaluated in this artifact, and no redaction accuracy is claimed.

The map's visual requirements can conflict with privacy. A repeated window pattern or storefront sign may be useful for recognition while also identifying a sensitive place. Replacing pixels with descriptors can reduce direct image exposure but does not prove that the resulting representation contains no location or identity information. The thesis consequently avoids calling

local descriptors or global embeddings inherently anonymous.

17.3 Location privacy and retrieval logs

A sequence of recognised places can become a movement history. Such a history may be more sensitive than an individual query because it reveals routines and associations. An engineering design should therefore separate transient inference from persistent analytics. A client can use a place hypothesis to render authorised content without retaining a timestamped record of every unsuccessful or successful observation.

A GPS prior should be optional and purpose-limited. Its accuracy and retention policy should be separate from the camera-processing policy. If location permission is denied, the application should expose a constrained visual-only mode or explain the functional limitation without covertly reconstructing a movement log. The current benchmark uses site metadata for evaluation labels, not a live tracking service.

The public legislative text of the GDPR articulates purpose limitation, data minimisation and storage limitation among its principles [29]. These principles inform the proposed design; the thesis does not certify compliance in any jurisdiction. Deployment would require review of the actual data flows, roles, lawful basis and applicable local law. A research dataset licence and an application privacy notice address different questions and cannot substitute for one another.

17.4 Consent and control over persistent content

The social concept needs more than permission from the person creating a post. A virtual object attached to another person's property or image may affect that person and later visitors. The application should define who can create public anchors at a place, whether owners or community managers can request removal, and how private or restricted locations are handled. Technical visibility of a surface does not itself establish permission to use it as a public message board.

Posts should have explicit visibility and expiry rules. Private content should be retrieved only after authorisation, even if the place is correctly recognised. Shared spatial coordinates can support several audiences without exposing all content to every visitor. The current JSON schema illustrates author and content identifiers, but it contains no authentication or access-control engine. Those fields are examples, not security guarantees.

Deletion should cover more than a row in the post database. Cached assets, thumbnails, search indexes and map associations may retain the content or its location. A proposed service should define how those copies are invalidated and what minimal audit information remains. The file-based prototype makes no distributed deletion claim, and its research outputs should not be mistaken for a retention policy for production users.

17.5 Moderation and misuse of place-bound posts

Persistent AR can be used for harassment, impersonation, targeted advertising, offensive content or deceptive navigation cues. Attaching a message to a real building may imply endorsement by the building's occupants. Even a harmless image can be abusive when repeatedly placed at a particular person's home. Moderation must therefore consider the relation among content, target place, audience and author, not only a classifier score on the uploaded asset.

A proposed moderation flow should support reporting, temporary hiding, appeal and removal. Limits on repeated posting and duplicate anchors can reduce spam, but limits should not be described as protection against all abuse. Human escalation is likely to be required for context-dependent cases. The present artifact implements none of these workflows and does not send any posts to an external service.

The original "digital garbage" metaphor should also be treated carefully in product design. It motivates playful spatial content, but uncontrolled accumulation can obscure the real environment or overwhelm an intended public place. User-controlled density, visibility and content categories are therefore important interface choices. These are requirements inferred from the application's persistence and public-space setting, rather than measured user-experience findings.

17.6 False localisation as an application hazard

The measured open-set diagnostic in Table 10.7 shows that geometric support can still produce a wrong-place acceptance. Such an error can display content on an unintended facade or suggest that a message belongs to the current location. A recogniser should therefore communicate uncertainty and avoid presenting a weakly supported anchor as authoritative spatial guidance.

This concern becomes more serious when AR content resembles instructions, warnings, official signage or emergency information. The current prototype is an image-post demonstration and should not be treated as a navigation or safety system. A future product should restrict misleading presentation and preserve a clear distinction between application content and real-world authority. The thesis makes no claim that the existing verifier is sufficient for safety-critical use.

Visual replay is another unresolved threat. A photograph or display showing a known place can trigger appearance-based recognition without the device being physically present there. Neither a correct descriptor rank nor a homography alone proves physical presence. A stronger presence claim would require additional signals and a threat model, and those signals should be evaluated without creating disproportionate location surveillance.

17.7 Map integrity and adversarial changes

A map can become incorrect through ordinary environmental change or malicious manipulation. An attacker might submit a misleading reference image, poison a shared place identity or attach content using an obsolete map relation. A proposed service should therefore authenticate map updates, version references and record who is authorised to modify them. The current local

database trusts its input files and is not a hardened ingestion service.

Reference fingerprints in the artifact help detect accidental stale caches, but a file hash is not proof that a scene is genuine. It identifies bytes, not their truthfulness or consent status. Similarly, the command log makes the research process auditable without certifying every dataset label. Security controls and scientific provenance are related forms of accountability but solve different problems.

17.8 Dataset and model licensing

The artifact separates imagery, software implementations, pretrained checkpoints and generated results. Each may carry different conditions. Official Aachen and RobotCar documentation, for example, specifies non-commercial academic restrictions; those datasets were not imported into the executed image set. HPatches includes source-specific attribution, and the Tokyo archive should not be assumed to grant unrestricted commercial redistribution.

The learned implementations are obtained from their official repositories and model hashes are recorded. Retaining a repository licence is necessary but may not resolve all terms associated with third-party weights or source datasets. The thesis therefore supplies provenance and links rather than a blanket assertion that the resulting application is commercially cleared. A production release should review every dependency and data source actually distributed.

17.9 Third-party platform privacy

If the proposed client uses ARCore Cloud Anchors or related geospatial services, it must consider the platform's current privacy disclosure requirements. Google's documentation specifically addresses disclosure of sensor-data processing for these APIs [30]. The experiments do not invoke those services, so no consent screen or remote-processing claim is fabricated in the artifact.

An ARKit world-map integration would likewise change the stored spatial state and sharing model. Native platform capability does not remove the application's responsibility for asset visibility, consent or retention. A clear architecture should document which processing happens on the device, which observations leave it, which identifiers are shared, and how a user can stop or remove persistent participation.

17.10 Research integrity

Ethical reporting also concerns the experiment itself. The thesis retains failed download checks, negative adaptation results, false accepted places and unavailable pose metrics. It does not turn a synthetic low-light transformation into a claimed real-night capture. It labels generated camera motion in the demo, distinguishes custom subsets from official benchmarks, and avoids presenting host CPU timings as phone measurements.

Author and institution details follow the supplied IIT Roorkee reference documents, and the

supervisor was explicitly selected by the author. Signatures and institutional acceptance are not manufactured; the certification remains unsigned for review. Bibliographic entries refer to real verified sources, and empirical figures derive from preserved outputs. These choices keep the artifact reviewable without manufacturing the social or institutional context around the research.

17.11 Responsible progression beyond the prototype

A limited pilot should first use a small consented set of places and non-sensitive content, with clear deletion and reporting mechanisms. It should measure wrong-place acceptance, tracking loss and content exposure alongside recognition recall. A later public system would need a broader place and device population, operational moderation and independent review of its data practices. The current evidence supports beginning that engineering evaluation; it does not support bypassing it.

Chapter 18

Conclusion and Future Work

18.1 Summary of the implemented work

This thesis converted Stitch's classical reference-overlay demonstration into a reproducible research artifact for place recognition and planar content anchoring. The original scripts were retained and executed through a bounded file-input adapter. A modular package adds global retrieval, learned local matching, shared geometric verification, explicit alpha rendering, a place/-content database, configuration and CLI operation. The report is built from saved predictions, data and checkpoint manifests, exact execution logs and generated figures and tables.

The evaluated data comprise 144 selected HPatches images with 120 ground-truth planar pairs, 96 real Tokyo photographs from 24 selected sites, and 90 seeded synthetic queries. These collections establish different evidence levels. The artifact reports true planar accuracy where homographies exist, place correctness where site labels exist, and unavailable metric pose rates where the necessary calibration and map are absent.

18.2 Answers to the research questions

18.2.1 Local correspondence under appearance change

The selected HPatches illumination comparison improves from 81.7% SIFT homography success to 100.0% with SuperPoint–LightGlue. This supports the use of pretrained local representations for the tested illumination changes. The viewpoint comparison yields 86.7% and 86.7%, respectively, and does not support a universal learned-method advantage. Geometric support and actual boundary accuracy must both be measured.

18.2.2 Global retrieval and candidate search

On the selected Tokyo day-to-night protocol, SIFT Recall@1 is 58.3%, while the learned local and CosPlace methods reach the observed saturated rate reported in the generated tables. The hierarchy correctly verifies 100.0% of those queries with a small shortlist. On the different-view daytime protocol, however, its correctly verified rate is 37.5%, showing that the global shortlist

can discard references recoverable by exhaustive local search.

The answer is therefore conditional: global retrieval is an effective search reduction when the useful view remains near the top of the global ranking. It is also an accuracy bottleneck when scene overlap or viewpoint reduces that ranking quality. The small custom Tokyo gallery and aligned cross-time views prevent direct comparison with official city-scale benchmark claims.

18.2.3 Verification and anchoring

Retrieval success is insufficient evidence for rendering. SIFT correctly verifies 33.3% of the selected day-to-night queries despite a higher retrieval recall, illustrating the difference between a ranked identity and accepted geometric support. The open-set diagnostic further shows that local verification can still accept a wrong candidate. The implemented rejection path is necessary, but a production identity or physical-presence guarantee requires stronger evaluation and controls.

The planar pipeline successfully connects a verified place record to its stored content and a rendered image overlay. It does not establish arbitrary three-dimensional persistence, occlusion correctness, world-scale pose or cross-device anchor consistency. Those require calibrated mapping and a native tracking integration beyond the current image-to-image homography model.

18.2.4 Short adaptation

The executed projection-head training does not provide a consistent held-out improvement over the pretrained global descriptor. Three fixed seeds expose variability, and synthetic night augmentation does not remove the mismatch between the limited training scenes and the evaluation conditions. The pretrained descriptor remains the primary pipeline choice. This negative result narrows the roadmap: broader data and a compression control are needed before attributing improvement to appearance adaptation.

18.2.5 Deployment trade-offs

The descriptor-only ONNX export and QDQ quantisation both execute. File size changes from 45.75 MB to 11.52 MB, and the recorded ONNX CPU descriptor latency changes from 24.04 ms to 17.32 ms. The difficult viewpoint split loses some retrieval accuracy, so quantisation must be validated through task outcomes as well as file size.

The complete learned pipeline's isolated host median is 1332.5 ms. It improves on exhaustive learned search but does not support per-display-frame recognition on this configuration. A mobile design should use occasional recognition to establish a map relation and native tracking to maintain it. No phone runtime, thermal or power result is claimed.

18.3 Principal limitations

The strongest limitation is coverage. The gallery is small, geographic diversity is narrow, and real seasonal/weather transitions were not evaluated. Synthetic transformations cannot replace those

observations. The optional training uses very few place identities, and its seed intervals do not represent the uncertainty of collecting a new city or device population.

The second limitation is geometric scope. A homography is appropriate for a plane-induced map, not a complete street scene. Tokyo supplies no independent homography or calibrated pose for the selected pairs. An accepted overlay there is consequently a demonstration of the implemented decision, not a quantified metric-localisation result. The requested translation-and-rotation thresholds remain unavailable.

The third limitation is deployment completeness. The package is a local CPU research implementation, with a compact on-disk database and image-post renderer. It has no authenticated multi-user backend, distributed deletion, map revision service or operational moderation. Export covers the global descriptor only. The proposed service and mobile sections define requirements rather than claim an existing product.

18.4 Staged future work

The first stage should broaden evaluation before changing model complexity. A consented multi-direction reference set and larger public galleries can test the observed viewpoint bottleneck. RobotCar, Extended CMU and MSLS can add genuine seasonal and weather transitions. Their acquisition and label conventions should be frozen before inference, following the same manifest and command-log discipline used here.

The second stage should add calibrated map-relative localisation. A sparse reconstruction with known intrinsics and validated 2D–3D associations would support PnP and the requested metric thresholds. Anchor regions should be represented in map or plane coordinates, with uncertainty and explicit conventions. This stage should evaluate boundary error, reprojection support and actual camera pose separately.

The third stage should establish native mobile feasibility. It should export or otherwise deploy the local feature and matching stages, profile their supported backend operators, and measure complete camera-to-render latency on representative devices. Recognition scheduling, tracking drift, loss recovery, memory pressure, thermal behaviour and power should be measured under sustained operation rather than inferred from a short warmed host run.

The fourth stage should address persistent social operation. Place/map revisions, access-controlled content, consent, deletion and moderation must be implemented together. A pilot should measure false content placement and user-visible failure behaviour, not only retrieval recall. The open-set results show why a visually plausible map cannot be treated as complete authorisation or identity evidence.

18.5 Closing assessment

The work provides a concrete, auditable path from Stitch's supplied demonstration to a learned planar-AR research system. Its contribution lies in executed integration, controlled comparison

and honest boundaries. The evidence supports improved illumination correspondence and useful coarse retrieval on the selected data, while exposing viewpoint, false-acceptance, adaptation and runtime limitations that must shape the next implementation.

Bibliography

[1] BitsJayMehta173. Stitch: Mobile AR/VR source repository. GitHub, inspected at commit 14a01a671d9b78751f817c4b8c2bc8c0e3d2f628, 2026. URL https://github.com/BitsJayMehta173/Stitch. Accessed 5 November 2026.

[2] Torsten Sattler, Will Maddern, Carl Toft, Akihiko Torii, Lars Hammarstrand, Erik Stenborg, Daniel Safari, Masatoshi Okutomi, Marc Pollefeys, Josef Sivic, Fredrik Kahl, and Tomas Pajdla. Benchmarking 6DOF outdoor visual localization in changing conditions. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8601–8610, 2018. URL https://openaccess.thecvf.com/content_cvpr_2018/papers/Sattler_Benchmarking_6DOF_Outdoor_CVPR_2018_paper.pdf.

[3] Vassileios Balntas, Karel Lenc, Andrea Vedaldi, and Krystian Mikolajczyk. HPatches: A benchmark and evaluation of handcrafted and learned local descriptors. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2017. URL https://arxiv.org/abs/1704.05939.

[4] Akihiko Torii, Relja Arandjelović, Josef Sivic, Masatoshi Okutomi, and Tomas Pajdla. 24/7 place recognition by view synthesis. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 1808–1817, 2015. URL https://openaccess.thecvf.com/content_cvpr_2015/papers/Torii_247_Place_Recognition_2015_CVPR_paper.pdf.

[5] David G. Lowe. Distinctive image features from scale-invariant keypoints. *International Journal of Computer Vision*, 60(2):91–110, 2004. doi: 10.1023/B:VISI.0000029664.99615.94. URL https://www.cs.ubc.ca/~lowe/papers/ijcv04.pdf.

[6] Marius Muja and David G. Lowe. Fast approximate nearest neighbors with automatic algorithm configuration. In *International Conference on Computer Vision Theory and Applications (VISAPP)*, pages 331–340, 2009. doi: 10.5220/0001787803310340. URL https://www.scitepress.org/PublishedPapers/2009/17878/17878.pdf.

[7] Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. SuperPoint: Self-supervised interest point detection and description. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2018. URL https://openaccess.thecvf.com/content_cvpr_2018_workshops/papers/w9/DeTone_SuperPoint_Self-Supervised_Interest_CVPR_2018_paper.pdf.

[8] Paul-Edouard Sarlin, Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. SuperGlue: Learning feature matching with graph neural networks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4938–4947, 2020. URL https://openaccess.thecvf.com/content_CVPR_2020/html/Sarlin_SuperGlue_Learning_Feature_Matching_With_Graph_Neural_Networks_CVPR_2020_paper.html.

[9] Philipp Lindenberger, Paul-Edouard Sarlin, and Marc Pollefeys. LightGlue: Local feature matching at light speed. In *IEEE/CVF International Conference on Computer Vision*, pages 17627–17638, 2023. URL https://openaccess.thecvf.com/content/ICCV2023/html/Lindenberger_LightGlue_Local_Feature_Matching_at_Light_Speed_ICCV_2023_paper.html.

[10] Computer Vision and Geometry Group. LightGlue: Official inference implementation, 2026. URL https://github.com/cvg/LightGlue. Accessed 5 November 2026; exact revision recorded in results/models.json.

[11] Jiaming Sun, Zehong Shen, Yuang Wang, Hujun Bao, and Xiaowei Zhou. LoFTR: Detector-free local feature matching with transformers. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8922–8931, 2021. URL https://openaccess.thecvf.com/content/CVPR2021/html/Sun_LoFTR_Detector-Free_Local_Feature_Matching_With_Transformers_CVPR_2021_paper.html.

[12] Relja Arandjelović, Petr Gronat, Akihiko Torii, Tomas Pajdla, and Josef Sivic. NetVLAD: CNN architecture for weakly supervised place recognition. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2016. URL https://www.cv-foundation.org/openaccess/content_cvpr_2016/papers/Arandjelovic_NetVLAD_CNN_Architecture_CVPR_2016_paper.pdf.

[13] Gabriele Berton, Carlo Masone, and Barbara Caputo. Rethinking visual geo-localization for large-scale applications. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. URL https://openaccess.thecvf.com/content/CVPR2022/papers/Berton_Rethinking_Visual_Geo-Localization_for_Large-Scale_Applications_CVPR_2022_paper.pdf.

[14] Gabriele Berton and contributors. CosPlace: Official model and pretrained weights, 2026. URL https://github.com/gmberton/CosPlace. Accessed 5 November 2026.

[15] Amar Ali-bey, Brahim Chaib-draa, and Philippe Giguère. MixVPR: Feature mixing for visual place recognition. In *IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 2998–3007, 2023. URL https://openaccess.thecvf.com/content/WACV2023/html/Ali-bey_MixVPR_Feature_Mixing_for_Visual_Place_Recognition_WACV_2023_paper.html.

[16] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. DINOv2: Learning robust visual features without supervision. arXiv:2304.07193, 2023. URL https://arxiv.org/abs/2304.07193.

[17] Richard Hartley and Andrew Zisserman. *Multiple View Geometry in Computer Vision*. Cambridge University Press, 2 edition, 2004. URL https://robots.ox.ac.uk/~vgg/hzbook/.

[18] Vincent Lepetit, Francesc Moreno-Noguer, and Pascal Fua. EPnP: An accurate $O(n)$ solution to the PnP problem. *International Journal of Computer Vision*, 81(2):155–166, 2009. doi: 10.1007/s11263-008-0152-6. URL https://www.iri.upc.edu/people/fmoreno/Publications/2009/pdf/Lepetit_ijcv2009.pdf.

[19] Martin A. Fischler and Robert C. Bolles. Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM*, 24(6):381–395, 1981. doi: 10.1145/358669.358692. URL https://www.sri.com/publication/artificial-intelligence-pubs/random-sample-consensus-a-paradigm-for-model-fitting-with-applications-to-image-analysis-and-automated-cartography-2/.

[20] Paul-Edouard Sarlin, Cesar Cadena, Roland Siegwart, and Marcin Dymczyk. From coarse to fine: Robust hierarchical localization at large scale. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12716–12725, 2019. URL https://openaccess.thecvf.com/content_CVPR_2019/html/Sarlin_From_Coarse_to_Fine_Robust_Hierarchical_Localization_at_Large_Scale_CVPR_2019_paper.html.

[21] HPatches authors. HPatches dataset documentation and source attribution, 2026. URL https://github.com/hpatches/hpatches-dataset. Accessed 5 November 2026.

[22] Visual Localization Benchmark maintainers. Benchmarking long-term visual localization: Datasets and evaluation, 2026. URL https://www.visuallocalization.net/datasets/. Accessed 5 November 2026.

[23] Frederik Warburg, Soren Hauberg, Manuel Lopez-Antequera, Pau Gargallo, Yubin Kuang, and Javier Civera. Mapillary street-level sequences: A dataset for lifelong place recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2626–2635, 2020. URL https://openaccess.thecvf.com/content_CVPR_2020/html/Warburg_Mapillary_Street-Level_Sequences_A_Dataset_for_Lifelong_Place_Recognition_CVPR_2020_paper.html.

[24] Google. Cloud anchors allow different users to share AR experiences, 2026. URL https://developers.google.com/ar/develop/cloud-anchors. Accessed 5 November 2026.

[25] Apple. Saving and loading world data, 2026. URL https://developer.apple.com/documentation/ARKit/saving-and-loading-world-data. Accessed 5 November 2026.

[26] Microsoft Research. Enabling interaction between mixed reality and robots via cloud-based localization, 2020. URL https://www.microsoft.com/en-us/research/blog/enabling-interaction-between-mixed-reality-and-robots-via-cloud-based-localization/. Page updated to note retirement of Azure Spatial Anchors in November 2024; accessed 5 November 2026.

[27] Gabriele Berton and contributors. Tokyo 24/7 dataset formatting and public query download, 2026. URL https://github.com/gmberton/VPR-datasets-downloader/blob/main/format_tokyo247.py. Accessed 5 November 2026.

[28] ONNX Runtime contributors. Quantize ONNX models, 2026. URL https://onnxruntime.ai/docs/performance/model-optimizations/quantization.html. Accessed 5 November 2026.

[29] European Parliament and Council of the European Union. Regulation (EU) 2016/679: General data protection regulation, 2016. URL https://eur-lex.europa.eu/eli/reg/2016/679/oj. Official legislative text, consulted 5 November 2026.

[30] Google. ARCore user privacy requirements, 2026. URL https://developers.google.com/ar/develop/privacy-requirements. Accessed 5 November 2026.

Appendix A

Reproduction Instructions and Command Ledger

A.1 Environment and setup

All commands below are issued from the delivered project root unless a directory change is shown. The setup script selects a supported CPython interpreter, creates a local virtual environment, installs the pinned requirements and local package, obtains the official learned-matcher source at its recorded revision, and downloads the pretrained checkpoints. The data directory, model directory and environment are excluded from source control.

```
bash scripts/setup.sh
make data
make baseline
make evaluate
make train
make export
make demo
make profile
make test
make thesis
```

The trained-head and export steps use the frozen data preparation. The demonstration and artifact generator consume the main evaluation results. The complete-query profile is executed after other heavy inference tasks finish. The output PDF is `report/thesis.pdf`; comparison clips and animation are in `demo/`. Exact arguments used in this run are preserved in `results/commands.jsonl` and in the ledger later in this appendix.

A.2 Direct CLI checks

```
.venv/bin/python scripts/run_logged.py .venv/bin/stitch index data/places.json
.venv/bin/python scripts/run_logged.py .venv/bin/stitch query data/places.json test.jpg results/raw/
    cli_query.png
.venv/bin/python scripts/run_logged.py .venv/bin/stitch video data/places.json video.mp4 results/raw/
    cli_video.mp4 --limit 8
```

```
.venv/bin/python scripts/run_logged.py .venv/bin/python scripts/download_data.py --verify-only
```

The query image is already represented in this demonstration database, so this CLI check is a functional content-retrieval check rather than a held-out accuracy result. The video command exercises decoding, EOF/limit handling, recognition and writing. Bounded file inputs replace the unavailable private DroidCam stream; no live phone capture is claimed.

A.3 Data integrity and restoration

`results/data_manifest.json` pins downloaded filenames, lengths and SHA256 hashes. `results/preparation.json` pins generated synthetic inputs and the frozen benchmark manifest. Existing valid collections are verified and skipped by the downloader. Missing collections can be restored from the recorded public sources and checked against the supplied manifest. A corrupt existing file fails explicitly, preventing unnoticed input replacement.

The original HPatches host and author mirror are documented in Chapter 5. Tokyo subset members are verified against ZIP CRC and decompressed size before their local hashes are recorded. Full-archive transfer failures remain in the command journal. Reproduction depends on the continued availability and terms of the original sources; their presence in this artifact is not a promise that a remote URL will remain unchanged.

A.4 Manual datasets for expanded evaluation

A.4.1 Aachen Day-Night v1.1

The official landing page is https://www.visuallocalization.net/datasets/. The inspected public archive root is https://data.ciirc.cvut.cz/public/projects/2020VisualLocalization/Aachen-Day-Night/. Follow both official README files, obtaining base images, the v1.1 extension, query intrinsics and the appropriate 3D reconstruction. The extension is not a substitute for every required base file. A proposed local layout is:

```
data/manual/aachen/
  images/
    database/
    query/
  queries/
  3D-models/
  aachen_v1_1/
```

Actual names should follow the archive README; an adapter should map them into explicit image, camera and 3D-point records. Query-pose evaluation may require the benchmark's evaluation procedure rather than public ground truth. The official image/model terms specify non-commercial academic restrictions. No Aachen pose score is supplied by the current artifact.

A.4.2 RobotCar Seasons

Use https://data.ciirc.cvut.cz/public/projects/2020VisualLocalization/RobotCar-Seasons/. Obtain the condition image archives, camera intrinsics/extrinsics, selected model and the correct version's split lists. A proposed adapter root is:

```
data/manual/robotcar/
  images/<condition>/<camera>/
  intrinsics/
  extrinsics/
  3D-models/
  robotcar_v2_train.txt
  robotcar_v2_test.txt
```

The saved official documentation describes the licensing and version distinctions. These images were not downloaded for the executed experiment. Their night and seasonal conditions could extend the present evidence, but must not be reported until acquisition, calibration and evaluation are actually run.

A.4.3 Extended CMU Seasons

Use https://data.ciirc.cvut.cz/public/projects/2020VisualLocalization/Extended-CMU-Seasons/. Each slice archive contains the corresponding images and map-related data; choose slices and conditions before inference and retain that selection. An expected adapter structure is:

```
data/manual/cmu/
  intrinsics.txt
  slice2/
    database/
    query/
    sparse/
    camera_poses/
```

Archive-internal directory names should be taken from the official README. The inspected source specifies attribution, non-commercial and share-alike restrictions. No CMU image or seasonal metric is included in this thesis.

A.4.4 MSLS and the full Tokyo street-view task

MSLS instructions are at https://github.com/mapillary/mapillary_sls and its linked download site https://www.mapillary.com/dataset/places. Preserve city, query/database and sequence metadata. The official structure uses `train_val/<city>/query/images/`, a corresponding database directory and associated CSV files. Test data and current ground-truth availability should follow the current official instructions rather than an assumed historical evaluation server.

For the official Tokyo street-view experiment, follow https://github.com/gmberton/VPR-datasets-downloader/blob/main/format_tokyo247.py and the original dataset's access instructions. That formatting source explains that the database must be obtained separately; the publicly downloaded query archive alone does not reproduce the official benchmark. A suitable

root retains street-view panorama crops, the dataset metadata and original queries in distinct directories.

Pittsburgh/SF-XL, Oxford/Paris and Google Landmarks are alternative future collections, not failed downloads in this run. They should be documented with their own licences and split protocols if used later. None is needed to reproduce the results already presented.

A.5 Build procedure

The installed TeX toolchain was used directly. The document uses the report class, numeric natbib citations, generated vector figures and a nomenclature index. From the project root, `make thesis` regenerates empirical artifacts and invokes latexmk. The thesis Makefile copies the completed PDF to the main deliverable path and runs the verification script. A clean TeX rebuild is available through the recorded clean/build commands.

Authoring and environment-inspection commands performed before the command runner existed are summarised in `results/bootstrap_commands.json`. Subsequent substantive executions are recorded in `results/commands.jsonl` with exact command text and outcomes, reproduced below. File edits are represented by the delivered source, not presented as experiment commands. The currently running build can appear as in progress when its own ledger is generated; the machine-readable journal records completion after the build returns.

A.6 Exact recorded execution ledger

Initial setup/inspection 1.

```
pwd && rg --files -g 'AGENTS.md' -g '!node_modules' -g '!venv' -g '!\.git' . /Users/bhattacharjee/.codex/
    attachments/96f3cc0c-84ef-4a82-a64b-b9fce597c78b 2>/dev/null
```

Initial setup/inspection 2.

```
cat '/Users/bhattacharjee/.codex/attachments/96f3cc0c-84ef-4a82-a64b-b9fce597c78b/Pasted text.txt'
```

Initial setup/inspection 3.

```
ls -la
```

Initial setup/inspection 4.

```
for p in /AGENTS.md /Users/AGENTS.md /Users/bhattacharjee/AGENTS.md /Users/bhattacharjee/Desktop/AGENTS.md;
    do if [ -f "$p" ]; then cat "$p"; fi; done
find . -name AGENTS.md -print
```

Initial setup/inspection 5.

```
python3 --version
which git pdflatex latexmk biber brew ffmpeg tectonic
 df -h .
sysctl -n machdep.cpu.brand_string hw.memsize
```

Initial setup/inspection 6.

```
git clone https://github.com/BitsJayMehta173/Stitch.git Stitch-research
```

Initial setup/inspection 7.

```
ls /opt/homebrew/bin/python* /usr/local/bin/python* /usr/bin/python* 2>/dev/null
uname -m
system_profiler SPHardwareDataType | head -22
kpsewhich algorithm.sty
kpsewhich algpseudocode.sty
kpsewhich siunitx.sty
```

Initial setup/inspection 8.

```
find /opt/homebrew/bin /usr/local/bin /usr/bin -maxdepth 1 -name 'python*' -print 2>/dev/null
```

Initial setup/inspection 9.

```
/usr/bin/python3 --version
/opt/homebrew/bin/python3.14 --version
ls /opt/homebrew/opt | rg 'python|uv'
find /Users/bhattacharjee/Desktop/internship/Stitch-research -maxdepth 2 -type f -not -path '*/.git/*' -
    print
```

Initial setup/inspection 10.

```
ls -la Stitch-research
curl -I --connect-timeout 10 --max-time 20 https://raw.githubusercontent.com/BitsJayMehta173/Stitch/main/
    tep3.py
```

Initial setup/inspection 11.

```
git rev-parse HEAD
rg --files
cat tep.py tep2.py tep3.py README.md
```

Initial setup/inspection 12.

```
/usr/bin/python3 -m venv .venv
mkdir -p src/stitch scripts configs tests data results/raw results/logs figures thesis/chapters thesis/
    figures thesis/tables demo vendor
.venv/bin/python -m pip --version
```

20261105T051728.588219Z: exit 0, 2.263 seconds.

```
.venv/bin/python -m pip install --upgrade pip
```

20261105T051730.887161Z: exit 0, 150.577 seconds.

```
.venv/bin/python -m pip install -r requirements.txt
```

20261105T051803.763940Z: exit 0, 14.033 seconds.

```
git clone --depth 1 https://github.com/cvg/LightGlue.git vendor/LightGlue
```

20261105T051817.840310Z: exit 0, 1.409 seconds.

```
git clone --depth 1 https://github.com/gmberton/CosPlace.git vendor/CosPlace
```

20261105T051830.718242Z: exit 6, 1.696 seconds.

```
curl -IL --connect-timeout 10 --max-time 25 https://icvl.ee.ic.ac.uk/vbalnt/hpatches/hpatches-sequences-
    release.tar.gz
```

20261105T051832.447559Z: exit 0, 3.881 seconds.

```
curl -IL --connect-timeout 10 --max-time 25 'http://www.ok.sc.e.titech.ac.jp/~torii/project/247/'
```

20261105T051859.042356Z: exit 28, 180.034 seconds.

```
curl -fL --connect-timeout 10 --max-time 180 -o data/tokyo_queries.zip https://data.ciirc.cvut.cz/public/
    projects/2015netVLAD/Tokyo247/queries/247query_subset_v2.zip
```

20261105T051946.478990Z: exit 0, 479.281 seconds.

```
curl -fL --connect-timeout 10 --max-time 1200 -o data/hpatches-sequences-release.zip https://huggingface.co/
    datasets/vbalnt/hpatches/resolve/main/hpatches-sequences-release.zip
```

20261105T052006.808242Z: exit 0, 1.099 seconds.

```
curl -sSIL --connect-timeout 10 --max-time 20 https://huggingface.co/datasets/vbalnt/hpatches/resolve/main/
    hpatches-sequences-release.zip
```

20261105T052007.946358Z: exit 0, 0.845 seconds.

```
curl -sS -I -H 'Range: bytes=0-99' --connect-timeout 10 --max-time 20 https://data.ciirc.cvut.cz/public/
    projects/2015netVLAD/Tokyo247/queries/247query_subset_v2.zip
```

20261105T052041.129322Z: exit 1, 40.940 seconds.

```
.venv/bin/python -c 'import sys,zipfile;sys.path.insert(0,'"'"'scripts'"'"');from remote_zip import
    RemoteZipIO;r=RemoteZipIO('"'"'https://data.ciirc.cvut.cz/public/projects/2015netVLAD/Tokyo247/queries
    /247query_subset_v2.zip'"'"');z=zipfile.ZipFile(r);print(len(z.namelist()));print('"'"'\n'"'"'.join(z.
    namelist()[:35]));print([(n,z.read(n)[:150]) for n in z.namelist() if n.endswith('"'"'.csv'"'"')][:1])'
```

20261105T052159.098463Z: exit 0, 2.430 seconds.

```
curl -fsSL --connect-timeout 10 --max-time 30 -o data/hpatches_hf_tree.json 'https://huggingface.co/api/
    datasets/JimmyFu/hpatches_sequences/tree/main?recursive=true'
```

20261105T052302.124472Z: exit 0, 15.673 seconds.

```
.venv/bin/python -c 'import sys,zipfile,json;sys.path.insert(0,'"'"'scripts'"'"');from remote_zip import
    RemoteZipIO;r=RemoteZipIO('"'"'https://data.ciirc.cvut.cz/public/projects/2015netVLAD/Tokyo247/queries
    /247query_subset_v2.zip'"'"');z=zipfile.ZipFile(r);print(z.read('"'"'247query_subset_v2/Readme.txt'"'"')
    .decode());n=next(n for n in z.namelist() if n.endswith('"'"'.csv'"'"'));print(n,z.read(n).decode());
    open('"'"'data/tokyo_zip_directory.json'"'"','"'"'w'"'"').write(json.dumps([dict(name=i.filename,offset=
    i.header_offset,size=i.file_size,compressed=i.compress_size,crc=i.CRC,method=i.compress_type) for i in z
    .infolist()],indent=2))'
```

20261105T052305.737685Z: exit 0, 129.830 seconds.

```
curl -fsSL --connect-timeout 10 --max-time 180 -o models_cosplace_download.pth https://github.com/gmberton/
    CosPlace/releases/download/v1.0/ResNet18_512_cosplace.pth
```

20261105T052317.888293Z: exit 0, 2.865 seconds.

```
.venv/bin/python -m pip install -e . --no-deps
```

20261105T052320.783618Z: exit 0, 2.437 seconds.

```
.venv/bin/python -m pip install -e vendor/LightGlue --no-deps
```

20261105T052454.115168Z: exit 0, 386.097 seconds.

```
.venv/bin/python scripts/download_data.py --dataset tokyo
```

20261105T052634.748331Z: exit 0, 61.876 seconds.

```
.venv/bin/python scripts/run_originals.py --limit 90
```

20261105T052649.085364Z: exit 0, 108.436 seconds.

```
.venv/bin/python scripts/download_models.py
```

20261105T052655.188712Z: exit 0, 31.692 seconds.

```
.venv/bin/python -c 'import cv2,json,platform,torch,psutil,importlib.metadata as m;from pathlib import Path;
    v=cv2.VideoCapture('"'"'video.mp4'"'"');d=dict(platform=platform.platform(),machine=platform.machine(),
    python=platform.python_version(),cpu=platform.processor(),physical_memory_bytes=psutil.virtual_memory().
    total,torch_mps_available=torch.backends.mps.is_available(),torch_cuda_available=torch.cuda.is_available
    (),packages={p:m.version(p) for p in ['"'"'numpy'"'"','"'"'opencv-contrib-python'"'"','"'"'torch
    '"'"','"'"'torchvision'"'"','"'"'kornia'"'"','"'"'scipy'"'"','"'"'matplotlib'"'"','"'"'Pillow'"'"','"'"'
    onnx'"'"','"'"'onnxruntime'"'"']},video=dict(width=v.get(3),height=v.get(4),fps=v.get(5),frames=v.get(7)
    ));v.release();Path('"'"'results/environment.json'"'"').write_text(json.dumps(d,indent=2));print(json.
    dumps(d,indent=2))'
```

20261105T052736.685439Z: exit 0, 4.905 seconds.

```
.venv/bin/python -m pytest -q
```

20261105T052950.790466Z: exit 0, 2.418 seconds.

```
.venv/bin/python scripts/download_data.py --dataset hpatches
```

20261105T053026.926415Z: exit 0, 5.494 seconds.

```
.venv/bin/python -c 'from stitch.features import LocalEngine;from stitch.io import read_image;from stitch.
    retrieval import GlobalEngine;import torch,time;torch.set_num_threads(2);im=read_image('"'"'reference.
    jpg'"'"')[0];e=LocalEngine('"'"'superpoint'"'"',512);a=e.extract(im);start=time.perf_counter();p,q,ids=e
    .match(a,a);print('"'"'SP points'"'"',len(a.points),'"'"'matches'"'"',len(p),'"'"'matching_ms'"'"',(time
    .perf_counter()-start)*1000);g=GlobalEngine();print('"'"'CosPlace dimension'"'"',g.extract(im).shape)'
```

20261105T053131.878032Z: exit 1, 0.067 seconds.

```
.venv/bin/python -m compileall -q src scripts tests
```

20261105T053145.439997Z: exit 0, 0.170 seconds.

```
.venv/bin/python -m compileall -q src scripts tests
```

20261105T053145.639983Z: exit 0, 2.712 seconds.

```
.venv/bin/python scripts/download_data.py --dataset hpatches
```

20261105T053148.401823Z: exit 0, 2.216 seconds.

```
.venv/bin/python scripts/prepare_data.py
```

20261105T053203.373611Z: exit 0, 968.671 seconds.

```
.venv/bin/python scripts/evaluate.py --learned-exhaustive
```

20261105T053231.430339Z: exit 0, 25.342 seconds.

```
.venv/bin/python scripts/train.py --steps 120
```

20261105T053256.841737Z: exit 0, 26.273 seconds.

```
.venv/bin/python scripts/export_models.py
```

20261105T053428.648382Z: exit 0, 2.723 seconds.

```
.venv/bin/python -m pytest -q
```

20261105T053431.416021Z: exit 0, 0.891 seconds.

```
.venv/bin/python -m pip freeze --all
```

20261105T053452.374058Z: exit 0, 1.796 seconds.

```
.venv/bin/python -c 'import cv2,json,numpy as np;from pathlib import Path;rows=json.loads(Path('"'"'data/
    tokyo/selected.json'"'"').read_text());panels=[];first=rows[0]['"'"'place_id'"'"'];selected=[r for r in
    rows if r['"'"'place_id'"'"']==first];print(selected);[(print(r['"'"'image'"'"'],r['"'"'condition'"'"'],
    float(cv2.imread('"'"'data/tokyo/'"'"'+r['"'"'image'"'"']).mean()))) for r in selected];panels=[cv2.
    resize(cv2.imread('"'"'data/tokyo/'"'"'+r['"'"'image'"'"']),(320,240)) for r in selected];cv2.imwrite
    ('"'"'figures/tokyo_inspection.jpg'"'"',np.hstack(panels))'
```

20261105T053456.339375Z: exit 0, 0.562 seconds.

```
curl -fsSL --connect-timeout 10 --max-time 30 -o results/sources/aachen_readme.md https://data.ciirc.cvut.cz
    /public/projects/2020VisualLocalization/Aachen-Day-Night/README_Aachen-Day-Night_v1_1.md
```

20261105T053456.950016Z: exit 0, 0.749 seconds.

```
curl -fsSL --connect-timeout 10 --max-time 30 -o results/sources/robotcar_readme.md https://data.ciirc.cvut.
    cz/public/projects/2020VisualLocalization/RobotCar-Seasons/README_RobotCar-Seasons.md
```

20261105T053457.777340Z: exit 0, 0.739 seconds.

```
curl -fsSL --connect-timeout 10 --max-time 30 -o results/sources/cmu_readme.md https://data.ciirc.cvut.cz/
    public/projects/2020VisualLocalization/Extended-CMU-Seasons/README_Extended-CMU-Seasons.md
```

20261105T053458.549454Z: exit 0, 0.401 seconds.

```
curl -fsSL --connect-timeout 10 --max-time 30 -o results/sources/hpatches_readme.md https://raw.
    githubusercontent.com/hpatches/hpatches-dataset/master/README.md
```

20261105T054650.922828Z: exit 0, 9.309 seconds.

```
.venv/bin/python tep.py
```

20261105T054652.047516Z: exit 0, 0.038 seconds.

```
.venv/bin/python -c 'import json;from pathlib import Path;d=json.loads(Path('"'"'results/training.json'"'"').
    read_text());print([(r['"'"'protocol'"'"'],r['"'"'variant'"'"'],r['"'"'augmentation'"'"'],r['"'"'seed
    '"'"'],round(r['"'"'recall1'"'"'],3)) for r in d['"'"'results'"'"'] if r['"'"'protocol'"'"'] in ['"'"'
    tokyo_day_to_day'"'"','"'"'hpatches_viewpoint'"'"']])'
```

20261105T054700.276524Z: exit 0, 5.598 seconds.

```
.venv/bin/stitch index data/places.json
```

20261105T054705.910108Z: exit 0, 28.593 seconds.

```
.venv/bin/stitch query data/places.json test.jpg results/raw/cli_query.png
```

20261105T054742.415068Z: exit 0, 16.932 seconds.

```
.venv/bin/python scripts/probe_data.py
```

20261105T054759.395798Z: exit 0, 2.074 seconds.

```
.venv/bin/python -c 'import sys,json;from pathlib import Path;sys.path.insert(0,'"'"'scripts'"'"');from
    download_data import member,TOKYO;rows=json.loads(Path('"'"'data/tokyo_zip_directory.json'"'"').
    read_text());i=next(r for r in rows if r['"'"'name'"'"'].endswith('"'"'/Readme.txt'"'"'));member(i,TOKYO
    ,Path.cwd()/'"'"'results/sources/tokyo_readme.txt'"'"')'
```

20261105T054953.631155Z: exit 1, 2.152 seconds.

```
.venv/bin/python scripts/qualitative.py
```

20261105T054954.748109Z: exit 0, 0.081 seconds.

```
.venv/bin/python -c 'import csv,json;from pathlib import Path;rs=list(csv.DictReader(Path('"'"'results/
    retrieval.csv'"'"').open()));print('"'"'\n'"'"'.join(str(r) for r in rs if r['"'"'protocol'"'"'].
    startswith('"'"'tokyo_'"'"') and r['"'"'method'"'"'] in ['"'"'A_SIFT'"'"','"'"'B_SP_LightGlue'"'"','"'"'
    C_CosPlace'"'"','"'"'D_full_top3'"'"'']));p=json.loads(Path('"'"'results/pairs.json'"'"').read_text());
    import numpy as np;print([(s,m,np.mean([r['"'"'homography_success'"'"'] for r in p if r['"'"'protocol
    '"'"']==s and r['"'"'method'"'"']==m]),np.mean([r['"'"'accepted'"'"'] for r in p if r['"'"'protocol
    '"'"']==s and r['"'"'method'"'"']==m])) for s in ['"'"'hpatches_illumination'"'"','"'"'
    hpatches_viewpoint'"'"'] for m in ['"'"'sift'"'"','"'"'superpoint'"'"']])'
```

20261105T054955.815465Z: exit 1, 1.022 seconds.

```
.venv/bin/python scripts/demo.py
```

20261105T054956.868306Z: exit 1, 1.010 seconds.

```
.venv/bin/python scripts/profile.py
```

20261105T054957.912795Z: exit 0, 118.743 seconds.

```
.venv/bin/python scripts/generate_artifacts.py
```

20261105T055242.908276Z: exit 0, 11.955 seconds.

```
.venv/bin/python scripts/qualitative.py
```

20261105T055254.934633Z: exit 0, 103.781 seconds.

```
.venv/bin/python scripts/demo.py
```

20261105T055438.792760Z: exit 0, 152.640 seconds.

```
.venv/bin/python scripts/benchmark_runtime.py
```

20261105T055441.193860Z: exit 0, 10.173 seconds.

```
/Users/bhattacharjee/Desktop/internship/Stitch-research/.venv/bin/python scripts/benchmark_runtime.py --
    method classical
```

20261105T055451.421071Z: exit 0, 84.772 seconds.

```
/Users/bhattacharjee/Desktop/internship/Stitch-research/.venv/bin/python scripts/benchmark_runtime.py --
    method local
```

20261105T055455.482303Z: exit 0, 0.045 seconds.

```
.venv/bin/python -c 'import json,csv;from pathlib import Path;p=json.loads(Path('"'"'results/predictions.
    json'"'"').read_text());print('"'"'Open-set'"'"',[(m,sum(r['"'"'accepted'"'"'] for r in p if r['"'"'
    protocol'"'"']=='"'"'open_set'"'"' and r['"'"'method'"'"']==m)) for m in ['"'"'A_SIFT'"'"','"'"'
    B_SP_LightGlue'"'"','"'"'D_full_top1'"'"','"'"'D_full_top3'"'"','"'"'D_full_top5'"'"','"'"'D_no_geometry
    '"'"']]);rows=list(csv.DictReader(Path('"'"'results/retrieval.csv'"'"').open()));print('"'"'Synthetic
    severe'"'"',[r for r in rows if r['"'"'protocol'"'"']=='"'"'synthetic_night_severe'"'"' and r['"'"'
    method'"'"'] in ['"'"'A_SIFT'"'"','"'"'B_SP_LightGlue'"'"','"'"'C_CosPlace'"'"','"'"'D_full_top3'"'"']])
    ;print('"'"'Ablation'"'"',[r for r in rows if r['"'"'protocol'"'"']=='"'"'tokyo_day_to_day'"'"'])'
```

20261105T055616.354179Z: exit 0, 6.809 seconds.

```
/Users/bhattacharjee/Desktop/internship/Stitch-research/.venv/bin/python scripts/benchmark_runtime.py --
    method retrieval
```

20261105T055623.199031Z: exit 0, 48.002 seconds.

```
/Users/bhattacharjee/Desktop/internship/Stitch-research/.venv/bin/python scripts/benchmark_runtime.py --
    method full
```

20261105T055711.499630Z: exit 0, 15.141 seconds.

```
.venv/bin/python scripts/generate_artifacts.py
```

20261105T060521.591059Z: exit 0, 0.166 seconds.

```
.venv/bin/python -c 'from pathlib import Path;import subprocess;Path('"'"'docs'"'"').mkdir(exist_ok=True);
    Path('"'"'docs/UPSTREAM_README.md'"'"').write_bytes(subprocess.check_output(['"'"'git'"'"','"'"'show
    '"'"','"'"'HEAD:README.md'"'"']))'
```

20261105T060521.796123Z: exit 0, 2.502 seconds.

```
.venv/bin/python -m pytest -q
```

20261105T060524.343033Z: exit 0, 39.402 seconds.

```
.venv/bin/stitch video data/places.json video.mp4 results/raw/cli_video.mp4 --limit 8
```

20261105T060527.912335Z: exit 0, 16.320 seconds.

```
bash scripts/setup.sh
```

20261105T060528.169115Z: exit 0, 1.718 seconds.

```
.venv/bin/python -m pip install --upgrade pip
```

20261105T060529.934447Z: exit 0, 0.542 seconds.

```
.venv/bin/python -m pip install -r requirements.txt
```

20261105T060530.633366Z: exit 0, 3.969 seconds.

```
.venv/bin/python -m pip install -e . --no-deps
```

20261105T060534.649113Z: exit 0, 3.841 seconds.

```
.venv/bin/python -m pip install -e vendor/LightGlue --no-deps
```

20261105T060538.608904Z: exit 0, 5.351 seconds.

```
.venv/bin/python scripts/download_models.py
```

20261105T060544.023207Z: exit 0, 0.205 seconds.

```
.venv/bin/python scripts/snapshot_environment.py
```

20261105T060603.812353Z: exit 0, 4.141 seconds.

```
make data
```

20261105T060604.722962Z: exit 0, 1.164 seconds.

```
.venv/bin/python scripts/download_data.py
```

20261105T060605.920926Z: exit 0, 2.028 seconds.

```
.venv/bin/python scripts/prepare_data.py
```

20261105T060752.805044Z: exit 2, 23.068 seconds.

```
make thesis
```

20261105T060752.894846Z: in progress at artifact generation.

```
/Library/Developer/CommandLineTools/usr/bin/make -C thesis thesis
```

20261105T060858.429350Z: exit 2, 19.297 seconds.

```
make thesis
```

20261105T060858.510106Z: exit 2, 19.212 seconds.

```
/Library/Developer/CommandLineTools/usr/bin/make -C thesis thesis
```

20261105T060927.786710Z: exit 0, 1.320 seconds.

```
.venv/bin/python scripts/download_data.py --verify-only
```

20261105T060929.156015Z: exit 0, 0.040 seconds.

```
.venv/bin/python -c 'import shutil;print({p:shutil.which(p) for p in ['"'"'gs'"'"','"'"'pdftoppm'"'"','"'"'
    mutool'"'"','"'"'latexmk'"'"','"'"'bibtex'"'"','"'"'makeindex'"'"']})'
```

20261105T061002.417428Z: exit 0, 0.212 seconds.

```
make clean
```

20261105T061002.667976Z: exit 2, 30.203 seconds.

```
make thesis
```

20261105T061002.720547Z: exit 2, 30.144 seconds.

```
/Library/Developer/CommandLineTools/usr/bin/make -C thesis thesis
```

20261105T061119.642113Z: exit 0, 0.958 seconds.

```
.venv/bin/python scripts/verify_thesis.py
```

20261105T061120.635443Z: exit 0, 5.394 seconds.

```
.venv/bin/python -c 'from pypdf import PdfReader;r=PdfReader('"'"'thesis/thesis.pdf'"'"');print([(i+1,p.
    extract_text()[:95].replace('"'"'\n'"'"','"'"' '"'"')) for i,p in enumerate(r.pages) if '"'"'Table
    8.1:'"'"' in p.extract_text() or '"'"'Chapter 1'"'"' in p.extract_text()[:100] or '"'"'Bibliography'"'"'
     in p.extract_text()[:100]]);print(r.outline[:3])'
```

20261105T061153.880326Z: exit 0, 0.892 seconds.

```
pdftoppm -f 67 -l 67 -scale-to 1300 -png -singlefile thesis/thesis.pdf results/raw/thesis_table_check
```

20261105T061154.809827Z: exit 0, 0.202 seconds.

```
pdftoppm -f 37 -l 37 -scale-to 1300 -png -singlefile thesis/thesis.pdf results/raw/thesis_architecture_check
```

20261105T061155.045666Z: exit 0, 0.237 seconds.

```
pdftoppm -f 78 -l 78 -scale-to 1300 -png -singlefile thesis/thesis.pdf results/raw/thesis_overlay_check
```

20261105T061155.320668Z: exit 0, 0.093 seconds.

```
pdftoppm -f 1 -l 1 -scale-to 1300 -png -singlefile thesis/thesis.pdf results/raw/thesis_title_check
```

20261105T061322.200930Z: exit 0, 0.145 seconds.

```
make clean
```

20261105T061322.384509Z: exit 0, 28.754 seconds.

```
make thesis
```

20261105T061322.430691Z: exit 0, 28.703 seconds.

```
/Library/Developer/CommandLineTools/usr/bin/make -C thesis thesis
```

20261105T061616.989356Z: exit 0, 109.893 seconds.

```
.venv/bin/python scripts/demo.py
```

20261105T061630.344476Z: exit 0, 0.151 seconds.

```
.venv/bin/python -c 'from pathlib import Path; import json; p=Path('"'"'results/demo.json'"'"'); d=json.
    loads(p.read_text()); print('"'"'Demo input:'"'"', *d['"'"'source_images'"'"']); print('"'"'Output
    :'"'"', Path('"'"'demo/comparison.mp4'"'"').resolve()); print('"'"'Video write in progress:'"'"', Path
    ('"'"'demo/comparison.mp4'"'"').stat().st_size, '"'"'bytes'"'"')'
```

20261105T061818.031121Z: exit 0, 0.379 seconds.

```
.venv/bin/python -c 'import cv2; c=cv2.VideoCapture('"'"'demo/comparison.mp4'"'"'); n=0; ok,f=c.read();
    assert ok, '"'"'Video cannot be decoded'"'"'; shape=f.shape; n=1; ok,f=c.read();
while ok:
 n+=1; ok,f=c.read()
c.release(); assert n==48, n; print('"'"'Verified fresh demo:'"'"', n, '"'"'frames;'"'"', shape[1], '"'"'x
    '"'"', shape[0])'
```

20261105T061818.441559Z: exit 1, 0.126 seconds.

```
/usr/bin/open demo/comparison.mp4
```

20261105T061832.531575Z: exit 0, 0.261 seconds.

```
/usr/bin/open -a 'QuickTime Player' demo/comparison.mp4
```

20261105T062500.396651Z: exit 0, 0.438 seconds.

```
.venv/bin/python -c 'from pypdf import PdfReader;from pathlib import Path;files=[Path('"'"'../PRONOB-
    BHATTACHARJEE-latex/PRONOB BHATTACHARJEE.pdf'"'"'),Path('"'"'../PRONOB-BHATTACHARJEE-Overleaf/main.pdf
    '"'"'),Path('"'"'../IITR_template.pdf'"'"')];
for path in files:
 r=PdfReader(path);print('"'"'\nFILE'"'"',path,'"'"'PAGES'"'"',len(r.pages),'"'"'SIZE'"'"',r.pages[0].
    mediabox);print('"'"'\n'"'"'.join((p.extract_text() or '"'"''"'"')[:6500] for p in r.pages[:6]))'
```

20261105T062547.765248Z: exit 0, 0.623 seconds.

```
.venv/bin/python -c 'from pathlib import Path;import zipfile;from PIL import Image;from pypdf import
    PdfReader;print('"'"'ASSETS'"'"');
for p in Path('"'"'..'"'"').glob('"'"'*'"'"'):
 if p.suffix.lower() in ['"'"'.png'"'"','"'"'.jpg'"'"']:print(p.name,Image.open(p).size)
for p in Path('"'"'..'"'"').glob('"'"'*.zip'"'"'):
 z=zipfile.ZipFile(p);print('"'"'ARCHIVE'"'"',p.name,len(z.namelist()));print([n for n in z.namelist() if n.
    endswith(('"'"'.sty'"'"','"'"'.cls'"'"','"'"'.map'"'"'))][:30])
r=PdfReader('"'"'../PRONOB-BHATTACHARJEE-Overleaf/main.pdf'"'"');print('"'"'THESIS PAGE TITLES'"'"');print
    ([(i+1,(r.pages[i].extract_text() or '"'"''"'"')[:180].replace('"'"'\n'"'"','"'"' '"'"')) for i in
    [7,8,9,10,11,12,13,14,15,20,35]]);print('"'"'PAGE1 IMAGES'"'"',[(v.name,len(v.data),v.image.size) for v
    in r.pages[0].images]);print('"'"'TEXT POSITIONS'"'"');
r.pages[0].extract_text(visitor_text=lambda t,cm,tm,f,s:print(repr(t[:85]),'"'"'xy'"'"',tm[4:6],'"'"'size
    '"'"',s) if t.strip() else None)'
```

20261105T062548.450208Z: exit 0, 0.200 seconds.

```
pdftoppm -f 1 -l 1 -scale-to 1300 -png -singlefile ../PRONOB-BHATTACHARJEE-Overleaf/main.pdf results/raw/
    reference_report_cover
```

20261105T062548.680032Z: exit 0, 0.219 seconds.

```
pdftoppm -f 14 -l 14 -scale-to 1300 -png -singlefile ../PRONOB-BHATTACHARJEE-Overleaf/main.pdf results/raw/
    reference_report_body
```

20261105T062548.928585Z: exit 0, 0.874 seconds.

```
pdftoppm -f 1 -l 1 -scale-to 1500 -png -singlefile ../IITR_template.pdf results/raw/reference_slide_title
```

20261105T062549.832160Z: exit 0, 1.149 seconds.

```
pdftoppm -f 4 -l 4 -scale-to 1500 -png -singlefile ../IITR_template.pdf results/raw/reference_slide_body
```

20261105T062654.091555Z: exit 0, 0.232 seconds.

```
.venv/bin/python -c 'from pypdf import PdfReader; r=PdfReader('"'"'../PRONOB-BHATTACHARJEE-Overleaf/main.pdf
    '"'"');
for i in [15,16]:
 print('"'"'PAGE'"'"',i+1);r.pages[i].extract_text(visitor_text=lambda t,cm,tm,f,s:print(repr(t[:70]),tm
    [4:6],s) if t.strip() and (tm[5]>590 or tm[5]<90) else None)
print('"'"'PDF modules'"'"');import importlib.util;print({n:bool(importlib.util.find_spec(n)) for n in ['"'"'
    pptx'"'"','"'"'fitz'"'"','"'"'reportlab'"'"']})'
```

20261105T062706.126122Z: exit 0, 0.214 seconds.

```
.venv/bin/python -c 'from pypdf import PdfReader;r=PdfReader('"'"'../PRONOB-BHATTACHARJEE-Overleaf/thesis-
    graphics.pdf'"'"');
for i in [0,1]:
 print('"'"'GRAPHICS'"'"',i+1);r.pages[i].extract_text(visitor_operand_before=lambda op,args,cm,tm:print(op,
    args) if op in [b'"'"'cm'"'"',b'"'"'Do'"'"'] else None)
r=PdfReader('"'"'../PRONOB-BHATTACHARJEE-Overleaf/main.pdf'"'"');r.pages[15].extract_text(visitor_text=
    lambda t,cm,tm,f,s:print(repr(t[:60]),tm[4:6],s) if t.strip() else None)'
```

20261105T063014.121353Z: exit 0, 2.735 seconds.

```
.venv/bin/python -m pip install python-pptx==1.0.2
```

20261105T063015.406488Z: exit 0, 0.276 seconds.

```
.venv/bin/python scripts/prepare_iitr_assets.py
```

20261105T063234.202515Z: exit 0, 0.033 seconds.

```
.venv/bin/python scripts/restyle_iitr_report.py
```

20261105T063234.266409Z: exit 0, 25.767 seconds.

```
make thesis
```

20261105T063234.342166Z: exit 0, 25.687 seconds.

```
/Library/Developer/CommandLineTools/usr/bin/make -C thesis thesis
```

20261105T063640.164469Z: exit 0, 0.560 seconds.

```
.venv/bin/python scripts/generate_presentation_data.py
```

20261105T063640.756639Z: exit 2, 6.796 seconds.

```
make -C presentation slides
```

20261105T063748.190935Z: exit 0, 0.200 seconds.

```
pdftoppm -f 1 -l 1 -scale-to 1300 -png -singlefile thesis/thesis.pdf results/raw/iitr_report_cover
```

20261105T063748.424041Z: exit 0, 0.130 seconds.

```
pdftoppm -f 2 -l 2 -scale-to 1300 -png -singlefile thesis/thesis.pdf results/raw/iitr_report_inner_cover
```

20261105T063748.584871Z: exit 0, 0.143 seconds.

```
pdftoppm -f 4 -l 4 -scale-to 1300 -png -singlefile thesis/thesis.pdf results/raw/iitr_report_declaration
```

20261105T063748.758544Z: exit 0, 0.191 seconds.

```
pdftoppm -f 20 -l 20 -scale-to 1300 -png -singlefile thesis/thesis.pdf results/raw/iitr_report_chapter
```

20261105T063826.276948Z: exit 0, 25.947 seconds.

```
make -C presentation slides
```

20261105T063852.288769Z: exit 0, 1.574 seconds.

```
pdftoppm -f 1 -l 2 -scale-to 1400 -png presentation/Stitch_IITR_Presentation.pdf results/raw/iitr_slide
```

20261105T063853.897155Z: exit 0, 0.357 seconds.

```
pdftoppm -f 15 -l 15 -scale-to 1400 -png -singlefile presentation/Stitch_IITR_Presentation.pdf results/raw/
    iitr_slide_results
```

20261105T063906.533229Z: exit 0, 0.215 seconds.

```
.venv/bin/python -c 'from pypdf import PdfReader;r=PdfReader('"'"'thesis/thesis.pdf'"'"');
for i in [0,1]:
 print('"'"'PAGE'"'"',i+1,r.pages[i].mediabox)
 def show(t,cm,tm,font,size):
  if t.strip():print(repr(t[:85]),'"'"'position'"'"',(round(tm[4]*cm[0]+tm[5]*cm[2]+cm[4],2),round(tm[4]*cm
     [1]+tm[5]*cm[3]+cm[5],2)))
 r.pages[i].extract_text(visitor_text=show)'
```

20261105T063906.841090Z: exit 0, 0.371 seconds.

```
pdftoppm -f 16 -l 16 -scale-to 1300 -png -singlefile ../PRONOB-BHATTACHARJEE-Overleaf/main.pdf results/raw/
    reference_report_chapter
```

20261105T064000.220624Z: exit 0, 0.308 seconds.

```
.venv/bin/python -c 'from pypdf import PdfReader;
for path,i in [('"'"'../PRONOB-BHATTACHARJEE-Overleaf/main.pdf'"'"',15),('"'"'thesis/thesis.pdf'"'"',19)]:
 print(path);r=PdfReader(path)
 def show(t,cm,tm,f,s):
  if t.strip() and tm[5]>490:print(repr(t[:65]),round(tm[4],3),round(tm[5],3),s)
 r.pages[i].extract_text(visitor_text=show)'
```

20261105T064016.391671Z: exit 0, 0.330 seconds.

```
.venv/bin/python -c 'from pypdf import PdfReader;
for path,i in [('"'"'../PRONOB-BHATTACHARJEE-Overleaf/main.pdf'"'"',16),('"'"'thesis/thesis.pdf'"'"',20)]:
 print(path);rows=[];PdfReader(path).pages[i].extract_text(visitor_text=lambda t,cm,tm,f,s:rows.append((t
    [:65],tm[4],tm[5],s)) if t.strip() else None);print(rows[:5]);print(rows[-2:])'
```

20261105T064112.752468Z: exit 0, 0.217 seconds.

```
.venv/bin/python -c 'from pypdf import PdfReader;r=PdfReader('"'"'presentation/Stitch_IITR_Presentation.pdf
    '"'"');print(r.pages[1].extract_text());
def f(t,cm,tm,font,size):
 if '"'"'INDIAN'"'"' in t:print(repr(t),cm,tm,size)
r.pages[1].extract_text(visitor_text=f)'
```

20261105T064113.000883Z: exit 0, 24.164 seconds.

```
make thesis
```

20261105T064113.045480Z: exit 0, 24.114 seconds.

```
/Library/Developer/CommandLineTools/usr/bin/make -C thesis thesis
```

20261105T064404.078519Z: exit 0, 0.461 seconds.

```
.venv/bin/python scripts/verify_iitr_documents.py
```

20261105T064404.596654Z: exit 0, 0.468 seconds.

```
.venv/bin/python scripts/package_iitr_documents.py
```

20261105T064447.228720Z: exit 0, 16.910 seconds.

```
.venv/bin/python scripts/check_document_packages.py
```

20261105T064448.337993Z: exit 0, 0.711 seconds.

```
pdftoppm -f 2 -l 2 -scale-to 1400 -png -singlefile presentation/Stitch_IITR_Presentation.pdf results/raw/
    iitr_slide_author_final
```

20261105T064449.080706Z: exit 0, 0.141 seconds.

```
.venv/bin/python -c 'from pypdf import PdfReader;r=PdfReader('"'"'thesis/thesis.pdf'"'"');p=r.pages[19];out
    =[];p.extract_text(visitor_text=lambda t,cm,tm,f,s:out.append((t[:50],tm[4],tm[5],s)) if t.strip() and
    tm[5]>490 else None);print(out);print('"'"'body bounds:'"'"',r.pages[19].mediabox)'
```

20261105T064504.180674Z: exit 0, 0.178 seconds.

```
.venv/bin/python scripts/snapshot_environment.py
```

20261105T064603.528739Z: exit 0, 1.219 seconds.

```
.venv/bin/python scripts/verify_iitr_documents.py
```

20261105T064604.779912Z: exit 0, 1.132 seconds.

```
.venv/bin/python scripts/verify_thesis.py
```

20261105T064605.947423Z: exit 0, 0.096 seconds.

```
.venv/bin/python -c 'from pathlib import Path;from pypdf import PdfReader;import json;d=Path('"'"'
    deliverables'"'"');print('"'"'\n'"'"'.join(str(p.resolve())+'"'"' | '"'"'+str(p.stat().st_size)+'"'"'
    bytes'"'"' for p in sorted(d.iterdir()) if p.is_file()));print(Path('"'"'results/
    standalone_packages_verification.json'"'"').read_text())'
```

20261105T064814.639158Z: exit 0, 2.171 seconds.

```
.venv/bin/python -m pytest -q
```

20261105T064922.537044Z: exit 1, 7.880 seconds.

```
.venv/bin/python -m pip download --no-deps --dest .cache/wheels opencv-python==4.11.0.86
```

20261105T064934.773406Z: exit 0, 6.122 seconds.

```
.venv/bin/python -m pip download --no-deps --dest .cache/wheels opencv-python==4.11.0.86
```

20261105T065059.529486Z: exit 0, 0.330 seconds.

```
.venv/bin/python -m pip uninstall -y opencv-contrib-python
```

20261105T065059.892978Z: exit 0, 0.437 seconds.

```
.venv/bin/python -m pip install --no-deps .cache/wheels/opencv_python-4.11.0.86-cp37-abi3-macosx_13_0_arm64.
    whl
```

20261105T065100.361237Z: exit 0, 0.268 seconds.

```
.venv/bin/python -m pip check
```

20261105T065147.072171Z: exit 0, 22.727 seconds.

```
.venv/bin/python -m pytest -q
```

20261105T065209.841463Z: exit 0, 0.062 seconds.

```
.venv/bin/python -m compileall -q src scripts tests
```

20261105T065209.932280Z: exit 0, 0.159 seconds.

```
.venv/bin/python scripts/snapshot_environment.py
```

20261105T065258.722422Z: exit 1, 1.701 seconds.

```
.venv/bin/python scripts/verify_research.py
```

20261105T065300.468906Z: exit 0, 3.647 seconds.

```
.venv/bin/stitch index data/places.json
```

20261105T065304.162382Z: exit 0, 20.794 seconds.

```
.venv/bin/stitch query data/places.json test.jpg results/raw/audit_cli_query.png
```

20261105T065325.002021Z: exit 0, 20.568 seconds.

```
.venv/bin/stitch video data/places.json video.mp4 results/raw/audit_cli_video.mp4 --limit 2
```

20261105T065353.751749Z: exit 0, 6.728 seconds.

```
.venv/bin/python scripts/verify_research.py
```

20261105T065433.490353Z: exit 0, 42.477 seconds.

```
make package
```

20261105T065433.536594Z: exit 0, 21.449 seconds.

```
/Library/Developer/CommandLineTools/usr/bin/make -C thesis thesis
```

20261105T065434.593113Z: exit 0, 0.248 seconds.

```
.venv/bin/python -
```

20261105T065455.024877Z: exit 0, 0.220 seconds.

```
.venv/bin/python scripts/prepare_iitr_assets.py
```

20261105T065455.276702Z: exit 0, 0.134 seconds.

```
.venv/bin/python scripts/generate_presentation_data.py
```

20261105T065455.441088Z: exit 0, 19.009 seconds.

```
/Library/Developer/CommandLineTools/usr/bin/make -C presentation slides
```

20261105T065514.494629Z: exit 0, 1.006 seconds.

```
.venv/bin/python scripts/verify_iitr_documents.py
```

20261105T065515.531302Z: exit 0, 0.432 seconds.

```
.venv/bin/python scripts/package_iitr_documents.py
```

20261105T065543.483631Z: exit 0, 1.004 seconds.

```
.venv/bin/python -
```

20261105T065550.691204Z: exit 0, 27.870 seconds.

```
make check-documents
```

20261105T065550.745068Z: exit 0, 8.323 seconds.

```
.venv/bin/python scripts/verify_research.py
```

20261105T065559.119409Z: exit 0, 0.883 seconds.

```
.venv/bin/python scripts/verify_thesis.py
```

20261105T065600.035437Z: exit 0, 1.063 seconds.

```
.venv/bin/python scripts/verify_iitr_documents.py
```

20261105T065601.130152Z: exit 0, 17.426 seconds.

```
.venv/bin/python scripts/check_document_packages.py
```

20261105T065713.174447Z: exit 1, 0.040 seconds.

```
.venv/bin/python -
```

20261105T065724.753702Z: exit 0, 17.102 seconds.

```
.venv/bin/python scripts/check_document_packages.py
```

20261105T065748.296746Z: exit 0, 0.056 seconds.

```
.venv/bin/python -m compileall -q src scripts tests
```

20261105T065748.435570Z: exit 0, 0.324 seconds.

```
.venv/bin/python -m pip check
```

20261105T171849.404819Z: exit 0, 0.872 seconds.

```
.venv/bin/python -
```

20261105T171917.747993Z: exit 0, 0.231 seconds.

```
.venv/bin/python -
```

20261105T172020.452180Z: exit 0, 3.390 seconds.

```
.venv/bin/python -m pytest -q
```

20261105T172023.874238Z: exit 0, 6.582 seconds.

```
.venv/bin/python scripts/verify_research.py
```

20261105T172109.500695Z: exit 0, 42.914 seconds.

```
make -j2 PYTHON=.venv/bin/python package
```

20261105T172109.551511Z: exit 0, 22.013 seconds.

```
/Library/Developer/CommandLineTools/usr/bin/make -C thesis PYTHON=/Users/bhattacharjee/Desktop/internship/
    Stitch-research/.venv/bin/python thesis
```

20261105T172131.599762Z: exit 0, 0.179 seconds.

```
.venv/bin/python scripts/prepare_iitr_assets.py
```

20261105T172131.807537Z: exit 0, 0.124 seconds.

```
.venv/bin/python scripts/generate_presentation_data.py
```

20261105T172131.964820Z: exit 0, 18.968 seconds.

```
/Library/Developer/CommandLineTools/usr/bin/make -C presentation PYTHON=/Users/bhattacharjee/Desktop/
    internship/Stitch-research/.venv/bin/python slides
```

20261105T172150.974376Z: exit 0, 0.999 seconds.

```
.venv/bin/python scripts/verify_iitr_documents.py
```

20261105T172152.003575Z: exit 0, 0.408 seconds.

```
.venv/bin/python scripts/package_iitr_documents.py
```

20261105T172222.235375Z: exit 0, 17.423 seconds.

```
.venv/bin/python scripts/verify_model_artifacts.py
```

20261105T172250.066037Z: exit 0, 47.782 seconds.

```
make check-documents
```

20261105T172250.181407Z: exit 0, 11.736 seconds.

```
.venv/bin/python scripts/verify_research.py
```

20261105T172301.957670Z: exit 0, 17.216 seconds.

```
.venv/bin/python scripts/verify_model_artifacts.py
```

20261105T172319.221315Z: exit 0, 0.755 seconds.

```
.venv/bin/python scripts/verify_thesis.py
```

20261105T172320.006398Z: exit 0, 1.075 seconds.

```
.venv/bin/python scripts/verify_iitr_documents.py
```

20261105T172321.118316Z: exit 0, 16.726 seconds.

```
.venv/bin/python scripts/check_document_packages.py
```

20261105T172324.634750Z: exit 0, 0.548 seconds.

```
.venv/bin/python -
```

20261105T172325.214992Z: exit 0, 0.050 seconds.

```
.venv/bin/python -m compileall -q src scripts tests
```

20261105T172325.294683Z: exit 0, 0.308 seconds.

```
.venv/bin/python -m pip check
```

20261105T172354.394797Z: exit 0, 21.212 seconds.

```
.venv/bin/stitch query data/places.json test.jpg results/raw/second_audit_cli_query.png
```

20261105T172415.653597Z: exit 0, 0.262 seconds.

```
.venv/bin/python scripts/snapshot_environment.py
```

20261105T205001.296062Z: exit 0, 51.011 seconds.

```
make thesis
```

20261105T205001.384604Z: exit 0, 50.918 seconds.

```
/Library/Developer/CommandLineTools/usr/bin/make -C thesis PYTHON=/Users/bhattacharjee/Desktop/internship/
    Stitch-research/.venv/bin/python thesis
```

20261105T205157.650794Z: exit 0, 3.916 seconds.

```
.venv/bin/python -
```

20261105T205201.608392Z: exit 0, 3.081 seconds.

```
.venv/bin/python -m pytest -q
```

20261105T205204.722463Z: exit 0, 67.738 seconds.

```
make package
```

20261105T205204.817739Z: exit 0, 45.750 seconds.

```
/Library/Developer/CommandLineTools/usr/bin/make -C thesis PYTHON=/Users/bhattacharjee/Desktop/internship/
    Stitch-research/.venv/bin/python thesis
```

20261105T205250.602065Z: exit 0, 0.191 seconds.

```
.venv/bin/python scripts/prepare_iitr_assets.py
```

20261105T205250.822360Z: exit 0, 0.128 seconds.

```
.venv/bin/python scripts/generate_presentation_data.py
```

20261105T205250.982915Z: exit 0, 19.021 seconds.

```
/Library/Developer/CommandLineTools/usr/bin/make -C presentation PYTHON=/Users/bhattacharjee/Desktop/
    internship/Stitch-research/.venv/bin/python slides
```

20261105T205310.047707Z: exit 0, 1.185 seconds.

```
.venv/bin/python scripts/verify_iitr_documents.py
```

20261105T205311.262414Z: exit 0, 1.195 seconds.

```
.venv/bin/python scripts/package_iitr_documents.py
```

20261105T205404.307368Z: exit 0, 0.616 seconds.

```
.venv/bin/python -c 'from scripts.generate_extended_evidence import write_manifest; result=write_manifest();
     print("Evidence assets hashed:",len(result["generated_outputs"]))'
```

20261105T205404.954443Z: exit 0, 56.487 seconds.

```
make check-documents
```

20261105T205405.046440Z: exit 0, 12.064 seconds.

```
.venv/bin/python scripts/verify_research.py
```

20261105T205417.155986Z: exit 0, 17.374 seconds.

```
.venv/bin/python scripts/verify_model_artifacts.py
```

20261105T205434.576635Z: exit 0, 4.240 seconds.

```
.venv/bin/python scripts/verify_thesis.py
```

20261105T205438.847011Z: exit 0, 1.198 seconds.

```
.venv/bin/python scripts/verify_iitr_documents.py
```

20261105T205440.074142Z: exit 0, 21.363 seconds.

```
.venv/bin/python scripts/check_document_packages.py
```

20261105T205542.620927Z: exit 0, 5.155 seconds.

```
latexmk -pdf -interaction=nonstopmode -halt-on-error -outdir=build main.tex
```

20261105T205543.738644Z: exit 0, 0.062 seconds.

```
.venv/bin/python -m compileall -q src scripts tests
```

20261105T205543.831580Z: exit 0, 0.338 seconds.

```
.venv/bin/python -m pip check
```

20261105T205552.631065Z: exit 0, 4.266 seconds.

```
.venv/bin/python scripts/verify_thesis.py
```

20261105T205556.927358Z: exit 0, 1.196 seconds.

```
.venv/bin/python scripts/verify_iitr_documents.py
```

20261105T205558.153498Z: exit 0, 1.159 seconds.

```
.venv/bin/python scripts/package_iitr_documents.py
```

20261105T205559.347193Z: exit 0, 21.175 seconds.

```
.venv/bin/python scripts/check_document_packages.py
```

20261105T205650.800177Z: exit 0, 4.185 seconds.

```
.venv/bin/python -
```

20261107T063039.615283Z: exit 0, 0.646 seconds.

```
.venv/bin/python -m pip install --upgrade pip
```

20261107T063040.292584Z: exit 0, 0.353 seconds.

```
.venv/bin/python -m pip install -r requirements.txt
```

20261107T063040.704200Z: exit 0, 1.951 seconds.

```
.venv/bin/python -m pip install -e . --no-deps
```

20261107T063042.686558Z: exit 0, 1.805 seconds.

```
.venv/bin/python -m pip install -e vendor/LightGlue --no-deps
```

20261107T063044.524714Z: exit 0, 3.830 seconds.

```
.venv/bin/python scripts/download_models.py
```

20261107T063048.387226Z: exit 0, 0.141 seconds.

```
.venv/bin/python scripts/snapshot_environment.py
```

20261107T063048.568397Z: exit 0, 1.207 seconds.

```
.venv/bin/python scripts/download_data.py
```

20261107T063049.813980Z: exit 0, 2.008 seconds.

```
.venv/bin/python scripts/prepare_data.py
```

20261107T103959.273112Z: exit 1, 2.785 seconds.

```
.venv/bin/python -m pytest -q
```

20261107T104017.187591Z: exit 0, 2.768 seconds.

```
.venv/bin/python -m pytest -q
```

20261107T104030.546778Z: exit 0, 12.277 seconds.

```
.venv/bin/python scripts/run_originals.py --limit 90
```

20261107T104042.888982Z: in progress at artifact generation.

```
.venv/bin/python scripts/evaluate.py --learned-exhaustive
```

20261107T183120.484098Z: exit 0, 0.695 seconds.

```
.venv/bin/python -m pip install --upgrade pip
```

20261107T183121.210974Z: exit 0, 0.341 seconds.

```
.venv/bin/python -m pip install -r requirements.txt
```

20261107T183121.610703Z: exit 0, 2.047 seconds.

```
.venv/bin/python -m pip install -e . --no-deps
```

20261107T183123.688979Z: exit 0, 1.844 seconds.

```
.venv/bin/python -m pip install -e vendor/LightGlue --no-deps
```

20261107T183125.566290Z: exit 0, 3.745 seconds.

```
.venv/bin/python scripts/download_models.py
```

20261107T183129.341507Z: exit 0, 0.136 seconds.

```
.venv/bin/python scripts/snapshot_environment.py
```

20261107T195701.490617Z: exit 0, 7.960 seconds.

```
.venv/bin/python -m pip install --upgrade pip
```

20261107T195709.485124Z: exit 0, 0.380 seconds.

```
.venv/bin/python -m pip install -r requirements.txt
```

20261107T195709.917630Z: exit 1, 7.951 seconds.

```
.venv/bin/python -m pip install -e . --no-deps
```

20261107T195736.805577Z: exit 0, 0.526 seconds.

```
.venv/bin/python -m pip install --upgrade pip
```

20261107T195737.363098Z: exit 0, 0.326 seconds.

```
.venv/bin/python -m pip install -r requirements.txt
```

20261107T195737.739104Z: exit 0, 1.933 seconds.

```
.venv/bin/python -m pip install -e . --no-deps
```

20261107T195739.703592Z: exit 0, 1.778 seconds.

```
.venv/bin/python -m pip install -e vendor/LightGlue --no-deps
```

20261107T195741.512731Z: exit 0, 2.967 seconds.

```
.venv/bin/python scripts/download_models.py
```

20261107T195744.510862Z: exit 0, 0.137 seconds.

```
.venv/bin/python scripts/snapshot_environment.py
```

20261109T073045.677679Z: exit 0, 0.004 seconds.

```
/Library/Developer/CommandLineTools/usr/bin/make -C report PYTHON=/Users/bhattacharjee/Desktop/internship/
    Stitch-research/.venv/bin/python thesis
```

20261109T074313.788133Z: exit 0, 3.131 seconds.

```
.venv/bin/python -m pytest -q
```

20261109T074323.382473Z: exit 0, 1.300 seconds.

```
.venv/bin/python scripts/download_data.py
```

20261109T074324.712374Z: exit 0, 1.613 seconds.

```
.venv/bin/python scripts/prepare_data.py
```

20261109T074326.362324Z: exit 0, 11.225 seconds.

```
.venv/bin/python scripts/run_originals.py --limit 90
```

20261109T074350.495586Z: exit 0, 893.074 seconds.

```
.venv/bin/python scripts/evaluate.py --learned-exhaustive
```

20261109T075918.479055Z: exit 0, 22.127 seconds.

```
.venv/bin/python scripts/train.py --steps 120
```

20261109T075940.750069Z: exit 0, 21.902 seconds.

```
.venv/bin/python scripts/export_models.py
```

20261109T080010.888464Z: exit 0, 101.332 seconds.

```
.venv/bin/python scripts/demo.py
```

20261109T080152.338377Z: exit 0, 164.440 seconds.

```
.venv/bin/python scripts/benchmark_runtime.py
```

20261109T080154.286035Z: exit 0, 9.125 seconds.

```
/Users/bhattacharjee/Desktop/internship/Stitch-research/.venv/bin/python scripts/benchmark_runtime.py --
    method classical
```

20261109T080203.449039Z: exit 0, 87.552 seconds.

```
/Users/bhattacharjee/Desktop/internship/Stitch-research/.venv/bin/python scripts/benchmark_runtime.py --
    method local
```

20261109T080331.147238Z: exit 0, 7.356 seconds.

```
/Users/bhattacharjee/Desktop/internship/Stitch-research/.venv/bin/python scripts/benchmark_runtime.py --
    method retrieval
```

20261109T080338.541730Z: exit 0, 57.976 seconds.

```
/Users/bhattacharjee/Desktop/internship/Stitch-research/.venv/bin/python scripts/benchmark_runtime.py --
    method full
```

20261109T080447.349271Z: in progress at artifact generation.

```
/Library/Developer/CommandLineTools/usr/bin/make -C report PYTHON=/Users/bhattacharjee/Desktop/internship/
    Stitch-research/.venv/bin/python thesis
```

Appendix B

Directory Structure and Data Identities

B.1 Delivered project structure

```
Stitch-research/
  src/stitch/
    app.py            augmentation.py cli.py
    database.py       depth.py        features.py
    geometry.py       io.py           pipeline.py
    pose3d.py         rendering.py    retrieval.py
  scripts/
    setup.sh             run_logged.py
    run_originals.py     download_data.py
    remote_zip.py        download_models.py
    prepare_data.py      evaluate.py
    train.py             export_models.py
    demo.py              benchmark_runtime.py
    qualitative.py       generate_artifacts.py
    probe_data.py        verify_thesis.py
    generate_extended_evidence.py
    verify_research.py   verify_model_artifacts.py
    prepare_iitr_assets.py
    generate_presentation_data.py
    export_presentation.py
    package_iitr_documents.py
    verify_iitr_documents.py
    check_document_packages.py
    restyle_iitr_report.py
    snapshot_environment.py
  configs/default.yaml
  tests/
  data/                  # ignored downloaded/generated inputs
  models/                # ignored weights, heads and ONNX graphs
  vendor/                # official source checkouts
  results/
    commands.jsonl       environment.json
    data_manifest.json   preparation.json
    pairs.json           predictions.json
    candidate_pairs.json retrieval.csv
```

```
  training.json        export.json
  profiles.json        statistics.json
  logs/                sources/       raw/
figures/               # plots derived from results
report/
  main.tex             preamble.tex
  frontmatter.tex      metadata.tex
  references.bib       chapters/
  figures/             tables/
  Makefile             .latexmkrc
  build/               thesis.pdf
demo/comparison.mp4     demo/comparison.gif
branding/project_metadata.tex
branding/reference_layout.json
presentation/           # original IITR theme, slides and assets
deliverables/           # PDFs, PowerPoint and standalone LaTeX ZIPs
README.md              requirements.txt
requirements-lock.txt  pyproject.toml  Makefile
tep.py  tep2.py  tep3.py # unchanged upstream scripts
```

B.2 Selected HPatches sequence identities

The following names are generated from the actual downloaded-subset manifest. Selection preceded inference and retained every target pair from each sequence. i_ajuntament

i_autannes
i_bologna
i_books
i_boutique
i_bridger
i_brooklyn
i_castle
i_chestnuts
i_contruction
i_crownday
i_crownnight
v_abstract
v_adam
v_apprentices
v_artisans
v_astronautis
v_azzola
v_bark
v_bees
v_beyus
v_bip
v_bird
v_birdwoman

B.3 Provenance chain

Image bytes are identified in the data manifest. Generated query bytes and their seeds are identified in preparation records. Model bytes and official implementation revisions are identified in the model manifest. The evaluation configuration pins the benchmark manifest hash. Raw pair and prediction rows support generated statistics, and `artifact_provenance.json` records the result-file hashes consumed by figure/table generation.

This chain separates input provenance from output interpretation. A filename or hash does not prove that a public label is correct, but it makes the precise source of a reported outcome inspectable. A future extension should preserve that distinction when adding metric maps or a service-side database.

Appendix C

Additional Qualitative Results

C.1 Selection and interpretation

The qualitative recorder uses the same working resolution, feature budget, matching seed and verification policy as the comparison. For each documented protocol, it prefers a positive pair accepted by the learned method but rejected by SIFT; if none exists, a fixed final-pair fallback is used. Coordinates, masks and transforms are saved in `results/qualitative_matches.json`. These figures are illustrative selections, not an unbiased sample of overall performance.

Figure C.1 complements the illumination example in the main chapter. RANSAC inlier status means compatibility with the fitted homography, not a manually established physical correspondence label. Up to eighty lines are displayed to preserve readability; full point arrays remain in the recorded artifact.

Figure C.2 shows that actual cross-time photographs, rather than only generated darkness, contribute to the comparison. The selection supports a visual explanation of the measured local-support difference. It cannot establish that all visible depths share one plane or that the resulting overlay is a calibrated world-space object.

Figure C.3 illustrates why daylight alone does not guarantee a useful anchor. The complete ranked predictions and accepted/rejected decisions, including wrong-place acceptances, are available for every query. Readers should inspect those aggregate records before drawing a method ranking from one figure.

C.2 Demo provenance

The comparison video is generated from the documented selected real daytime and night photographs, with small known two-dimensional motion applied to form query clips. This is explicitly labelled in the frames. The demo executes classical and full recognition independently on those queries and stores frame-level decisions and measured latency. No ground-truth homography is substituted when a recogniser fails.

The animation is a downsampled presentation of the same recorded rendered frames. Its playback rate is chosen for viewing and does not represent inference speed. The source-video

Figure C.1: Actual matching on the selected HPatches viewpoint example. Sparse visual support and boundary extrapolation should be considered alongside the aggregate corner-error results.

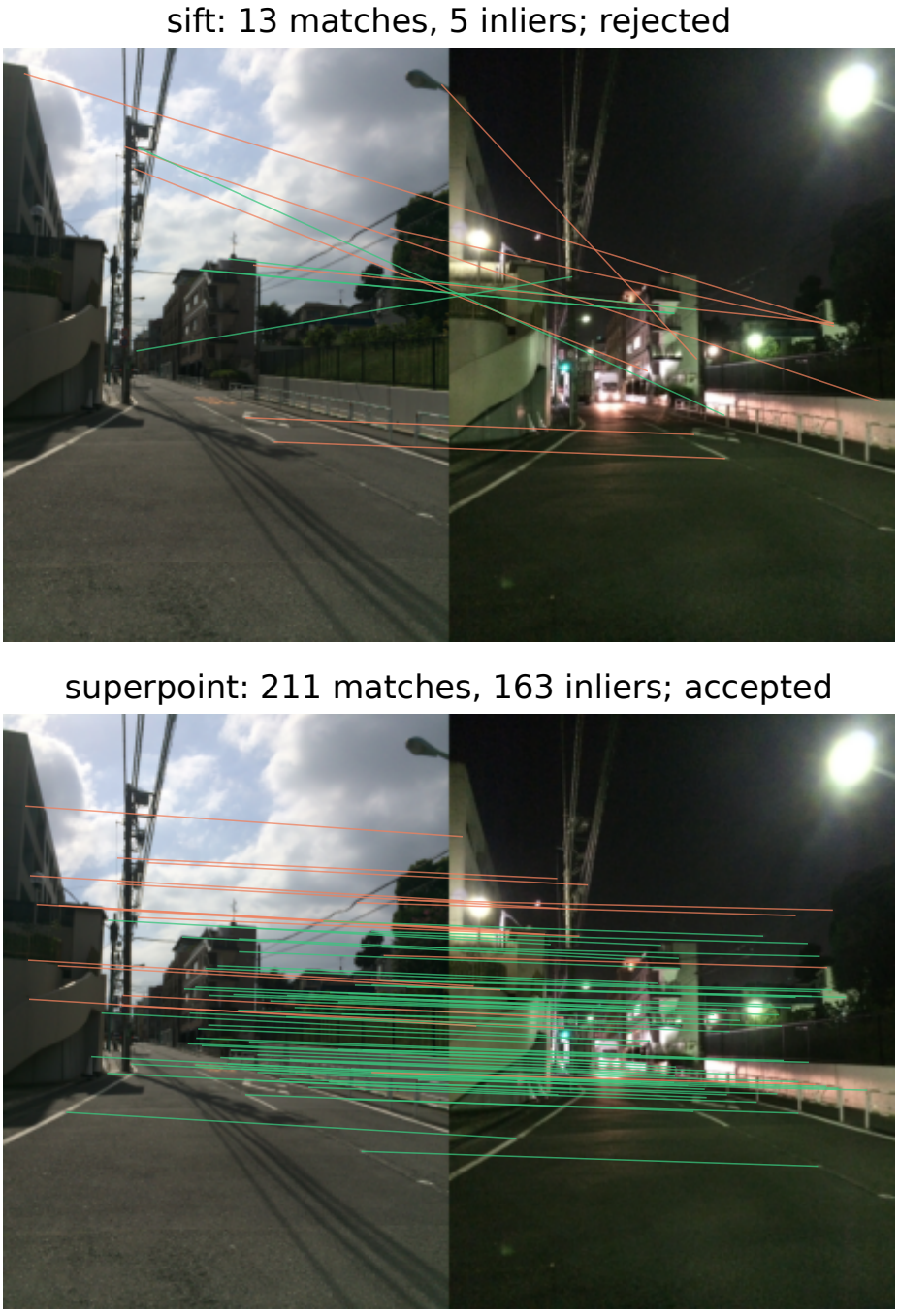

Figure C.2: Executed local correspondences across a real selected Tokyo day/night pair. The fitted map is not evaluated as a metric pose or ground-truth planar alignment.

Figure C.3: Actual matching for the selected daytime direction-change pair. This protocol tests reduced scene overlap despite unchanged daylight labels.

regression and CLI video output remain separate artifacts, because real temporal input and motion synthesised from still images provide different evidence.

Appendix D

Key Implementation Listings

D.1 Robust geometry and coordinate utilities

Listing D.1 contains the delivered geometric implementation. The listing is included directly from the package source, so it cannot drift through manual transcription.

Listing D.1: Robust homography, corner-error and camera-centre utilities.

```
"""Robust planar geometry. All homographies map reference to query pixels."""
from dataclasses import dataclass
from typing import Optional, Tuple
import cv2
import numpy as np

@dataclass
class Verification:
    H: Optional[np.ndarray]
    inlier_mask: np.ndarray
    inliers: int
    ratio: float
    accepted: bool

def project(points: np.ndarray, H: np.ndarray) -> np.ndarray:
    """Apply a projective map, returning NaNs for points at infinity."""
    xy = np.asarray(points, dtype=np.float64).reshape(-1, 2)
    H = np.asarray(H,dtype=np.float64)
    # Homographies are defined only up to scale. Use a stable representative
    # before testing the denominator against an absolute pixel-domain epsilon.
    scale = H[2,2] if H[2,2] != 0 else np.max(np.abs(H))
    if not np.isfinite(H).all() or not np.isfinite(scale) or scale == 0:
        return np.full((len(xy),2),np.nan)
    with np.errstate(over='ignore',invalid='ignore'):
        H = H / scale
    if not np.isfinite(H).all():
        return np.full((len(xy),2),np.nan)
    homogeneous = np.column_stack((xy, np.ones(len(xy)))) @ H.T
    denom = homogeneous[:, 2:3]
    return np.divide(homogeneous[:, :2], denom,
                     out=np.full((len(xy), 2), np.nan), where=np.abs(denom)>1e-10)

def corners(shape: Tuple[int, ...]) -> np.ndarray:
```

```
    h, w = shape[:2]
    return np.array([[0,0], [w-1,0], [w-1,h-1], [0,h-1]], dtype=np.float64)

def valid_homography(H: Optional[np.ndarray], shape: Optional[Tuple[int,...]] = None) -> bool:
    """Reject singular, nonfinite, horizon-crossing and collapsed projective maps."""
    if H is None or H.shape != (3,3) or not np.isfinite(H).all():
        return False
    if H[2,2] == 0:
        return False
    with np.errstate(over='ignore',invalid='ignore'):
        H = H / H[2,2]
    if not np.isfinite(H).all():
        return False
    if np.linalg.matrix_rank(H) < 3 or np.linalg.cond(H) > 1e12:
        return False
    if shape is not None:
        c = corners(shape)
        den = np.column_stack([c, np.ones(4)]) @ H[2]
        if np.any(np.abs(den)<1e-8) or not (np.all(den>0) or np.all(den<0)):
            return False
        polygon = project(c, H).astype(np.float32)
        if not np.isfinite(polygon).all() or not cv2.isContourConvex(polygon):
            return False
        if abs(cv2.contourArea(polygon)) < 16:
            return False
    return True

def valid_anchor(anchor: np.ndarray) -> bool:
    """An ordered anchor must describe a finite, convex quadrilateral."""
    polygon=np.asarray(anchor,dtype=np.float64)
    if polygon.shape!=(4,2) or not np.isfinite(polygon).all():
        return False
    polygon=polygon.astype(np.float32)
    return bool(cv2.isContourConvex(polygon) and abs(cv2.contourArea(polygon))>1e-8)

def verify(src: np.ndarray, dst: np.ndarray, threshold: float = 5.0,
           min_inliers: int = 12, min_ratio: float = 0.20,
           shape: Optional[Tuple[int,...]] = None) -> Verification:
    """Estimate H with RANSAC; retain failure rows instead of hiding them."""
    empty = np.zeros(len(src), dtype=bool)
    if len(src) < 4 or len(src) != len(dst):
        return Verification(None, empty, 0, 0.0, False)
    try:
        H, mask = cv2.findHomography(np.float32(src).reshape(-1,1,2),
                                     np.float32(dst).reshape(-1,1,2), cv2.RANSAC,
                                     threshold, maxIters=3000, confidence=0.999)
    except cv2.error:
        return Verification(None, empty, 0, 0.0, False)
    mask = mask.ravel().astype(bool) if mask is not None else empty
    n = int(mask.sum())
    ratio = n / len(src)
    accepted = n >= min_inliers and ratio >= min_ratio and valid_homography(H, shape)
    return Verification(H, mask, n, ratio, bool(accepted))

def corner_error(H: Optional[np.ndarray], ground_truth: np.ndarray, shape: Tuple[int,...]) -> float:
    """Mean four-corner error in resized query pixels; infinity on failure."""
    if not valid_homography(H, shape):
```

```
        return float('inf')
    c = corners(shape)
    return float(np.linalg.norm(project(c,H)-project(c,ground_truth), axis=1).mean())

def pose_errors(R: np.ndarray, t: np.ndarray, R_gt: np.ndarray, t_gt: np.ndarray) -> Tuple[float,float]:
    """Camera-center distance in map units and SO(3) error in degrees."""
    center = -R.T @ t.reshape(3)
    center_gt = -R_gt.T @ t_gt.reshape(3)
    angle = np.degrees(np.arccos(np.clip((np.trace(R @ R_gt.T)-1)/2, -1, 1)))
    return float(np.linalg.norm(center-center_gt)), float(angle)
```

D.2 Local extraction and matching

Listing D.2 shows how both feature families share a correspondence contract while retaining model-specific tensors for learned assignment.

Listing D.2: Classical and learned local-feature interface.

```
"""Classical and pretrained sparse features with a shared pixel convention."""
from dataclasses import dataclass
from pathlib import Path
from typing import Any, Dict, Tuple
import sys
import cv2
import numpy as np
import torch

ROOT = Path(__file__).resolve().parents[2]
torch.hub.set_dir(str(ROOT/'models'/'torch'/'hub'))
sys.path.insert(0, str(ROOT/'vendor'/'LightGlue'))

@dataclass
class Features:
    points: np.ndarray
    descriptors: np.ndarray
    raw: Dict[str, Any]

class LocalEngine:
    """Extract/match SIFT+FLANN or SuperPoint+LightGlue, on the same images."""
    def __init__(self, method: str = 'sift', max_keypoints: int = 1024,
                 ratio: float = 0.70, device: str = 'cpu', enhance: bool = False,
                 depth_confidence: float = .95, width_confidence: float = .99):
        if method not in ('sift', 'superpoint'):
            raise ValueError(method)
        self.method, self.device, self.ratio = method, device, ratio
        self.enhance = enhance
        if method == 'sift':
            self.extractor = cv2.SIFT_create(nfeatures=max_keypoints)
            self.matcher = cv2.FlannBasedMatcher(dict(algorithm=1, trees=5), dict(checks=50))
        else:
            from lightglue import SuperPoint, LightGlue
            self.extractor = SuperPoint(max_num_keypoints=max_keypoints).eval().to(device)
            self.matcher = LightGlue(features='superpoint', flash=False,
                                     depth_confidence=depth_confidence,
                                     width_confidence=width_confidence).eval().to(device)
```

```

    @torch.inference_mode()
    def extract(self, image: np.ndarray) -> Features:
        gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
        if self.enhance:
            gray = cv2.createCLAHE(2.0,(8,8)).apply(gray)
        if self.method == 'sift':
            points, descriptors = self.extractor.detectAndCompute(gray, None)
            xy = np.array([p.pt for p in points], np.float32).reshape(-1,2)
            descriptors = descriptors if descriptors is not None else np.empty((0,128),np.float32)
            return Features(xy, descriptors, {})
        tensor = torch.from_numpy(gray.copy()).float()[None]/255
        feats = self.extractor.extract(tensor.to(self.device), resize=None)
        return Features(feats['keypoints'][0].cpu().numpy(),
                        feats['descriptors'][0].cpu().numpy(), feats)

    @torch.inference_mode()
    def match(self, a: Features, b: Features) -> Tuple[np.ndarray,np.ndarray,np.ndarray]:
        if len(a.points) == 0 or len(b.points) < 2:
            empty = np.empty((0,2),np.float32)
            return empty, empty.copy(), np.empty((0,2),int)
        if self.method == 'sift':
            nearest = self.matcher.knnMatch(a.descriptors, b.descriptors, k=2)
            pairs = [(p[0].queryIdx,p[0].trainIdx) for p in nearest
                     if len(p)==2 and p[0].distance < self.ratio*p[1].distance]
            indices = np.asarray(pairs, dtype=int).reshape(-1,2)
        else:
            out = self.matcher({'image0':a.raw,'image1':b.raw})
            indices = out['matches'][0].cpu().numpy().reshape(-1,2)
        return a.points[indices[:,0]], b.points[indices[:,1]], indices
```

D.3 Hierarchical place/content query

Listing D.3 implements selection, rejection and content rendering through the shared database.

Listing D.3: Implemented query and posted-content pipeline.

```
"""Coarse retrieval, local verification and content rendering."""
from dataclasses import dataclass
import math
from typing import Any, Dict, List, Optional
import time
import cv2
import numpy as np
from .database import PlaceDatabase
from .features import Features, LocalEngine
from .geometry import Verification, verify
from .io import read_image
from .rendering import render, render_projected_quad
from .retrieval import GlobalEngine, rank
from .pose3d import camera_matrix, estimate_pose, project_points

@dataclass
class Prediction:
    place_id: Optional[str]
    reference: Optional[str]
```

```
    verification: Optional[Verification]
    candidates: List[Dict[str,Any]]
    elapsed_ms: float
    content: List[Dict[str,Any]]
    pose: Any = None
    query_matrix: Optional[np.ndarray] = None

class Pipeline:
    def __init__(self, database: PlaceDatabase, config: Dict[str,Any], method: str='full'):
        if method not in ('classical','local','full','depth3d'):
            raise ValueError('Unknown pipeline method: '+method)
        if config['top_k'] < 1:
            raise ValueError('top_k must be positive')
        self.database,self.config,self.method=database,config,method
        self.local=LocalEngine('sift' if method in ('classical','depth3d') else 'superpoint',
                               config['max_keypoints'],config['sift_ratio'],config['device'],
                               depth_confidence=config.get('lightglue_depth_confidence',.95),
                               width_confidence=config.get('lightglue_width_confidence',.99))
        self.global_engine=None if method in ('classical','local','depth3d') else GlobalEngine(
            config['cosplace_weights'],config['device'],size=config['cosplace_size'])
        if self.global_engine:
            self.descriptors=database.load_descriptors(self.global_engine,config['max_side'])
        self.images=[]
        self.features=[]
        self.depth_maps=[]
        self.reference_matrices=[]
        has_3d_placement=False
        fov = float(config.get('camera_fov_x_deg', 70.0))
        for row in database.rows:
            image,_=read_image(database.resolve(row['reference']),config['max_side'])
            self.images.append(image)
            self.features.append(self.local.extract(image))
            contents = row.get('content', [])
            needs_3d = any(content.get('anchor_3d_camera_m') is not None for content in contents)
            has_3d_placement = has_3d_placement or needs_3d
            depth_path = next((content.get('depth_map') for content in contents
                               if content.get('depth_map')), None)
            depth_map = None
            if depth_path:
                depth_map = np.load(database.resolve(depth_path), allow_pickle=False).astype(np.float32)
                if depth_map.ndim != 2 or not np.isfinite(depth_map).all() or np.any(depth_map <= 0):
                    raise ValueError(f"Invalid saved depth map for '{row['reference']}'")
            if needs_3d and depth_map is None:
                raise ValueError(f"3D overlay for '{row['reference']}' has no saved depth map")
            if method != 'depth3d' and needs_3d:
                raise ValueError("This database has 3D overlay anchors; choose the depth3d scan mode")
            metadata = next((content.get('capture_metadata', {}) for content in contents
                             if content.get('capture_metadata')), {})
            reference_fov = metadata.get('camera_fov_x_deg')
            if (needs_3d and reference_fov is not None
                    and not math.isclose(float(reference_fov), fov, rel_tol=0.0, abs_tol=1e-3)):
                raise ValueError(
                    f"3D reference '{row['reference']}' was captured with a {reference_fov:g}ř "
                    f"horizontal FOV; set camera_fov_x_deg to the same calibrated value for scanning")
            matrix = metadata.get('reference_camera_matrix')
            if matrix is not None:
                matrix = np.asarray(matrix, dtype=np.float64).reshape(3, 3)
```

```
            elif depth_map is not None:
                matrix = camera_matrix(depth_map.shape, fov)
            else:
                matrix = camera_matrix(image.shape, fov)
            self.depth_maps.append(depth_map)
            self.reference_matrices.append(matrix)
        if method == 'depth3d' and not has_3d_placement:
            raise ValueError('No depth-backed 3D placements are registered; create one in the 3D placement tab')

    def query(self,image: np.ndarray, latitude: Optional[float]=None,
              longitude: Optional[float]=None) -> Prediction:
        if (latitude is None) != (longitude is None):
            raise ValueError('Query location needs both latitude and longitude')
        if latitude is not None and not (math.isfinite(latitude) and math.isfinite(longitude)
                                         and -90 <= latitude <= 90 and -180 <= longitude <= 180):
            raise ValueError('Query coordinates are outside valid latitude/longitude ranges')
        start=time.perf_counter()
        feats=self.local.extract(image)
        candidates=np.arange(len(self.features))
        if self.global_engine:
            candidates=rank(self.global_engine.extract(image),self.descriptors)[:self.config['top_k']]
            if latitude is not None:
                nearby=sorted((i for i,row in enumerate(self.database.rows)
                               if row.get('latitude') is not None),
                              key=lambda i:self.database.distance_m(latitude,longitude,
                                  self.database.rows[i]['latitude'],self.database.rows[i]['longitude']))
                candidates=np.asarray(list(dict.fromkeys([*candidates.tolist(),
                                                          *nearby[:self.config['top_k']]])))
        query_matrix = camera_matrix(image.shape, float(self.config.get('camera_fov_x_deg', 70.0)))
        rows=[]
        best=None
        for i in candidates:
            a,b,indices=self.local.match(self.features[i],feats)
            planar=verify(a,b,self.config['ransac_px'],self.config['min_inliers'],
                          self.config['min_inlier_ratio'],self.images[i].shape)
            row=self.database.rows[i]
            needs_3d=any(content.get('anchor_3d_camera_m') is not None
                         for content in row.get('content', []))
            pose=None
            if self.method == 'depth3d' and needs_3d and self.depth_maps[i] is not None:
                raw_h,raw_w=self.depth_maps[i].shape[:2]
                feature_h,feature_w=self.images[i].shape[:2]
                reference_scale=np.asarray([raw_w/feature_w,raw_h/feature_h],np.float32)
                raw_reference_points=a*reference_scale
                pose=estimate_pose(
                    raw_reference_points,b,self.depth_maps[i],self.reference_matrices[i],
                    query_matrix,self.config['min_inliers'],self.config['min_inlier_ratio'],
                    self.config['ransac_px'],float(self.config.get('depth_max_m',80.0)))
                v=Verification(planar.H,pose.inlier_mask,pose.inliers,pose.ratio,pose.accepted)
            else:
                v=planar
            distance=(self.database.distance_m(latitude,longitude,row['latitude'],row['longitude'])
                      if latitude is not None and row.get('latitude') is not None else None)
            rows.append({'index':int(i),'place_id':row['place_id'],'reference':row['reference'],
                         'name':row.get('name'),
                         'latitude':row.get('latitude'),'longitude':row.get('longitude'),
```

```
                        'distance_m':distance,
                        'inliers':v.inliers,'ratio':v.ratio,'accepted':v.accepted,
                        'geometry':'pnp' if pose is not None else 'homography'})
            old_distance=(self.database.distance_m(latitude,longitude,
                self.database.rows[best[0]]['latitude'],self.database.rows[best[0]]['longitude'])
                if best is not None and latitude is not None and
                self.database.rows[best[0]].get('latitude') is not None else float('inf'))
            new_distance=distance if distance is not None else float('inf')
            if v.accepted and (best is None or (v.inliers,v.ratio,-new_distance)>
                               (best[1].inliers,best[1].ratio,-old_distance)):
                best=(int(i),v,pose)
        elapsed=(time.perf_counter()-start)*1000
        if best is None:
            return Prediction(None,None,None,rows,elapsed,[])
        row=self.database.rows[best[0]]
        return Prediction(row['place_id'],row['reference'],best[1],rows,elapsed,row['content'],
                          best[2],query_matrix if best[2] is not None else None)

    def render(self,frame: np.ndarray,prediction: Prediction) -> np.ndarray:
        if prediction.place_id is None:
            return frame.copy()
        index=next(i for i,r in enumerate(self.database.rows)
                   if r['place_id']==prediction.place_id and r['reference']==prediction.reference)
        output=frame.copy()
        for content in prediction.content:
            asset=cv2.imread(str(self.database.resolve(content['asset'])),cv2.IMREAD_UNCHANGED)
            if asset is None:
                continue
            anchor_3d=content.get('anchor_3d_camera_m')
            if anchor_3d is not None:
                if prediction.pose is None or prediction.query_matrix is None:
                    continue
                quad=project_points(anchor_3d,prediction.pose.rvec,prediction.pose.tvec,
                                    prediction.query_matrix)
                if quad is not None:
                    output=render_projected_quad(output,asset,quad,content.get('opacity',.8))
                continue
            anchor=np.array(content['anchor_normalized'],float)
            h,w=self.images[index].shape[:2]
            anchor=anchor*np.array([w-1,h-1])
            output=render(output,asset,prediction.verification.H,self.images[index].shape,anchor,
                          content.get('opacity',.8))
        return output
```

D.4 Explicit-alpha content rendering

Listing D.4 provides the compositing path tested with opaque black and transparent assets.

Listing D.4: Reference-plane anchor composition and premultiplied alpha.

```
"""Render posted content with an explicit alpha channel and anchor polygon."""
from typing import Optional
import cv2
import numpy as np
from .geometry import corners, valid_anchor, valid_homography

```

```
def render(frame: np.ndarray, overlay: np.ndarray, H: Optional[np.ndarray],
           reference_shape: tuple, anchor: Optional[np.ndarray] = None,
           opacity: float = 0.8) -> np.ndarray:
    """Warp content coordinates into a reference anchor, then into the query."""
    if not valid_homography(H, reference_shape):
        return frame.copy()
    H = H / H[2,2]
    if not 0 <= opacity <= 1:
        raise ValueError('opacity must be in [0,1]')
    if overlay.ndim==2:
        overlay=cv2.cvtColor(overlay,cv2.COLOR_GRAY2BGR)
    if overlay.ndim!=3 or overlay.shape[2] not in (3,4) or min(overlay.shape[:2])<2:
        raise ValueError('Overlay must be a grayscale, BGR or BGRA image of at least 2x2 pixels')
    anchor = corners(reference_shape) if anchor is None else np.asarray(anchor)
    if not valid_anchor(anchor):
        raise ValueError('Anchor must be a finite convex quadrilateral with ordered corners')
    content_to_ref = cv2.getPerspectiveTransform(corners(overlay.shape).astype(np.float32),
                                                 anchor.astype(np.float32))
    transform = H @ content_to_ref
    color = overlay[:,:,:3]
    alpha = overlay[:,:,3]/255.0 if overlay.shape[2] == 4 else np.ones(overlay.shape[:2])
    alpha = alpha.astype(np.float32) * opacity
    # Premultiplication avoids dark fringes and preserves valid black content.
    premult = color.astype(np.float32) * alpha[:,:,None]
    size = (frame.shape[1], frame.shape[0])
    warped = cv2.warpPerspective(premult, transform, size)
    mask = cv2.warpPerspective(alpha, transform, size)
    return np.clip(frame*(1-mask[:,:,None]) + warped, 0, 255).astype(np.uint8)


def render_projected_quad(frame: np.ndarray, overlay: np.ndarray, query_quad: np.ndarray,
                          opacity: float = 0.8) -> np.ndarray:
    """Project an overlay onto four already-projected 3D plane corners."""
    if not 0 <= opacity <= 1:
        raise ValueError('Opacity must be in [0,1]')
    if overlay.ndim == 2:
        overlay = cv2.cvtColor(overlay, cv2.COLOR_GRAY2BGR)
    if overlay.ndim != 3 or overlay.shape[2] not in (3, 4) or min(overlay.shape[:2]) < 2:
        raise ValueError('Overlay must be a grayscale, BGR or BGRA image of at least 2x2 pixels')
    quad = np.asarray(query_quad, dtype=np.float32).reshape(4, 2)
    polygon = np.rint(quad).astype(np.int32).reshape(-1, 1, 2)
    if (not np.isfinite(quad).all() or not cv2.isContourConvex(polygon)
            or abs(cv2.contourArea(polygon)) < 4):
        return frame.copy()

    height, width = overlay.shape[:2]
    source = np.float32([[0, 0], [width-1, 0], [width-1, height-1], [0, height-1]])
    transform = cv2.getPerspectiveTransform(source, quad)
    color = overlay[:, :, :3].astype(np.float32)
    alpha = (overlay[:, :, 3].astype(np.float32) / 255.0
             if overlay.shape[2] == 4 else np.ones((height, width), dtype=np.float32))
    alpha *= float(opacity)
    premultiplied = color * alpha[:, :, None]
    size = (frame.shape[1], frame.shape[0])
    warped_color = cv2.warpPerspective(premultiplied, transform, size)
    warped_alpha = cv2.warpPerspective(alpha, transform, size)[:, :, None]
    base = frame.astype(np.float32)
```

```
    return np.clip(base * (1.0 - warped_alpha) + warped_color, 0, 255).astype(np.uint8)
```

D.5 Executable configuration

Listing D.5 is read by the CLI and comparison. Other executed hyperparameters are kept with their scripts and the run configuration artifact.

Listing D.5: Delivered default configuration.

```
seed: 17
device: cpu
threads: 2
max_side: 640
# Approximate horizontal camera FOV. Calibrate this value for the source camera
# before relying on metric 3D placement. It is used with zero lens distortion.
camera_fov_x_deg: 70.0
# Suggested total yaw sweep for additional reference captures.
capture_angle_sweep_deg: 30.0
# Optional outdoor metric monocular depth model; fetched on first 3D placement.
depth_model_id: depth-anything/Depth-Anything-V2-Metric-Outdoor-Small-hf
depth_max_m: 80.0
max_keypoints: 512
sift_ratio: 0.70
ransac_px: 5.0
min_inliers: 12
min_inlier_ratio: 0.20
top_k: 3
cosplace_size: 224
cosplace_weights: models/cosplace_r18_512.pth
lightglue_depth_confidence: 0.95
lightglue_width_confidence: 0.99
```

Appendix E

Complete Oracle-Pair Result Register

This register retains every positive pair for both local methods, including rejections. It is generated from `results/pairs.json`; the scene atlases provide the corresponding images. Error is mean corner error in working pixels. The Tokyo rows have no independent homography ground truth, so their error cells are unavailable. The register is an audit view of the existing experiment and adds no new test observations.

E.1 hpatches_illumination

Table E.1: All oracle-pair results for hpatches_illumination.

Query	Method	Matches	Inliers	Ratio	Error	Accept
i_ajuntament_2	SIFT	202	176	0.87	0.50	yes
i_ajuntament_2	SP+LG	402	366	0.91	0.71	yes
i_ajuntament_3	SIFT	31	9	0.29	–	no
i_ajuntament_3	SP+LG	289	256	0.89	0.85	yes
i_ajuntament_4	SIFT	14	5	0.36	–	no
i_ajuntament_4	SP+LG	217	193	0.89	0.93	yes
i_ajuntament_5	SIFT	35	12	0.34	–	no
i_ajuntament_5	SP+LG	218	179	0.82	3.86	yes
i_ajuntament_6	SIFT	15	7	0.47	–	no
i_ajuntament_6	SP+LG	220	193	0.88	2.33	yes
i_autannes_2	SIFT	127	122	0.96	0.39	yes
i_autannes_2	SP+LG	337	335	0.99	0.54	yes
i_autannes_3	SIFT	64	60	0.94	6.79	yes
i_autannes_3	SP+LG	315	305	0.97	0.83	yes
i_autannes_4	SIFT	25	19	0.76	1.25	yes
i_autannes_4	SP+LG	223	173	0.78	3.11	yes
i_autannes_5	SIFT	14	9	0.64	2.38	no
i_autannes_5	SP+LG	229	221	0.97	1.46	yes
i_autannes_6	SIFT	68	57	0.84	1.56	yes
i_autannes_6	SP+LG	317	315	0.99	0.61	yes
i_bologna_2	SIFT	82	80	0.98	0.80	yes
i_bologna_2	SP+LG	368	366	0.99	0.64	yes
i_bologna_3	SIFT	48	43	0.90	1.85	yes
i_bologna_3	SP+LG	350	349	1.00	0.38	yes
i_bologna_4	SIFT	49	48	0.98	0.99	yes

Query	Method	Matches	Inliers	Ratio	Error	Accept
i_bologna_4	SP+LG	348	336	0.97	0.65	yes
i_bologna_5	SIFT	52	48	0.92	2.28	yes
i_bologna_5	SP+LG	332	261	0.79	1.82	yes
i_bologna_6	SIFT	41	35	0.85	1.03	yes
i_bologna_6	SP+LG	325	308	0.95	1.31	yes
i_books_2	SIFT	215	188	0.87	0.76	yes
i_books_2	SP+LG	427	421	0.99	0.20	yes
i_books_3	SIFT	174	164	0.94	0.68	yes
i_books_3	SP+LG	434	429	0.99	0.24	yes
i_books_4	SIFT	212	202	0.95	0.89	yes
i_books_4	SP+LG	442	438	0.99	0.31	yes
i_books_5	SIFT	253	234	0.92	0.09	yes
i_books_5	SP+LG	443	441	1.00	0.21	yes
i_books_6	SIFT	296	267	0.90	0.33	yes
i_books_6	SP+LG	434	431	0.99	0.27	yes
i_boutique_2	SIFT	171	157	0.92	0.27	yes
i_boutique_2	SP+LG	389	371	0.95	0.32	yes
i_boutique_3	SIFT	55	51	0.93	0.29	yes
i_boutique_3	SP+LG	316	272	0.86	0.45	yes
i_boutique_4	SIFT	76	70	0.92	0.68	yes
i_boutique_4	SP+LG	340	317	0.93	0.51	yes
i_boutique_5	SIFT	17	7	0.41	–	no
i_boutique_5	SP+LG	261	248	0.95	0.66	yes
i_boutique_6	SIFT	13	8	0.62	0.57	no
i_boutique_6	SP+LG	265	226	0.85	0.80	yes
i_bridger_2	SIFT	129	127	0.98	0.43	yes
i_bridger_2	SP+LG	348	339	0.97	0.29	yes
i_bridger_3	SIFT	104	99	0.95	0.42	yes
i_bridger_3	SP+LG	328	318	0.97	0.45	yes
i_bridger_4	SIFT	93	89	0.96	0.23	yes
i_bridger_4	SP+LG	345	341	0.99	0.39	yes
i_bridger_5	SIFT	62	56	0.90	1.25	yes
i_bridger_5	SP+LG	319	308	0.97	0.61	yes
i_bridger_6	SIFT	4	4	1.00	–	no
i_bridger_6	SP+LG	201	172	0.86	1.27	yes
i_brooklyn_2	SIFT	156	140	0.90	0.71	yes
i_brooklyn_2	SP+LG	336	296	0.88	0.48	yes
i_brooklyn_3	SIFT	246	232	0.94	1.73	yes
i_brooklyn_3	SP+LG	398	396	0.99	1.62	yes
i_brooklyn_4	SIFT	18	15	0.83	2.53	yes
i_brooklyn_4	SP+LG	310	295	0.95	1.03	yes
i_brooklyn_5	SIFT	261	244	0.93	2.00	yes
i_brooklyn_5	SP+LG	412	411	1.00	1.96	yes
i_brooklyn_6	SIFT	134	121	0.90	2.45	yes
i_brooklyn_6	SP+LG	339	322	0.95	1.85	yes
i_castle_2	SIFT	18	12	0.67	3.85	yes
i_castle_2	SP+LG	286	282	0.99	1.30	yes
i_castle_3	SIFT	52	49	0.94	0.96	yes
i_castle_3	SP+LG	363	359	0.99	0.83	yes
i_castle_4	SIFT	62	53	0.85	1.19	yes
i_castle_4	SP+LG	324	299	0.92	1.14	yes
i_castle_5	SIFT	16	10	0.62	5.27	no
i_castle_5	SP+LG	301	295	0.98	0.64	yes
i_castle_6	SIFT	15	12	0.80	1.02	yes
i_castle_6	SP+LG	291	290	1.00	0.48	yes
i_chestnuts_2	SIFT	158	153	0.97	0.25	yes

Query	Method	Matches	Inliers	Ratio	Error	Accept
i_chestnuts_2	SP+LG	378	317	0.84	0.73	yes
i_chestnuts_3	SIFT	155	150	0.97	0.36	yes
i_chestnuts_3	SP+LG	377	370	0.98	0.23	yes
i_chestnuts_4	SIFT	162	155	0.96	0.25	yes
i_chestnuts_4	SP+LG	348	342	0.98	0.28	yes
i_chestnuts_5	SIFT	123	121	0.98	0.42	yes
i_chestnuts_5	SP+LG	334	317	0.95	0.31	yes
i_chestnuts_6	SIFT	56	56	1.00	0.63	yes
i_chestnuts_6	SP+LG	303	273	0.90	0.49	yes
i_contruction_2	SIFT	424	424	1.00	0.05	yes
i_contruction_2	SP+LG	462	462	1.00	0.07	yes
i_contruction_3	SIFT	416	414	1.00	0.05	yes
i_contruction_3	SP+LG	450	449	1.00	0.28	yes
i_contruction_4	SIFT	292	276	0.95	0.32	yes
i_contruction_4	SP+LG	393	392	1.00	0.15	yes
i_contruction_5	SIFT	76	70	0.92	0.46	yes
i_contruction_5	SP+LG	270	253	0.94	0.46	yes
i_contruction_6	SIFT	30	29	0.97	1.11	yes
i_contruction_6	SP+LG	239	236	0.99	0.54	yes
i_crownday_2	SIFT	26	20	0.77	1.29	yes
i_crownday_2	SP+LG	284	272	0.96	0.48	yes
i_crownday_3	SIFT	43	37	0.86	2.11	yes
i_crownday_3	SP+LG	329	325	0.99	0.63	yes
i_crownday_4	SIFT	90	82	0.91	0.60	yes
i_crownday_4	SP+LG	351	350	1.00	0.37	yes
i_crownday_5	SIFT	191	181	0.95	0.68	yes
i_crownday_5	SP+LG	387	373	0.96	0.76	yes
i_crownday_6	SIFT	27	22	0.81	1.60	yes
i_crownday_6	SP+LG	279	274	0.98	0.29	yes
i_crownnight_2	SIFT	89	79	0.89	0.28	yes
i_crownnight_2	SP+LG	346	343	0.99	0.47	yes
i_crownnight_3	SIFT	75	65	0.87	0.78	yes
i_crownnight_3	SP+LG	350	308	0.88	0.77	yes
i_crownnight_4	SIFT	63	54	0.86	1.45	yes
i_crownnight_4	SP+LG	340	327	0.96	0.20	yes
i_crownnight_5	SIFT	37	34	0.92	0.33	yes
i_crownnight_5	SP+LG	326	318	0.98	0.37	yes
i_crownnight_6	SIFT	5	0	0.00	–	no
i_crownnight_6	SP+LG	144	119	0.83	1.95	yes

E.2 hpatches_viewpoint

Table E.2: All oracle-pair results for hpatches_viewpoint.

Query	Method	Matches	Inliers	Ratio	Error	Accept
v_abstract_2	SIFT	243	226	0.93	0.38	yes
v_abstract_2	SP+LG	376	373	0.99	0.20	yes
v_abstract_3	SIFT	208	181	0.87	0.61	yes
v_abstract_3	SP+LG	360	353	0.98	0.37	yes
v_abstract_4	SIFT	108	86	0.80	2.60	yes
v_abstract_4	SP+LG	234	215	0.92	1.96	yes
v_abstract_5	SIFT	145	130	0.90	1.50	yes

Query	Method	Matches	Inliers	Ratio	Error	Accept
v_abstract_5	SP+LG	287	279	0.97	0.67	yes
v_abstract_6	SIFT	69	49	0.71	3.80	yes
v_abstract_6	SP+LG	208	189	0.91	1.80	yes
v_adam_2	SIFT	237	231	0.97	0.90	yes
v_adam_2	SP+LG	376	375	1.00	0.53	yes
v_adam_3	SIFT	178	173	0.97	0.86	yes
v_adam_3	SP+LG	382	377	0.99	0.79	yes
v_adam_4	SIFT	64	54	0.84	2.58	yes
v_adam_4	SP+LG	259	259	1.00	1.08	yes
v_adam_5	SIFT	147	135	0.92	1.66	yes
v_adam_5	SP+LG	328	288	0.88	1.72	yes
v_adam_6	SIFT	119	102	0.86	1.28	yes
v_adam_6	SP+LG	320	320	1.00	1.07	yes
v_apprentices_2	SIFT	313	302	0.96	0.38	yes
v_apprentices_2	SP+LG	394	392	0.99	0.22	yes
v_apprentices_3	SIFT	240	223	0.93	0.92	yes
v_apprentices_3	SP+LG	321	320	1.00	0.44	yes
v_apprentices_4	SIFT	118	104	0.88	1.93	yes
v_apprentices_4	SP+LG	287	282	0.98	0.41	yes
v_apprentices_5	SIFT	54	45	0.83	4.00	yes
v_apprentices_5	SP+LG	266	258	0.97	0.34	yes
v_apprentices_6	SIFT	44	41	0.93	0.96	yes
v_apprentices_6	SP+LG	280	279	1.00	0.47	yes
v_artisans_2	SIFT	132	125	0.95	0.42	yes
v_artisans_2	SP+LG	246	243	0.99	0.35	yes
v_artisans_3	SIFT	91	87	0.96	1.20	yes
v_artisans_3	SP+LG	213	209	0.98	0.93	yes
v_artisans_4	SIFT	91	84	0.92	4.22	yes
v_artisans_4	SP+LG	259	248	0.96	1.54	yes
v_artisans_5	SIFT	9	8	0.89	–	no
v_artisans_5	SP+LG	119	109	0.92	–	no
v_artisans_6	SIFT	117	116	0.99	0.40	yes
v_artisans_6	SP+LG	262	260	0.99	0.36	yes
v_astronautis_2	SIFT	165	162	0.98	1.35	yes
v_astronautis_2	SP+LG	312	307	0.98	1.02	yes
v_astronautis_3	SIFT	57	51	0.89	5.82	yes
v_astronautis_3	SP+LG	254	236	0.93	1.08	yes
v_astronautis_4	SIFT	91	83	0.91	1.10	yes
v_astronautis_4	SP+LG	315	305	0.97	0.72	yes
v_astronautis_5	SIFT	25	21	0.84	2.80	yes
v_astronautis_5	SP+LG	259	203	0.78	7.21	yes
v_astronautis_6	SIFT	165	160	0.97	1.12	yes
v_astronautis_6	SP+LG	363	298	0.82	3.44	yes
v_azzola_2	SIFT	192	164	0.85	1.99	yes
v_azzola_2	SP+LG	261	217	0.83	1.58	yes
v_azzola_3	SIFT	46	33	0.72	3.56	yes
v_azzola_3	SP+LG	214	199	0.93	2.63	yes
v_azzola_4	SIFT	283	278	0.98	0.91	yes
v_azzola_4	SP+LG	254	249	0.98	0.64	yes
v_azzola_5	SIFT	76	60	0.79	5.79	yes
v_azzola_5	SP+LG	179	168	0.94	1.83	yes
v_azzola_6	SIFT	9	6	0.67	20.65	no
v_azzola_6	SP+LG	132	107	0.81	3.76	yes
v_bark_2	SIFT	116	116	1.00	0.41	yes
v_bark_2	SP+LG	251	240	0.96	0.66	yes
v_bark_3	SIFT	73	73	1.00	0.67	yes

Query	Method	Matches	Inliers	Ratio	Error	Accept
v_bark_3	SP+LG	11	4	0.36	–	no
v_bark_4	SIFT	57	53	0.93	0.41	yes
v_bark_4	SP+LG	17	5	0.29	451.46	no
v_bark_5	SIFT	49	47	0.96	0.54	yes
v_bark_5	SP+LG	48	41	0.85	1.38	yes
v_bark_6	SIFT	25	23	0.92	0.46	yes
v_bark_6	SP+LG	21	5	0.24	495.20	no
v_bees_2	SIFT	312	307	0.98	0.43	yes
v_bees_2	SP+LG	368	368	1.00	0.70	yes
v_bees_3	SIFT	345	341	0.99	0.23	yes
v_bees_3	SP+LG	402	401	1.00	0.47	yes
v_bees_4	SIFT	309	304	0.98	0.44	yes
v_bees_4	SP+LG	345	342	0.99	0.90	yes
v_bees_5	SIFT	164	161	0.98	0.84	yes
v_bees_5	SP+LG	272	231	0.85	2.14	yes
v_bees_6	SIFT	174	166	0.95	1.86	yes
v_bees_6	SP+LG	337	325	0.96	0.96	yes
v_beyus_2	SIFT	102	93	0.91	0.58	yes
v_beyus_2	SP+LG	243	217	0.89	0.98	yes
v_beyus_3	SIFT	15	9	0.60	43.75	no
v_beyus_3	SP+LG	212	189	0.89	1.51	yes
v_beyus_4	SIFT	241	230	0.95	0.23	yes
v_beyus_4	SP+LG	290	281	0.97	0.32	yes
v_beyus_5	SIFT	22	14	0.64	5.91	yes
v_beyus_5	SP+LG	203	158	0.78	2.00	yes
v_beyus_6	SIFT	43	37	0.86	1.80	yes
v_beyus_6	SP+LG	232	211	0.91	1.34	yes
v_bip_2	SIFT	188	173	0.92	0.32	yes
v_bip_2	SP+LG	227	227	1.00	0.38	yes
v_bip_3	SIFT	149	140	0.94	0.20	yes
v_bip_3	SP+LG	182	180	0.99	0.68	yes
v_bip_4	SIFT	154	146	0.95	0.38	yes
v_bip_4	SP+LG	193	187	0.97	0.36	yes
v_bip_5	SIFT	95	88	0.93	1.86	yes
v_bip_5	SP+LG	24	7	0.29	5.71	no
v_bip_6	SIFT	82	78	0.95	0.56	yes
v_bip_6	SP+LG	11	4	0.36	–	no
v_bird_2	SIFT	217	206	0.95	0.38	yes
v_bird_2	SP+LG	201	190	0.95	0.36	yes
v_bird_3	SIFT	138	129	0.93	1.20	yes
v_bird_3	SP+LG	188	173	0.92	1.99	yes
v_bird_4	SIFT	126	120	0.95	2.20	yes
v_bird_4	SP+LG	216	214	0.99	1.65	yes
v_bird_5	SIFT	60	53	0.88	14.30	yes
v_bird_5	SP+LG	198	172	0.87	4.15	yes
v_bird_6	SIFT	6	4	0.67	–	no
v_bird_6	SP+LG	145	126	0.87	26.82	yes
v_birdwoman_2	SIFT	23	22	0.96	0.94	yes
v_birdwoman_2	SP+LG	51	49	0.96	1.47	yes
v_birdwoman_3	SIFT	36	26	0.72	4.65	yes
v_birdwoman_3	SP+LG	89	82	0.92	2.11	yes
v_birdwoman_4	SIFT	92	84	0.91	0.79	yes
v_birdwoman_4	SP+LG	96	91	0.95	0.91	yes
v_birdwoman_5	SIFT	249	240	0.96	0.29	yes
v_birdwoman_5	SP+LG	222	219	0.99	0.28	yes
v_birdwoman_6	SIFT	51	50	0.98	0.80	yes

Query	Method	Matches	Inliers	Ratio	Error	Accept
v_birdwoman_6	SP+LG	110	106	0.96	0.98	yes

E.3 tokyo_day_to_day

Table E.3: All oracle-pair results for tokyo_day_to_day.

Query	Method	Matches	Inliers	Ratio	Error	Accept
t000_day_to_day	SIFT	80	62	0.78	–	yes
t000_day_to_day	SP+LG	116	106	0.91	–	yes
t001_day_to_day	SIFT	74	69	0.93	–	yes
t001_day_to_day	SP+LG	170	129	0.76	–	yes
t002_day_to_day	SIFT	92	73	0.79	–	yes
t002_day_to_day	SP+LG	125	105	0.84	–	yes
t003_day_to_day	SIFT	19	10	0.53	–	no
t003_day_to_day	SP+LG	77	73	0.95	–	yes
t004_day_to_day	SIFT	32	28	0.88	–	yes
t004_day_to_day	SP+LG	81	78	0.96	–	yes
t005_day_to_day	SIFT	74	60	0.81	–	yes
t005_day_to_day	SP+LG	168	166	0.99	–	yes
t006_day_to_day	SIFT	58	40	0.69	–	yes
t006_day_to_day	SP+LG	154	137	0.89	–	yes
t007_day_to_day	SIFT	44	11	0.25	–	no
t007_day_to_day	SP+LG	36	34	0.94	–	yes
t008_day_to_day	SIFT	80	67	0.84	–	yes
t008_day_to_day	SP+LG	171	163	0.95	–	yes
t009_day_to_day	SIFT	121	100	0.83	–	yes
t009_day_to_day	SP+LG	204	186	0.91	–	yes
t010_day_to_day	SIFT	15	6	0.40	–	no
t010_day_to_day	SP+LG	21	9	0.43	–	no
t011_day_to_day	SIFT	2	0	0.00	–	no
t011_day_to_day	SP+LG	12	5	0.42	–	no
t012_day_to_day	SIFT	20	15	0.75	–	yes
t012_day_to_day	SP+LG	81	56	0.69	–	yes
t013_day_to_day	SIFT	57	44	0.77	–	yes
t013_day_to_day	SP+LG	92	70	0.76	–	yes
t014_day_to_day	SIFT	0	0	0.00	–	no
t014_day_to_day	SP+LG	36	6	0.17	–	no
t015_day_to_day	SIFT	6	4	0.67	–	no
t015_day_to_day	SP+LG	12	5	0.42	–	no
t016_day_to_day	SIFT	10	8	0.80	–	no
t016_day_to_day	SP+LG	33	7	0.21	–	no
t017_day_to_day	SIFT	0	0	0.00	–	no
t017_day_to_day	SP+LG	25	10	0.40	–	no
t018_day_to_day	SIFT	3	0	0.00	–	no
t018_day_to_day	SP+LG	27	6	0.22	–	no
t019_day_to_day	SIFT	7	4	0.57	–	no
t019_day_to_day	SP+LG	18	6	0.33	–	no
t020_day_to_day	SIFT	13	6	0.46	–	no
t020_day_to_day	SP+LG	54	44	0.81	–	yes
t021_day_to_day	SIFT	0	0	0.00	–	no
t021_day_to_day	SP+LG	26	7	0.27	–	no
t022_day_to_day	SIFT	6	4	0.67	–	no

Query	Method	Matches	Inliers	Ratio	Error	Accept
t022_day_to_day	SP+LG	28	7	0.25	–	no
t023_day_to_day	SIFT	1	0	0.00	–	no
t023_day_to_day	SP+LG	34	6	0.18	–	no

E.4 tokyo_day_to_night

Table E.4: All oracle-pair results for tokyo_day_to_night.

Query	Method	Matches	Inliers	Ratio	Error	Accept
t000_day_to_night	SIFT	13	5	0.38	–	no
t000_day_to_night	SP+LG	211	163	0.77	–	yes
t001_day_to_night	SIFT	22	17	0.77	–	yes
t001_day_to_night	SP+LG	247	199	0.81	–	yes
t002_day_to_night	SIFT	35	32	0.91	–	yes
t002_day_to_night	SP+LG	249	176	0.71	–	yes
t003_day_to_night	SIFT	25	15	0.60	–	yes
t003_day_to_night	SP+LG	233	168	0.72	–	yes
t004_day_to_night	SIFT	11	5	0.45	–	no
t004_day_to_night	SP+LG	229	211	0.92	–	yes
t005_day_to_night	SIFT	17	5	0.29	–	no
t005_day_to_night	SP+LG	191	172	0.90	–	yes
t006_day_to_night	SIFT	22	8	0.36	–	no
t006_day_to_night	SP+LG	253	243	0.96	–	yes
t007_day_to_night	SIFT	15	8	0.53	–	no
t007_day_to_night	SP+LG	125	87	0.70	–	yes
t008_day_to_night	SIFT	4	4	1.00	–	no
t008_day_to_night	SP+LG	261	245	0.94	–	yes
t009_day_to_night	SIFT	16	12	0.75	–	yes
t009_day_to_night	SP+LG	227	190	0.84	–	yes
t010_day_to_night	SIFT	7	5	0.71	–	no
t010_day_to_night	SP+LG	210	198	0.94	–	yes
t011_day_to_night	SIFT	1	0	0.00	–	no
t011_day_to_night	SP+LG	171	141	0.82	–	yes
t012_day_to_night	SIFT	27	22	0.81	–	yes
t012_day_to_night	SP+LG	245	231	0.94	–	yes
t013_day_to_night	SIFT	12	9	0.75	–	no
t013_day_to_night	SP+LG	193	170	0.88	–	yes
t014_day_to_night	SIFT	16	8	0.50	–	no
t014_day_to_night	SP+LG	150	125	0.83	–	yes
t015_day_to_night	SIFT	12	10	0.83	–	no
t015_day_to_night	SP+LG	111	39	0.35	–	yes
t016_day_to_night	SIFT	23	12	0.52	–	no
t016_day_to_night	SP+LG	176	167	0.95	–	yes
t017_day_to_night	SIFT	7	7	1.00	–	no
t017_day_to_night	SP+LG	169	164	0.97	–	yes
t018_day_to_night	SIFT	40	34	0.85	–	yes
t018_day_to_night	SP+LG	167	152	0.91	–	yes
t019_day_to_night	SIFT	12	5	0.42	–	no
t019_day_to_night	SP+LG	180	178	0.99	–	yes
t020_day_to_night	SIFT	28	23	0.82	–	yes
t020_day_to_night	SP+LG	190	139	0.73	–	yes
t021_day_to_night	SIFT	71	62	0.87	–	yes

Query	Method	Matches	Inliers	Ratio	Error	Accept
t021_day_to_night	SP+LG	207	201	0.97	–	yes
t022_day_to_night	SIFT	12	10	0.83	–	no
t022_day_to_night	SP+LG	179	162	0.91	–	yes
t023_day_to_night	SIFT	16	7	0.44	–	no
t023_day_to_night	SP+LG	154	135	0.88	–	yes

E.5 tokyo_night_to_day

Table E.5: All oracle-pair results for tokyo_night_to_day.

Query	Method	Matches	Inliers	Ratio	Error	Accept
t000_night_to_day	SIFT	10	5	0.50	–	no
t000_night_to_day	SP+LG	211	168	0.80	–	yes
t001_night_to_day	SIFT	23	17	0.74	–	yes
t001_night_to_day	SP+LG	247	176	0.71	–	yes
t002_night_to_day	SIFT	36	30	0.83	–	yes
t002_night_to_day	SP+LG	249	202	0.81	–	yes
t003_night_to_day	SIFT	27	20	0.74	–	yes
t003_night_to_day	SP+LG	233	167	0.72	–	yes
t004_night_to_day	SIFT	11	5	0.45	–	no
t004_night_to_day	SP+LG	229	202	0.88	–	yes
t005_night_to_day	SIFT	8	4	0.50	–	no
t005_night_to_day	SP+LG	191	178	0.93	–	yes
t006_night_to_day	SIFT	16	7	0.44	–	no
t006_night_to_day	SP+LG	253	244	0.96	–	yes
t007_night_to_day	SIFT	4	4	1.00	–	no
t007_night_to_day	SP+LG	125	88	0.70	–	yes
t008_night_to_day	SIFT	5	0	0.00	–	no
t008_night_to_day	SP+LG	261	256	0.98	–	yes
t009_night_to_day	SIFT	18	13	0.72	–	yes
t009_night_to_day	SP+LG	227	189	0.83	–	yes
t010_night_to_day	SIFT	13	6	0.46	–	no
t010_night_to_day	SP+LG	210	185	0.88	–	yes
t011_night_to_day	SIFT	3	0	0.00	–	no
t011_night_to_day	SP+LG	171	138	0.81	–	yes
t012_night_to_day	SIFT	27	18	0.67	–	yes
t012_night_to_day	SP+LG	245	234	0.96	–	yes
t013_night_to_day	SIFT	13	10	0.77	–	no
t013_night_to_day	SP+LG	193	167	0.87	–	yes
t014_night_to_day	SIFT	16	7	0.44	–	no
t014_night_to_day	SP+LG	150	140	0.93	–	yes
t015_night_to_day	SIFT	6	5	0.83	–	no
t015_night_to_day	SP+LG	111	47	0.42	–	yes
t016_night_to_day	SIFT	14	9	0.64	–	no
t016_night_to_day	SP+LG	176	167	0.95	–	yes
t017_night_to_day	SIFT	10	8	0.80	–	no
t017_night_to_day	SP+LG	169	164	0.97	–	yes
t018_night_to_day	SIFT	42	33	0.79	–	yes
t018_night_to_day	SP+LG	167	162	0.97	–	yes
t019_night_to_day	SIFT	4	4	1.00	–	no
t019_night_to_day	SP+LG	180	180	1.00	–	yes
t020_night_to_day	SIFT	27	22	0.81	–	yes

Query	Method	Matches	Inliers	Ratio	Error	Accept
t020_night_to_day	SP+LG	190	140	0.74	–	yes
t021_night_to_day	SIFT	64	60	0.94	–	yes
t021_night_to_day	SP+LG	207	200	0.97	–	yes
t022_night_to_day	SIFT	8	4	0.50	–	no
t022_night_to_day	SP+LG	179	171	0.96	–	yes
t023_night_to_day	SIFT	4	4	1.00	–	no
t023_night_to_day	SP+LG	154	137	0.89	–	yes

E.6 tokyo_day_to_sunset

Table E.6: All oracle-pair results for tokyo_day_to_sunset.

Query	Method	Matches	Inliers	Ratio	Error	Accept
t000_day_to_sunset	SIFT	41	31	0.76	–	yes
t000_day_to_sunset	SP+LG	268	194	0.72	–	yes
t001_day_to_sunset	SIFT	153	132	0.86	–	yes
t001_day_to_sunset	SP+LG	369	280	0.76	–	yes
t002_day_to_sunset	SIFT	129	119	0.92	–	yes
t002_day_to_sunset	SP+LG	350	315	0.90	–	yes
t003_day_to_sunset	SIFT	27	13	0.48	–	yes
t003_day_to_sunset	SP+LG	196	101	0.52	–	yes
t004_day_to_sunset	SIFT	171	144	0.84	–	yes
t004_day_to_sunset	SP+LG	336	328	0.98	–	yes
t005_day_to_sunset	SIFT	107	98	0.92	–	yes
t005_day_to_sunset	SP+LG	298	282	0.95	–	yes
t006_day_to_sunset	SIFT	58	43	0.74	–	yes
t006_day_to_sunset	SP+LG	302	275	0.91	–	yes
t007_day_to_sunset	SIFT	55	37	0.67	–	yes
t007_day_to_sunset	SP+LG	297	280	0.94	–	yes
t008_day_to_sunset	SIFT	41	37	0.90	–	yes
t008_day_to_sunset	SP+LG	325	318	0.98	–	yes
t009_day_to_sunset	SIFT	106	76	0.72	–	yes
t009_day_to_sunset	SP+LG	329	307	0.93	–	yes
t010_day_to_sunset	SIFT	9	6	0.67	–	no
t010_day_to_sunset	SP+LG	223	207	0.93	–	yes
t011_day_to_sunset	SIFT	10	7	0.70	–	no
t011_day_to_sunset	SP+LG	205	128	0.62	–	yes
t012_day_to_sunset	SIFT	128	113	0.88	–	yes
t012_day_to_sunset	SP+LG	332	316	0.95	–	yes
t013_day_to_sunset	SIFT	128	119	0.93	–	yes
t013_day_to_sunset	SP+LG	274	255	0.93	–	yes
t014_day_to_sunset	SIFT	13	9	0.69	–	no
t014_day_to_sunset	SP+LG	184	173	0.94	–	yes
t015_day_to_sunset	SIFT	25	11	0.44	–	no
t015_day_to_sunset	SP+LG	165	69	0.42	–	yes
t016_day_to_sunset	SIFT	14	5	0.36	–	no
t016_day_to_sunset	SP+LG	164	110	0.67	–	yes
t017_day_to_sunset	SIFT	15	10	0.67	–	no
t017_day_to_sunset	SP+LG	168	113	0.67	–	yes
t018_day_to_sunset	SIFT	44	33	0.75	–	yes
t018_day_to_sunset	SP+LG	211	201	0.95	–	yes
t019_day_to_sunset	SIFT	6	4	0.67	–	no

Query	Method	Matches	Inliers	Ratio	Error	Accept
t019_day_to_sunset	SP+LG	178	171	0.96	–	yes
t020_day_to_sunset	SIFT	31	25	0.81	–	yes
t020_day_to_sunset	SP+LG	184	140	0.76	–	yes
t021_day_to_sunset	SIFT	110	103	0.94	–	yes
t021_day_to_sunset	SP+LG	256	251	0.98	–	yes
t022_day_to_sunset	SIFT	86	78	0.91	–	yes
t022_day_to_sunset	SP+LG	206	198	0.96	–	yes
t023_day_to_sunset	SIFT	13	6	0.46	–	no
t023_day_to_sunset	SP+LG	172	149	0.87	–	yes

E.7 synthetic_day

Table E.7: All oracle-pair results for synthetic_day.

Query	Method	Matches	Inliers	Ratio	Error	Accept
repo_reference_day_17	SIFT	338	321	0.95	0.12	yes
repo_reference_day_17	SP+LG	465	456	0.98	0.40	yes
repo_reference_day_23	SIFT	333	311	0.93	0.14	yes
repo_reference_day_23	SP+LG	463	436	0.94	0.53	yes
repo_reference_day_42	SIFT	340	326	0.96	0.36	yes
repo_reference_day_42	SP+LG	455	455	1.00	0.26	yes
repo_test_day_17	SIFT	266	251	0.94	0.19	yes
repo_test_day_17	SP+LG	404	358	0.89	0.78	yes
repo_test_day_23	SIFT	208	194	0.93	0.32	yes
repo_test_day_23	SP+LG	410	405	0.99	0.38	yes
repo_test_day_42	SIFT	208	195	0.94	0.09	yes
repo_test_day_42	SP+LG	391	375	0.96	0.36	yes
i_ajuntament_day_17	SIFT	285	267	0.94	0.36	yes
i_ajuntament_day_17	SP+LG	426	418	0.98	0.16	yes
i_ajuntament_day_23	SIFT	303	282	0.93	0.17	yes
i_ajuntament_day_23	SP+LG	427	415	0.97	0.28	yes
i_ajuntament_day_42	SIFT	279	256	0.92	0.25	yes
i_ajuntament_day_42	SP+LG	426	423	0.99	0.20	yes
i_autannes_day_17	SIFT	217	215	0.99	0.12	yes
i_autannes_day_17	SP+LG	385	384	1.00	0.27	yes
i_autannes_day_23	SIFT	216	214	0.99	0.51	yes
i_autannes_day_23	SP+LG	386	386	1.00	0.24	yes
i_autannes_day_42	SIFT	152	149	0.98	0.27	yes
i_autannes_day_42	SP+LG	381	350	0.92	0.32	yes
v_abstract_day_17	SIFT	300	287	0.96	0.17	yes
v_abstract_day_17	SP+LG	441	440	1.00	0.16	yes
v_abstract_day_23	SIFT	266	254	0.95	0.21	yes
v_abstract_day_23	SP+LG	427	362	0.85	0.16	yes
v_abstract_day_42	SIFT	265	250	0.94	0.09	yes
v_abstract_day_42	SP+LG	432	432	1.00	0.18	yes
v_adam_day_17	SIFT	306	300	0.98	0.07	yes
v_adam_day_17	SP+LG	438	414	0.95	0.13	yes
v_adam_day_23	SIFT	281	278	0.99	0.18	yes
v_adam_day_23	SP+LG	439	426	0.97	0.08	yes
v_adam_day_42	SIFT	233	224	0.96	0.18	yes
v_adam_day_42	SP+LG	425	424	1.00	0.15	yes

E.8 synthetic_night

Table E.8: All oracle-pair results for synthetic_night.

Query	Method	Matches	Inliers	Ratio	Error	Accept
repo_reference_night_17	SIFT	312	298	0.96	0.30	yes
repo_reference_night_17	SP+LG	442	422	0.95	0.64	yes
repo_reference_night_23	SIFT	307	285	0.93	0.52	yes
repo_reference_night_23	SP+LG	445	419	0.94	0.59	yes
repo_reference_night_42	SIFT	329	308	0.94	0.45	yes
repo_reference_night_42	SP+LG	428	421	0.98	0.51	yes
repo_test_night_17	SIFT	103	87	0.84	5.97	yes
repo_test_night_17	SP+LG	315	285	0.90	0.95	yes
repo_test_night_23	SIFT	88	64	0.73	1.00	yes
repo_test_night_23	SP+LG	337	320	0.95	0.82	yes
repo_test_night_42	SIFT	131	114	0.87	0.37	yes
repo_test_night_42	SP+LG	321	294	0.92	0.58	yes
i_ajuntament_night_17	SIFT	186	147	0.79	0.58	yes
i_ajuntament_night_17	SP+LG	413	359	0.87	0.72	yes
i_ajuntament_night_23	SIFT	198	163	0.82	0.53	yes
i_ajuntament_night_23	SP+LG	411	390	0.95	0.32	yes
i_ajuntament_night_42	SIFT	185	147	0.79	0.83	yes
i_ajuntament_night_42	SP+LG	407	396	0.97	0.30	yes
i_autannes_night_17	SIFT	42	37	0.88	1.00	yes
i_autannes_night_17	SP+LG	319	301	0.94	0.91	yes
i_autannes_night_23	SIFT	49	44	0.90	0.98	yes
i_autannes_night_23	SP+LG	332	316	0.95	0.95	yes
i_autannes_night_42	SIFT	39	32	0.82	0.75	yes
i_autannes_night_42	SP+LG	324	277	0.85	1.12	yes
v_abstract_night_17	SIFT	219	210	0.96	0.11	yes
v_abstract_night_17	SP+LG	397	395	0.99	0.43	yes
v_abstract_night_23	SIFT	227	207	0.91	0.17	yes
v_abstract_night_23	SP+LG	393	390	0.99	0.14	yes
v_abstract_night_42	SIFT	204	190	0.93	0.19	yes
v_abstract_night_42	SP+LG	402	401	1.00	0.26	yes
v_adam_night_17	SIFT	32	19	0.59	0.63	yes
v_adam_night_17	SP+LG	392	388	0.99	0.21	yes
v_adam_night_23	SIFT	29	17	0.59	4.90	yes
v_adam_night_23	SP+LG	390	389	1.00	0.29	yes
v_adam_night_42	SIFT	22	11	0.50	0.59	no
v_adam_night_42	SP+LG	380	378	0.99	0.20	yes

E.9 synthetic_night_severe

Table E.9: All oracle-pair results for synthetic_night_severe.

Query	Method	Matches	Inliers	Ratio	Error	Accept
repo_reference_night_severe_17	SIFT	205	176	0.86	0.33	yes
repo_reference_night_severe_17	SP+LG	392	390	0.99	0.84	yes
repo_reference_night_severe_23	SIFT	188	170	0.90	0.91	yes
repo_reference_night_severe_23	SP+LG	393	375	0.95	0.44	yes

Query	Method	Matches	Inliers	Ratio	Error	Accept
repo_reference_night_severe_42	SIFT	180	163	0.91	0.26	yes
repo_reference_night_severe_42	SP+LG	388	385	0.99	0.52	yes
repo_test_night_severe_17	SIFT	21	12	0.57	–	no
repo_test_night_severe_17	SP+LG	117	111	0.95	2.68	yes
repo_test_night_severe_23	SIFT	23	10	0.43	–	no
repo_test_night_severe_23	SP+LG	123	104	0.85	3.35	yes
repo_test_night_severe_42	SIFT	20	10	0.50	–	no
repo_test_night_severe_42	SP+LG	135	117	0.87	1.19	yes
i_ajuntament_night_severe_17	SIFT	0	0	0.00	–	no
i_ajuntament_night_severe_17	SP+LG	301	279	0.93	0.77	yes
i_ajuntament_night_severe_23	SIFT	38	0	0.00	–	no
i_ajuntament_night_severe_23	SP+LG	309	295	0.95	1.09	yes
i_ajuntament_night_severe_42	SIFT	5	0	0.00	–	no
i_ajuntament_night_severe_42	SP+LG	305	263	0.86	1.42	yes
i_autannes_night_severe_17	SIFT	19	11	0.58	–	no
i_autannes_night_severe_17	SP+LG	170	140	0.82	1.22	yes
i_autannes_night_severe_23	SIFT	3	0	0.00	–	no
i_autannes_night_severe_23	SP+LG	179	118	0.66	2.17	yes
i_autannes_night_severe_42	SIFT	9	5	0.56	83.14	no
i_autannes_night_severe_42	SP+LG	164	145	0.88	1.43	yes
v_abstract_night_severe_17	SIFT	92	75	0.82	0.71	yes
v_abstract_night_severe_17	SP+LG	277	256	0.92	0.48	yes
v_abstract_night_severe_23	SIFT	104	64	0.62	0.87	yes
v_abstract_night_severe_23	SP+LG	268	236	0.88	0.67	yes
v_abstract_night_severe_42	SIFT	94	74	0.79	0.35	yes
v_abstract_night_severe_42	SP+LG	262	224	0.85	0.76	yes
v_adam_night_severe_17	SIFT	0	0	0.00	–	no
v_adam_night_severe_17	SP+LG	184	132	0.72	2.01	yes
v_adam_night_severe_23	SIFT	0	0	0.00	–	no
v_adam_night_severe_23	SP+LG	180	152	0.84	0.84	yes
v_adam_night_severe_42	SIFT	0	0	0.00	–	no
v_adam_night_severe_42	SP+LG	164	137	0.84	3.17	yes

E.10 synthetic_warm_night

Table E.10: All oracle-pair results for synthetic_warm_night.

Query	Method	Matches	Inliers	Ratio	Error	Accept
repo_reference_warm_night_17	SIFT	304	289	0.95	0.33	yes
repo_reference_warm_night_17	SP+LG	441	438	0.99	0.57	yes
repo_reference_warm_night_23	SIFT	314	290	0.92	0.46	yes
repo_reference_warm_night_23	SP+LG	439	437	1.00	0.38	yes
repo_reference_warm_night_42	SIFT	318	295	0.93	0.45	yes
repo_reference_warm_night_42	SP+LG	425	419	0.99	0.34	yes
repo_test_warm_night_17	SIFT	77	72	0.94	0.79	yes
repo_test_warm_night_17	SP+LG	297	289	0.97	0.35	yes
repo_test_warm_night_23	SIFT	70	54	0.77	0.47	yes
repo_test_warm_night_23	SP+LG	326	260	0.80	1.57	yes
repo_test_warm_night_42	SIFT	103	93	0.90	0.58	yes
repo_test_warm_night_42	SP+LG	309	300	0.97	0.42	yes
i_ajuntament_warm_night_17	SIFT	144	117	0.81	0.67	yes
i_ajuntament_warm_night_17	SP+LG	398	384	0.96	0.21	yes

Query	Method	Matches	Inliers	Ratio	Error	Accept
i_ajuntament_warm_night_23	SIFT	154	129	0.84	0.67	yes
i_ajuntament_warm_night_23	SP+LG	408	397	0.97	0.37	yes
i_ajuntament_warm_night_42	SIFT	147	109	0.74	0.88	yes
i_ajuntament_warm_night_42	SP+LG	410	404	0.99	0.16	yes
i_autannes_warm_night_17	SIFT	31	23	0.74	1.20	yes
i_autannes_warm_night_17	SP+LG	314	308	0.98	0.32	yes
i_autannes_warm_night_23	SIFT	37	31	0.84	0.97	yes
i_autannes_warm_night_23	SP+LG	327	293	0.90	0.35	yes
i_autannes_warm_night_42	SIFT	30	24	0.80	0.99	yes
i_autannes_warm_night_42	SP+LG	312	305	0.98	0.75	yes
v_abstract_warm_night_17	SIFT	209	201	0.96	0.11	yes
v_abstract_warm_night_17	SP+LG	392	375	0.96	0.44	yes
v_abstract_warm_night_23	SIFT	226	208	0.92	0.11	yes
v_abstract_warm_night_23	SP+LG	386	385	1.00	0.18	yes
v_abstract_warm_night_42	SIFT	190	177	0.93	0.15	yes
v_abstract_warm_night_42	SP+LG	396	395	1.00	0.22	yes
v_adam_warm_night_17	SIFT	32	13	0.41	2.08	yes
v_adam_warm_night_17	SP+LG	391	341	0.87	0.45	yes
v_adam_warm_night_23	SIFT	45	26	0.58	–	no
v_adam_warm_night_23	SP+LG	384	369	0.96	0.34	yes
v_adam_warm_night_42	SIFT	37	12	0.32	–	no
v_adam_warm_night_42	SP+LG	375	375	1.00	0.19	yes

E.11 synthetic_fog

Table E.11: All oracle-pair results for synthetic_fog.

Query	Method	Matches	Inliers	Ratio	Error	Accept
repo_reference_fog_17	SIFT	256	234	0.91	0.41	yes
repo_reference_fog_17	SP+LG	454	402	0.89	0.79	yes
repo_reference_fog_23	SIFT	263	248	0.94	0.27	yes
repo_reference_fog_23	SP+LG	445	437	0.98	0.18	yes
repo_reference_fog_42	SIFT	258	238	0.92	0.81	yes
repo_reference_fog_42	SP+LG	441	435	0.99	0.56	yes
repo_test_fog_17	SIFT	88	65	0.74	0.23	yes
repo_test_fog_17	SP+LG	362	361	1.00	0.39	yes
repo_test_fog_23	SIFT	66	51	0.77	0.27	yes
repo_test_fog_23	SP+LG	386	363	0.94	0.73	yes
repo_test_fog_42	SIFT	89	73	0.82	0.18	yes
repo_test_fog_42	SP+LG	355	287	0.81	1.26	yes
i_ajuntament_fog_17	SIFT	107	87	0.81	0.90	yes
i_ajuntament_fog_17	SP+LG	421	402	0.95	0.29	yes
i_ajuntament_fog_23	SIFT	117	99	0.85	0.49	yes
i_ajuntament_fog_23	SP+LG	426	419	0.98	0.25	yes
i_ajuntament_fog_42	SIFT	97	79	0.81	0.65	yes
i_ajuntament_fog_42	SP+LG	421	372	0.88	0.51	yes
i_autannes_fog_17	SIFT	49	39	0.80	1.64	yes
i_autannes_fog_17	SP+LG	364	360	0.99	0.29	yes
i_autannes_fog_23	SIFT	47	38	0.81	4.74	yes
i_autannes_fog_23	SP+LG	355	338	0.95	0.71	yes
i_autannes_fog_42	SIFT	51	38	0.75	4.90	yes
i_autannes_fog_42	SP+LG	347	341	0.98	0.33	yes

Query	Method	Matches	Inliers	Ratio	Error	Accept
v_abstract_fog_17	SIFT	190	168	0.88	0.27	yes
v_abstract_fog_17	SP+LG	404	404	1.00	0.24	yes
v_abstract_fog_23	SIFT	185	166	0.90	0.28	yes
v_abstract_fog_23	SP+LG	411	408	0.99	0.13	yes
v_abstract_fog_42	SIFT	171	157	0.92	0.26	yes
v_abstract_fog_42	SP+LG	408	408	1.00	0.17	yes
v_adam_fog_17	SIFT	47	35	0.74	0.40	yes
v_adam_fog_17	SP+LG	415	413	1.00	0.31	yes
v_adam_fog_23	SIFT	34	33	0.97	0.41	yes
v_adam_fog_23	SP+LG	411	410	1.00	0.25	yes
v_adam_fog_42	SIFT	27	23	0.85	0.38	yes
v_adam_fog_42	SP+LG	395	395	1.00	0.16	yes